



Universidade do Minho
Escola de Ciências

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Proof Search in Natural Deduction with Partial Proof Terms



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**Proof Search in Natural Deduction
with Partial Proof Terms**

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Work developed under the supervision of

Professor José Carlos Soares do Espírito Santo

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*"I wish that Heaven had visiting hours
So I could just swing by and ask your advice.*

...

*Can we just talk a while until my worries disappear?
I'd tell you that I'm scared of turnin' out a failure
You'd say: "Remember that the answer's in the love we create".*

...

*And I would ask them if I could take you home
But I know what they'd say, that it's for the best
So I will live life the way you taught me.*

...

And everyone I love will know exactly who you are."

À minha mãe, Olinda.

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Statement of Integrity

I hereby declare having conducted this academic work with integrity and I confirm that I have not used plagiarism or any form of undue use of information or falsification of results along the process leading to its elaboration.

I further declare that I have fully acknowledged the Code of Ethical Conduct of the University of Minho.

Resumo

O objetivo desta tese é melhorar a modelação do raciocínio matemático através da dedução natural, seguindo três princípios: o raciocínio é a procura de provas; a procura de provas obtém provas em dedução natural; e a procura de provas é feita diretamente em dedução natural.

Para atingir este objetivo, propomos uma extensão do Isomorfismo de Curry-Howard através do conceito de termo de prova parcial. Os termos de prova parciais expressam lacunas em derivações incompletas com a ajuda de sequentes formais, isto é, sequentes que ocorrem como componentes próprios da sintaxe dos termos de prova. Especificamos os procedimentos de procura de provas como sistemas de reescrita que atuam sobre termos de prova parciais e estendemos os sistemas de prova convencionais para acomodar estes termos.

A metodologia é ilustrada através de estudos de caso nas lógicas intuicionista e clássica proposicionais, com os cálculos de sequentes LJT e LKT , cujo procedimento de procura de provas é a focagem, e dois sistemas de dedução natural bidirecional, NJT e NKT , cujo procedimento de procura de provas segue as ideias do Cálculo de Intercalação de Sieg (IC), mas é focado num sentido específico da dedução natural. Demonstramos a equivalência entre a focagem no cálculo de sequentes e a intercalação focada na dedução natural, para as lógicas consideradas. De seguida, generalizamos o nosso sistema de procura de provas para NJT de forma a capturar o IC sem restrições.

Esta tese contribui para a área da Procura de Provas Idealizada, um meio-termo entre a Teoria da Demonstração em Lógica e a Demonstração Interativa de Teoremas em Ciências da Computação. Neste contexto, as considerações de controlo, efetividade ou implementação são postas de lado, enquanto o estudo teórico das provas parciais (i.e., incompletas) e a sua conversão em provas completas é central.

Palavras-chave Dedução Natural, Focagem, Intercalação, Procura de Provas, Termo de Prova Parcial.

Abstract

The goal of this thesis is to improve the modeling of mathematical reasoning through natural deduction, following three principles: reasoning is proof search; proof search obtains natural deduction proofs; and proof search is made directly in natural deduction.

To achieve this goal, we propose an extension of the Curry-Howard Isomorphism through the concept of partial proof terms, which are proof terms expressing gaps in incomplete derivations with the help of formal sequents, that is, sequents occurring as proper components of the syntax of proof terms. We specify proof search procedures as rewriting systems acting on partial proof terms, and extend conventional proof systems to accommodate these terms.

The methodology is illustrated through case studies in propositional intuitionistic and classical logics, with the sequent calculi LJT and LKT , whose proof search procedure is focusing, and two bidirectional natural deduction systems, NJT and NKT , whose proof search procedure follows the ideas of the Sieg's Intercalation Calculus (IC), but is focused in a sense specific to natural deduction. We prove the equivalence between focusing in sequent calculus and focused intercalation in natural deduction, for both intuitionistic and classical logics. We then generalize our proof search system for NJT in order to capture IC without restrictions.

This thesis contributes to Idealized Proof Search, a middle ground between Proof Theory in Logic and Interactive Theorem Proving in Computer Science. In this context, considerations of control, effectiveness, or implementation are set aside, while the theoretical study of partial (i.e., incomplete) proofs and their conversion into finished proofs is central.

Keywords Focusing, Intercalation, Natural Deduction, Partial Proof Term, Proof Search.

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List of Abbreviations

LJT	Focused Sequent Calculus for Intuitionistic Logic
LJT_{∂}	Partial Focused Sequent Calculus for Intuitionistic Logic
NJT	Focused Natural Deduction for Intuitionistic Logic
NJT_{∂}	Partial Focused Natural Deduction for Intuitionistic Logic
LKT	Focused Sequent Calculus for Classical Logic
LKT_{∂}	Partial Focused Sequent Calculus for Classical Logic
NKT	Focused Natural Deduction for Classical Logic
NKT_{∂}	Partial Focused Natural Deduction for Classical Logic
IC	Sieg's Intercalation Calculus
$\mathbb{NJT}_{\partial}$	Intercalation Calculus with Partial Proof Terms for Intuitionistic Logic
IH	Induction Hypothesis

Chapter 1

Introduction

1.1 Objectives

In structural proof theory, natural deduction is regarded as the proof system that more closely formalizes the reasoning found in mathematical proofs [32], but developments in proof search and theorem proving are rarely based on this deductive system. The goal of this thesis is to improve the modeling of mathematical reasoning through natural deduction, following three principles: (1) reasoning is proof search; (2) proof search obtains natural deduction proofs; and (3) proof search is made directly in natural deduction.

To achieve this, an extension of the Curry-Howard Isomorphism [57] is proposed through the concept of partial proof terms, which are proof terms expressing gaps in incomplete derivations with the help of formal sequents, that is, sequents occurring as proper components of the syntax of proof terms. These formal proof objects encode both what has already been proved and, via formal sequents, what remains to be proved. Proof search procedures are formalized as rewriting systems operating on partial proof terms, with the search space represented by reduction graphs. Consequently, a proof system with proof terms has to be extended to accommodate partial proof terms and incomplete proofs, resulting in a calculus of partial derivations.

Extending the language of proof terms to partial proof terms generalizes a proof system $\mathcal{S}$ to a system $\mathcal{S}_\partial$. The extended system comprises: (1) a rewriting system that formalizes a proof search procedure; and (2) a typing system for partial proof terms, which is a logical system handling partial sequents such as $\Xi \vdash t : \sigma$, where σ is a sequent of $\mathcal{S}$, t is a partial proof term, and Ξ is the list of formal sequents present in t . These partial sequents implement the fundamental concept of proof state, which includes a goal sequent, a record of the search history, and a list of proof obligations. The rewriting and typing systems are linked by a subject reduction theorem [57], which asserts that the goal sequent is preserved as the partial proof term reduces. The ultimate objective is to establish that proof search is characterized by the rewriting system: there is a proof in $\mathcal{S}$ of σ represented by t if and only if $\underline{\sigma} \rightarrow t$, where $\underline{\sigma}$ denotes

the formal version of σ .

This work contributes to the development of Idealized Proof Search, a research area that occupies a middle ground between Proof Theory in Logic and Theorem Proving in Computer Science. The objective in this area is to formalize and compare concrete proof search objects and procedures at a conceptual and theoretical level, disregarding efficiency and implementation concerns. Idealized Proof Search aims to complement Proof Theory with an approach more centered in reasoning, while Interactive Theorem Proving is a source of motivation and a possible field of application.

1.2 Background

Proof Theory and Proof Search in Sequent Calculi

Structural proof theory [47] began with Gentzen [31], who, in 1935, introduced two logical systems: Sequent Calculus and Natural Deduction. These systems based on inference rules represented a significant shift from the previously dominant axiomatic approaches. Natural Deduction was designed as a “formalism that reflects as accurately as possible the actual logical reasoning involved in mathematical proofs”. In contrast, Sequent Calculus was developed, apparently due to difficulties in proving normalization for classical logic [50]. However, this reason was never given for the creation of the system – Gentzen justifies the creation of Sequent Calculus as an attempt to formulate a deductive system in which derivations do not contain assumption formulas, as in Natural Deduction [32]. This explicit structure facilitates the management of assumptions, particularly through sequents, and structural rules such as cut, weakening, and contraction. The explicitness of the Sequent Calculus clarifies the symmetry between assumptions and conclusions, a feature that is less evident in Natural Deduction. This quality makes Sequent Calculus particularly suitable for meta-theoretical analysis, such as cut elimination, and, with particular relevance to the present study, proof search.

Consequently, for several decades, the study of proof search, driven by the demands of implementation and automation, primarily relied on sequent calculi [30]. Within proof theory, in [47] we can find some sequent calculi specifically developed for bottom-up proof search, such as the G3 systems. These calculi consist solely of invertible inference rules, allowing the search for derivations to be purely mechanical. Further improvements to these systems are described in [11], which addresses the issue of non-terminating root-first proof search in G3/LJ caused by loops generated by the left-implication rule.

In the context of classical linear logic, Andreoli [2] introduced focusing, a technique that structures proof search in sequent calculi by dividing proofs into phases of invertible and non-invertible rules. This

method reduces nondeterministic choices in sequent calculi. The core idea is to alternate between an asynchronous phase, in which all invertible rules are applied, and a synchronous phase, in which a formula is selected, and all its non-invertible rules are applied under “focus”. Herbelin [36, 35, 7] established the connection between focusing and computation by extending the Curry-Howard isomorphism to the sequent calculus through the $\bar{\lambda}$ -calculus and the permutation-free LJT calculi for intuitionistic logic. Dyckhoff, Pinto, and Lengrand [11, 13, 15, 14, 10, 12] further advanced the integration of focusing into intuitionistic sequent calculi and proof search by analyzing rule permutability and search strategies, aligning them with the focusing discipline and eliminating redundant permutations. Liang and Miller [37] transformed focusing from a linear-logic technique into a methodology applicable to multiple logics, through their polarized, focused calculi LJF , LKF , and LKU , which encompass several other focusing proof systems.

In contrast to these proof-theoretic developments, on the side of theorem provers, almost all fundamental work in this area was based on Resolution [29], a method proposed by J. Robinson in the 1960s that became quite popular in automated theorem proving. This method proves a formula's validity by establishing that its negation is unsatisfiable and consists of a single rule (the resolution rule) [59, 30]. But this method is not unique: Semantic Tableaux was also proposed by Smullyan in the 1960s [29].

However, proof search strategies based on Sequent Calculus are primarily designed to achieve completeness and efficiency, rather than to emulate human mathematical reasoning. Consequently, these strategies diverge from Gentzen's original intention to develop a system that mirrors natural reasoning processes [32].

Proof Search in Natural Deduction

Sieg and Derakhshan [56] argue that a human-centered proof search “aims to capture structures of ordinary mathematical arguments and to discover efficient strategies that are used (implicitly) in their construction”. Furthermore, they assert that the resulting proofs “should be indistinguishable from good mathematical writing”. Sieg suggests that a human-centered approach is best achieved by grounding reasoning in Natural Deduction, thereby renewing the need for a theory of reasoning grounded in the latter proof system.

Natural Deduction experienced renewed interest through the work of Prawitz [51] and the rediscovery of normalization. However, it was long regarded as unsuitable for proof search [29]. This perception stemmed from its Modus Ponens rule and the purported absence of the subformula property, which made Natural Deduction appear incompatible with backward search – a method considered essential, particularly for automated proof search [56].

Sieg addressed these limitations by introducing his “technically convenient and heuristically motivated” Intercalation Calculus [54, 55]. The core concept of this calculus is, during proof search, to progressively close the gap between assumptions and the conclusion (or goal formula) by systematically applying elimination rules from above and inverted introduction rules from below. Formulas are intercalated between the assumptions and the conclusion, ultimately resulting in a normal proof if the goal formula is a theorem, or a counter-example otherwise [55]. However, this calculus has been implemented only in educational contexts through Sieg’s AProS project [53] and has not influenced theorem provers. Progress in interactive theorem proving rarely builds upon theoretical developments that extend the theory of natural deduction [34].

Nevertheless, further research has addressed proof search in natural deduction. Ferrari and Fiorentini [28, 27] developed two natural deduction calculi optimized for proof search in classical and intuitionistic propositional logics: *Nbu* for intuitionistic logic and *Ncr* for classical logic. These calculi enable goal-oriented proof search that avoids backtracking and loop checking, remain equivalent to standard natural deduction, and achieve performances competitive with sequent and tableaux-based provers. Miller [38] designed a Proof Search Framework (PSF) to support proof search across a variety of logics, capturing dynamic aspects of inference that standard natural deduction does not. Mints [40] introduced the ADC Method for intuitionistic propositional natural deduction, which describes proof search in two distinct phases. In the Analysing (A) phase, all possible introduction rules are applied bottom-up to decompose the goal. In the Direct Chaining (DC) phase, the context Γ is saturated top-down with the conclusions of all applicable elimination rules. However, this method is not complete, but it was used to study the complexity of proof search across various fragments of natural deduction.

The Curry-Howard Isomorphism and its Extensions to Proof Search

Concurrently, research has explored interpreting Natural Deduction as a language of “programs” [36]. In this context, as formalized by the Curry-Howard Isomorphism, Natural Deduction provides the proof side of the correspondence, while the simply typed λ -calculus [6] and its extensions, with their language of proof terms, represent the program side. This perspective contributed to the development of functional programming, and the creation of languages that integrate proofs and programs.

On the one hand, the area of theorem proving has inspired the development of type theories with meta-variables and explicit substitutions [46], which are indeed extensions of the Curry-Howard paradigm useful for the study of proof search. Muñoz [43, 44, 45] introduced a λ -calculus featuring explicit substitutions and typed meta-variables for dependent type systems, including the Calculus of Constructions. In this

framework, incomplete proofs are represented as proof terms with meta-variables, which serve as typed symbols within the context, and substitutions for both variables and meta-variables are made explicit in the term language, allowing the modeling of proof reuse and the delay of choices in unification and proof search. Geuvers and Jojgov [33] subsequently developed the concepts of open derivations and open proof terms, which are derivations in natural deduction that may include a generic “claim” inference. These were developed for higher-order logic as formal tools to model delayed choices in proof search or the reuse of proofs, although they do not provide an explicit formalization of a search process. Espirito Santo, Matthes, and Pinto [21, 23, 20] introduced “coinductive proof search” by using proof terms with meta-variables X^σ , with X denoting a fixed-point variable typed with a sequent σ , enabling the representation of the entire search space. This approach does not yet provide a method for expressing proof-search procedures.

On the other hand, two main rewriting processes are related to proof systems: normalization (or cut elimination) and proof search. Languages of proof terms (variations and extensions of typed λ -terms) provide a well-known representation of proofs, which is particularly useful and used, even mandatory, in the expression of the computational interpretation of normalization, and its link with functional programming. In contrast, proof search, with works like Uniform Proofs, which interpret the logical connectives as “simple and fixed search instructions” [39], offers a foundation to logic programming [39], but proof terms are comparatively less employed in the specification and study of proof search procedures. In fact, no such use is found: in studies of proof search which consistently employ proof systems with proof terms [10, 13, 15]; in studies of inhabitation problems (search for λ -terms) in type theory [57, 52, 58, 1, 4]; in works like [42], even though the author makes great use of proof terms elsewhere [41].

1.3 Outline

This thesis is divided into 6 chapters and organized as follows:

Chapter 2: Partial Proofs and Partial Proof Terms

In Chapter 2, we establish the foundation for the remainder of this thesis. We begin with the focused sequent calculus LJT [35] for Intuitionistic Propositional Logic. Proof search in this system follows the focusing discipline [37], characterized by alternating inversion and focusing phases and by minimizing nondeterminism. This system serves as the basis for introducing the main tool of this work, partial proof terms, and related concepts, such as formal sequents. A deductive system, LJT_∂ , is defined to accommodate partial derivations. Finally, a rewriting system on partial proof terms is introduced and proved to

specify proof search in LJT according to the focusing discipline.

Chapter 3: A Focused Strategy in Natural Deduction

In Chapter 3, we apply the methodology of partial proof terms to study proof search in natural deduction for Intuitionistic Propositional Logic, within NJT , a bidirectional natural deduction system inspired by bidirectional typing [5, 9, 13] and developed in [19, 16]. Proof search in NJT refines concepts initially proposed for the Intercalation Calculus [54, 55], which itself constitutes a contribution. A deductive system, NJT_{∂} , is introduced to represent partial derivations, and a rewriting system on partial proof terms is defined to specify proof search in NJT . After formalizing the proof search procedures, a central theorem is established: proof search in the natural deduction system NJT following intercalation is isomorphic to proof search in the sequent calculus LJT following the focusing discipline.

Chapter 4: Focused Proof Search in Classical Logic

In Chapter 4, we extend the results of the previous chapters to classical logic, using the same methodology. The focused sequent calculus LKT , originating in linear logic [8] but not previously developed as a sequent calculus with proof terms, and the bidirectional natural deduction system NKT with alternative conclusions, following the tradition of Parigot's classical natural deduction and the corresponding $\lambda\mu$ -calculus [49], are considered. We provide precise definitions of the systems LKT and NKT and their associated search procedures, as well as the deductive systems for representing partial derivations. The methodology of partial proof terms is shown to provide an adequate representation of the two search procedures. Using this representation, we prove that the correspondence of the proof search procedures observed in the intuitionistic case extends to the classical case.

Chapter 5: An Intercalation Calculus with Partial Proof Terms

In Chapter 5, we revisit the original formulation Sieg's Intercalation Calculus [55]. In this search system, the bottom-up application of introduction inference rules from conclusion and the top-down application of elimination inference rules from the hypotheses are mixed in arbitrary order to yield normal derivations in ordinary natural deduction. We adapt the system to yield η -expanded derivations. We then confirm that partial proof terms can adequately represent this search procedure, and we obtain a formalization that is a generalization of NJT_{∂} , with the search procedure producing total proof terms and the corresponding derivations in NJT .

Chapter 6: Conclusions and Future Work

In Chapter 6, we summarize our results and provide a detailed comparison with the related work previously mentioned in the introduction. Some directions for future research, arising from the results and methodologies of this thesis, are also presented.

Chapter 2

Partial Proofs and Partial Proof Terms

In this chapter, we show that the Curry-Howard paradigm of representation of proofs can be extended in a new way to facilitate the theoretical study of idealized proof search. Our starting point is the paradigmatic situation of bottom-up proof search in the sequent calculus. In this process we deal with incomplete derivations where some leaves may be sequents we want to prove, but whose proof is still totally unspecified. Such sequents are “holes” in the incomplete derivation. We propose that, within a proof term representation, such holes are represented by the sequents themselves. In this approach, sequents are treated as first-class components within proof terms, and we refer to these occurrences as formal sequents. We immediately see the usefulness of such a step: in the sequent calculus, formal sequents can serve as redexes in rewrite rules that formalize the steps of the proof search process. Consequently, proof search procedures can be specified as rewriting systems acting on partial proof terms, and logical systems can be extended to accommodate partial derivations, where sequents express proof states, including the proof goal, a record of the search history, and outstanding proof obligations.

The results discussed in this chapter were partially presented in the paper [25].

2.1 Sequent Calculus LJT

To introduce the methodology of partial proof terms, we start by recalling the sequent calculus LJT [35, 13, 15] for implicational intuitionistic logic and use it as our simple first case study.

We start with the definition of formulas.

Definition 2.1 (Formulas). Let $\mathcal{V}_P$ be the countable set of propositional variables (also called atoms), denoted by p, q , etc., and $\mathcal{A}_P = \mathcal{V}_P \cup \{\supset\} \cup \{(\cdot, \cdot)\}$. The set of all words in $\mathcal{A}_P$ is represented by $\mathcal{A}_P^*$. The set of formulas, represented by $\mathcal{F}_P$, is defined inductively on $\mathcal{A}_P^*$ by:

$$A, B ::= p \mid (A \supset B)$$

By the Curry-Howard correspondence, formulas can be seen as simple types. By convention, the outermost brackets of each formula can be omitted.

LJT handles two kinds of sequents.

Definition 2.2 (Usual Sequent of LJT). A usual sequent of LJT is an expression of the form

$$\Gamma \vdash A ,$$

ranged over by σ , where Γ is a finite set, possibly empty, of pairs $(x : B)$ and A is a formula. We can think of Γ as a multiset of formulas, with no variable occurring in two distinct pairs. Γ is called the context, antecedent, or left-hand side (LHS) of the sequent, while A is the conclusion, the consequent, or the right-hand side (RHS) of the sequent.

Definition 2.3 (Focused Sequent of LJT). A focused sequent of LJT is an expression of the form

$$\Gamma; A \vdash p ,$$

ranged over by τ , where Γ is a finite set, possibly empty, of pairs $(x : B)$, A is a formula, and p is a propositional variable. To the distinguished formula A in the left-hand side of the sequent we call the **focus** of the sequent.

Definition 2.4 (Inference Rules of LJT). The set of inference rules of LJT is defined as:

$$\frac{}{\Gamma; p \vdash p} AX \quad \frac{\Gamma, x : A; A \vdash p}{\Gamma, x : A \vdash p} CTR \quad \frac{\Gamma, x : A \vdash B}{\Gamma \vdash A \supset B} RI \quad \frac{\Gamma \vdash A \quad \Gamma; B \vdash p}{\Gamma; A \supset B \vdash p} LI$$

These rules are the axiom, AX , the contraction rule, CTR , the right introduction rule for implication, RI , and the left introduction rule for implication, LI ; and together they determine how the sequents are derived in LJT .

In a derivation of LJT , the formula in the focus of a sequent is either the atom p resulting from an application of the AX , or the implication introduced by LI . This naturally applies to the focused formula B in the right premise of LI , which places constraints on how this right premise is derived and affects the permutability of inferences. Indeed, derivations in LJT are not only cut-free, but also permutation-free [13].

Example 2.1. Let $(p \supset p) \supset p \supset p$ be the type of Church numerals. The derivation of the Church numeral $\bar{1}$ in LJT , in the empty context, is the following:

$$\frac{\frac{\frac{\frac{\frac{}{f : p \supset p, x : p; p \vdash p} AX}{f : p \supset p, x : p \vdash p} CTR}{f : p \supset p, x : p; p \supset p \vdash p} CTR}{\frac{f : p \supset p, x : p \vdash p}{f : p \supset p \vdash p \supset p} RI} RI}{\vdash (p \supset p) \supset p \supset p} RI$$

2.2 LJT with Proof Terms

We now extend the Sequent Calculus LJT with proof terms.

Definition 2.5 (Proof Terms of LJT). Let $\mathcal{V}_\lambda$ be the countable set of term variables, denoted by x , y , etc., and $\mathcal{A}_\lambda = \mathcal{V}_\lambda \cup \{\lambda, :, \hat{\cdot}, \star, ::\} \cup \{(\cdot, \cdot)\} \cup \mathcal{F}_P$. The set of all words in $\mathcal{A}_\lambda$ is represented by $\mathcal{A}_\lambda^*$. The set of proof terms of LJT , denoted by LJT_λ , is inductively defined on $\mathcal{A}_\lambda^*$ by:

$$\begin{aligned} \text{(Proof terms)} \quad t, u &::= (\lambda(x : A) t) \mid (x \hat{\cdot} l) \\ \text{(Proof lists)} \quad l &::= \star \mid (u :: l). \end{aligned}$$

By convention, the outermost brackets of each term can be omitted, and the proof term $(\lambda(x : A) t)$ abbreviated to $\lambda x^A t$. Next, we omit in this term the outermost brackets of t and add a “.” as follows: $\lambda x^A.t$. We presented here the formal definition of the set of proof terms of LJT with all the details, but in the remainder of this work, we will define sets of proof terms solely by their grammar.

LJT with proof terms handles two kinds of sequents. For reasons to become clear soon, these sequents are called total sequents.

Definition 2.6 (Term Sequent of LJT with Proof Terms). A term sequent of LJT with proof terms is an expression of the form

$$\Gamma \vdash t : A ,$$

where Γ is a finite set, possibly empty, of pairs $(x : B)$, t is a proof term, and A is a formula. Here, we can think of Γ as a typing context.

Definition 2.7 (List Sequent of LJT with Proof Terms). A list sequent of LJT with proof terms is an expression of the form

$$\Gamma; l : A \vdash p ,$$

where Γ is a finite set, possibly empty, of pairs $(x : B)$, l is a proof list, A is a formula, and p is an atom. The formula A is the focus of the sequent.

Definition 2.8 (Inference Rules of LJT with Proof Terms). The set of inference rules of LJT with proof terms is defined as:

$$\begin{array}{c} \frac{}{\Gamma; \star : p \vdash p} AX \quad \frac{\Gamma, x : A; l : A \vdash p}{\Gamma, x : A \vdash x \hat{l} : p} CTR \\[10pt] \frac{\Gamma, x : A \vdash t : B}{\Gamma \vdash \lambda x^A. t : A \supset B} RI \quad \frac{\Gamma \vdash u : A \quad \Gamma; l : B \vdash p}{\Gamma; u :: l : A \supset B \vdash p} LI \end{array}$$

If we read formulas as simple types, these rules can be read as rules for assigning types to proof terms. In this thesis, we will not insist on this reading. Nevertheless, each proof term constructor witnesses the use of a certain inference rule. This means that, if $\Gamma \vdash t : A$ is derivable, then t encodes a derivation of $\Gamma \vdash A$.

Example 2.2. Adding proof terms to the derivation of Example 2.1, we have the following derivation in LJT with proof terms:

$$\begin{array}{c} \frac{\frac{\frac{}{f : p \supset p, x : p; \star : p \vdash p} AX}{f : p \supset p, x : p \vdash x \hat{\star} : p} CTR}{f : p \supset p, x : p; (x \hat{\star}) :: \star : p \supset p \vdash p} CTR \quad \frac{\frac{\frac{}{f : p \supset p, x : p; \star : p \vdash p} AX}{f : p \supset p, x : p \vdash x \hat{\star} : p} CTR}{f : p \supset p, x : p \vdash f \hat{((x \hat{\star}) :: \star)} : p} RI \\[10pt] \frac{f : p \supset p \vdash \lambda x^p. f \hat{((x \hat{\star}) :: \star)} : p \supset p}{\vdash \lambda f^{(p \supset p)}. \lambda x^p. f \hat{((x \hat{\star}) :: \star)} : (p \supset p) \supset p \supset p} RI \end{array}$$

Thus, the proof term $\lambda f^{(p \supset p)}. \lambda x^p. f \hat{((x \hat{\star}) :: \star)}$, which represents the Church numeral $\bar{1}$, encodes the derivation of the sequent $\vdash (p \supset p) \supset p \supset p$.

Weakening rules are admissible in LJT with proof terms.

Proposition 2.1 (Admissibility of Weakening in LJT). The weakening rules

$$\frac{\Gamma \vdash t : B}{\Gamma, x : A \vdash t : B} LW \quad \frac{\Gamma; l : B \vdash p}{\Gamma, x : A; l : B \vdash p} LW_f$$

are admissible in LJT with proof terms.

Proof. The admissibility of LW and LW_f is proved by a simultaneous induction on the derivations of $\Gamma \vdash t : B$ and $\Gamma; l : B \vdash p$.

- Case AX . Suppose $\Gamma; \star : p \vdash p$. Applying the AX rule, $\Gamma, x : A; \star : p \vdash p$.
- Case CTR . Suppose $\Gamma, y : C \vdash y\hat{l} : p$ is obtained from $\Gamma, y : C; l : C \vdash p$. By IH, $\Gamma, y : C, x : A; l : C \vdash p$. Applying the CTR rule, $\Gamma, y : C, x : A \vdash y\hat{l} : p$.
- Case RI . Suppose $\Gamma \vdash \lambda y^C.M : C \supset D$ is obtained from $\Gamma, y : C \vdash M : D$. By IH, $\Gamma, y : C, x : A \vdash M : D$. Applying the RI rule, $\Gamma, x : A \vdash \lambda y^C.M : C \supset D$.
- Case LI . Suppose $\Gamma; u :: l : C \supset D \vdash p$ is obtained from $\Gamma \vdash u : C$ and $\Gamma; l : D \vdash p$. By IH, $\Gamma, x : A \vdash u : C$ and $\Gamma, x : A; l : D \vdash p$. Applying the LI rule, $\Gamma, x : A; u :: l : C \supset D \vdash p$.

□

This interplay between inference rules and proof terms becomes clearer when we observe that admissible inference rules are witnessed by admissible proof term operations, as stated in the following theorem.

Theorem 2.1 (Admissible cut rules of LJT). Consider the equations

$$\begin{aligned}
 [u/x](\lambda y^C.t) &= \lambda y^C.[u/x]t & [u/x]\star &= \star \\
 [u/x](x\hat{l}) &= u@[u/x]l & [u/x](t :: l) &= ([u/x]t) :: ([u/x]l) \\
 [u/x](y\hat{l}) &= y\hat{[u/x]l} \quad (y \neq x)
 \end{aligned}$$

$$\begin{aligned}
 (\lambda y^C.t)@(u :: l) &= ([u/y]t)@l \\
 (x\hat{l})@\star &= x\hat{l}
 \end{aligned}$$

In LJT one has:

- (1) For all A, t : if $\Gamma \vdash u : A$ and $\Gamma, x : A \vdash t : B$, then the equations define the proof term $[u/x]t$ and $\Gamma \vdash [u/x]t : B$.
- (2) For all A, l : if $\Gamma \vdash u : A$ and $\Gamma, x : A; l : C \vdash p$, then the equations define the proof list $[u/x]l$ and $\Gamma; [u/x]l : C \vdash p$.
- (3) For all A : if $\Gamma \vdash u : A$ and $\Gamma; l : A \vdash p$, then the equations define the proof term $u@l$ and $\Gamma \vdash u@l : p$.

As a consequence, the following cut rules are admissible in LJT :

$$\frac{\Gamma \vdash u : A \quad \Gamma, x : A \vdash t : B}{\Gamma \vdash [u/x]t : B} \quad (1) \quad \frac{\Gamma \vdash u : A \quad \Gamma, x : A; l : B \vdash p}{\Gamma; [u/x]l : B \vdash p} \quad (2)$$

$$\frac{\Gamma \vdash u : A \quad \Gamma; l : A \vdash p}{\Gamma \vdash u@l : p} \quad (3)$$

Proof. For all A , t and l , the size $|A|$ is defined by recursion on A , as follows: $|p| = 1$, $|A \supset B| = 1 + |A| + |B|$; and the sizes $|t|$ and $|l|$ are defined by simultaneous structural recursion on t and l as follows: $|\lambda x^A.t| = 1 + |t|$, $|x\hat{l}| = 1 + |l|$, $|\star| = 1$, and $|u :: l| = 1 + |u| + |l|$. Items (1), (2) and (3) are proved by simultaneous induction on: $(|A|, |t|)$, $(|A|, |l|)$ and $(|A|, 0)$. The pairs are ordered lexicographically.

In (1),

- Case $t = \lambda y^C.t'$. Suppose $\Gamma \vdash u : A$ and $\Gamma, x : A \vdash t : B$. We show that the above equations define the proof term $\lambda y^C.[u/x]t'$ and that $\Gamma \vdash \lambda y^C.[u/x]t' : B$. By inversion of the inference rule RI , $B = C \supset D$, and $\Gamma, x : A, y : C \vdash t' : D$. By admissibility of left weakening LW , $\Gamma, y : C \vdash u : A$. Now, $(|A|, |t'|) < (|A|, |t|)$, because $|t'| < |t| = 1 + |t'|$. By IH(1), the above equations define the proof term $[u/x]t'$ and $\Gamma, y : C \vdash [u/x]t' : D$. Hence, $\lambda y^C.[u/x]t'$ is defined. Applying the inference rule RI , $\Gamma \vdash \lambda y^C.[u/x]t' : C \supset D$.
- Case $t = x\hat{l}$. Suppose $\Gamma \vdash u : A$ and $\Gamma, x : A \vdash t : B$. We show that the above equations define the proof term $u@[u/x]l$ and that $\Gamma \vdash u@[u/x]l : B$. By inversion of the inference rule CTR , $\Gamma, x : A; l : A \vdash B$. Now $(|A|, |l|) < (|A|, |t|)$, because $|l| < |t| = 1 + |l|$. By IH(2), the above equations define the proof list $[u/x]l$ and $\Gamma; [u/x]l : A \vdash B$. Also, $(|A|, 0) < (|A|, |t|)$, because $0 < |t|$. By IH(3) the above equations define $u@[u/x]l$ and $\Gamma \vdash u@[u/x]l : B$.
- Case $t = y\hat{l}$, with $y \neq x$. Suppose $\Gamma \vdash u : A$ and $\Gamma, x : A \vdash t : B$. We show that the above equations define the proof term $y\hat{[u/x]l}$ and that $\Gamma \vdash y\hat{[u/x]l} : B$. By inversion of the inference rule CTR , $(y : A) \in \Gamma$, B is the atom p and $\Gamma, x : A; l : A \vdash p$. Now $(|A|, |l|) < (|A|, |t|)$, because $|l| < |t| = 1 + |l|$. By IH(2), the above equations define the proof list $[u/x]l$ and $\Gamma; [u/x]l : A \vdash p$. Hence, $y\hat{[u/x]l}$ is defined. Applying the inference rule CTR , $\Gamma \vdash y\hat{[u/x]l} : p$.

In (2),

- Case $l = \star$. Suppose $\Gamma \vdash u : A$ and $\Gamma, x : A; l : B \vdash p$. Hence $[u/x]\star = \star$ is defined. It remains to prove $\Gamma, x : A; \star : p \vdash p$. By the rule AX , $B = p$ and we are done.
- Case $l = t :: l'$. Suppose $\Gamma \vdash u : A$ and $\Gamma, x : A; l : B \vdash p$. We show that the above equations define the proof list $([u/x]t) :: ([u/x]l')$ and that $\Gamma; ([u/x]t) :: ([u/x]l') : B \vdash p$. By inversion of the inference rule LI , $B = C \supset D$ and $\Gamma, x : A \vdash t : C$ and $\Gamma, x : A; l' : D \vdash p$. Now $(|A|, |t|) < (|A|, |l|)$, because $|t| < |l| = 1 + |t| + |l'|$. By IH(1), the above equations define the proof term $[u/x]t$ and $\Gamma \vdash [u/x]t : C$. Also, $(|A|, |l'|) < (|A|, |l|)$, because $|l'| < |l| = 1 + |t| + |l'|$. By IH(2), the above equations define the proof list $[u/x]l'$ and $\Gamma; [u/x]l' : D \vdash p$. Hence, $([u/x]t) :: ([u/x]l')$ is defined. Applying the inference rule LI , $\Gamma; ([u/x]t) :: ([u/x]l') : C \supset D \vdash p$.

In (3),

- Case $A = B \supset C$. Suppose $\Gamma \vdash u : A$ and $\Gamma; l : A \vdash p$. Then $u = \lambda y^B.t$ and $l = t' :: l'$. We show that the above equations define the proof term $([t'/y]t)@l'$ and that $\Gamma \vdash ([t'/y]t)@l' : p$. By inversion of the inference rule RI , $\Gamma, y : B \vdash t : C$, and $\Gamma \vdash t' : B$ and $\Gamma; l' : C \vdash p$, given that $A = B \supset C$. Now $(|B|, |t|) < (|A|, 0)$, because $|B| < |A| = 1 + |B| + |C|$. By IH(1), the above equations define the proof term $[t'/y]t$ and $\Gamma \vdash [t'/y]t : C$. Also, $(|C|, 0) < (|A|, 0)$, because $|C| < |A| = 1 + |B| + |C|$. By IH(3), the above equations define the proof term $([t'/y]t)@l'$ and $\Gamma \vdash ([t'/y]t)@l' : p$.
- Case $A = p$. Suppose $\Gamma \vdash u : A$ and $\Gamma; l : A \vdash p$. Then, $p=q$, $l = \star$ and $u = x^l$. Hence, $(x^l)@ \star = x^l$ is defined. By the AX rule, $A = q$. Thus, $\Gamma \vdash (x^l)@ \star : q$.

□

Now, we can prove the admissibility of contraction in LJT , with the help of Theorem 2.1 and the following lemma.

Lemma 2.1. For every formula A , there is a proof term $t_{x,A}$ such that $\Gamma, x : A \vdash t_{x,A} : A$.

Proof. The proof is by induction on A . Given that a formula A has the form $A = \vec{B} \supset p$, with $\vec{B} = B_1, \dots, B_n$, we have that $\Gamma, x : A \vdash \lambda y_1^{B_1} \dots y_n^{B_n}. x^{\wedge} [t_{y_1, B_1}, \dots, t_{y_n, B_n}] : A$, where t_{y_i, B_i} are given by IH.

□

Proposition 2.2 (Admissibility of Contraction in LJT). The contraction rules

$$\frac{\Gamma, x_1 : A, x_2 : A \vdash t : B}{\Gamma, x : A \vdash [t_{x,A}/x_1][t_{x,A}/x_2]t : B} \quad \frac{\Gamma, x_1 : A, x_2 : A; l : B \vdash p}{\Gamma, x : A; [t_{x,A}/x_1][t_{x,A}/x_2]l : B \vdash p}$$

are admissible in LJT .

Proof. For the first rule, suppose $\Gamma, x_1 : A, x_2 : A \vdash t : B$. By Lemma 2.1, $\Gamma, x : A \vdash t_{x,A} : A$. Then:

$$\frac{\Gamma, x : A \vdash t_{x,A} : A \quad \frac{\Gamma, x_1 : A, x_2 : A \vdash t : B}{\Gamma, x_1 : A, x_2 : A, x : A \vdash t : B} LW}{\Gamma, x_1 : A, x : A \vdash [t_{x,A}/x_2]t : B} LW \quad (1)$$

$$\frac{\Gamma, x : A \vdash t_{x,A} : A \quad \Gamma, x_1 : A, x : A \vdash [t_{x,A}/x_2]t : B}{\Gamma, x : A \vdash [t_{x,A}/x_1][t_{x,A}/x_2]t : B} (1)$$

For the second rule, the proof is analogous – we assume $\Gamma, x_1 : A, x_2 : A; l : B \vdash p$ and, instead of LW and the admissible rule (1), we apply LW_f and the admissible rule (2). □

2.3 Proof Search in LJT

Proof search in LJT proceeds by bottom-up application of inference rules and obeys the focusing discipline. As described in the diagram of the Figure 1, first, we eagerly invert RI to decompose implications in the right-hand side; when the right-hand side becomes atomic, we decide the formula in the left-hand side to focus on by applying CTR ; and as long as the focus is an implication, we apply LI , keeping the focus on the succedent formula, and launching sub-problems. When the focus becomes atomic, if it's equal to the atom in the right-hand side, we finish with AX .

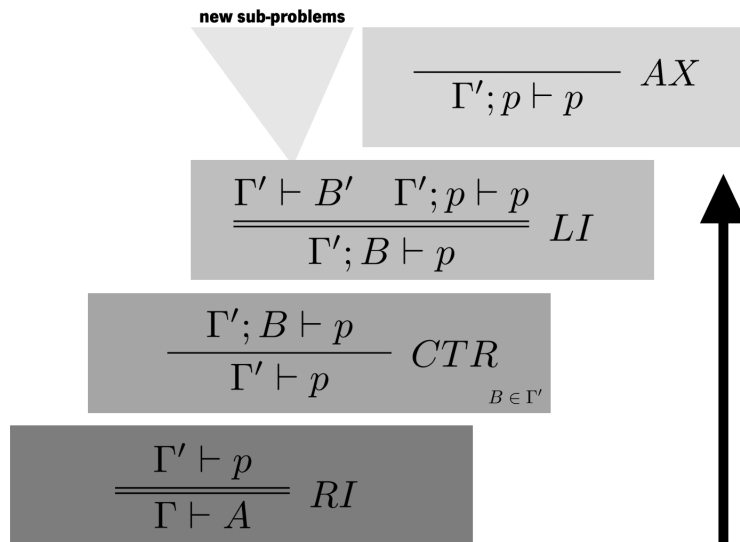


Figure 1: Focusing in LJT

Example 2.3. Keeping with the sequent $\vdash (p \supset p) \supset p \supset p$ from the previous examples, we show an example of proof search in LJT described informally.

$$\begin{array}{ccc}
\frac{}{\vdash (p \supset p) \supset p \supset p} \partial \uparrow & \longrightarrow & \frac{\frac{}{f : (p \supset p) \vdash p \supset p} \partial \uparrow}{\vdash (p \supset p) \supset p \supset p} RI \\
& & \downarrow \\
\frac{\frac{\frac{\frac{}{\Gamma; p \supset p \vdash p} \partial \uparrow}{\Gamma \vdash p} CTR}{f : (p \supset p) \vdash p \supset p} RI}{\vdash (p \supset p) \supset p \supset p} RI & \longleftarrow & \frac{\frac{\frac{}{f : (p \supset p), x : p \vdash p} \partial \uparrow}{f : (p \supset p) \vdash p \supset p} RI}{\vdash (p \supset p) \supset p \supset p} RI \\
& & \downarrow \\
\frac{\frac{\frac{\frac{}{\Gamma \vdash p} \partial \uparrow}{\Gamma; p \supset p \vdash p} LI}{\Gamma \vdash p} CTR}{f : (p \supset p) \vdash p \supset p} RI}{\vdash (p \supset p) \supset p \supset p} RI & \longrightarrow & \frac{\frac{\frac{\frac{}{\Gamma \vdash p} \partial \uparrow}{\Gamma; p \vdash p} AX}{\Gamma; p \supset p \vdash p} LI}{\Gamma \vdash p} CTR}{f : (p \supset p) \vdash p \supset p} RI}{\vdash (p \supset p) \supset p \supset p} RI \\
& & \downarrow \\
\frac{\frac{\frac{\frac{}{\Gamma; p \vdash p} AX}{\Gamma \vdash p} CTR}{\Gamma; p \supset p \vdash p} LI}{\Gamma \vdash p} CTR}{f : (p \supset p) \vdash p \supset p} RI}{\vdash (p \supset p) \supset p \supset p} RI & \longleftarrow & \frac{\frac{\frac{\frac{}{\Gamma; p \vdash p} \partial \uparrow}{\Gamma \vdash p} CTR}{\Gamma; p \supset p \vdash p} LI}{\Gamma \vdash p} CTR}{f : (p \supset p) \vdash p \supset p} RI}{\vdash (p \supset p) \supset p \supset p} RI
\end{array}$$

Notice how the ∂ -inferences denote gaps in the derivation. We signal the bottom-up applications of the ∂ -inference rules by the symbol $\uparrow$. Γ contains the assumptions $(f : p \supset p)$ and $(x : p)$.

Our goal is to describe this process formally, using a rewriting system, as this formalization will allow us to establish a clear correspondence between the normal forms and the typable proof terms of the LJT . In other words, the normal forms obtained through the rewriting process should accurately represent the valid derivations within the LJT proof system.

However, to address the dynamics of proof search, we need to broaden the traditional notion of a proof term.

Definition 2.9 (Partial Proof Term). A partial proof term is a term which contains gaps or unsolved parts. The distinctive feature of partial proof terms is that they can contain sequents, which explicitly signal the places at which a proof search problem still needs to be solved. A more formal definition will be presented later.

Example 2.4. A partial proof term can be

$$\lambda f^{(p \supset p)}. \lambda x^p. (\underline{f : (p \supset p), x : p \vdash p}) .$$

Definition 2.10 (Formal Sequent). A formal sequent is a sequent occurring in a partial proof term, which functions as a marker that indicates where a derivation is missing. Formal sequents are distinguished from the sequents of LJT because the former always appear underlined.

Example 2.5. In the partial proof term from Example 2.4, there is one occurrence of the formal sequent

$$\underline{f : (p \supset p), x : p \vdash p} .$$

There are two kinds of formal sequents in the partial proof terms of LJT .

Definition 2.11 (Formal Term Sequent). A formal term sequent is the underlined version of a term sequent

$$\underline{\Gamma \vdash A} ,$$

ranged over by $\underline{\sigma}$.

Definition 2.12 (Formal List Sequent). A formal list sequent is the underlined version of a list sequent

$$\underline{\Gamma; A \vdash p} ,$$

ranged over by $\underline{\tau}$.

The rewriting system will act on partial proof terms. However, in addition to the rewriting system, there is a typing system for the partial proof terms. This typing system classifies each partial proof term according to the sequent it proves and the unproven sequents it exhibits. This allows us to track not only the progress of proof construction but also the remaining proof obligations during the search procedure. We denote this new set of artifacts by LJT_{∂}

Definition 2.13 (Proof Terms of LJT_∂). The proof terms of LJT_∂ – or partial proof terms of LJT – are inductively defined by the following grammar:

$$\begin{aligned} \text{(Partial proof terms)} \quad t, u &::= \lambda x^A. t \mid x \hat{l} \mid \underline{\sigma} \\ \text{(Partial proof lists)} \quad l &::= \star \mid u :: l \mid \underline{\tau} \end{aligned}$$

where $\underline{\sigma}$ ranges over *formal term sequents*, and similarly $\underline{\tau}$ ranges over *formal list sequents*. A partial proof term without occurrences of formal sequents is called a total term.

Definition 2.14 (Lists of formal sequents). Lists of formal sequents are ranged over by Ξ , and written $[\underline{\sigma}, \underline{\tau}, \underline{\tau}']$ for instance. For such lists, we use ϵ to denote the empty list and $\Xi_1 @ \Xi_2$ to denote concatenation.

The proof system of LJT_∂ handles two kinds of sequents, called partial sequents.

Definition 2.15 (Partial Term Sequent of LJT_∂). A partial term sequent of LJT_∂ is an expression of the form

$$\Xi \Vdash t : \sigma ,$$

where Ξ is a list of formal sequents, possibly empty, t is a partial proof term and σ is the goal term sequent.

Definition 2.16 (Partial List Sequent of LJT_∂). A partial list sequent of LJT_∂ is an expression of the form

$$\Xi \Vdash l : \tau ,$$

where Ξ is a list of formal sequents, possibly empty, l is a partial proof list and τ is the goal list sequent.

Definition 2.17 (Inference Rules of LJT_∂). The inference rules of LJT_∂ are the following:

$$\begin{aligned} &\overline{[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma} \partial \quad \overline{[\underline{\tau}] \Vdash \underline{\tau} : \tau} \partial \\ &\overline{\epsilon \Vdash \star : (\Gamma; p \vdash p)} AX \quad \frac{\Xi \Vdash l : (\Gamma, x : A; A \vdash p)}{\Xi \Vdash x \hat{l} : (\Gamma, x : A \vdash p)} CTR \\ &\frac{\Xi \Vdash t : (\Gamma, x : A \vdash B)}{\Xi \Vdash \lambda x^A. t : (\Gamma \vdash A \supset B)} RI \quad \frac{\Xi_1 \Vdash u : (\Gamma \vdash A) \quad \Xi_2 \Vdash l : (\Gamma; B \vdash p)}{\Xi_1 @ \Xi_2 \Vdash u :: l : (\Gamma; A \supset B \vdash p)} LI \end{aligned}$$

There is an inference rule of LJT_{∂} for each inference rule of LJT and, furthermore, two additional rules, the ∂ -rules, are introduced. The rules CTR and RI are just modified presentations of the already studied rules; the rule AX says $\star$ represents a proof term for the sequent $\Gamma; p \vdash p$ without further proof obligations; rule LI concatenates the proof obligations of the two branches; and the first of the ∂ -rules says $\underline{\sigma}$ will be converted to a proof of σ , once the proof obligation $\underline{\sigma}$ is fulfilled - similarly for the other ∂ -rule.

This deductive system gives origin to what we call partial derivations.

Definition 2.18 (Partial Derivations LJT_{∂}). The set of partial derivations of LJT_{∂} is the smallest set of sequent trees closed for the inference rules of Definition 2.17.

Example 2.6. Recall the informal derivation of the sequent $\vdash (p \supset p) \supset p \supset p$ in Example 2.3. The partial derivation, after the first two proof search steps, is:

$$\frac{\frac{\frac{[(f : (p \supset p), x : p \vdash p)] \Vdash f : (p \supset p), x : p \vdash p : (f : (p \supset p), x : p \vdash p)}{[(f : (p \supset p), x : p \vdash p)] \Vdash \lambda x^p. (f : (p \supset p), x : p \vdash p) : (f : (p \supset p) \vdash p \supset p)} \partial}{[(f : (p \supset p), x : p \vdash p)] \Vdash \lambda x^p. (f : (p \supset p), x : p \vdash p) : (f : (p \supset p) \vdash p \supset p)} RI}{[(f : (p \supset p), x : p \vdash p)] \Vdash \lambda f^{(p \supset p)}. \lambda x^p. (f : (p \supset p), x : p \vdash p) : (\vdash (p \supset p) \supset p \supset p)} RI$$

Example 2.7. The proof state represented in the Example 2.6 is:

$$[(f : (p \supset p), x : p \vdash p)] \Vdash \lambda f^{(p \supset p)}. \lambda x^p. (f : (p \supset p), x : p \vdash p) : (\vdash (p \supset p) \supset p \supset p)$$

Notice that:

- the list of proof obligations is a single list containing only one formal sequent,

$$\underline{f : (p \supset p), x : p \vdash p},$$

which corresponds to the formal sequent in the partial proof term;

- the goal sequent,

$$\vdash (p \supset p) \supset p \supset p,$$

is the one we intended to prove at the beginning of the proof search; and

- the partial proof term,

$$\lambda f^{(p \supset p)}. \lambda x^p. (f : (p \supset p), x : p \vdash p),$$

encodes a record of the search: two applications of the rule RI and the formal sequent signaling the derivation that is missing.

A partial sequent $\Xi \Vdash t : \sigma$ derivable in LJT_{∂} represents a proof state found in the process of proof search in LJT : Ξ contains a list of proof obligations, t is a record of the search and σ is the goal sequent. The formal sequents in Ξ are those occurring in t .

We now formalize, in the following proposition, the idea that derivability in LJT is partial derivability from the empty list of proof obligations in LJT_{∂} .

Proposition 2.3 (Conservativity).

- (1) LJT derives $\Gamma \vdash t : A$ iff LJT_{∂} derives $\epsilon \Vdash t : (\Gamma \vdash A)$.
- (2) LJT derives $\Gamma; l : A \vdash p$ iff LJT_{∂} derives $\epsilon \Vdash l : (\Gamma; A \vdash p)$.

Proof. The two “only if” statements are proved by a simultaneous induction on the derivations of $\Gamma \vdash t : A$ and $\Gamma; l : A \vdash p$.

- Case *AX*. Suppose $\Gamma; \star : p \vdash p$. Hence $l = \star$ and $A = p$. By the *AX* rule, $\epsilon \Vdash \star : (\Gamma; p \vdash p)$ in LJT_{∂} .
- Case *CTR*. Suppose $\Gamma, x : A \vdash x^{\wedge} l : p$ is obtained from $\Gamma, x : A; l : A \vdash p$. By IH, $\epsilon \Vdash l : (\Gamma, x : A; A \vdash p)$. Applying the *CTR* rule, $\epsilon \Vdash x^{\wedge} l : (\Gamma, x : A \vdash p)$ in LJT_{∂} .
- Case *RI*. Suppose $\Gamma \vdash \lambda x^A. t : A \supset B$ is obtained from $\Gamma, x : A \vdash t : B$. By IH, $\epsilon \Vdash t : (\Gamma, x : A \vdash B)$. Applying the *RI* rule, $\epsilon \Vdash \lambda x^A. t : (\Gamma \vdash A \supset B)$ in LJT_{∂} .
- Case *LI*. Suppose $\Gamma; u :: l : A \supset B \vdash p$ is obtained from $\Gamma \vdash u : A$ and $\Gamma; l : B \vdash p$. By IH, $\epsilon \Vdash u : (\Gamma \vdash A)$ and $\epsilon \Vdash l : (\Gamma; B \vdash p)$. Applying the *LI* rule, $\epsilon \Vdash u :: l : (\Gamma; A \supset B \vdash p)$ in LJT_{∂} , since $\epsilon @ \epsilon = \epsilon$.

For the “if” statements, we prove:

- (1) If LJT_{∂} derives $\Xi \Vdash t : (\Gamma \vdash A)$, with $t \in LJ\mathcal{T}$, then $\Xi = \epsilon$ and LJT derives $\Gamma \vdash t : A$.
- (2) If LJT_{∂} derives $\Xi \Vdash l : (\Gamma; A \vdash p)$, with $l \in LJ\mathcal{T}$, then $\Xi = \epsilon$ and LJT derives $\Gamma; l : A \vdash p$.

The proof is by simultaneous induction on the derivations of $\Xi \Vdash t : (\Gamma \vdash A)$ and $\Xi \Vdash l : (\Gamma; A \vdash p)$.

- Case ∂ . Suppose $[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma$. Since $\underline{\sigma} \notin LJ\mathcal{T}$, the statement is trivially true. Similar reasoning applies to $[\underline{\tau}] \Vdash \underline{\tau} : \tau$.
- Case *AX*. Suppose $\epsilon \Vdash \star : (\Gamma; p \vdash p)$, with $\star \in LJ\mathcal{T}$. By the *AX* rule, $\Gamma; \star : p \vdash p$ in LJT .

- Case *CTR*. Suppose $\Xi \Vdash x\hat{l} : (\Gamma; A \vdash p)$, with $x\hat{l} \in LJT$, is obtained from $\Xi \Vdash l : (\Gamma, x : A; A \vdash p)$. Hence, $l \in LJT$. By IH, $\Xi = \epsilon$ and $\Gamma, x : A; l : A \vdash p$. Applying the *CTR* rule, $\Gamma, x : A \vdash x\hat{l} : p$ in *LJT*.
- Case *RI*. Suppose $\Xi \Vdash \lambda x^A.t : (\Gamma \vdash A \supset B)$, with $\lambda x^A.t \in LJT$, is obtained from $\Xi \Vdash t : (\Gamma, x : A \vdash B)$. Hence, $t \in NJT$. By IH, $\Xi = \epsilon$ and $\Gamma, x : A \vdash t : B$. Applying the *RI* rule, $\Gamma \vdash \lambda x^A.t : A \supset B$ in *LJT*.
- Case *LI*. Suppose $\Xi \Vdash u :: l : (\Gamma; A \supset B \vdash p)$, with $u :: l \in LJT$, is obtained from $\Xi_1 \Vdash u : (\Gamma \vdash A)$ and $\Xi_2 \Vdash l : (\Gamma; B \vdash p)$. Notice $\Xi = \Xi_1 @ \Xi_2$. Hence $u, l \in LJT$. By IH, $\Xi_1 = \Xi_2 = \epsilon$, and $\Gamma \vdash u : A$ and $\Gamma; l : B \vdash p$. Hence, $\Xi = \epsilon$. Applying the *LI* rule, $\Gamma; u :: l : A \supset B \vdash p$ in *LJT*.

□

The derivations of LJT_{∂} are partial because they contain occurrences of special rules named ∂ , which is where proof obligations are created. Proof obligations can be fulfilled by reducing t according to reduction rules given in the following definition.

Definition 2.19 (Reduction Rules of LJT_{∂}). The reduction rules for proof search in *LJT* are the following:

$$\begin{aligned}
(IR) \quad & \Gamma \vdash A \supset B \rightarrow \lambda x^A.(\Gamma, x : A \vdash B) \\
(SFL) \quad & \Gamma, x : A \vdash p \rightarrow x\hat{ }(\Gamma, x : A; A \vdash p) \\
(KFL) \quad & \Gamma; A \supset B \vdash p \rightarrow (\Gamma \vdash A) :: (\Gamma; B \vdash p) \\
(FFL) \quad & \Gamma; p \vdash p \rightarrow \star
\end{aligned}$$

There is a reduction rule for each inference rule of the proof system *LJT* and their names stand for: invert right (*IR*), start focus left (*SFL*), keep focus left (*KFL*) and finish focus left (*FFL*). These rules play a crucial role, as they define how to use the inference rule during proof search. In this approach, redexes are formal sequents and the reduction process not only transform these redexes, but also builds the record of the proof search. If all the proof obligations in Ξ are fulfilled (i.e. reduced away), the partial proof term t is converted into a proof term of the sequent σ . Similar considerations apply to $\Xi \Vdash l : \tau$.

We now explain how the reduction rules presented in Def. 2.19 describe the focusing discipline. Recall the rules in Definitions 2.4 and 2.8 and the diagram in Figure 1. The goal is to prove the sequent $\Gamma \vdash A$.

- As long as A is not atomic, we apply *RI* bottom-up: this is the inversion phase, which decomposes the implication in the right-hand side formula. The inversion phase is implemented by the reduction

rule IR : the redex sequent evolves to a more determined proof term, reflecting the application of RI ;

- Once a sequent $\Gamma' \vdash p$ is found, we need to move our focus to the left-hand side. This corresponds to the bottom-up application of rule CTR , which starts a focus on the left-hand side, by choosing a formula in Γ' to focus on. This is implemented by the reduction rule SFL , which brings our search problem to the form $\Gamma'; B \vdash p$.
- As long as the formula B in the left focus is not atomic, we apply LI bottom-up. This is implemented by the reduction rule KFL , which keeps the left-hand side focus in the right premiss, and launches a search sub-problem in the left premiss. The redex sequent evolves to a more determined proof list.
- If in that right premiss we ever reach a sequent $\Gamma'; p \vdash p$, with the left-hand side focus matching the right-hand side atom, this branch of the search succeeds, and we may produce the proof term $\star$: that is what rule FFL does, finishing the left-focusing phase.

The non-determinism of the process is entirely due to rule SFL , since that is the only point at which a choice - which formula to focus on - is made.

The reduction rules define a binary relation on partial proof terms and another on partial proof lists. The compatible closure of those binary relations, denoted $\rightarrow$, is a pair of two binary relations, again one on proof terms and one on proof lists. The closure rules producing the compatible closure say: reduction in t is reduction in $\lambda x^A.t$; reduction in u is reduction in $u :: l$; and reduction in l is reduction in x^l and $u :: l$. We use $\rightarrow$ to denote the reflexive and transitive closure of $\rightarrow$.

Example 2.8. The formalization of the proof search in the Example 2.3 is now presented. Again, Γ contains the assumptions $(f : p \supset p)$ and $(x : p)$.

$$\begin{array}{ll}
& \frac{}{\vdash (p \supset p) \supset p \supset p} \\
\rightarrow_{IR} & \lambda f^{(p \supset p)}. \frac{}{\vdash (p \supset p) \supset p \supset p} \\
\rightarrow_{IR} & \lambda f^{(p \supset p)}. \lambda x^p. \frac{}{\vdash (p \supset p), x : p \vdash p} \\
\rightarrow_{SFL} & \lambda f^{(p \supset p)}. \lambda x^p. f^{\wedge}(\Gamma; p \supset p \vdash p) \\
\rightarrow_{KFL} & \lambda f^{(p \supset p)}. \lambda x^p. f^{\wedge}((\Gamma \vdash p) :: (\Gamma; p \vdash p)) \\
\rightarrow_{FFL} & \lambda f^{(p \supset p)}. \lambda x^p. f^{\wedge}((\Gamma \vdash p) :: \star) \\
\rightarrow_{SFL} & \lambda f^{(p \supset p)}. \lambda x^p. f^{\wedge}((x^{\wedge}(\Gamma; p \vdash p)) :: \star) \\
\rightarrow_{FFL} & \lambda f^{(p \supset p)}. \lambda x^p. f^{\wedge}((x^{\wedge} \star) :: \star)
\end{array}$$

We know that, if $\Gamma \vdash t : A$ is derivable in LJT , then the proof term t encodes the successful result of a proof search for $\Gamma \vdash A$ (a derivation of this sequent). The next result says that, if $\Xi \Vdash t : \sigma$ is derivable in LJT_∂ , then the partial proof term t records the search that starts from $\underline{\sigma}$.

Proposition 2.4 (Record of search).

1. If $\Xi \Vdash t : \sigma$ then $\underline{\sigma} \rightarrow t$.
2. If $\Xi \Vdash l : \tau$ then $\underline{\tau} \rightarrow l$.

Proof. By simultaneous induction on the derivations of $\Xi \Vdash t : \sigma$ and $\Xi \Vdash l : \tau$.

- Case ∂ . Suppose $\underline{\sigma} \Vdash \underline{\sigma} : \sigma$. Then, $\underline{\sigma} \rightarrow \underline{\sigma}$, because of the reflexivity of $\rightarrow$. Similar reasoning applies to $\underline{\tau} \Vdash \underline{\tau} : \tau$.
- Case AX . Suppose $\epsilon \Vdash \star : (\Gamma; p \vdash p)$. Then, $\underline{\Gamma; p \vdash p} \rightarrow \star$, by FFL .
- Case CTR . Suppose $\Xi \Vdash x^l : (\Gamma, x : A \vdash p)$ is obtained from $\Xi \Vdash l : (\Gamma, x : A; A \vdash p)$. Then, $\underline{\Gamma, x : A \vdash p} \rightarrow x^l(\underline{\Gamma, x : A; A \vdash p}) \rightarrow x^l$, where the first step is by SFL and the rest of the reduction is by IH. Notice how the form of t dictates (and records) the search from $\underline{\Gamma, x : A \vdash p}$.
- Case RI . Suppose $\Xi \Vdash \lambda x^A.t : (\Gamma \vdash A \supset B)$ is obtained from $\Xi \Vdash t : (\Gamma, x : A \vdash B)$. Then, $\underline{\Gamma \vdash A \supset B} \rightarrow \lambda x^A.(\underline{\Gamma, x : A \vdash B}) \rightarrow \lambda x^A.t$, where the first step is by IR and the rest of the reduction is by IH.
- Case LI . Suppose $\Xi \Vdash u :: l : (\Gamma; A \supset B \vdash p)$ is obtained from $\Xi \Vdash u : (\Gamma \vdash A)$ and $\Xi \Vdash l : (\Gamma; B \vdash p)$. Then, $\underline{\Gamma; A \supset B \vdash p} \rightarrow (\underline{\Gamma \vdash A}) :: (\underline{\Gamma; B \vdash p}) \rightarrow u :: l$, where the first step is by KFL and the rest of the reduction is by IH.

□

Relation $\rightarrow$ determines a rewriting relation, also denoted $\rightarrow$, on lists of formal sequents.

Definition 2.20 (Reduction Rules for Lists of Formal Sequents in LJT_∂). Let Ξ, Ξ' be lists of proof obligations. $\Xi \rightarrow \Xi'$ is inductively defined by:

$$\begin{array}{lll}
(IR) & [\underline{\Gamma \vdash A \supset B}] & \rightarrow [\underline{\Gamma, x : A \vdash B}] \\
(SFL) & [\underline{\Gamma, x : A \vdash p}] & \rightarrow [\underline{\Gamma, x : A; A \vdash p}] \\
(KFL) & [\underline{\Gamma; A \supset B \vdash p}] & \rightarrow [\underline{\Gamma \vdash A; \Gamma; B \vdash p}] \\
(FFL) & [\underline{\Gamma; p \vdash p}] & \rightarrow \epsilon
\end{array}$$

and

$$\frac{\Xi_1 \rightarrow \Xi'_1}{\Xi_1 @ \Xi_2 \rightarrow \Xi'_1 @ \Xi_2} \quad \frac{\Xi_2 \rightarrow \Xi'_2}{\Xi_1 @ \Xi_2 \rightarrow \Xi_1 @ \Xi'_2}$$

Each rule of Definition 2.19 determines another one, that rewrites a formal sequent either into another formal sequent (the case of rules IR , SFL), or two other formal sequents (the case of rule KFL), or to no formal sequent (the case of rule FFL) - the latter just erases the redex. With these associated rules we can rewrite lists of proof obligations.

Proposition 2.5 (Subject reduction for LJT_∂).

1. If $\Xi \Vdash t : \sigma$ and $t \rightarrow t'$, then there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash t' : \sigma$.
2. If $\Xi \Vdash l : \tau$ and $l \rightarrow l'$, then there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash l' : \tau$.

Proof. By simultaneous induction on $t \rightarrow t'$ and $l \rightarrow l'$.

Base Cases:

- Case IR . Suppose $t = \Gamma \vdash A \supset B \rightarrow \lambda x^A.(\Gamma, x : A \vdash B) = t'$ and $\Xi \Vdash t : \sigma$. By the first ∂ -rule, $\Xi = [\Gamma \vdash A \supset B]$ and $\sigma = \Gamma \vdash A \supset B$. Using IR , $\Xi \rightarrow [\Gamma, x : A \vdash B]$. We now prove $[\Gamma, x : A \vdash B] \Vdash \lambda x^A.(\Gamma, x : A \vdash B) : (\Gamma \vdash A \supset B)$. By the first ∂ -rule, $[\Gamma, x : A \vdash B] \Vdash \Gamma, x : A \vdash B : (\Gamma, x : A \vdash B)$. Applying the RI rule, $[\Gamma, x : A \vdash B] \Vdash \lambda x^A.(\Gamma, x : A \vdash B) : (\Gamma \vdash A \supset B)$.
- Case SFL . Suppose $t = \Gamma, x : A \vdash p \rightarrow x^\wedge(\Gamma, x : A; A \vdash p) = t'$ and $\Xi \Vdash t : \sigma$. By the first ∂ -rule, $\Xi = [\Gamma, x : A \vdash p]$ and $\sigma = \Gamma, x : A \vdash p$. Using SFL , $\Xi \rightarrow [\Gamma, x : A; A \vdash p]$. We now prove $[\Gamma, x : A; A \vdash p] \Vdash x^\wedge(\Gamma, x : A; A \vdash p) : (\Gamma, x : A \vdash p)$. By the second ∂ -rule, $[\Gamma, x : A; A \vdash p] \Vdash \Gamma, x : A; A \vdash p : (\Gamma, x : A; A \vdash p)$. Applying the CTR rule, $[\Gamma, x : A; A \vdash p] \Vdash x^\wedge(\Gamma, x : A; A \vdash p) : (\Gamma, x : A \vdash p)$.
- Case KFL . Suppose $l = \Gamma; A \supset B \vdash p \rightarrow (\Gamma \vdash A) :: (\Gamma; B \vdash p) = l'$ and $\Xi \Vdash l : \tau$. By the second ∂ -rule, $\Xi = [\Gamma; A \supset B \vdash p]$ and $\tau = \Gamma; A \supset B \vdash p$. Using KFL , $\Xi \rightarrow [\Gamma \vdash A; \Gamma; B \vdash p]$. We now prove $[\Gamma \vdash A; \Gamma; B \vdash p] \Vdash (\Gamma \vdash A) :: (\Gamma; B \vdash p) : (\Gamma; A \supset B \vdash p)$. By the first ∂ -rule, $[\Gamma \vdash A] \Vdash \Gamma \vdash A : (\Gamma \vdash A)$ and, by the second ∂ -rule, $[\Gamma; B \vdash p] \Vdash \Gamma; B \vdash p : (\Gamma; B \vdash p)$. Applying the LI rule, $[\Gamma \vdash A; \Gamma; B \vdash p] \Vdash (\Gamma \vdash A) :: (\Gamma; B \vdash p) : (\Gamma; A \supset B \vdash p)$.

- Case FFL . Suppose $l = \underline{\Gamma; p \vdash p} \rightarrow \star = l'$ and $\Xi \Vdash l : \tau$. By the second ∂ -rule, $\Xi = [\underline{\Gamma; p \vdash p}]$ and $\tau = \Gamma; p \vdash p$. Using FFL , $\Xi \rightarrow \epsilon$. We now prove $\epsilon \Vdash \star : (\Gamma; p \vdash p)$, by just applying the AX rule.

Inductive Cases:

- Suppose $\lambda x^A.t \rightarrow \lambda x^A.t'$, with $t \rightarrow t'$ and $\Xi \Vdash \lambda x^A.t : (\Gamma \vdash A \supset B)$. By inversion of the inference rule RI , $\Xi \Vdash t : (\Gamma, x : A \vdash B)$. By IH, there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash t' : (\Gamma, x : A \vdash B)$. Applying the RI rule, $\Xi' \Vdash \lambda x^A.t' : (\Gamma \vdash A \supset B)$.
- Suppose $x \hat{l} \rightarrow x \hat{l}'$, with $l \rightarrow l'$ and $\Xi \Vdash x \hat{l} : (\Gamma, x : A \vdash p)$. By inversion of the inference rule CTR , $\Xi \Vdash l : (\Gamma, x : A; A \vdash p)$. By IH, there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash l' : (\Gamma, x : A; A \vdash p)$. Applying the CTR rule, $\Xi' \Vdash x \hat{l}' : (\Gamma, x : A \vdash p)$.
- Suppose $u :: l \rightarrow u' :: l$, with $u \rightarrow u'$ and $\Xi \Vdash u :: l : (\Gamma; A \supset B \vdash p)$. By inversion of the inference rule LI , $\Xi_1 \Vdash u : (\Gamma \vdash A)$ and $\Xi_2 \Vdash l : (\Gamma; B \vdash p)$ and $\Xi = \Xi_1 @ \Xi_2$. By IH, there is Ξ'_1 such that $\Xi_1 \rightarrow \Xi'_1$ and $\Xi'_1 \Vdash u' : (\Gamma \vdash A)$. Applying the LI rule, $\Xi'_1 @ \Xi_2 \Vdash u' :: l : (\Gamma; A \supset B \vdash p)$ and $\Xi_1 @ \Xi_2 \rightarrow \Xi'_1 @ \Xi_2$.
- Suppose $u :: l \rightarrow u :: l'$, with $l \rightarrow l'$ and $\Xi \Vdash u :: l : (\Gamma; A \supset B \vdash p)$. By inversion of the inference rule LI , $\Xi_1 \Vdash u : (\Gamma \vdash A)$ and $\Xi_2 \Vdash l : (\Gamma; B \vdash p)$, and $\Xi = \Xi_1 @ \Xi_2$. By IH, there is Ξ'_2 such that $\Xi_2 \rightarrow \Xi'_2$ and $\Xi'_2 \Vdash l' : (\Gamma; B \vdash p)$. Applying the LI rule, $\Xi_1 @ \Xi'_2 \Vdash u :: l' : (\Gamma; A \supset B \vdash p)$ and $\Xi_1 @ \Xi_2 \rightarrow \Xi_1 @ \Xi'_2$.

□

Every formal sequent reduces, except for the blocked sequents $\vdash p$ and $\underline{\Gamma; p \vdash q}$ with $p \neq q$. On the other hand, a proof term or list must contain a formal sequent in order to reduce. Hence proof terms or lists of LJT are normal forms.

The following theorem thus confirms that the methodology of partial proof terms for proof search developed in this chapter behaves as intended: proof search succeeds if and only if the formal sequent normalizes to a total term.

Theorem 2.2 (Proof search as normalization).

1. For all $t \in LJ\mathcal{T}$: $LJ\mathcal{T}$ derives $\Gamma \vdash t : A$ iff $\underline{\sigma} \twoheadrightarrow t$, where $\underline{\sigma} = \underline{\Gamma \vdash A}$.
2. For all $l \in LJ\mathcal{T}$: $LJ\mathcal{T}$ derives $\Gamma; l : A \vdash p$ iff $\underline{\tau} \twoheadrightarrow l$, where $\underline{\tau} = \underline{\Gamma; A \vdash p}$.

Proof. We just prove the first statement, the second is entirely analogous. Suppose LJT derives $\Gamma \vdash t : A$. By Conservativity (Prop. 2.3), LJT_{∂} derives $\epsilon \Vdash t : (\Gamma \vdash A)$. By Record of Search (Prop. 2.4), $\Gamma \vdash A \rightarrow t$. For the converse implication, suppose $\underline{\sigma} \rightarrow t$, with $\sigma = (\Gamma \vdash A)$. By the first ∂ -rule, $[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma$. By Subject Reduction (Prop. 2.5) and the assumption, there is Ξ , such that, $[\underline{\sigma}] \rightarrow \Xi$ and $\Xi \Vdash t : \sigma$. Since t is a total term, Ξ is actually empty, so LJT_{∂} derives $\epsilon \Vdash t : \sigma$. By Conservativity, LJT derives $\Gamma \vdash t : A$. □

Chapter 3

A Focused Strategy in Natural Deduction

In the previous chapter, we presented partial proof terms as a new tool in the study of proof search, and we used the sequent calculus LJT for implicational intuitionistic logic as our simple first case study.

We now apply these ideas to NJT , a bidirectional natural deduction system for intuitionistic implicational logic, inspired by bidirectional typing [5, 9, 13] and developed in [19, 16]. Proof search in NJT is a refinement of ideas initially proposed for the Intercalation Calculus [54, 55]. Here, a sequent may be derived top-down, even be valid because it is obtained by a complete proof, and yet be considered a point of incompleteness in the proof, because such a sequent may be involved in closing a gap opened by a previous bottom-up phase. For the proof term representation of such situations, we need formal sequents, not standing on their own as representation of “holes”, but rather as components of more complex expressions recording the top-down derivation. We say that a formal sequent represents a partiality point, and speak again of partial proof terms, proofs, or derivations.

After formalizing the proof search procedures according to our methodology, we prove a theorem: proof search according to focusing in LJT is isomorphic to proof search in NJT . The theorem is of interest because it shows that a natural deduction system matches a sequent calculus optimized for proof search. Perhaps more interesting is the ease with which the procedures are formalized, and the result is proved.

The content of this chapter was partially presented in the paper [25].

3.1 Natural Deduction NJT

The natural deduction system here named NJT is implicit in [13] and developed as a system in close correspondence with LJT in [19, 16] – but with emphasis on normalization and its correspondence with cut-elimination. As in the previous chapter, we will pay some attention to proof terms, as representing η -extended proofs in NJT involves some subtleties.

NJT handles two kinds of sequents.

Definition 3.1 (Sequents of NJT). NJT handles sequents $\Gamma \vdash A$, ranged over by σ , and $\Gamma \triangleright A$, ranged over by ρ .

In the latter form, we will say the formula A to the right of $\triangleright$ is the focus of the sequent.

Definition 3.2 (Inference Rules of NJT). The set of inference rules of NJT is defined as:

$$\frac{}{\Gamma, x : A \triangleright A} A \quad \frac{\Gamma \triangleright p}{\Gamma \vdash p} C \quad \frac{\Gamma, x : A \vdash B}{\Gamma \vdash A \supset B} I \quad \frac{\Gamma \triangleright A \supset B \quad \Gamma \vdash A}{\Gamma \triangleright B} E$$

These rules are the assumption rule, A ; the coercion rule, C ; the introduction rule for implication, I ; and the elimination rule for implication, E . The assumption, introduction and elimination rules are standard, except that some of the formulas are required to be in the focus.

In a derivation of NJT , the formulas in the focus constitute the elimination part [51] of a branch, starting with an assumption, containing a sequence of formulas figuring as the main premiss of elimination inferences, down to an atom p . Rule C configures the turning point between phases, as it starts the introduction part of the branch, which can be grown downwards through rule I .

Example 3.1. Let $(p \supset p) \supset p \supset p$ be the type of Church numerals. The derivation of the Church numeral $\bar{1}$ in NJT , in the empty context, is the following:

$$\frac{\frac{\frac{f : p \supset p, x : p \triangleright p \supset p}{f : p \supset p, x : p \vdash p} A \quad \frac{\frac{f : p \supset p, x : p \triangleright p}{f : p \supset p, x : p \vdash p} C}{\frac{f : p \supset p, x : p \triangleright p}{f : p \supset p, x : p \vdash p} E} \quad \frac{\frac{f : p \supset p, x : p \triangleright p}{f : p \supset p, x : p \vdash p} C}{\frac{f : p \supset p, x : p \vdash p}{f : p \supset p \vdash p \supset p} I} \quad \frac{\frac{f : p \supset p \vdash p \supset p}{\vdash (p \supset p) \supset p \supset p} I$$

3.2 NJT with Proof Terms

We now extend the natural deduction NJT to include proof terms.

Definition 3.3 (Proof Terms of NJT). Proof terms M, N and neutral terms H of NJT are defined by the following grammar:

$$\begin{array}{ll} \text{(Proof terms)} & M, N ::= \lambda x^A.M \mid \text{app}(H) \\ \text{(Neutral terms)} & H ::= x \mid HN \end{array}$$

This is a well-known inductive definition of the β -normal λ -terms, which are either λ -abstractions with β -normal bodies, or applicative terms $\text{app}(xN_1 \cdots N_m)$ with β -normal arguments N_i . If $H = xN_1 \cdots N_m$, then x is the head variable of H and $\text{app}(H)$.

NJT with proof terms handles two kinds of total sequents.

Definition 3.4 (Total Sequents of NJT with proof terms). NJT with proof terms handles proof term sequents $\Gamma \vdash M : A$ and neutral term sequents $\Gamma \triangleright H : A$.

Definition 3.5 (Inference Rules of NJT with proof terms). The inference rules of NJT with proof terms are the following:

$$\frac{\Gamma, x : A \vdash M : B}{\Gamma \vdash \lambda x^A.M : A \supset B} I \quad \frac{\Gamma \triangleright H : p}{\Gamma \vdash \text{app}(H) : p} C$$

$$\frac{}{\Gamma, x : A \triangleright x : A} A \quad \frac{\Gamma \triangleright H : A \supset B \quad \Gamma \vdash N : A}{\Gamma \triangleright HN : B} E$$

As a system for assigning types to λ -terms, this is perhaps the simplest bidirectional typing system [9].

Example 3.2. Adding proof terms to the derivation of Example 3.1, we have the following derivation in NJT with proof terms:

$$\frac{\frac{\frac{}{f : p \supset p, x : p \triangleright f : p \supset p} A \quad \frac{\frac{f : p \supset p, x : p \triangleright x : p}{f : p \supset p, x : p \vdash \text{app}(x) : p} C}{f : p \supset p, x : p \vdash \text{app}(f(\text{app}(x))) : p} C}{f : p \supset p, x : p \triangleright f(\text{app}(x)) : p} C}{f : p \supset p \vdash \lambda x^p.\text{app}(f(\text{app}(x))) : p \supset p} I}{\vdash \lambda f^{(p \supset p)}.\lambda x^p.\text{app}(f(\text{app}(x))) : (p \supset p) \supset p \supset p} I$$

Thus, the proof term $\lambda f^{(p \supset p)}.\lambda x^p.\text{app}(f(\text{app}(x)))$ encodes the derivation of the sequent $\vdash (p \supset p) \supset p \supset p$ in NJT .

Weakening rules are admissible in NJT with proof terms.

Proposition 3.1 (Admissibility of Weakening in NJT). The weakening rules

$$\frac{\Gamma \vdash M : B}{\Gamma, x : A \vdash M : B} W \quad \frac{\Gamma \triangleright H : B}{\Gamma, x : A \triangleright H : B} W_f$$

are admissible in NJT .

Proof. The admissibility of W and W_f is proved by a simultaneous induction on the derivations of $\Gamma \vdash B$ and $\Gamma \triangleright B$.

- Case A . Suppose $\Gamma, y : B \triangleright y : B$. Applying the A rule, $\Gamma, y : B, x : A \triangleright y : B$.
- Case C . Suppose $\Gamma \vdash \text{app}(H) : p$ is obtained from $\Gamma \triangleright H : p$. By IH, $\Gamma, x : A \triangleright H : p$. Applying the C rule, $\Gamma, x : A \vdash \text{app}(H) : p$.
- Case I . Suppose $\Gamma \vdash \lambda y^C.M : C \supset D$ is obtained from $\Gamma, y : C \vdash M : D$. By IH, $\Gamma, y : C, x : A \vdash M : D$. Applying the I rule, $\Gamma, x : A \vdash \lambda y^C.M : C \supset D$.
- Case E . Suppose $\Gamma \triangleright HN : B$ is obtained from $\Gamma \vdash N : C$ and $\Gamma \triangleright N : C \supset B$. By IH, $\Gamma, x : A \vdash N : C$ and $\Gamma, x : A \triangleright H : C \supset B$. Applying the E rule, $\Gamma, x : A \triangleright HN : B$.

□

In order to state the next theorem, we develop some auxiliary derived concepts of NJT .

First we introduce extended neutral terms.

Definition 3.6 (Extended Neutral Terms of NJT). An extended neutral term is either a neutral term H , or a proof term considered as a neutral term, called a head term and denoted $hd(M)$. Extended neutral terms are ranged over by $\mathbb{H}$.

Notice that we are not extending NJT to a system NJT^+ where the class of neutral terms is defined by $H ::= hd(M) \mid x \mid HN$. Such extension is well known [19, 16] and contains a representation of all of the λ -terms, including those not in β -normal form. Indeed, in such extension, the form $hd(M)$ is the location of non-normality. Here we are just introducing concepts to elaborate on NJT .

Next we introduce the auxiliary form of sequent for extended neutral terms.

Definition 3.7 (Extended Neutral Term Sequents of NJT). An extended neutral term sequent is an expression of the form $\Gamma \blacktriangleright \mathbb{H} : A$, which is an abbreviation defined by the following rules:

$$\frac{\Gamma \triangleright H : A}{\Gamma \blacktriangleright H : A} \quad \frac{\Gamma \vdash M : A}{\Gamma \blacktriangleright hd(M) : A}$$

Finally we extend the form $\text{app}(H)$ to extended neutral terms.

Definition 3.8 (Extension of $\text{app}(H)$ to extended neutral terms in NJT). Given an extended neutral term $\mathbb{H}$, $\text{app}(\llbracket \mathbb{H} \rrbracket)$, is the proof term of NJT defined by:

$$\text{app}(\llbracket H \rrbracket) = \text{app}(H) \quad \text{app}(\llbracket hd(M) \rrbracket) = M$$

Proposition 3.2. The inference rules

$$\frac{\Gamma \blacktriangleright H : p}{\Gamma \vdash \text{app}(\lfloor H \rfloor) : p} \quad \frac{\Gamma \blacktriangleright \text{hd}(M) : A}{\Gamma \vdash \text{app}(\lfloor \text{hd}(M) \rfloor) : A}$$

are admissible in NJT .

Proof. For the first admissible rule, suppose $\Gamma \blacktriangleright H : p$. By Definition 3.7, $\Gamma \triangleright H : p$. Applying the inference rule C , $\Gamma \vdash \text{app}(H) : p$. By Definition 3.8, $\Gamma \vdash \text{app}(\lfloor H \rfloor) : p$. For the second admissible rule, suppose $\Gamma \blacktriangleright \text{hd}(M) : A$. By Definition 3.7, $\Gamma \vdash M : A$. By Definition 3.8, $\Gamma \vdash \text{app}(\lfloor \text{hd}(M) \rfloor) : A$. $\square$

The next theorem contains an extension of the form HN to $\mathbb{H}@N$.

Theorem 3.1 (Admissible inference rules of NJT). Consider the equations

$$\begin{aligned} [N/x](\lambda y^C.M) &= \lambda y^C.[N/x]M & H@N &= HN \\ [N/x](\text{app}(H)) &= \text{app}(\lfloor [N/x]H \rfloor) & (\text{hd}(M))@N &= \text{hd}(M@N) \\ [N/x]x &= \text{hd}(N) & (\lambda y^A.M)@N &= [N/y]M \\ [N/x]y &= y \quad (y \neq x) \\ [N/x](HN') &= ([N/x]H)@([N/x]N') \end{aligned}$$

In NJT one has:

- (1) For all A, M : if $\Gamma \vdash N : A$ and $\Gamma, x : A \vdash M : B$, then the equations define the proof term $[N/x]M$ and $\Gamma \vdash [N/x]M : B$.
- (2) For all A, H : if $\Gamma \vdash N : A$ and $\Gamma, x : A \triangleright H : B$, then the equations define the extended neutral term $[N/x]H$ and $\Gamma \blacktriangleright [N/x]H : B$; moreover, if $[N/x]H$ is a head term, then x is the head variable of H .
- (3) For all $A, B, \mathbb{H}$: if $\Gamma \blacktriangleright \mathbb{H} : A \supset B$ and $\Gamma \vdash N : A$, then the equations define the extended neutral term $\mathbb{H}@N$ and $\Gamma \blacktriangleright \mathbb{H}@N : B$.
- (4) For all A, B, M : if $\Gamma \vdash M : A \supset B$ and $\Gamma \vdash N : A$, then the equations define the proof term $M@N$ and $\Gamma \vdash M@N : B$.

As a consequence, the following inference rules are admissible in NJT :

$$\frac{\Gamma \vdash N : A \quad \Gamma, x : A \vdash M : B}{\Gamma \vdash [N/x]M : B} \quad (1)$$

$$\frac{\Gamma \blacktriangleright \mathbb{H} : A \supset B \quad \Gamma \vdash N : A}{\Gamma \blacktriangleright \mathbb{H}@N : B} \quad (3)$$

$$\frac{\Gamma \vdash N : A \quad \Gamma, x : A \triangleright H : B}{\Gamma \blacktriangleright [N/x]H : B} \quad (2)$$

$$\frac{\Gamma \vdash M : A \supset B \quad \Gamma \vdash N : A}{\Gamma \vdash M@N : B} \quad (4)$$

Proof. For all A , M and H , the size $|A|$ is defined in Theorem 2.1; and the sizes $|M|$ and $|H|$ are defined by simultaneous structural recursion on M and H as follows: $|\lambda x^A.M| = 1 + |M|$, $|\text{app}(H)| = 1 + |H|$, $|x| = 1$, and $|HN| = |H| + |N|$. We also put $|hd(M)| = 1 + |M|$. Items (1), (2), (3) and (4) are proved by simultaneous induction on: $(|A|, |M|)$, $(|A|, |H|)$, $(f(\mathbb{H}, A, B), |\mathbb{H}|)$, and $(|A| + |B|, |M|)$, where $f(H, A, B) = 0$ and $f(hd(M), A, B) = |A| + |B|$. The pairs are ordered lexicographically.

In (1),

- Case $M = \lambda y^C.M'$. Suppose $\Gamma \vdash N : A$ and $\Gamma, x : A \vdash M : B$. We show that the above equations define the proof term $\lambda y^C.[N/x]M'$ and that $\Gamma \vdash \lambda y^C.[N/x]M' : B$. By inversion of the inference rule I , $B = C \supset D$, and $\Gamma, x : A, y : C \vdash M' : D$. By admissibility of weakening W , $\Gamma, y : C \vdash N : A$. Now, $(|A|, |M'|) < (|A|, |M|)$, because $|M'| < |M| = 1 + |M'|$. By IH(1), the above equations define the proof term $[N/x]M'$ and $\Gamma, y : C \vdash [N/x]M' : D$. Hence, $\lambda y^C.[N/x]M'$ is defined. Applying the inference rule I , $\Gamma \vdash \lambda y^C.[N/x]M' : C \supset D$.
- Case $M = \text{app}(H)$. Suppose $\Gamma \vdash N : A$ and $\Gamma, x : A \vdash M : B$. We show that the above equations define the proof term $\text{app}([N/x]H)$ and that $\Gamma \vdash \text{app}([N/x]H) : B$. By inversion of the inference rule C , $B = p$, and $\Gamma, x : A \triangleright H : p$. Now, $(|A|, |H|) < (|A|, |M|)$, because $|H| < |M| = 1 + |H|$. By IH(2), the above equations define the extended neutral term $[N/x]H$ and $\Gamma \blacktriangleright [N/x]H : p$. Hence, $\text{app}([N/x]H)$ is defined. Applying the admissible rule C for extended neutral terms, $\Gamma \vdash \text{app}([N/x]H) : p$.

In (2),

- Case $H = x$. Suppose $\Gamma \vdash N : A$ and $\Gamma, x : A \triangleright H : B$. Given that $[N/x]H = hd(N)$ is a head term, we observe that x is the head variable of H . We show that $\Gamma \blacktriangleright hd(N) : B$. By the A rule, $B = A$. Applying the admissible rule for introducing extended neutral terms from Definition 3.7, $\Gamma \blacktriangleright hd(N) : A$.
- Case $H = y$, with $y \neq x$. Suppose $\Gamma \vdash N : A$ and $\Gamma, x : A \triangleright H : B$. We show that

$\Gamma \blacktriangleright y : B$. Since $(y : B) \in \Gamma$, by the A rule, we obtain $\Gamma \triangleright y : B$. Applying the admissible rule for introducing extended neutral terms from Definition 3.7, $\Gamma \blacktriangleright y : B$.

- Case $H = H'N'$. Suppose $\Gamma \vdash N : A$ and $\Gamma, x : A \triangleright H : B$. We show that the above equations define the extended neutral term $([N/x]H')@([N/x]N')$ and that $\Gamma \blacktriangleright ([N/x]H')@([N/x]N') : B$. By inversion of the inference rule E , $\Gamma, x : A \vdash N' : C$ and $\Gamma, x : A \triangleright H' : C \supset B$. Now, $(|A|, |N'|) < (|A|, |H|)$, because $|N'| < |H| = |N'| + |H'|$. By IH(1), the above equations define the proof term $[N/x]N'$ and $\Gamma \vdash [N/x]N' : C$. Also, $(|A|, |H'|) < (|A|, |H|)$, because $|H'| < |H| = |N'| + |H'|$. By IH(2), the above equations define the extended neutral term $[N/x]H'$ and $\Gamma \blacktriangleright [N/x]H' : C \supset B$. Now, we have two cases: if $[N/x]H'$ is a head term, then, by IH(2), x is the head variable of H' . From $\Gamma, x : A \triangleright H' : C \supset B$, it follows that $|C \supset B| \leq |A|$, hence $|C| + |B| < |A|$. Thus, $(f([N/x]H', C, B), |[N/x]H'|) < (|A|, |H|)$, because $f([N/x]H', C, B) = |C| + |B| < |A|$. Otherwise, $(f([N/x]H', C, B), |[N/x]H'|) < (|A|, |H|)$, because $f([N/x]H', C, B) = 0 < |A|$. So, by IH(3), the above equations define the extended neutral term $([N/x]H')@([N/x]N')$ and $\Gamma \blacktriangleright ([N/x]H')@([N/x]N') : B$. By the above equations, $([N/x]H')@([N/x]N')$ is a head term only if $[N/x]H'$ is a head term; and the latter holds only if x is the head variable of H' (by IH(2)), which implies that x is the head variable of $H'N'$.

In (3),

- Case $\mathbb{H} = H$. Suppose $\Gamma \blacktriangleright \mathbb{H} : A \supset B$ and $\Gamma \vdash N : A$. The extended neutral term HN is defined and we show that $\Gamma \blacktriangleright HN : B$. By inversion of the inference rules in Definition 3.7, $\Gamma \triangleright H : A \supset B$. Applying the inference rule E , $\Gamma \triangleright HN : B$. Applying the rule for introducing extended neutral terms from Definition 3.7, $\Gamma \blacktriangleright HN : B$.
- Case $\mathbb{H} = hd(M)$. Suppose $\Gamma \blacktriangleright \mathbb{H} : A \supset B$ and $\Gamma \vdash N : A$. We show that the above equations define the extended neutral term $hd(M@N)$ and that $\Gamma \blacktriangleright hd(M@N) : B$. By inversion of the inference rules in Definition 3.7, $\Gamma \vdash M : A \supset B$. Now, $(|A| + |B|, |M|) < (f(\mathbb{H}, A, B), |\mathbb{H}|)$, because $|M| < |\mathbb{H}| = 1 + |M|$. By IH(4), the above equations define the proof term $M@N$ and $\Gamma \vdash M@N : B$. Hence, $hd(M@N)$ is defined. Applying the rule for introducing extended neutral terms from Definition 3.7, $\Gamma \blacktriangleright hd(M@N) : B$.

In (4),

- Case $M = \lambda y^A.M'$. Suppose $\Gamma \vdash M : A \supset B$ and $\Gamma \vdash N : A$. We show that the above equations define the proof term $[N/y]M'$ and that $\Gamma \vdash [N/y]M' : B$. By inversion of the

inference rule I , $\Gamma, y : A \vdash M' : B$. Now, $(|A|, |M'|) < (|A| + |B|, |M|)$, because $|A| < |A| + |B|$. By IH(1), the above equations define the proof term $[N/y]M'$ and $\Gamma \vdash [N/y]M' : B$.

□

3.3 Proof Search in NJT

Proof search in NJT follows the generic discipline of the intercalation calculus [54, 55], according to which derivations are built by “intercalating” formulas in the gap between assumptions and the conclusion. This is done in NJT in a particularly disciplined way: first, as in focusing, implications in the right-hand side are decomposed by the bottom-up application of rule I . When an atom p is computed, we change direction and decompose implications by top-down application of rule E , starting from an assumption chosen by rule A . The hope is that the atom which will be computed in this way is the same atom p which was computed by the bottom-up phase.

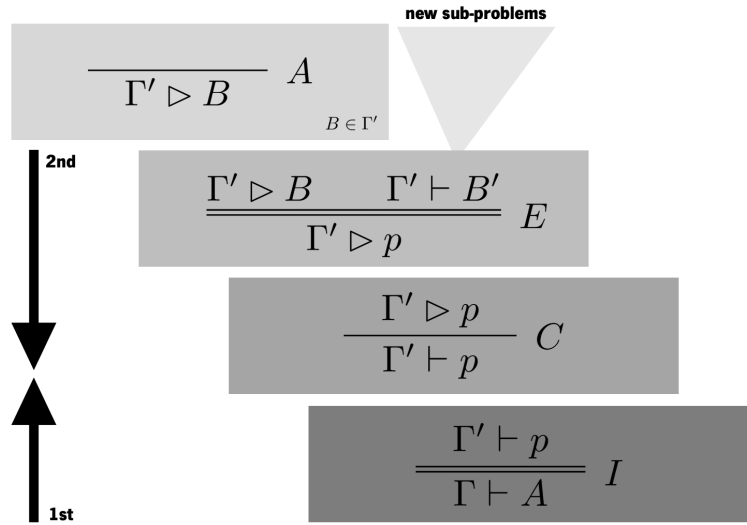


Figure 2: Intercalation in NJT

Example 3.3. Keeping with the sequent $\vdash (p \supset p) \supset p \supset p$ from the previous examples, we find an example of proof search in NJT described informally.

$$\frac{}{\vdash (p \supset p) \supset p \supset p} \partial \uparrow \quad \longrightarrow \quad \frac{\frac{f : (p \supset p) \vdash p \supset p}{\vdash (p \supset p) \supset p \supset p} I}{\vdash (p \supset p) \supset p \supset p} \partial \uparrow$$

↓

$$\begin{array}{ccc}
& & \downarrow \\
\frac{\frac{\frac{\overline{\Gamma \triangleright p \supset p} \quad A}{\Gamma \vdash p} \partial \downarrow}{f : (p \supset p) \vdash p \supset p} I}{\vdash (p \supset p) \supset p \supset p} I & \longleftarrow & \frac{\frac{\frac{f : (p \supset p), x : p \vdash p}{f : (p \supset p) \vdash p \supset p} I}{\vdash (p \supset p) \supset p \supset p} I}{\vdash (p \supset p) \supset p \supset p} I \partial \uparrow \\
& & \downarrow \\
\frac{\frac{\frac{\overline{\Gamma \triangleright p \supset p} \quad A}{\Gamma \vdash p} \partial \downarrow}{f : (p \supset p) \vdash p \supset p} I}{\vdash (p \supset p) \supset p \supset p} I & \longrightarrow & \frac{\frac{\frac{\overline{\Gamma \triangleright p \supset p} \quad A}{\Gamma \vdash p} \partial \uparrow}{\Gamma \triangleright p \quad C} E}{f : (p \supset p) \vdash p \supset p} I \\
& & \downarrow \\
\frac{\frac{\frac{\overline{\Gamma \triangleright p \supset p} \quad A}{\Gamma \vdash p} C}{\Gamma \triangleright p \quad C} E}{f : (p \supset p) \vdash p \supset p} I & \longleftarrow & \frac{\frac{\frac{\overline{\Gamma \triangleright p \supset p} \quad A}{\Gamma \vdash p} \partial \downarrow}{\Gamma \triangleright p \quad C} E}{f : (p \supset p) \vdash p \supset p} I
\end{array}$$

Here the ∂ -inferences denote two kinds of gaps in the derivation: in the figure, we signal applications of the bottom-up (resp. top-down) ∂ -inference rule by the symbol $\uparrow$ (resp. $\downarrow$). Γ contains the assumptions $(f : p \supset p)$ and $(x : p)$.

We now describe the process of proof search in NJT formally. Starting from NJT , we obtain the system NJT_{∂} using the strategy of partial proof terms introduced in the previous chapter with LJT , although there will be noteworthy differences. This is a sign that the methodology is not specific to a particular proof system.

There are two kinds of formal sequents appearing in the proof terms of NJT_{∂} .

Definition 3.9 (Formal Sequents of NJT_{∂}). NJT_{∂} handles formal proof term sequents, which are the underlined version of a proof term sequent, $\underline{\Gamma \vdash A}$, ranged over $\underline{\sigma}$; and formal neutral term sequents, which are the underlined version of a neutral term sequent, $\underline{\Gamma \triangleright A}$, ranged over $\underline{\rho}$.

Definition 3.10 (Proof Terms of NJT_∂). Proof terms for NJT_∂ , which are the partial proof terms for NJT , are generated as follows:

$$\begin{aligned} \text{(Partial proof terms)} \quad M, N &::= \lambda x^A.M \mid \text{app}(H) \mid \text{app}(H, \underline{\rho}, p) \mid \underline{\sigma} \\ \text{(Partial neutral terms)} \quad H &::= x \mid HN \end{aligned}$$

where $\underline{\sigma}$ ranges over formal terms sequents, and $\underline{\rho}$ ranges over formal neutral sequents.

Notice the unexpected placement of formal neutral sequents $\underline{\rho}$, not as partial neutral terms, but as part of partial proof terms of the form $\text{app}(H, \underline{\rho}, p)$ and the hybrid form of such formal sequents, carrying more information than a neutral sequent ρ , namely a partial neutral term H and an atom p . Partial neutral terms thus keep the form $xN_1 \cdots N_m$ and their partiality comes from the fact that the N_i 's are partial proof terms.

The proof system of NJT_∂ handles two kinds of partial sequents.

Definition 3.11 (Partial Sequents of NJT_∂). NJT_∂ handles partial proof term sequents $\Xi \Vdash M : \sigma$ and partial neutral term sequents $\Xi \Vdash H : \rho$.

In NJT_∂ , Ξ is a list of proof obligations, that can be of two forms: formal sequents $\underline{\sigma}$ and pairs $(\underline{\rho}, p)$, which denote the gap between a formula that has already been proven in the neutral term sequent $\underline{\rho}$ during the top-down phase; and p , the atom that was put “on hold” when we moved from the bottom-up phase to this top-down phase.

Definition 3.12 (Inference Rules of NJT_∂). The set of inference rules of NJT_∂ is defined as:

$$\begin{aligned} \frac{}{[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma} \partial \quad & \frac{\Xi \Vdash H : \rho \quad \rho = (\Gamma \triangleright A)}{\Xi @ [(\underline{\rho}, p)] \Vdash \text{app}(H, \underline{\rho}, p) : (\Gamma \vdash p)} \partial \\ \frac{}{\epsilon \Vdash x : (\Gamma, x : A \triangleright A)} A \quad & \frac{\Xi \Vdash H : (\Gamma \triangleright p)}{\Xi \Vdash \text{app}(H) : (\Gamma \vdash p)} C \\ \frac{\Xi \Vdash M : (\Gamma, x : A \vdash B)}{\Xi \Vdash \lambda x^A.M : (\Gamma \vdash A \supset B)} I \quad & \frac{\Xi_1 \Vdash H : (\Gamma \triangleright A \supset B) \quad \Xi_2 \Vdash N : (\Gamma \vdash A)}{\Xi_1 @ \Xi_2 \Vdash HN : (\Gamma \triangleright B)} E \end{aligned}$$

There is a partial inference rule for each inference rule of NJT and two ∂ -rules that collect the proof obligations. However, the second rule is more complex, since the formal neutral context $\underline{\rho}$ comes in a package $\text{app}(H, \underline{\rho}, p)$, together with its history, encoded in the term H (this has to be checked, hence

the premiss of the rule) and the desired atom p . The partiality at this point comes from the gap between the formula A already obtained top-down and the atom. If $A = p$, one can derive without partiality using rule C , and $\text{app}(H, \underline{\rho}, p)$ can be reduced to $\text{app}(H)$. Rules I and E need no further comment. Rule A says that making an assumption does not bring any proof obligation.

Example 3.4. Recall the informal derivation of the sequent $\vdash (p \supset p) \supset p \supset p$ in Example 3.3. The partial derivation, after the first three proof search steps, is:

$$\frac{\frac{\frac{\overline{\epsilon \Vdash f : (\Gamma \triangleright p \supset p)} A}{[(\Gamma \triangleright p \supset p), p] \Vdash \text{app}(f, (\Gamma \triangleright p \supset p), p) : (\Gamma \vdash p)} \partial}{[(\Gamma \triangleright p \supset p), p] \Vdash \lambda x^p. \text{app}(f, (\Gamma \triangleright p \supset p), p) : (f : (p \supset p) \vdash p \supset p)} I}{[(\Gamma \triangleright p \supset p), p] \Vdash \lambda f^{(p \supset p)}. \lambda x^p. \text{app}(f, (\Gamma \triangleright p \supset p), p) : (\vdash (p \supset p) \supset p \supset p)} I$$

Here, Γ contains the assumptions $(f : p \supset p)$ and $(x : p)$ and the ∂ -rule is signaling the gap between the formula $p \supset p$, selected from the context Γ , through the inference rule A ; and p , the atom obtained in the bottom-up phase.

Example 3.5. The proof state represented in Example 3.4 is:

$$[(\Gamma \triangleright p \supset p), p] \Vdash \lambda f^{(p \supset p)}. \lambda x^p. \text{app}(f, (\Gamma \triangleright p \supset p), p) : (\vdash (p \supset p) \supset p \supset p) .$$

Notice that:

- the list of proof obligations contains a single tuple,

$$((\Gamma \triangleright p \supset p), p),$$

which represents the gap between the formula $p \supset p$, derived in the top-down phase; and p . Also, both the formal sequent and the atom p are the ones occurring in the partial proof term;

- the goal sequent,

$$\vdash (p \supset p) \supset p \supset p,$$

is the one we intended to prove at the beginning of the proof search; and

- the partial proof term,

$$\lambda f^{(p \supset p)}. \lambda x^p. \text{app}(f, (\Gamma \triangleright p \supset p), p),$$

encodes a record of the search: two applications of the inference rule I , followed by an application of the inference rule A , and the formal sequent together with the atom, signaling the derivation that is missing between them.

A partial sequent $\Xi \Vdash M : \sigma$ derivable in NJT_∂ represents a proof state found in the process of proof search in NJT : Ξ contains a list of proof obligations – formal sequents $\underline{\sigma}$ and pairs $(\underline{\rho}, p)$ – and σ is the goal sequent. The formal sequents and the atoms in Ξ are those occurring in M .

The following proposition states that a sequent is derivable in NJT exactly when it is derivable in NJT_∂ from an empty list of proof obligations.

Proposition 3.3 (Conservativity).

- (1) NJT derives $\Gamma \vdash M : A$ iff NJT_∂ derives $\epsilon \Vdash M : (\Gamma \vdash A)$.
- (2) NJT derives $\Gamma \triangleright H : A$ iff NJT_∂ derives $\epsilon \Vdash H : (\Gamma \triangleright A)$.

Proof. The two “only if” statements are proved by a simultaneous induction on the derivations of $\Gamma \vdash M : A$ and $\Gamma \triangleright H : A$.

- Case *A*. Suppose $\Gamma, x : A \triangleright x : A$. Hence $H = x$. By the *A* rule, $\epsilon \Vdash x : (\Gamma, x : A \triangleright A)$ in NJT_∂ .
- Case *C*. Suppose $\Gamma \vdash \text{app}(H) : p$ is obtained from $\Gamma \triangleright H : p$. By IH, $\epsilon \Vdash H : (\Gamma \triangleright p)$. Applying the *C* rule, $\epsilon \Vdash \text{app}(H) : (\Gamma \vdash p)$ in NJT_∂ .
- Case *I*. Suppose $\Gamma \vdash \lambda x^A.M : A \supset B$ is obtained from $\Gamma, x : A \vdash M : B$. By IH, $\epsilon \Vdash M : (\Gamma, x : A \vdash B)$. Applying the *I* rule, $\epsilon \Vdash \lambda x^A.M : (\Gamma \vdash A \supset B)$ in NJT_∂ .
- Case *E*. Suppose $\Gamma \triangleright HN : B$ is obtained from $\Gamma \triangleright H : A \supset B$ and $\Gamma \vdash N : A$. By IH, $\epsilon \Vdash H : (\Gamma \triangleright A \supset B)$ and $\epsilon \Vdash N : (\Gamma \vdash A)$. Applying the *E* rule, $\epsilon \Vdash HN : (\Gamma \triangleright B)$ in NJT_∂ , since $\epsilon @ \epsilon = \epsilon$.

For the “if” statements, we prove:

- (1) If NJT_∂ derives $\Xi \Vdash M : (\Gamma \vdash A)$, with $M \in NJT$, then $\Xi = \epsilon$ and NJT derives $\Gamma \vdash M : A$.
- (2) If NJT_∂ derives $\Xi \Vdash H : (\Gamma \triangleright A)$, with $H \in NJT$, then $\Xi = \epsilon$ and NJT derives $\Gamma \triangleright H : A$.

The proof is by simultaneous induction on the derivations of $\Xi \Vdash M : (\Gamma \vdash A)$ and $\Xi \Vdash H : (\Gamma \triangleright A)$.

- Case ∂ . Suppose $\underline{\sigma} \Vdash \underline{\sigma} : \sigma$. Since $\underline{\sigma} \notin NJT$, the statement is trivially true. Similar reasoning applies to $\Xi @ (\underline{\rho}, p) \Vdash \text{app}(H, \underline{\rho}, p) : (\Gamma \vdash p)$.

- Case *A*. Suppose $\epsilon \Vdash x : (\Gamma, x : A \triangleright A)$, with $x \in NJT$. By the *A* rule, $\Gamma, x : A \triangleright x : A$ in NJT .
- Case *C*. Suppose $\Xi \Vdash \text{app}(H) : (\Gamma \vdash p)$, with $\text{app}(H) \in NJT$, is obtained from $\Xi \Vdash H : (\Gamma \triangleright p)$. Hence, $H \in NJT$. By IH, $\Xi = \epsilon$ and $\Gamma \triangleright H : p$. Applying the *C* rule, $\Gamma \vdash \text{app}(H) : p$ in NJT .
- Case *I*. Suppose $\Xi \Vdash \lambda x^A.M : (\Gamma \vdash A \supset B)$, with $\lambda x^A.M \in NJT$, is obtained from $\Xi \Vdash M : (\Gamma, x : A \vdash B)$. Hence $M \in NJT$. By IH, $\Xi = \epsilon$ and $\Gamma, x : A \vdash M : B$. Applying the *I* rule, $\Gamma \vdash \lambda x^A.M : A \supset B$ in NJT .
- Case *E*. Suppose $\Xi \Vdash HN : (\Gamma \triangleright B)$, with $HN \in NJT$, is obtained from $\Xi_1 \Vdash H : (\Gamma \triangleright A \supset B)$ and $\Xi_2 \Vdash N : (\Gamma \vdash A)$. Notice $\Xi = \Xi_1 @ \Xi_2$. Hence $H, N \in NJT$. By IH, $\Xi_1 = \Xi_2 = \epsilon$ and $\Gamma \triangleright H : A \supset B$ and $\Gamma \vdash N : A$. Hence $\Xi = \epsilon$. Applying the *E* rule, $\Gamma \triangleright HN : B$ in NJT .

□

Proof obligations can be eliminated by reducing a partial proof term M according to reduction rules given in the following definition.

Definition 3.13 (Reduction Rules of NJT_∂). The reduction rules for proof search in NJT are the following:

$$\begin{aligned}
(IR) \quad & \Gamma \vdash A \supset B \rightarrow \lambda x^A.(\Gamma, x : A \vdash B) \\
(SFD) \quad & \Gamma, x : A \vdash p \rightarrow \text{app}(x, \Gamma, x : A \triangleright A, p) \\
(KFD) \quad & \text{app}(H, \Gamma \triangleright A \supset B, p) \rightarrow \text{app}(H(\Gamma \vdash A), \Gamma \triangleright B, p) \\
(FFD) \quad & \text{app}(H, \Gamma \triangleright p, p) \rightarrow \text{app}(H)
\end{aligned}$$

Each inference rule in the proof system NJT is associated with a corresponding reduction rule: invert right (*IR*), start focus down (*SFD*), keep focus down (*KFD*), and finish focus down (*FFD*). In NJT_∂ , the partial derivations are represented by partial proof terms, and the rewriting of partial derivations in the search process is governed by these reduction rules. If all the proof obligations in Ξ are fulfilled (i.e., reduced away), the partial proof term M is converted into a proof term of the sequent σ . On the other hand, during the elimination phase, if all the proof obligations in Ξ , in particular (ρ, p) , are reduced away, the gap between $\rho = \Gamma \triangleright A$ and the formal sequent $\Gamma \vdash p$, that was left waiting, is closed and the proof term $\text{app}(H, \rho, p)$ is converted into the proof term $\text{app}(H)$, representing a proof of the latter sequent.

We now explain how the reduction rules presented in Definition 3.13 describe bidirectional proof search in NJT - see [18] for a similar search process. Recall the rules in Definitions 3.2 and 3.5 and the diagram in Figure 2. Our goal is to prove $\Gamma \vdash A$.

- As long as A is not atomic, we apply I bottom-up: this is the inversion phase, which decomposes the right-hand side formula. The inversion phase is implemented by rule IR : the redex sequent evolves to a more determined proof term, reflecting the bottom-up application of I .
- Once a sequent $\Gamma' \vdash p$ is found, we have a sequent that can be the conclusion of rule C only. But we will not apply C bottom-up. Instead, we leave the sequent $\Gamma' \vdash p$ on hold and change the direction of the search, starting the *focus-down* phase.
- The kick-off of the focus-down phase is done by rule A , which chooses a formula in Γ' to be under focus, that is, to the right of the symbol $\triangleright$. This is implemented by reduction rule SFD and converts our search problem to the generic form $\text{app}(H, \underline{\rho}, p)$, with $\underline{\rho} = \underline{\Gamma'} \triangleright B$.
- As long as the formula B under the $\triangleright$ -focus is not atomic, we apply E in the top-down direction. This is implemented by the reduction rule KFD , which simplifies the focus, evolves the term H , and launches a search sub-problem corresponding to the right premiss of E .
- When the focus becomes atomic, the focus-down succeeds if the obtained atom is equal to the atom p left waiting. In that case, we may conclude the proof by applying the coercion rule C , reconnecting to the derivation produced before, and left on hold, in the inversion phase. In the reduction system, this is implemented by rule FFD : we collect the obtained neutral term H to produce the proof term $\text{app}(H)$.

The non-determinism of the process is entirely due to the rule SFD , which allows us to choose which formula of Γ to focus on.

The reduction rules define a binary relation on partial proof terms and another on partial neutral terms. The compatible closure of those binary relations, denoted $\rightarrow$, is a pair of two binary relations, again one on proof terms and one on neutral terms. The closure rules producing the compatible closure say: reduction in M is reduction in $\lambda x^A.M$, reduction in N is reduction in HN ; reduction in H is reduction in $\text{app}(H)$ and HN . We use $\rightarrow$ to denote the reflexive and transitive closure of $\rightarrow$.

Example 3.6. The formalization of the proof search in the Example 3.3 is now presented. Again, Γ

contains the assumptions $(f : p \supset p)$ and $(x : p)$.

$$\begin{array}{ll}
& \frac{}{\vdash (p \supset p) \supset p \supset p} \\
\rightarrow_{IR} & \lambda f^{(p \supset p)}. \underline{(f : (p \supset p) \vdash p \supset p)} \\
\rightarrow_{IR} & \lambda f^{(p \supset p)}. \lambda x^p. \underline{(f : (p \supset p), x : p \vdash p)} \\
\rightarrow_{SFD} & \lambda f^{(p \supset p)}. \lambda x^p. \text{app}(f, (\Gamma \triangleright p \supset p), p) \\
\rightarrow_{KFD} & \lambda f^{(p \supset p)}. \lambda x^p. \text{app}(f(\Gamma \vdash p), (\Gamma \triangleright p), p) \\
\rightarrow_{FFD} & \lambda f^{(p \supset p)}. \lambda x^p. \text{app}(f(\Gamma \vdash p)) \\
\rightarrow_{SFD} & \lambda f^{(p \supset p)}. \lambda x^p. \text{app}(f(\text{app}(x, (\Gamma \triangleright p), p))) \\
\rightarrow_{FFD} & \lambda f^{(p \supset p)}. \lambda x^p. \text{app}(f(\text{app}(x)))
\end{array}$$

If the sequent $\Gamma \vdash M : A$ is derivable in NJT , then M encodes a derivation of $\Gamma \vdash A$, representing a successful proof search for the sequent. The next result states that if $\Xi \Vdash M : \sigma$ is derivable in NJT_∂ , then the partial proof term M records the search starting from $\underline{\sigma}$.

Proposition 3.4 (Record of search).

- (1) If $\Xi \Vdash M : \sigma$ then $\underline{\sigma} \twoheadrightarrow M$.
- (2) If $\Xi \Vdash H : (\Gamma \triangleright A)$ then $\underline{\Gamma \vdash p} \twoheadrightarrow \text{app}(H, \underline{\Gamma \triangleright A}, p)$.

Proof. By simultaneous induction on the derivations of $\Xi \Vdash M : \sigma$ and $\Xi \Vdash H : (\Gamma \triangleright A)$.

- Case (first) ∂ -rule. Suppose $[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma$. Then, $\underline{\sigma} \twoheadrightarrow \underline{\sigma}$, because of the reflexivity of $\twoheadrightarrow$.
- Case (second) ∂ -rule. Suppose $\Xi \Vdash \text{app}(H, (\Gamma \triangleright A), p) : (\Gamma \vdash p)$ is obtained from $\Xi' \Vdash H : (\Gamma \triangleright A)$. Then $\underline{\Gamma \vdash p} \twoheadrightarrow \text{app}(H, (\Gamma \triangleright A), p)$, by IH.
- Case A . Suppose $\epsilon \Vdash x : (\Gamma, x : A \triangleright A)$. Then, $\underline{\Gamma, x : A \vdash p} \twoheadrightarrow \text{app}(x, \underline{\Gamma, x : A \triangleright A}, p)$, by SFD .
- Case C . Suppose $\Xi \Vdash \text{app}(H) : (\Gamma \vdash p)$ is obtained from $\Xi \Vdash H : (\Gamma \triangleright p)$. Then, $\underline{\Gamma \vdash p} \twoheadrightarrow \text{app}(H, \underline{\Gamma \triangleright p}, p) \rightarrow \text{app}(H)$, where the last step is by FFD and the rest of the reduction is by IH.
- Case I . Suppose $\Xi \Vdash \lambda x^A. M : (\Gamma \vdash A \supset B)$ is obtained from $\Xi \Vdash M : (\Gamma, x : A \vdash B)$. Then, $\underline{\Gamma \vdash A \supset B} \twoheadrightarrow \lambda x^A. \underline{(\Gamma, x : A \vdash B)} \twoheadrightarrow \lambda x^A. M$, where the first step is by IR and the rest of the reduction is by IH.

- Case E . Suppose $\Xi_1 @ \Xi_2 \Vdash HN : (\Gamma \triangleright B)$ is obtained from $\Xi_1 \Vdash H : (\Gamma \triangleright A \supset B)$ and $\Xi_2 \Vdash N : (\Gamma \vdash A)$. Then, $\Gamma \vdash p \rightarrow \text{app}(H, \Gamma \triangleright A \supset B, p) \rightarrow \text{app}(H(\Gamma \vdash A), \Gamma \triangleright B, p) \rightarrow \text{app}(HN, \Gamma \triangleright B, p)$, where the second step is by KFD and the rest of the reduction is by IH.

□

The relation $\rightarrow$ determines a rewriting relation, also denoted $\rightarrow$, in the lists Ξ of formal sequents and tuples of a formal sequent and an atom.

Definition 3.14 (Reduction Rules for Ξ in NJT_∂). Let Ξ, Ξ' be lists of proof obligations. $\Xi \rightarrow \Xi'$ is inductively defined by

$$\begin{aligned}
(IR) \quad & [\Gamma \vdash A \supset B] \rightarrow [\Gamma, x : A \vdash B] \\
(SFD) \quad & [\Gamma, x : A \vdash p] \rightarrow [(\Gamma, x : A \triangleright A, p)] \\
(KFD) \quad & [(\Gamma \triangleright A \supset B, p)] \rightarrow [\Gamma \vdash A, (\Gamma \triangleright B, p)] \\
(FFD) \quad & [(\Gamma \triangleright p, p)] \rightarrow \epsilon
\end{aligned}$$

and

$$\frac{\Xi_1 \rightarrow \Xi'_1}{\Xi_1 @ \Xi_2 \rightarrow \Xi'_1 @ \Xi_2} \quad \frac{\Xi_2 \rightarrow \Xi'_2}{\Xi_1 @ \Xi_2 \rightarrow \Xi_1 @ \Xi'_2}$$

Each rule rewrites a formal sequent either into another formal sequent (the case of rule IR) or to a pair comprising a formal sequent and an atom (the case of rule SFD); or rewrites a pair of a formal sequent and an atom either into another pair and a formal sequent (the case of rule KFD), or to no formal sequent (the case of rule FFD) - the latter just erases the redex. With these associated rules we can rewrite Ξ 's, as described in Definition 3.14.

Proposition 3.5 (Subject reduction for NJT_∂).

1. If $\Xi \Vdash M : \sigma$ and $M \rightarrow M'$, then there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash M' : \sigma$.
2. If $\Xi \Vdash H : \rho$ and $H \rightarrow H'$, then there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash H' : \rho$

Proof. The proof is by simultaneous induction on $M \rightarrow M'$ and $H \rightarrow H'$.

Base Cases:

- Case IR . Suppose $M = \Gamma \vdash A \supset B \rightarrow \lambda x^A.(\Gamma, x : A \vdash B) = M'$ and $\Xi \Vdash M : \sigma$. By the first ∂ -rule, $\Xi = [\Gamma \vdash A \supset B]$ and $\sigma = \Gamma \vdash A \supset B$. Using IR , $\Xi \rightarrow [\Gamma, x : A \vdash B]$.

We now prove $[\Gamma, x : A \vdash B] \Vdash \lambda x^A.(\Gamma, x : A \vdash B) : (\Gamma \vdash A \supset B)$. By the first ∂ -rule, $[\Gamma, x : A \vdash B] \Vdash \Gamma, x : A \vdash B : (\Gamma, x : A \vdash B)$. Applying the I rule, $[\Gamma, x : A \vdash B] \Vdash \lambda x^A.(\Gamma, x : A \vdash B) : (\Gamma \vdash A \supset B)$.

- Case *SFD*. Suppose $M = \Gamma, x : A \vdash p \rightarrow \text{app}(x, \Gamma, x : A \triangleright A, p) = M'$ and $\Xi \Vdash M : \sigma$. By the first ∂ -rule, $\Xi = [\Gamma, x : A \vdash p]$ and $\sigma = \Gamma, x : A \vdash p$. Using *SFD*, $\Xi \rightarrow [(\Gamma, x : A \triangleright A, p)]$. We now prove $[(\Gamma, x : A \triangleright A, p)] \Vdash \text{app}(x, \Gamma, x : A \triangleright A, p) : (\Gamma, x : A \vdash p)$. By $A, \epsilon \Vdash x : (\Gamma, x : A \triangleright A)$. Applying the second ∂ -rule, $[(\Gamma, x : A \triangleright A, p)] \Vdash \text{app}(x, \Gamma, x : A \triangleright A, p) : (\Gamma, x : A \vdash p)$.
- Case *KFD*. Suppose $M = \text{app}(H, \Gamma \triangleright A \supset B, p) \rightarrow \text{app}(H(\Gamma \vdash A), \Gamma \triangleright B, p) = M'$ and $\Xi \Vdash M : \sigma$. By the second ∂ -rule, $\Xi = \Xi_1 @ [(\Gamma \triangleright A \supset B, p)]$ and $\sigma = \Gamma \vdash p$. Using *KFD*, $\Xi \rightarrow \Xi_1 @ [(\Gamma \triangleright B, p), \Gamma \vdash A]$. We now prove $\Xi_1 @ [\Gamma \vdash A, (\Gamma \triangleright B, p)] \Vdash \text{app}(H(\Gamma \vdash A), \Gamma \triangleright B, p) : (\Gamma \vdash p)$. By the second ∂ -rule, $\Xi_1 \Vdash H : (\Gamma \triangleright A \supset B)$ and, by the first ∂ -rule, $[\Gamma \vdash A] \Vdash \Gamma \vdash A : (\Gamma \vdash A)$. Applying the E rule, $\Xi_1 @ [\Gamma \vdash A] \Vdash H(\Gamma \vdash A) : (\Gamma \triangleright B)$. Applying the second ∂ -rule, $\Xi_1 @ [\Gamma \vdash A] @ [(\Gamma \triangleright B, p)] \Vdash \text{app}(H(\Gamma \vdash A), \Gamma \triangleright B, p) : (\Gamma \vdash p)$.
- Case *FFD*. Suppose $M = \text{app}(H, \Gamma \triangleright p, p) \rightarrow \text{app}(H) = M'$ and $\Xi \Vdash M : \sigma$. By the second ∂ -rule, $\Xi = \Xi_1 @ [(\Gamma \triangleright p, p)]$ and $\sigma = \Gamma \vdash p$. Using *FFD*, $\Xi \rightarrow \Xi_1$. We now prove $\Xi_1 \Vdash \text{app}(H) : (\Gamma \vdash p)$. By the second ∂ -rule, $\Xi_1 \Vdash H : (\Gamma \triangleright p)$. Applying the C rule, $\Xi_1 \Vdash \text{app}(H) : (\Gamma \vdash p)$.

Inductive Cases:

- Suppose $\lambda x^A.M \rightarrow \lambda x^A.M'$, with $M \rightarrow M'$ and $\Xi \Vdash \lambda x^A.M : (\Gamma \vdash A \supset B)$. By the inference rule I , $\Xi \Vdash M : (\Gamma, x : A \vdash B)$. By IH, there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash M' : (\Gamma, x : A \vdash B)$. Applying the I rule, $\Xi' \Vdash \lambda x^A.M' : (\Gamma \vdash A \supset B)$.
- Suppose $\text{app}(H) \rightarrow \text{app}(H')$, with $H \rightarrow H'$ and $\Xi \Vdash \text{app}(H) : (\Gamma \vdash p)$. By the inference rule C , $\Xi \Vdash H : (\Gamma \triangleright p)$. By IH, there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash H' : (\Gamma \triangleright p)$. Applying the C rule, $\Xi' \Vdash \text{app}(H') : (\Gamma \vdash p)$.
- Suppose $\text{app}(H, \underline{\rho}, p) \rightarrow \text{app}(H', \underline{\rho}, p)$, with $H \rightarrow H'$ and $\Xi \Vdash \text{app}(H, \underline{\rho}, p) : (\Gamma \vdash p)$. By the second ∂ -rule, $\Xi = \Xi_0 @ [(\underline{\rho}, p)]$, and $\Xi_0 \Vdash H : \rho$. By IH, there is Ξ' such that $\Xi_0 \rightarrow \Xi'$ and $\Xi' \Vdash H' : \rho$. Applying the second ∂ -rule, $\Xi' @ [(\underline{\rho}, p)] \Vdash \text{app}(H', \underline{\rho}, p) : (\Gamma \vdash p)$. From $\Xi_0 \rightarrow \Xi'$, we get $\Xi \rightarrow \Xi' @ [(\underline{\rho}, p)]$.

- Suppose $HN \rightarrow HN'$, with $N \rightarrow N'$ and $\Xi \Vdash HN : (\Gamma \triangleright B)$. By the inference rule E , $\Xi = \Xi_1 @ \Xi_2$ and $\Xi_1 \Vdash H : (\Gamma \triangleright A \supset B)$ and $\Xi_2 \Vdash N : (\Gamma \vdash A)$. By IH, there is Ξ'_2 such that $\Xi_2 \rightarrow \Xi'_2$ and $\Xi'_2 \Vdash N' : (\Gamma \vdash A)$. Applying the E rule, $\Xi_1 @ \Xi'_2 \Vdash HN' : (\Gamma \triangleright B)$. From $\Xi_2 \rightarrow \Xi'_2$, we get $\Xi \rightarrow \Xi_1 @ \Xi'_2$.
- Suppose $HN \rightarrow H'N$, with $H \rightarrow H'$ and $\Xi \Vdash HN : (\Gamma \triangleright B)$. By the inference rule E , $\Xi = \Xi_1 @ \Xi_2$, and $\Xi_1 \Vdash H : (\Gamma \triangleright A \supset B)$ and $\Xi_2 \Vdash N : (\Gamma \vdash A)$. By IH, there is Ξ'_1 such that $\Xi_1 \rightarrow \Xi'_1$ and $\Xi'_1 \Vdash H' : (\Gamma \triangleright A \supset B)$. Applying the E rule, $\Xi'_1 @ \Xi_2 \Vdash H'N : (\Gamma \triangleright B)$. From $\Xi_1 \rightarrow \Xi'_1$, we get $\Xi \rightarrow \Xi'_1 @ \Xi_2$.

□

Every formal term sequent reduces, except for the blocked sequents $\vdash p$; and every $\text{app}(H, \underline{\rho}, p)$ reduces, except in the case $\text{app}(H, \underline{\Gamma \triangleright q}, p)$ with $p \neq q$. On the other hand, a proof or head term must contain a formal sequent in order to reduce. Hence proof or neutral terms of NJT are normal forms.

Finally, the following theorem states that proof search succeeds in NJT_∂ if and only if the formal sequent normalizes. Thus, the methodology of partial proof terms represents, as expected, the proof search in this system of natural deduction.

Theorem 3.2 (Proof search as normalization).

1. For all $M \in NJT$: NJT derives $\Gamma \vdash M : A$ iff $\underline{\Gamma \vdash A} \rightarrow M$.
2. For all $H \in NJT$: NJT derives $\Gamma \triangleright H : A$ iff $\underline{\Gamma \vdash p} \rightarrow \text{app}(H, \underline{\Gamma \triangleright A}, p)$.

Proof. Similar to the proof of Theorem 2.2. This time we prove the second statement - the first is analogous. The “only if” implication follows from Conservativity (Prop. 3.3) and Record of Search (Prop. 3.4). As to the “if” implication, let $\underline{\sigma} = \underline{\Gamma \vdash p}$ and suppose $\underline{\sigma} \rightarrow \text{app}(H, \underline{\Gamma \triangleright A}, p)$. By the first ∂ -rule, $[\underline{\sigma}] \Vdash \underline{\sigma} : (\Gamma \vdash p)$. By the assumption and subject reduction, there is Ξ , such that $[\underline{\sigma}] \rightarrow \Xi$ and $\Xi \Vdash \text{app}(H, \underline{\Gamma \triangleright A}, p) : (\Gamma \vdash p)$. Since H is total, $\Xi = [\underline{\Gamma \triangleright A}]$. Since the second ∂ -rule is invertible, we obtain $\epsilon \Vdash H : (\Gamma \triangleright A)$. By conservativity, NJT derives $\Gamma \triangleright H : A$. □

3.4 Focusing vs Bidirectionality

In the previous chapter and sections, we showed that partial proof terms are an adequate tool to represent the search procedures of LJT and NJT , respectively. We now make it evident that we can use such a

representation as a basis to investigate proof search by showing that the two proof search procedures are isomorphic.

First, we present the bijection between LJT_{∂} -terms and NJT_{∂} -terms, which extends the bijection between LJT -terms and NJT -terms obtained in [13]. Notice that in [13] there is no treatment of bidirectional natural deduction NJT , but instead a presentation of the β -normal λ -terms as a grammar with two syntactic classes, which is the same, modulo notation, as the grammar of NJT -terms.

Definition 3.15 (Translations Θ and Ψ). Let us name as $L\text{-Terms}$ and $Lists$ the two sets of expressions defined simultaneously by the grammar of LJT_{∂} , and analogously use the names $N\text{-Terms}$ and $Neutrals$ for NJT_{∂} . The map $\Theta : L\text{-Terms} \rightarrow N\text{-Terms}$ is defined together with $\Theta : Neutrals \times Lists \rightarrow N\text{-Terms}$ by simultaneous recursion on $t \in L\text{-Terms}$ and $l \in Lists$, and the map $\Psi : N\text{-Terms} \rightarrow L\text{-Terms}$ is defined together with $\Psi : Neutrals \times Lists \rightarrow L\text{-Terms}$ by simultaneous recursion on $M \in N\text{-Terms}$ and $H \in Neutrals$, as:

$$\begin{aligned}
\Theta(\lambda x^A.t) &= \lambda x^A.\Theta(t) & \Psi(\lambda x^A.M) &= \lambda x^A.\Psi(M) \\
\Theta(x\hat{l}) &= \Theta(x, l) & \Psi(\text{app}(H)) &= \Psi(H, \star) \\
\Theta(\underline{\sigma}) &= \underline{\sigma} & \Psi(\underline{\sigma}) &= \underline{\sigma} \\
\Psi(\text{app}(H, \underline{\Gamma \triangleright A}, p)) &= \Psi(H, \underline{\Gamma}; A \vdash p) \\
\\
\Theta(H, u :: l) &= \Theta(H\Theta(u), l) \\
\Theta(H, \star) &= \text{app}(H) & \Psi(HN, l) &= \Psi(H, \Psi(N) :: l) \\
\Theta(H, \underline{\Gamma}; A \vdash p) &= \text{app}(H, \underline{\Gamma \triangleright A}, p) & \Psi(x, l) &= x\hat{l}
\end{aligned}$$

To help us prove the next theorem, we extend the definitions of Θ and Ψ to formal sequents $\underline{\sigma}$, $\underline{\tau}$ and $\underline{\rho}$, and to lists of proof obligations Ξ .

Definition 3.16 (Translations Θ and Ψ – Formal Sequents). The extension of the translations Θ and Ψ to formal sequents is defined as:

$$\begin{aligned}
\Theta(\underline{\sigma}) &= \underline{\sigma} & \Psi(\underline{\sigma}) &= \underline{\sigma} \\
\Theta(\underline{\Gamma}; A \vdash p) &= (\underline{\Gamma \triangleright A}, p) & \Psi((\underline{\Gamma \triangleright A}, p)) &= \underline{\Gamma}; A \vdash p
\end{aligned}$$

Definition 3.17 (Translations Θ and Ψ – Lists of Proof Obligations). The extension of the translations Θ and Ψ to lists of proof obligations is defined as:

$$\begin{aligned}\Theta(\epsilon) &= \epsilon & \Psi(\epsilon) &= \epsilon \\ \Theta([\underline{\sigma}]@ \Xi) &= [\Theta(\underline{\sigma})]@ \Theta(\Xi) & \Psi([\underline{\sigma}]@ \Xi) &= [\Psi(\underline{\sigma})]@ \Psi(\Xi) \\ \Theta([\underline{\tau}]@ \Xi) &= [\Theta(\underline{\tau})]@ \Theta(\Xi) & \Psi([\underline{(\rho, p)}]@ \Xi) &= [\Psi(\underline{(\rho, p)})]@ \Psi(\Xi)\end{aligned}$$

The following theorem states that Θ and Ψ preserve typing and derivability between the two proof systems LJT_{∂} and NJT_{∂} .

Theorem 3.3 (Soundness).

1. If $\Xi \Vdash t : \sigma$ in LJT_{∂} then $\Theta(\Xi) \Vdash \Theta(t) : \sigma$ in NJT_{∂} .
2. If $\Xi \Vdash M : \sigma$ in NJT_{∂} then $\Psi(\Xi) \Vdash \Psi(M) : \sigma$ in LJT_{∂} .

Proof. The first item is proved together with:

- If $\Xi \Vdash l : \tau$ in LJT_{∂} , with $\tau = (\Gamma; A \vdash p)$, and $\Xi' \Vdash H : (\Gamma \triangleright A)$ in NJT_{∂} , then $\Xi' @ \Theta(\Xi) \Vdash \Theta(H, l) : (\Gamma \vdash p)$ in NJT_{∂} .

By simultaneous induction on the derivations of $\Xi \Vdash t : \sigma$ and $\Xi \Vdash l : \tau$.

- Case of first ∂ -rule. Suppose $[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma$ in LJT_{∂} . By the definitions of Θ , $\Theta(\underline{\sigma}) = \underline{\sigma}$ and $\Theta([\underline{\sigma}]) = [\underline{\sigma}]$. By the first ∂ -rule, $[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma$ in NJT_{∂} .
- Case of second ∂ -rule. Suppose $[\underline{\tau}] \Vdash \underline{\tau} : \tau$ in LJT_{∂} , with $\underline{\tau} = \underline{\Gamma; A \vdash p}$, and $\Xi' \Vdash H : (\Gamma \triangleright A)$ in NJT_{∂} . By the definitions of Θ , $\Theta(H, \underline{\tau}) = \text{app}(H, (\underline{\Gamma \triangleright A}), p)$ and $\Theta([\underline{\tau}]) = [(\underline{\Gamma \triangleright A}, p)]$. By the second ∂ -rule, $\Xi' @ [(\underline{\Gamma \triangleright A}, p)] \Vdash \text{app}(H, (\underline{\Gamma \triangleright A}), p) : (\Gamma \vdash p)$ in NJT_{∂} .
- Case AX . Suppose $\epsilon \Vdash \star : (\Gamma; p \vdash p)$ in LJT_{∂} and $\Xi' \Vdash H : (\Gamma \triangleright p)$ in NJT_{∂} . By the definitions of Θ , $\Theta(H, \star) = \text{app}(H)$ and $\Theta(\epsilon) = \epsilon$. Applying the C rule, $\Xi' \Vdash \text{app}(H) : (\Gamma \vdash p)$ in NJT_{∂} .
- Case CTR . Suppose $\Xi \Vdash x\hat{l} : (\Gamma, x : A \vdash p)$ is obtained from $\Xi \Vdash l : (\Gamma, x : A; A \vdash p)$ in LJT_{∂} . By the definition of Θ , $\Theta(x\hat{l}) = \Theta(x, l)$. We now prove $\Theta(\Xi) \Vdash \Theta(x, l) : (\Gamma, x : A \vdash p)$. Applying the A rule, $\epsilon \Vdash x : (\Gamma, x : A \triangleright A)$, in NJT_{∂} . By IH, $\Theta(\Xi) \Vdash \Theta(x, l) : (\Gamma, x : A \vdash p)$ in NJT_{∂} .

- Case *RI*. Suppose $\Xi \Vdash \lambda x^A.t : (\Gamma \vdash A \supset B)$ is obtained from $\Xi \Vdash t : (\Gamma, x : A \vdash B)$ in LJT_∂ . By the definition of Θ , $\Theta(\lambda x^A.t) = \lambda x^A.\Theta(t)$. We now prove $\Theta(\Xi) \Vdash \lambda x^A.\Theta(t) : (\Gamma \vdash A \supset B)$. By IH, $\Theta(\Xi) \Vdash \Theta(t) : (\Gamma, x : A \vdash B)$. Applying the *I* rule, $\Theta(\Xi) \Vdash \lambda x^A.\Theta(t) : (\Gamma \vdash A \supset B)$ in NJT_∂ .
- Case *LI*. Suppose $\Xi_1 @ \Xi_2 \Vdash u :: l : (\Gamma; A \supset B \vdash p)$ is obtained from $\Xi_1 \Vdash u : (\Gamma \vdash A)$ and $\Xi_2 \Vdash l : (\Gamma; B \vdash p)$ in LJT_∂ and $\Xi' \Vdash H : (\Gamma \triangleright A \supset B)$ in NJT_∂ . By the definition of Θ , $\Theta(H, u :: l) = \Theta(H\Theta(u), l)$. We now prove $\Xi' @ (\Theta(\Xi_1) @ \Theta(\Xi_2)) \Vdash \Theta(H\Theta(u), l) : (\Gamma \vdash p)$. By IH, $\Theta(\Xi_1) \Vdash \Theta(u) : (\Gamma \vdash A)$ in NJT_∂ . Applying the *E* rule, $\Xi' @ \Theta(\Xi_1) \Vdash H\Theta(u) : (\Gamma \triangleright B)$. By IH, $(\Xi' @ \Theta(\Xi_1)) @ \Theta(\Xi_2) \Vdash \Theta(H\Theta(u), l) : (\Gamma \vdash p)$ in NJT_∂ . Clearly $\Xi' @ (\Theta(\Xi_1) @ \Theta(\Xi_2)) = (\Xi' @ \Theta(\Xi_1)) @ \Theta(\Xi_2)$.

The second item is proved together with:

- If $\Xi \Vdash H : \rho$ in NJT_∂ , with $\rho = (\Gamma \triangleright A)$, and $\Xi' \Vdash l : (\Gamma; A \vdash p)$ in LJT_∂ then $\Psi(\Xi) @ \Xi' \Vdash \Psi(H, l) : (\Gamma \vdash p)$ in LJT_∂ .

The proof is made by simultaneous induction on the derivations of $\Xi \Vdash M : \sigma$ and $\Xi \Vdash H : \rho$.

- Case of first ∂ -rule. Suppose $[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma$ in NJT_∂ . By the definitions of Ψ , $\Psi(\underline{\sigma}) = \underline{\sigma}$ and $\Psi([\underline{\sigma}]) = [\underline{\sigma}]$. By the first ∂ -rule, $[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma$ in LJT_∂ .
- Case of second ∂ -rule. Suppose $\Xi @ [(\underline{\rho}, p)] \Vdash \text{app}(H, \underline{\rho}, p) : (\Gamma \vdash p)$ is obtained from $\Xi \Vdash H : \rho$, where $\underline{\rho} = \underline{\Gamma \triangleright A}$. By the definitions of Ψ , $\Psi(\text{app}(H, \underline{\rho}, p)) = \psi(H, (\underline{\Gamma; A \vdash p}))$ and $\Psi(\Xi @ [(\underline{\rho}, p)]) = \Psi(\Xi) @ [\underline{\Gamma; A \vdash p}]$. We now prove $\Psi(\Xi) @ [\underline{\Gamma; A \vdash p}] \Vdash \psi(H, (\underline{\Gamma; A \vdash p})) : (\Gamma \vdash p)$. By the second ∂ -rule, $[\underline{\Gamma; A \vdash p}] \Vdash \underline{\Gamma; A \vdash p} : (\Gamma; A \vdash p)$ in LJT_∂ . By IH, $\Psi(\Xi) @ [\underline{\Gamma; A \vdash p}] \Vdash \psi(H, (\underline{\Gamma; A \vdash p})) : (\Gamma \vdash p)$ in LJT_∂ .
- Case *A*. Suppose $\epsilon \Vdash x : (\Gamma, x : A \triangleright A)$ in NJT_∂ and $\Xi \Vdash l : (\Gamma, x : A; A \vdash p)$ in LJT_∂ . By the definitions of Ψ , $\Psi(x, l) = x \hat{\cdot} l$ and $\Psi(\epsilon) = \epsilon$. Applying *CTR* rule, $\Xi \Vdash x \hat{\cdot} l : (\Gamma, x : A \vdash p)$ in LJT_∂ .
- Case *C*. Suppose $\Xi \Vdash \text{app}(H) : (\Gamma \vdash p)$ is obtained from $\Xi \Vdash H : (\Gamma \triangleright p)$ in NJT_∂ . By the definition of Ψ , $\Psi(\text{app}(H)) = \Psi(H, \star)$. We now prove $\Psi(\Xi) \Vdash \Psi(H, \star) : (\Gamma \vdash p)$. By the *AX* rule, $\epsilon \Vdash \star : (\Gamma; p \vdash p)$, in LJT_∂ . By IH, $\Psi(\Xi) \Vdash \Psi(H, \star) : (\Gamma \vdash p)$ in LJT_∂ .
- Case *I*. Suppose $\Xi \Vdash \lambda x^A.M : (\Gamma \vdash A \supset B)$ is obtained from $\Xi \Vdash M : (\Gamma, x : A \vdash B)$ in NJT_∂ . By the definition of Ψ , $\Psi(\lambda x^A.M) = \lambda x^A.\Psi(M)$. We now prove $\Psi(\Xi) \Vdash$

$\lambda x^A. \Psi(M) : (\Gamma \vdash A \supset B)$. By IH, $\Psi(\Xi) \Vdash \Psi(M) : (\Gamma, x : A \vdash B)$. Applying the *RI* rule, $\Psi(\Xi) \Vdash \lambda x^A. \Psi(M) : (\Gamma \vdash A \supset B)$ in LJT_∂ .

- Case *E*. Suppose $\Xi_1 @ \Xi_2 \Vdash HN : (\Gamma \triangleright B)$ is obtained from $\Xi_1 \Vdash H : (\Gamma \triangleright A \supset B)$ and $\Xi_2 \Vdash N : (\Gamma \vdash A)$ in NJT_∂ and $\Xi \Vdash l : (\Gamma; B \vdash p)$ in LJT_∂ . By the definition of Ψ , $\Psi(HN, l) = \Psi(H, \Psi(N) :: l)$. We now prove $(\Psi(\Xi_1) @ \Psi(\Xi_2)) @ \Xi \Vdash \Psi(H, \Psi(N) :: l) : (\Gamma \vdash p)$. By IH, $\Psi(\Xi_2) \Vdash \Psi(N) : (\Gamma \vdash A)$ in LJT_∂ . Applying *LI* rule, $\Psi(\Xi_2) @ \Xi \Vdash \Psi(N) :: l : (\Gamma; A \supset B \vdash p)$. By IH, $\Psi(\Xi_1) @ (\Psi(\Xi_2) @ \Xi) \Vdash \Psi(H, \Psi(N) :: l) : (\Gamma \vdash p)$ in LJT_∂ . Clearly, $(\Psi(\Xi_1) @ \Psi(\Xi_2)) @ \Xi = \Psi(\Xi_1) @ (\Psi(\Xi_2) @ \Xi)$

□

Before proving the final theorem of this chapter, which establishes the isomorphism between the terms and reductions of LJT_∂ and NJT_∂ , we prove the following lemma, which will be useful in such a proof.

Lemma 3.1.

1. If $l \rightarrow l'$ then $\Psi(H, l) \rightarrow \Psi(H, l')$.
2. If $H \rightarrow H'$ then $\Theta(H, l) \rightarrow \Theta(H', l)$.

Proof. The first item is proved by induction on H and the second item is proved by induction on l .

In (1),

- Case $H = x$. Suppose $l \rightarrow l'$. Then, $\Psi(x, l) = x \hat{\cdot} l \rightarrow x \hat{\cdot} l' = \Psi(x, l')$. The reduction step is the assumption.
- Case $H = H'N$. Suppose $l \rightarrow l'$. Then $\Psi(H'N, l) = \Psi(H', \Psi(N) :: l) \rightarrow \Psi(H', \Psi(N) :: l') = \Psi(H'N, l')$. The reduction step is by IH, given that $\Psi(N) :: l \rightarrow \Psi(N) :: l'$.

In (2),

- Case $l = \star$. Suppose $H \rightarrow H'$. Then, $\Theta(H, l) = \Theta(H, \star) = \text{app}(H) \rightarrow \text{app}(H') = \Theta(H', \star) = \Theta(H', l)$. The reduction step is by the assumption.
- Case $l = u :: l'$. Suppose $H \rightarrow H'$. Then, $\Theta(H, l) = \Theta(H, u :: l') = \Theta(H\Theta(u), l') \rightarrow \Theta(H'\Theta(u), l') = \Theta(H', u :: l') = \Theta(H', l)$. The reduction step is by IH, given that $H\Theta(u) \rightarrow H'\Theta(u)$.

- Case $l = \underline{\Gamma; A \vdash p}$. Suppose $H \rightarrow H'$. Then, $\Theta(H, l) = \Theta(H, \underline{\Gamma; A \vdash p}) = \text{app}(H, (\underline{\Gamma \triangleright A}), p) \rightarrow \text{app}(H', (\underline{\Gamma \triangleright A}), p) = \Theta(H', \underline{\Gamma; A \vdash p}) = \Theta(H', l)$. The reduction step is by the assumption.

□

Finally, we prove the isomorphism between LJT_∂ and NJT_∂ .

Theorem 3.4 ($LJT_\partial \cong NJT_\partial$).

1. $\Theta(\Psi(M)) = M$ and $\Psi(\Theta(t)) = t$.
2. $t \rightarrow t'$ in LJT_∂ iff $\Theta(t) \rightarrow \Theta(t')$ in NJT_∂ .
3. $M \rightarrow M'$ in NJT_∂ iff $\Psi(M) \rightarrow \Psi(M')$ in LJT_∂ .

Proof. The proof of the first item is an extension of the proof found in [13]. The statement $\Theta(\Psi(M)) = M$ is proved together with $\Theta(\Psi(H, l)) = \Theta(H, l)$. The proof is by simultaneous induction on M and H .

- Case $M = \lambda x^A.M'$.

$$\begin{aligned} \Theta(\Psi(\lambda x^A.M')) &= \Theta(\lambda x^A.\Psi(M')) \quad (\text{def. of } \Psi) \\ &= \lambda x^A.\Theta(\Psi(M')) \quad (\text{def. of } \Theta) \\ &= \lambda x^A.M' \quad (\text{IH}); \end{aligned}$$

- Case $M = \text{app}(H)$.

$$\begin{aligned} \Theta(\Psi(\text{app}(H))) &= \Theta(\Psi(H, \star)) \quad (\text{def. of } \Psi) \\ &= \Theta(H, \star) \quad (\text{IH}) \\ &= \text{app}(H) \quad (\text{def. of } \Theta); \end{aligned}$$

- Case $M = \underline{\sigma}$.

$$\begin{aligned} \Theta(\Psi(\underline{\sigma})) &= \Theta(\underline{\sigma}) \quad (\text{def. of } \Psi) \\ &= \underline{\sigma} \quad (\text{def. of } \Theta); \end{aligned}$$

- Case $M = \text{app}(H, (\underline{\Gamma \triangleright A}), p)$.

$$\begin{aligned} \Theta(\Psi(\text{app}(H, (\underline{\Gamma \triangleright A}), p))) &= \Theta(\Psi(H, (\underline{\Gamma; A \vdash p}))) \quad (\text{def. of } \Psi) \\ &= \Theta(H, (\underline{\Gamma; A \vdash p})) \quad (\text{IH}) \\ &= \text{app}(H, (\underline{\Gamma \triangleright A}), p) \quad (\text{def. of } \Theta); \end{aligned}$$

- Case $H = x$.

$$\begin{aligned}\Theta(\Psi(x, l)) &= \Theta(x \hat{l}) \quad (\text{def. of } \Psi) \\ &= \Theta(x, l) \quad (\text{def. of } \Theta);\end{aligned}$$

- Case $H = H'N$.

$$\begin{aligned}\Theta(\Psi(H'N, l)) &= \Theta(\Psi(H', \Psi(N) :: l)) \quad (\text{def. of } \Psi) \\ &= \Theta(H', \Psi(N) :: l) \quad (\text{IH}) \\ &= \Theta(H' \Theta(\Psi(N)), l) \quad (\text{def. of } \Theta) \\ &= \Theta(H'N, l) \quad (\text{IH}).\end{aligned}$$

On the other hand, $\Psi(\Theta(t)) = t$ is proved together with $\Psi(\Theta(H, l)) = \Psi(H, l)$ and the proof is made by simultaneous induction on t and l .

- Case $t = \lambda x^A.t'$.

$$\begin{aligned}\Psi(\Theta(\lambda x^A.t')) &= \Psi(\lambda x^A.\Theta(t')) \quad (\text{def. of } \Theta) \\ &= \lambda x^A.\Psi(\Theta(t')) \quad (\text{def. of } \Psi) \\ &= \lambda x^A.t' \quad (\text{IH});\end{aligned}$$

- Case $t = x \hat{l}$.

$$\begin{aligned}\Psi(\Theta(x \hat{l})) &= \Psi(\Theta(x, l)) \quad (\text{def. of } \Theta) \\ &= \Psi(x, l) \quad (\text{IH}) \\ &= x \hat{l} \quad (\text{def. of } \Psi);\end{aligned}$$

- Case $t = \underline{\sigma}$.

$$\begin{aligned}\Psi(\Theta(\underline{\sigma})) &= \Psi(\underline{\sigma}) \quad (\text{def. of } \Theta) \\ &= \underline{\sigma} \quad (\text{def. of } \Psi);\end{aligned}$$

- Case $l = \star$.

$$\begin{aligned}\Psi(\Theta(H, \star)) &= \Psi(\text{app}(H)) \quad (\text{def. of } \Theta) \\ &= \Psi(H, \star) \quad (\text{def. of } \Psi);\end{aligned}$$

- Case $l = u :: l'$.

$$\begin{aligned}\Psi(\Theta(H, u :: l')) &= \Psi(\Theta(H \Theta(u), l')) \quad (\text{def. of } \Theta) \\ &= \Psi(H \Theta(u), l') \quad (\text{IH}) \\ &= \Psi(H, \Psi(\Theta(u)) :: l') \quad (\text{def. of } \Psi) \\ &= \Psi(H, u :: l') \quad (\text{IH});\end{aligned}$$

- Case $l = \underline{\Gamma; A \vdash p}$.

$$\begin{aligned}\Psi(\Theta(H, (\underline{\Gamma; A \vdash p}))) &= \Psi(\text{app}(H, (\underline{\Gamma \triangleright A}), p)) \quad (\text{def. of } \Theta) \\ &= \Psi(H, (\underline{\Gamma; A \vdash p})) \quad (\text{def. of } \Psi).\end{aligned}$$

Using the first item, we obtain the “if” statements of second and third items from the “only if” statements: in item (2), suppose $\Theta(t) \rightarrow \Theta(t')$. Then $t = \Psi(\Theta(t)) \rightarrow \Psi(\Theta(t')) = t'$, so $t \rightarrow t'$. Similar reasoning applies to item (3). It remains to prove the “only if” statements.

The “only if” statement of item (2) is proved together with: if $l \rightarrow l'$ in LJT_∂ then $\Theta(H, l) \rightarrow \Theta(H, l')$ in NJT_∂ , for all H . The proof is by simultaneous induction on $t \rightarrow t'$ and $l \rightarrow l'$.

Base Cases:

- Case IR . Suppose $t = \underline{\Gamma \vdash A \supset B} \rightarrow \lambda x^A.(\underline{\Gamma, x : A \vdash B}) = t'$. Then:

$$\begin{aligned}\Theta(\underline{\Gamma \vdash A \supset B}) &= \underline{\Gamma \vdash A \supset B} \quad (\text{def. of } \Theta) \\ &\rightarrow \lambda x^A.(\underline{\Gamma, x : A \vdash B}) \quad (IR) \\ &= \lambda x^A.\Theta(\underline{\Gamma, x : A \vdash B}) \quad (\text{def. of } \Theta) \\ &= \Theta(\lambda x^A.(\underline{\Gamma, x : A \vdash B})) \quad (\text{def. of } \Theta);\end{aligned}$$

- Case SFL . Suppose $t = \underline{\Gamma, x : A \vdash p} \rightarrow x^\wedge(\underline{\Gamma, x : A; A \vdash p}) = t'$. Then

$$\begin{aligned}\Theta(\underline{\Gamma, x : A \vdash p}) &= \underline{\Gamma, x : A \vdash p} \quad (\text{def. of } \Theta) \\ &\rightarrow \text{app}(x, (\underline{\Gamma, x : A \triangleright A}), p) \quad (SFD) \\ &= \Theta(x, (\underline{\Gamma, x : A; A \vdash p})) \quad (\text{def. of } \Theta) \\ &= \Theta(x^\wedge(\underline{\Gamma, x : A; A \vdash p})) \quad (\text{def. of } \Theta);\end{aligned}$$

- Case KFL . Suppose $l = \underline{\Gamma; A \supset B \vdash p} \rightarrow (\underline{\Gamma \vdash A}) :: (\underline{\Gamma; B \vdash p}) = l'$. Then

$$\begin{aligned}\Theta(H, (\underline{\Gamma; A \supset B \vdash p})) &= \text{app}(H, (\underline{\Gamma \triangleright A \supset B}), p) \quad (\text{def. of } \Theta) \\ &\rightarrow \text{app}(H(\underline{\Gamma \vdash A}), (\underline{\Gamma \triangleright B}), p) \quad (KFD) \\ &= \Theta(H(\underline{\Gamma \vdash A}), (\underline{\Gamma; B \vdash p})) \quad (\text{def. of } \Theta) \\ &= \Theta(H(\Theta(\underline{\Gamma \vdash A})), (\underline{\Gamma; B \vdash p})) \quad (\text{def. of } \Theta) \\ &= \Theta(H, (\underline{\Gamma \vdash A}) :: (\underline{\Gamma; B \vdash p})) \quad (\text{def. of } \Theta);\end{aligned}$$

- Case FFL . Suppose $l = \underline{\Gamma; p \vdash p} \rightarrow \star = l'$. Then

$$\begin{aligned}\Theta(H, (\underline{\Gamma; p \vdash p})) &= \text{app}(H, (\underline{\Gamma \triangleright p}), p) \quad (\text{def. of } \Theta) \\ &\rightarrow \text{app}(H) \quad (FFD) \\ &= \Theta(H, \star) \quad (\text{def. of } \Theta).\end{aligned}$$

Inductive Cases:

- Suppose $\lambda x^A.t \rightarrow \lambda x^A.t'$, with $t \rightarrow t'$. Then

$$\begin{aligned}\Theta(\lambda x^A.t) &= \lambda x^A.\Theta(t) \quad (\text{def. of } \Theta) \\ &\rightarrow \lambda x^A.\Theta(t') \quad (\text{IH}) \\ &= \Theta(\lambda x^A.t') \quad (\text{def. of } \Theta);\end{aligned}$$

- Suppose $x\hat{l} \rightarrow x\hat{l}'$, with $l \rightarrow l'$. Then

$$\begin{aligned}\Theta(x\hat{l}) &= \Theta(x, l) \quad (\text{def. of } \Theta) \\ &\rightarrow \Theta(x, l') \quad (\text{IH}) \\ &= \Theta(x\hat{l}') \quad (\text{def. of } \Theta);\end{aligned}$$

- Suppose $u :: l \rightarrow u' :: l$, with $u \rightarrow u'$. Then

$$\begin{aligned}\Theta(H, u :: l) &= \Theta(H\Theta(u), l) \quad (\text{def. of } \Theta) \\ &\rightarrow \Theta(H\Theta(u'), l) \quad (\text{IH \& Lemma 3.1}) \\ &= \Theta(H, u' :: l) \quad (\text{def. of } \Theta);\end{aligned}$$

- Suppose $u :: l \rightarrow u :: l'$, with $l \rightarrow l'$. Then

$$\begin{aligned}\Theta(H, u :: l) &= \Theta(H\Theta(u), l) \quad (\text{def. of } \Theta) \\ &\rightarrow \Theta(H\Theta(u), l') \quad (\text{IH}) \\ &= \Theta(H, u :: l') \quad (\text{def. of } \Theta).\end{aligned}$$

The “only if” statement of item (3) is proved together with: if $H \rightarrow H'$ in NJT_∂ then $\Psi(H, l) \rightarrow \Psi(H', l)$ in LJT_∂ , for all l . The proof is by simultaneous induction on $M \rightarrow M'$ and $H \rightarrow H'$.

Base Cases:

- Case *IR*. Suppose $M = \Gamma \vdash A \supset B \rightarrow \lambda x^A.(\Gamma, x : A \vdash B) = M'$. Then:

$$\begin{aligned}\Psi(\Gamma \vdash A \supset B) &= \Gamma \vdash A \supset B \quad (\text{def. of } \Psi) \\ &\rightarrow \lambda x^A.\Gamma, x : A \vdash B \quad (\text{IR}) \\ &= \lambda x^A.\Psi(\Gamma, x : A \vdash B) \quad (\text{def. of } \Psi) \\ &= \Psi(\lambda x^A.(\Gamma, x : A \vdash B)) \quad (\text{def. of } \Psi);\end{aligned}$$

- Case *SFD*. Suppose $M = \Gamma, x : A \vdash p \rightarrow \text{app}(x, \Gamma, x : A \triangleright A, p) = M'$. Then:

$$\begin{aligned}\Psi(\Gamma, x : A \vdash p) &= \Gamma, x : A \vdash p \quad (\text{def. of } \Psi) \\ &\rightarrow x\hat{(\Gamma, x : A; A \vdash p)} \quad (\text{SFL}) \\ &= \Psi(x, (\Gamma, x : A; A \vdash p)) \quad (\text{def. of } \Psi) \\ &= \Psi(\text{app}(x, \Gamma, x : A \triangleright A, p)) \quad (\text{def. of } \Psi);\end{aligned}$$

- Case *KFD*. Suppose $M = \text{app}(H, \underline{\Gamma \triangleright A \supset B}, p) \rightarrow \text{app}(H(\underline{\Gamma \vdash A}), \underline{\Gamma \triangleright B}, p) = M'$.

Then:

$$\begin{aligned}
\Psi(\text{app}(H, \underline{\Gamma \triangleright A \supset B}, p)) &= \Psi(H, (\underline{\Gamma}; A \supset B \vdash p)) && \text{(def. of } \Psi) \\
&\rightarrow \Psi(H, (\underline{\Gamma \vdash A}) :: (\underline{\Gamma}; B \vdash p)) && \text{(IH \& Lemma 3.1)} \\
&= \Psi(H, \Psi(\underline{\Gamma \vdash A}) :: (\underline{\Gamma}; B \vdash p)) && \text{(def. of } \Psi) \\
&= \Psi(H(\underline{\Gamma \vdash A}), (\underline{\Gamma}; B \vdash p)) && \text{(def. of } \Psi) \\
&= \Psi(\text{app}(H(\underline{\Gamma \vdash A}), \underline{\Gamma \triangleright B}, p)) && \text{(def. of } \Psi);
\end{aligned}$$

- Case *FFD*. Suppose $M = \text{app}(H, \underline{\Gamma \triangleright p}, p) \rightarrow \text{app}(H) = M'$. Then:

$$\begin{aligned}
\Psi(\text{app}(H, \underline{\Gamma \triangleright p}, p)) &= \Psi(H, (\underline{\Gamma}; p \vdash p)) && \text{(def. of } \Psi) \\
&\rightarrow \Psi(H, \star) && \text{(IH \& Lemma 3.1)} \\
&= \Psi(\text{app}(H)) && \text{(def. of } \Psi).
\end{aligned}$$

Inductive Cases:

- Suppose $\lambda x^A.M \rightarrow \lambda x^A.M'$, with $M \rightarrow M'$. Then

$$\begin{aligned}
\Psi(\lambda x^A.M) &= \lambda x^A.\Psi(M) && \text{(def. of } \Psi) \\
&\rightarrow \lambda x^A.\Psi(M') && \text{(IH)} \\
&= \Psi(\lambda x^A.M') && \text{(def. of } \Psi);
\end{aligned}$$

- Suppose $\text{app}(H) \rightarrow \text{app}(H')$, with $H \rightarrow H'$. Then

$$\begin{aligned}
\Psi(\text{app}(H)) &= \Psi(H, \star) && \text{(def. of } \Psi) \\
&\rightarrow \Psi(H', \star) && \text{(IH)} \\
&= \Psi(\text{app}(H')) && \text{(def. of } \Psi);
\end{aligned}$$

- Suppose $\text{app}(H, \underline{\Gamma \triangleright A}, p) \rightarrow \text{app}(H', \underline{\Gamma \triangleright A}, p)$, with $H \rightarrow H'$. Then

$$\begin{aligned}
\Psi(\text{app}(H, \underline{\Gamma \triangleright A}, p)) &= \Psi(H, \underline{\Gamma}; A \vdash p) && \text{(def. of } \Psi) \\
&\rightarrow \Psi(H', \underline{\Gamma}; A \vdash p) && \text{(IH)} \\
&= \Psi(\text{app}(H', \underline{\Gamma \triangleright A}, p)) && \text{(def. of } \Psi);
\end{aligned}$$

- Suppose $HN \rightarrow HN'$, with $N \rightarrow N'$. Then

$$\begin{aligned}
\Psi(HN, l) &= \Psi(H, \Psi(N) :: l) && \text{(def. of } \Psi) \\
&\rightarrow \Psi(H, \Psi(N') :: l) && \text{(IH \& Lemma 3.1)} \\
&= \Psi(HN', l) && \text{(def. of } \Psi);
\end{aligned}$$

- Suppose $HN \rightarrow H'N$, with $H \rightarrow H'$. Then

$$\begin{aligned}
\Psi(HN, l) &= \Psi(H, \Psi(N) :: l) && \text{(def. of } \Psi) \\
&\rightarrow \Psi(H', \Psi(N) :: l) && \text{(IH)} \\
&= \Psi(H'N, l) && \text{(def. of } \Psi).
\end{aligned}$$

□

The previous theorem says that reduction of $t \in LJT_\partial$ is isomorphic to reduction of $M = \Theta(t) \in NJT_\partial$. If we consider the particular case of $t = M = \underline{\sigma}$, and given that, according to Theorems 2.2 and 3.2, reduction of $\underline{\sigma}$ in LJT_∂ (resp. NJT_∂) implements proof search in LJT (resp. NJT), we may conclude:

Corollary 3.1. Proof search in LJT is isomorphic to proof search in NJT .

Chapter 4

Focused Proof Search in Classical Logic

In this chapter, we investigate whether the correspondence between focusing in LJT and intercalation in NJT can be lifted to classical logic, using the same tools of our previous chapters. In classical logic, the precise definition of the proof systems and associated search procedures is less off-the-shelf. Sequent calculus LKT has its origin in linear logic [8] but has not been developed as a sequent calculus with proof terms; and the intercalation calculus of [54] is not a natural deduction system in the tradition of Parigot's classical natural deduction and the corresponding $\lambda\mu$ -calculus [49] – a tradition we will adopt in this work. The precise definitions of systems LKT and NKT and associated search procedures are already a contribution. Then, we confirm that the methodology of partial proof terms can give an adequate representation of the two search procedures. Finally, using this representation, we can prove that the correspondence of procedures previously observed in the intuitionistic case indeed lifts to the classical case.

The results discussed in this chapter were partially presented in the paper [26].

4.1 Preliminaries

In this chapter, we will assume the following definition of formulas, including absurdity.

Definition 4.1 (Formulas). Let $\mathcal{V}_P$ be the countable set of propositional variables, denoted by p, q , etc., and $\mathcal{A}_P = \mathcal{V}_P \cup \{\supset\} \cup \{(\, , \,)\}$. The set of all words in $\mathcal{A}_P$ is represented by $\mathcal{A}_P^*$. The set of formulas, represented by $\mathcal{F}_P$, is defined inductively on $\mathcal{A}_P^*$ by:

$$A, B ::= p \mid \perp \mid (A \supset B)$$

By convention, the outermost brackets of each formula can be omitted.

In contrast to intuitionistic logic, classical sequents will have sets in both the antecedent (on the left-hand side) and the consequent (on the right-hand side).

Definition 4.2 (Context). Let x be a variable and A a formula. Contexts Γ appear on the left-hand side of the sequents and range over sets of pairs (x, A) , written $(x : A)$, understood as assumptions. In Γ , each variable x occurs in at most one assumption.

Definition 4.3 (Succedent Context). Let a be a co-variable and A a formula. Succedent contexts Δ appear on the right-hand side of the sequents and range over sets of pairs (a, A) , written $(a : A)$, understood as alternative conclusions. In Δ , each co-variable a occurs in at most one alternative conclusion.

4.2 Sequent Calculus LKT

In this section, we consider the focused, cut-free sequent calculus LKT for classical propositional logic, which is a variant of the T -fragment of $\bar{\lambda}\mu\tilde{\mu}$ [7], and it is also the classical version of LJT [35, 25], presented in Chapter 2.

The sequent calculus LKT handles three kinds of sequents.

Definition 4.4 (Sequents of LKT). LKT handles sequents $\Gamma \vdash A \mid \Delta$ and $\Gamma; A \vdash \Delta$ and $\Gamma \vdash \Delta$, ranged over by σ , τ , and ζ , respectively. In the first form, the sequent shows a distinguished formula, in the right-hand side, which is not a focus, but just a selected conclusion; in the second form, the sequent shows a distinguished formula A in the left-hand side, which, as in LJT , we call the focus of the sequent; and, in the third form, there is no distinguished formula.

Definition 4.5 (Inference Rules of LKT). The inference rules of LKT are the following:

$$\begin{array}{c} \frac{\Gamma, x : A \vdash B \mid \Delta}{\Gamma \vdash A \supset B \mid \Delta} RI \quad \frac{\Gamma \vdash a : A, \Delta}{\Gamma \vdash A \mid \Delta} ACT \\[2ex] \frac{\Gamma, x : A; A \vdash \Delta}{\Gamma, x : A \vdash \Delta} LCTR \quad \frac{\Gamma, x : A \vdash B \mid a : A \supset B, \Delta}{\Gamma \vdash a : A \supset B, \Delta} RCTR \\[2ex] \frac{}{\Gamma; p \vdash a : p, \Delta} AX \quad \frac{}{\Gamma; \perp \vdash \Delta} L\perp \quad \frac{\Gamma \vdash A \mid \Delta \quad \Gamma; B \vdash \Delta}{\Gamma; A \supset B \vdash \Delta} LI \end{array}$$

These rules are the axiom AX ; the right and left introduction rules (RI , LI) for implication, the left introduction rule for absurdity ($L\perp$) and three structural rules:

1. the activation rule ACT , which selects a formula in the right-hand side;

Definition 4.6 (Proof Terms of LKT). Proof terms t, u , commands c , and proof lists l of LKT are defined by the following grammar:

$$\begin{aligned} \text{(Proof terms)} \quad t, u &::= \lambda x^A.t \mid \mu a^A.c \\ \text{(Commands)} \quad c &::= \langle x \mid l \rangle \mid \langle \lambda x^A.t \mid a \rangle \\ \text{(Proof lists)} \quad l &::= a \mid \text{abort} \mid u :: l. \end{aligned}$$

LKT with proof terms handles three kinds of total sequents.

Definition 4.7 (Total Sequents of LKT with proof terms). LKT with proof terms handles proof term sequents $\Gamma \vdash t : A \mid \Delta$, proof list sequents $\Gamma; l : A \vdash \Delta$ and command sequents $\Gamma \vdash^c \Delta$.

Definition 4.8 (Inference rules of LKT with proof terms). The set of inference rules of LKT with proof terms is:

$$\begin{aligned} &\frac{\Gamma, x : A \vdash t : B \mid \Delta}{\Gamma \vdash \lambda x^A.t : A \supset B \mid \Delta} RI \quad \frac{\Gamma \vdash^c a : A, \Delta}{\Gamma \vdash \mu a^A.c : A \mid \Delta} ACT \\ &\frac{\Gamma, x : A; l : A \vdash \Delta}{\Gamma, x : A \mid \frac{\langle x \mid l \rangle}{\Delta}} LCTR \quad \frac{\Gamma, x : A \vdash t : B \mid a : A \supset B, \Delta}{\Gamma \mid \frac{\langle \lambda x^A.t \mid a \rangle}{a : A \supset B, \Delta}} RCTR \\ &\frac{}{\Gamma; a : p \vdash a : p, \Delta} AX \quad \frac{}{\Gamma; \text{abort} : \perp \vdash \Delta} L\perp \quad \frac{\Gamma \vdash u : A \mid \Delta \quad \Gamma; l : B \vdash \Delta}{\Gamma; u :: l : A \supset B \vdash \Delta} LI \end{aligned}$$

Each proof constructor witnesses the use of a certain inference. Therefore, if $\Gamma \vdash t : A \mid \Delta$ is derivable, t encodes the derivation of $\Gamma \vdash A \mid \Delta$.

Example 4.2. Adding proof terms to the derivation of Example 4.1, we have the following derivation in LKT with proof terms:

$$\begin{aligned} &\frac{x : (p \supset q) \supset p, y : p; b : p \vdash c : q, \Delta}{x : (p \supset q) \supset p, y : p \mid \frac{\langle y \mid b \rangle}{c : q, \Delta}} AX \\ &\frac{x : (p \supset q) \supset p, y : p \mid \frac{\langle y \mid b \rangle}{c : q, \Delta}}{x : (p \supset q) \supset p, y : p \vdash \mu c^q. \langle y \mid b \rangle : q \mid \Delta} LCTR \\ &\frac{x : (p \supset q) \supset p, y : p \vdash \mu c^q. \langle y \mid b \rangle : q \mid \Delta}{x : (p \supset q) \supset p, y : p \vdash \lambda y^p. (\mu c^q. \langle y \mid b \rangle) : p \supset q \mid \Delta} ACT \\ &\frac{x : (p \supset q) \supset p, y : p \vdash \lambda y^p. (\mu c^q. \langle y \mid b \rangle) : p \supset q \mid \Delta}{x : (p \supset q) \supset p; b : p \vdash \Delta} RI \\ &\frac{x : (p \supset q) \supset p; b : p \vdash \Delta}{x : (p \supset q) \supset p; ((\lambda y^p. (\mu c^q. \langle y \mid b \rangle)) :: b) : (p \supset q) \supset p \vdash \Delta} LI \\ &\frac{x : (p \supset q) \supset p; ((\lambda y^p. (\mu c^q. \langle y \mid b \rangle)) :: b) : (p \supset q) \supset p \vdash \Delta}{x : (p \supset q) \supset p \mid \frac{\langle x \mid ((\lambda y^p. (\mu c^q. \langle y \mid b \rangle)) :: b) \rangle}{b : p, a : ((p \supset q) \supset p) \supset p}} LCTR \\ &\frac{x : (p \supset q) \supset p \mid \frac{\langle x \mid ((\lambda y^p. (\mu c^q. \langle y \mid b \rangle)) :: b) \rangle}{b : p, a : ((p \supset q) \supset p) \supset p}}{x : (p \supset q) \supset p \vdash \mu b^p. \langle x \mid ((\lambda y^p. (\mu c^q. \langle y \mid b \rangle)) :: b) \rangle : p \mid a : ((p \supset q) \supset p) \supset p} ACT \\ &\frac{x : (p \supset q) \supset p \vdash \mu b^p. \langle x \mid ((\lambda y^p. (\mu c^q. \langle y \mid b \rangle)) :: b) \rangle : p \mid a : ((p \supset q) \supset p) \supset p}{\vdash \mu a^{((p \supset q) \supset p) \supset p}. \langle \lambda x^{(p \supset q) \supset p}. \mu b^p. \langle x \mid ((\lambda y^p. (\mu c^q. \langle y \mid b \rangle)) :: b) \rangle \mid a \rangle : ((p \supset q) \supset p) \supset p \mid} RCTR \\ &\frac{\vdash \mu a^{((p \supset q) \supset p) \supset p}. \langle \lambda x^{(p \supset q) \supset p}. \mu b^p. \langle x \mid ((\lambda y^p. (\mu c^q. \langle y \mid b \rangle)) :: b) \rangle \mid a \rangle : ((p \supset q) \supset p) \supset p \mid}{\vdash \mu a^{((p \supset q) \supset p) \supset p}. \langle \lambda x^{(p \supset q) \supset p}. \mu b^p. \langle x \mid ((\lambda y^p. (\mu c^q. \langle y \mid b \rangle)) :: b) \rangle \mid a \rangle : ((p \supset q) \supset p) \supset p \mid} ACT \end{aligned}$$

where $\Delta = \{b : p, a : ((p \supset q) \supset p) \supset p\}$.

Thus, the proof term $\mu a^{((p \supset q) \supset p) \supset p}. \langle \lambda x^{(p \supset q) \supset p}. \mu b^p. \langle x \mid ((\lambda y^p. (\mu c^q c. \langle y \mid b \rangle)) :: b) \rangle \mid a \rangle$ encodes the derivation of the sequent $\vdash ((p \supset q) \supset p) \supset p$, witnessing that Peirce's Law is a theorem.

Weakening rules are admissible in LKT with proof terms.

Proposition 4.1 (Admissibility of Left Weakening in LKT). The left weakening rules

$$\frac{\Gamma \vdash t : B \mid \Delta}{\Gamma, x : A \vdash t : B \mid \Delta} \quad \frac{\Gamma; l : B \vdash \Delta}{\Gamma, x : A; l : B \vdash \Delta} \quad \frac{\Gamma \vdash^c \Delta}{\Gamma, x : A \vdash^c \Delta}$$

are admissible in LKT .

Proof. By simultaneous induction on the derivations of $\Gamma \vdash t : B \mid \Delta$, $\Gamma; l : B \vdash \Delta$, and $\Gamma \vdash^c \Delta$.

- Case AX . Suppose $\Gamma; a : p \vdash a : p, \Delta$. Applying the AX rule, $\Gamma, x : A; a : p \vdash a : p, \Delta$.
- Case $L\perp$. Suppose $\Gamma; \text{abort} : \perp \vdash \Delta$. Applying the $L\perp$ rule, $\Gamma, x : A; \text{abort} : \perp \vdash \Delta$.
- Case ACT . Suppose $\Gamma \vdash \mu b^B.c : B \mid \Delta$ is obtained from $\Gamma \vdash^c b : B, \Delta$. By IH, $\Gamma, x : A \vdash^c b : B, \Delta$. Applying the ACT rule, $\Gamma, x : A \vdash \mu b^B.c : B \mid \Delta$.
- Case $LCTR$. Suppose $\Gamma, y : B \vdash^{\langle y \mid l \rangle} \Delta$ is obtained from $\Gamma, y : B; l : B \vdash \Delta$. By IH, $\Gamma, y : B, x : A; l : B \vdash \Delta$. Applying the $LCTR$ rule, $\Gamma, y : B, x : A \vdash^{\langle y \mid l \rangle} \Delta$.
- Case $RCTR$. Suppose $\Gamma \vdash^{\langle \lambda y^B.t \mid b \rangle} b : B \supset C, \Delta$ is obtained from $\Gamma, y : B \vdash t : C \mid b : B \supset C, \Delta$. By IH, $\Gamma, y : B, x : A \vdash t : C \mid b : B \supset C, \Delta$. Applying the $RCTR$ rule, $\Gamma, x : A \vdash^{\langle \lambda y^B.t \mid b \rangle} b : B \supset C, \Delta$.
- Case RI . Suppose $\Gamma \vdash \lambda y^B.t : B \supset C \mid \Delta$ is obtained from $\Gamma, y : B \vdash t : C \mid \Delta$. By IH, $\Gamma, y : B, x : A \vdash t : C \mid \Delta$. Applying the RI rule, $\Gamma, x : A \vdash \lambda y^B.t : B \supset C \mid \Delta$.
- Case LI . Suppose $\Gamma; u :: l : C \supset D \vdash \Delta$ is obtained from $\Gamma \vdash u : C \mid \Delta$ and $\Gamma; l : D \vdash \Delta$. By IH, $\Gamma, x : A \vdash u : C \mid \Delta$ and $\Gamma, x : A; l : D \vdash \Delta$. Applying the LI rule, $\Gamma, x : A; u :: l : C \supset D \vdash \Delta$.

□

Proposition 4.2 (Admissibility of Right Weakening in LKT). The right weakening rules

$$\frac{\Gamma \vdash t : B \mid \Delta}{\Gamma \vdash t : B \mid \Delta, a : A} \quad \frac{\Gamma; l : B \vdash \Delta}{\Gamma; l : B \vdash \Delta, a : A} \quad \frac{\Gamma \vdash^c \Delta}{\Gamma \vdash^c \Delta, a : A}$$

are admissible in LKT .

Proof. By simultaneous induction on the derivations of $\Gamma \vdash t : A \mid \Delta$, $\Gamma; l : A \vdash \Delta$, and $\Gamma \vdash^c \Delta$.

- Case *AX*. Suppose $\Gamma; b : p \vdash b : p, \Delta$. Applying the *AX* rule, $\Gamma; b : p \vdash b : p, \Delta, a : A$.
- Case *L $\perp$* . Suppose $\Gamma; \text{abort} : \perp \vdash \Delta$. Applying the *L $\perp$* rule, $\Gamma; \text{abort} : \perp \vdash \Delta, a : A$.
- Case *ACT*. Suppose $\Gamma \vdash \mu b^B.c : B \mid \Delta$ is obtained from $\Gamma \vdash^c b : B, \Delta$. By IH, $\Gamma \vdash^c b : B, \Delta, a : A$. Applying the *ACT* rule, $\Gamma \vdash \mu b^B.c : B \mid \Delta, a : A$.
- Case *LCTR*. Suppose $\Gamma, y : B \vdash^{\langle y \mid l \rangle} \Delta$ is obtained from $\Gamma, y : B; l : B \vdash \Delta$. By IH, $\Gamma, y : B; l : B \vdash \Delta, a : A$. Applying the *LCTR* rule, $\Gamma, y : B \vdash^{\langle y \mid l \rangle} \Delta, a : A$.
- Case *RCTR*. Case *RCTR*. Suppose $\Gamma \vdash^{\langle \lambda y^B.t \mid b \rangle} b : B \supset C, \Delta$ is obtained from $\Gamma, y : B \vdash t : C \mid b : B \supset C, \Delta$. By IH, $\Gamma, y : B \vdash t : C \mid b : B \supset C, \Delta, a : A$. Applying the *RCTR* rule, $\Gamma \vdash^{\langle \lambda y^B.t \mid b \rangle} b : B \supset C, \Delta, a : A$.
- Case *RI*. Suppose $\Gamma \vdash \lambda y^B.t : B \supset C \mid \Delta$ is obtained from $\Gamma, y : B \vdash t : C \mid \Delta$. By IH, $\Gamma, y : B \vdash t : C \mid \Delta, a : A$. Applying the *RI* rule, $\Gamma \vdash \lambda y^B.t : B \supset C \mid \Delta, a : A$.
- Case *LI*. Suppose $\Gamma; u :: l : C \supset D \vdash \Delta$ is obtained from $\Gamma \vdash u : C \mid \Delta$ and $\Gamma; l : D \vdash \Delta$. By IH, $\Gamma \vdash u : C \mid \Delta, a : A$ and $\Gamma; l : D \vdash \Delta, a : A$. Applying the *LI* rule, $\Gamma; u :: l : C \supset D \vdash \Delta, a : A$.

□

Admissible inference rules are witnessed by admissible proof-term operations.

Theorem 4.1 (Admissible cut rules of *LKT*). Consider the equations

$$\begin{aligned}
[u/x](\lambda y^A.t) &= \lambda y^A.[u/x]t & [l/a]\langle x \mid l' \rangle &= \langle x \mid [l/a]l' \rangle \\
[u/x](\mu a^A.c) &= \mu a^A.[u/x]c & [u :: l/a]\langle \lambda x^A.t \mid a \rangle &= ([u/x]([u :: l/a]t))@l \\
& & [l/a]\langle \lambda x^A.t \mid b \rangle &= \langle \lambda x^A.[l/a]t \mid b \rangle \quad (b \neq a) \\
[u/x]\langle x \mid l \rangle &= u@[u/x]l \\
[u/x]\langle y \mid l \rangle &= \langle y \mid [u/x]l \rangle \quad (y \neq x) & [l/a]a &= l \\
[u/x]\langle \lambda y^A.t \mid a \rangle &= \langle \lambda y^A.[u/x]t \mid a \rangle & [l/a]b &= b \quad (b \neq a) \\
& & [l/a]\text{abort} &= \text{abort} \\
[u/x]a &= a & [l/a](u :: l') &= ([l/a]u) :: ([l/a]l') \\
[u/x]\text{abort} &= \text{abort} \\
[u/x](t :: l) &= ([u/x]t) :: ([u/x]l) & [l/a](\lambda x^A.t) &= \lambda x^A.[l/a]t \\
& & [l/a](\mu b^B.c) &= \mu b^B.[l/a]c
\end{aligned}$$

$$\begin{aligned}
(\lambda y^A.t)@(u :: l) &= ([u/y]t)@l \\
(\mu a^A.c)@l &= [l/a]c
\end{aligned}$$

In LKT one has:

- (1) For all A, t : if $\Gamma \vdash u : A \mid \Delta$ and $\Gamma, x : A \vdash t : B \mid \Delta$, then the equations define the proof term $[u/x]t$ and $\Gamma \vdash [u/x]t : B \mid \Delta$.
- (2) For all A, c : if $\Gamma \vdash u : A \mid \Delta$ and $\Gamma, x : A \vdash^c \Delta$, then the equations define the command $[u/x]c$ and $\Gamma \vdash^{[u/x]c} \Delta$.
- (3) For all A, l : if $\Gamma \vdash u : A \mid \Delta$ and $\Gamma, x : A; l : B \vdash \Delta$, then the equations define the proof list $[u/x]l$ and $\Gamma; [u/x]l : B \vdash \Delta$.
- (4) For all A, u : if $\Gamma \vdash u : A \mid \Delta$ and $\Gamma; l : A \vdash \Delta$, then the equations define the command $u@l$ and $\Gamma \vdash^{u@l} \Delta$.
- (5) For all A, c : if $\Gamma \vdash^c \Delta, a : A$ and $\Gamma; l : A \vdash \Delta$, then the equations define the command $[l/a]c$ and $\Gamma \vdash^{[l/a]c} \Delta$.
- (6) For all A, l' : if $\Gamma; l' : B \vdash \Delta, a : A$ and $\Gamma; l : A \vdash \Delta$, then the equations define the proof list $[l/a]l'$ and $\Gamma; [l/a]l' : B \vdash \Delta$.
- (7) For all A, u : if $\Gamma \vdash u : B \mid \Delta, a : A$ and $\Gamma; l : A \vdash \Delta$, then the equations define the proof term $[l/a]u$ and $\Gamma \vdash [l/a]u : B \mid \Delta$.

As consequence, the following cut rules:

$$\frac{\Gamma \vdash u : A \mid \Delta \quad \Gamma, x : A \vdash t : B \mid \Delta}{\Gamma \vdash [u/x]t : B \mid \Delta} \quad (1) \qquad \frac{\Gamma \vdash^c \Delta, a : A \quad \Gamma; l : A \vdash \Delta}{\Gamma \vdash^{[l/a]c} \Delta} \quad (5)$$

$$\frac{\Gamma \vdash u : A \mid \Delta \quad \Gamma, x : A \vdash^c \Delta}{\Gamma \vdash^{[u/x]c} \Delta} \quad (2) \qquad \frac{\Gamma; l' : B \vdash \Delta, a : A \quad \Gamma; l : A \vdash \Delta}{\Gamma; [l/a]l' : B \vdash \Delta} \quad (6)$$

$$\frac{\Gamma \vdash u : A \mid \Delta \quad \Gamma, x : A; l : B \vdash \Delta}{\Gamma; [u/x]l : B \vdash \Delta} \quad (3) \qquad \frac{\Gamma \vdash u : B \mid \Delta, a : A \quad \Gamma; l : A \vdash \Delta}{\Gamma \vdash [l/a]u : B \mid \Delta} \quad (7)$$

$$\frac{\Gamma \vdash u : A \mid \Delta \quad \Gamma; l : A \vdash \Delta}{\Gamma \vdash^{u@l} \Delta} \quad (4)$$

are admissible in LKT .

Proof. For all A, t, c and l , the size $|A|$ is defined by recursion on A , as follows: $|p| = 1$, $|\perp| = 1$, $|A \supset B| = 1 + |A| + |B|$; and the sizes $|t|$, $|c|$ and $|l|$ are defined by simultaneous structural recursion on A, t, c and l as follows: $|\lambda x^A.t| = 1 + |t|$, $|\mu a^A.c| = 1 + |c|$, $|\langle x | l \rangle| = 2 + |l|$, $|\langle \lambda x^A.t | a \rangle| = 3 + |t|$, $|a| = 1$, $|\text{abort}| = 1$ and $|u :: l| = 1 + |u| + |l|$. The proof of items (1) to (7) is made by simultaneous induction on the lexicographically ordered tuples: $(|A|, 1, |t|)$, $(|A|, 1, |c|)$, $(|A|, 1, |l|)$, $(|A|, 0, |u|)$, $(|A|, 0, |c|)$, $(|A|, 0, |l'|)$ and $(|A|, 0, |u|)$.

In (1),

- Case $t = \lambda y^C.t'$. Suppose $\Gamma \vdash u : A \mid \Delta$ and $\Gamma, x : A \vdash t : B \mid \Delta$. We show that the above equations define the proof term $\lambda y^C.[u/x]t'$ and that $\Gamma \vdash \lambda y^C.[u/x]t' : B \mid \Delta$. By inversion of the inference rule RI , $B = C \supset D$ and $\Gamma, x : A, y : C \vdash t' : D \mid \Delta$. By admissibility of left weakening, $\Gamma, y : C \vdash u : A \mid \Delta$. Now, $(|A|, 1, |t'|) < (|A|, 1, |t|)$, because $|t'| < |t| = 1 + |t'|$. By IH(1), the above equations define the proof term $[u/x]t'$ and $\Gamma, y : C \vdash [u/x]t' : D \mid \Delta$. Hence, $\lambda y^C.[u/x]t'$ is defined. Applying the inference rule RI , $\Gamma \vdash \lambda y^C.[u/x]t' : C \supset D \mid \Delta$.
- Case $t = \mu b^B.c$. Suppose $\Gamma \vdash u : A \mid \Delta$ and $\Gamma, x : A \vdash t : B \mid \Delta$. We show that the above equations define the proof term $\mu b^B.[u/x]c$ and that $\Gamma \vdash \mu b^B.[u/x]c : B \mid \Delta$. By inversion of the inference rule ACT , $\Gamma, x : A \vdash^c b : B, \Delta$. By admissibility of right weakening, $\Gamma \vdash u : A \mid b : B, \Delta$. Now $(|A|, 1, |c|) < (|A|, 1, |\mu b^B.c|)$, because $|c| < |\mu b^B.c| = 1 + |c|$. By IH(2), the above equations define the command $[u/x]c$ and $\Gamma \vdash^{\frac{[u/x]c}{b}} b : B, \Delta$. Hence, $\mu b^B.[u/x]c$ is defined. Applying the inference rule ACT , $\Gamma \vdash \mu b^B.[u/x]c : B \mid \Delta$.

In (2),

- Case $c = \langle x | l \rangle$. Suppose $\Gamma \vdash u : A \mid \Delta$ and $\Gamma, x : A \vdash^c \Delta$. We show that the above equations define the command $u@[u/x]l$ and that $\Gamma \vdash^{\frac{u@[u/x]l}{\Delta}} \Delta$. By inversion of the inference rule $LCTR$, $\Gamma, x : A; l : A \vdash \Delta$. Now $(|A|, 1, |l|) < (|A|, 1, |\langle x | l \rangle|)$, because $|l| < |\langle x | l \rangle| = 2 + |l|$. By IH(3), the above equations define the proof list $[u/x]l$ and $\Gamma; [u/x]l : A \vdash \Delta$. Now, $(|A|, 0, |u|) < (|A|, 1, |\langle x | l \rangle|)$, because $0 < 1$. By IH(4), the above equations define the command $u@[u/x]l$ and $\Gamma \vdash^{\frac{u@[u/x]l}{\Delta}} \Delta$.
- Case $c = \langle y | l \rangle$, with $y \neq x$. Suppose $\Gamma, y : B \vdash u : A \mid \Delta$ and $\Gamma, y : B, x : A \vdash^c \Delta$. We show that the above equations define the command $\langle y | [u/x]l \rangle$ and that $\Gamma, y : B \vdash^{\frac{\langle y | [u/x]l \rangle}{\Delta}} \Delta$.

By inversion of the inference rule $LCTR$, $\Gamma, y : B, x : A; l : B \vdash \Delta$. Now $(|A|, 1, |l|) < (|A|, 1, |\langle y \mid l \rangle|)$, because $|l| < |\langle y \mid l \rangle| = 2 + |l|$. By IH(3), the above equations define the proof list $[u/x]l$ and $\Gamma, y : B; [u/x]l : B \vdash \Delta$. Hence, $\langle y \mid [u/x]l \rangle$ is defined. Applying the inference rule $LCTR$, $\Gamma, y : B \mid \frac{\langle y \mid [u/x]l \rangle}{\Delta}$.

- Case $c = \langle \lambda y^C. t \mid a \rangle$. Suppose $\Gamma \vdash u : A \mid \Delta, a : C \supset D$ and $\Gamma, x : A \mid \Delta, a : C \supset D$. We show that the above equations define the command $\langle \lambda y^C. [u/x]t \mid a \rangle$ and that $\Gamma \mid \frac{\langle \lambda y^C. [u/x]t \mid a \rangle}{\Delta, a : C \supset D}$. By inversion of the inference rule $RCTR$, $\Gamma, x : A, y : C \vdash t : D \mid \Delta, a : C \supset D$. By admissibility of left weakening, $\Gamma, y : C \vdash u : A \mid \Delta, a : C \supset D$. Now, $(|A|, 1, |t|) < (|A|, 1, |c|)$, because $|t| < |c| = 3 + |t|$. By IH(1), the above equations define the proof term $[u/x]t$ and $\Gamma, y : C \vdash [u/x]t : D \mid \Delta, a : C \supset D$. Hence, $\langle \lambda y^C. [u/x]t \mid a \rangle$ is defined. Applying the inference rule $RCTR$, $\Gamma \mid \frac{\langle \lambda y^C. [u/x]t \mid a \rangle}{\Delta, a : C \supset D}$.

In (3),

- Case $l = a$. Suppose $\Gamma \vdash u : A \mid \Delta$ and $\Gamma, x : A; l : B \vdash \Delta$. We show that $\Gamma; a : B \vdash \Delta$. By the AX rule, $B = p$ and $a : p \in \Delta$. Applying the AX rule, $\Gamma; a : p \vdash \Delta$.
- Case $l = \text{abort}$. Suppose $\Gamma \vdash u : A \mid \Delta$ and $\Gamma, x : A; l : B \vdash \Delta$. We show that $\Gamma; \text{abort} : B \vdash \Delta$. By the $L\perp$ rule, $B = \perp$ and, applying the same rule, $\Gamma; \text{abort} : \perp \vdash \Delta$. Thus, the above equations define the proof list abort .
- Case $l = t :: l'$. Suppose $\Gamma \vdash u : A \mid \Delta$ and $\Gamma, x : A; l : B \vdash \Delta$. We show that the above equations define the proof list $([u/x]t) :: ([u/x]l')$ and that $\Gamma; ([u/x]t) :: ([u/x]l') : B \vdash \Delta$. By inversion of the inference rule LI , $B = C \supset D$ and $\Gamma, x : A \vdash t : C \mid \Delta$ and $\Gamma, x : A; l' : D \vdash \Delta$. Now $(|A|, 1, |t|) < (|A|, 1, |l|)$, because $|t| < |l| = 1 + |t| + |l'|$. By IH(1), the above equations define the proof term $[u/x]t$ and $\Gamma \vdash [u/x]t : C \mid \Delta$. Also, $(|A|, 1, |l'|) < (|A|, 1, |l|)$, because $|l'| < |l| = 1 + |t| + |l'|$. By IH(3), the above equations define the proof list $[u/x]l'$ and $\Gamma; [u/x]l' : D \vdash \Delta$. Hence, $([u/x]t) :: ([u/x]l')$ is defined. Applying the inference rule LI , $\Gamma; ([u/x]t) :: ([u/x]l') : C \supset D \vdash \Delta$.

In (4),

- Case $u = \lambda y^B. t'$. Suppose $\Gamma \vdash u : A \mid \Delta$ and $\Gamma; l : A \vdash \Delta$. By inversion of the inference rules RI and LI , $A = B \supset C$ and $l = t :: l'$. We show that the above equations define the command $([t/y]t')@l'$ and that $\Gamma \mid \frac{([t/y]t')@l'}{\Delta}$. Again, by inversion of the inference rules RI and LI , $\Gamma, y : B \vdash t' : C \mid \Delta$, and $\Gamma \vdash t : B \mid \Delta$ and $\Gamma; l' : C \vdash \Delta$. Now $(|B|, 1, |t'|) <$

$(|A|, 0, |u|)$, because $|B| < |A| = 1 + |B| + |C|$. By IH(1), the above equations define the proof term $[t/y]t'$ and $\Gamma \vdash [t/y]t' : C \mid \Delta$. Also, $(|C|, 0, |[t/y]t'|) < (|A|, 0, |u|)$, because $|C| < |A| = 1 + |B| + |C|$. By IH(4), the above equations define the command $([t/y]t')@l'$ and $\Gamma \vdash \frac{([t/y]t')@l'}{ } \Delta$.

- Case $u = \mu a^A.c$. Suppose $\Gamma \vdash u : A \mid \Delta$ and $\Gamma; l : A \vdash \Delta$. We show that the above equations define the command $[l/a]c$ and that $\Gamma \vdash \frac{[l/a]c}{ } \Delta$. By inversion of the inference rule ACT , $\Gamma \vdash^c a : A, \Delta$. Now $(|A|, 0, |c|) < (|A|, 0, |u|)$, because $|c| < |u| = 1 + |c|$. By IH(5), the above equations define the command $[l/a]c$ and $\Gamma \vdash \frac{[l/a]c}{ } \Delta$.

In (5),

- Case $c = \langle x \mid l' \rangle$. Suppose $\Gamma \vdash^c \Delta, a : A$ and $\Gamma; l : A \vdash \Delta$. We show that the above equations define the command $\langle x \mid [l/a]l' \rangle$ and that $\Gamma \vdash \frac{\langle x \mid [l/a]l' \rangle}{ } \Delta$. By inversion of the inference rule $LCTR$, $\Gamma; l' : B \vdash \Delta, a : A$, with $x : B \in \Gamma$. Now $(|A|, 0, |l'|) < (|A|, 0, |c|)$, because $|l'| < |c| = 2 + |l'|$. By IH(6), the above equations define the proof list $[l/a]l'$ and $\Gamma; [l/a]l' : B \vdash \Delta$. Hence, $\langle x \mid [l/a]l' \rangle$ is defined. Applying the inference rule $LCTR$, $\Gamma \vdash \frac{\langle x \mid [l/a]l' \rangle}{ } \Delta$.
- Case $c = \langle \lambda x^B.t \mid a \rangle$. Suppose $\Gamma \vdash^c \Delta, a : A$ and $\Gamma; l : A \vdash \Delta$. By inversion of the inference rules RI and LI , $A = B \supset C$ and $l = u :: l'$. We show that the above equations define the command $([u/x]([u :: l'/a]t))@l'$ and that $\Gamma \vdash \frac{([u/x]([u :: l'/a]t))@l'}{ } \Delta$. Again, by inversion of the inference rules RI and LI , $\Gamma, x : B \vdash t : C \mid a : B \supset C, \Delta$ and $\Gamma \vdash u : B \mid \Delta$ and $\Gamma; l' : C \vdash \Delta$. By the admissibility of LW , $\Gamma, x : B; l : A \vdash \Delta$. Now $(|A|, 0, |t|) < (|A|, 0, |c|)$, because $|t| < |c| = 3 + |t|$. By IH(7), the above equations define the proof term $[u :: l'/a]t$ and $\Gamma, x : B \vdash [u :: l'/a]t : C \mid \Delta$. Again, $(|B|, 1, |u|) < (|A|, 0, |c|)$, because $|B| < |A| = 1 + |B| + |C|$. By IH(1), the above equations define the proof term $[u/x]([u :: l'/a]t)$ and $\Gamma \vdash [u/x]([u :: l'/a]t) : C \mid \Delta$. Finally, $(|C|, 0, |[u/x]([u :: l'/a]t)|) < (|A|, 0, |c|)$, because $|C| < |A| = 1 + |B| + |C|$. By IH(4), the above equations define the command $([u/x]([u :: l'/a]t))@l'$ and $\Gamma \vdash \frac{([u/x]([u :: l'/a]t))@l'}{ } \Delta$.
- Case $c = \langle \lambda x^B.t \mid b \rangle$, with $b \neq a$. Suppose $\Gamma \vdash^c \Delta, a : A$ and $\Gamma; l : A \vdash \Delta$. We show that the above equations define the command $\langle \lambda x^B.[l/a]t \mid b \rangle$ and that $\Gamma \vdash \frac{\langle \lambda x^B.[l/a]t \mid b \rangle}{ } \Delta$. By inversion of the inference rule $RCTR$, $A = B \supset C$ and $\Gamma, x : B \vdash t : C \mid \Delta, a : A$, with $(b : B \supset C) \in \Delta$. By the admissibility of left weakening, $\Gamma, x : B; l : A \vdash \Delta$. Now, $(|A|, 0, |t|) < (|A|, 0, |c|)$, because $|t| < |c| = 3 + |t|$. By IH(7), the above equations define the

proof term $[l/a]t$ and $\Gamma, x : B \vdash [l/a]t : C \mid \Delta$. Hence, $\langle \lambda x^B.[l/a]t \mid b \rangle$ is defined. Applying the inference rule $RCTR$, $\Gamma \mid \frac{\langle \lambda x^B.[l/a]t \mid b \rangle}{\Delta}$.

In (6),

- Case $l' = a$. Suppose $\Gamma; l' : B \vdash \Delta, a : A$ and $\Gamma; l : A \vdash \Delta$. We show that $\Gamma; l : B \vdash \Delta$. By the AX rule, $A = B$. Hence, $\Gamma; l : B \vdash \Delta$.
- Case $l' = b$, with $b \neq a$. Suppose $\Gamma; l' : B \vdash \Delta, a : A$ and $\Gamma; l : A \vdash \Delta$. We show that $\Gamma; b : B \vdash \Delta$. By the AX rule, $B = p$, with $(b : p) \in \Delta$. Thus, applying the inference AX , $\Gamma; b : p \vdash \Delta$.
- Case $l' = \text{abort}$. Suppose $\Gamma; l' : B \vdash \Delta, a : A$ and $\Gamma; l : A \vdash \Delta$. We show that $\Gamma; \text{abort} : B \vdash \Delta$. By the $L\perp$ rule, $B = \perp$. Thus, applying the inference rule $L\perp$, $\Gamma; \text{abort} : \perp \vdash \Delta$, and so the above equations define the proof list abort .
- Case $l' = u :: l''$. Suppose $\Gamma; l' : B \vdash \Delta, a : A$ and $\Gamma; l : A \vdash \Delta$. We show that the above equations define the proof list $([l/a]u) :: ([l/a]l'')$ and that $\Gamma; ([l/a]u) :: ([l/a]l'') : B \vdash \Delta$. By inversion of the inference rule LI , $B = C \supset D$ and $\Gamma \vdash u : C \mid \Delta, a : A$ and $\Gamma; l'' : D \vdash \Delta, a : A$. Now, $(|A|, 0, |u|) < (|A|, 0, |l'|)$, because $|u| < |l'| = 1 + |u| + |l''|$. By IH(7), the above equations define the proof term $[l/a]u$ and $\Gamma \vdash [l/a]u : C \mid \Delta$. Again, $(|A|, 0, |l''|) < (|A|, 0, |l'|)$, because $|l''| < |l'| = 1 + |u| + |l''|$. By IH(6), the above equations define the proof list $[l/a]l''$ and $\Gamma; [l/a]l'' : D \vdash \Delta$. Hence, $([l/a]u) :: ([l/a]l'')$ is defined. Applying the inference rule LI , $\Gamma; ([l/a]u) :: ([l/a]l'') : C \supset D \vdash \Delta$.

In (7),

- Case $u = \lambda x^C.t$. Suppose $\Gamma \vdash u : B \mid \Delta, a : A$ and $\Gamma; l : A \vdash \Delta$. We show that the above equations define the proof term $\lambda x^C.[l/a]t$ and that $\Gamma \vdash \lambda x^C.[l/a]t : B \mid \Delta$. By inversion of the inference rule RI , $B = C \supset D$ and $\Gamma, x : C \vdash t : D \mid \Delta, a : A$. By admissibility of LW , $\Gamma, x : C; l : A \vdash \Delta$. Now, $(|A|, 0, |t|) < (|A|, 0, |u|)$, because $|t| < |u| = 1 + |t|$. By IH(7), the above equations define the proof term $[l/a]t$ and $\Gamma, x : C \vdash [l/a]t : D \mid \Delta$. Hence, $\lambda x^C.[l/a]t$ is defined. Applying the inference rule RI , $\Gamma \vdash \lambda x^C.[l/a]t : C \supset D \mid \Delta$.
- Case $u = \mu b^B.c$. Suppose $\Gamma \vdash u : B \mid \Delta, a : A$ and $\Gamma; l : A \vdash \Delta$. We show that the above equations define the proof term $\mu b^B.[l/a]c$ and $\Gamma \vdash \mu b^B.[l/a]c : B \mid \Delta$. By inversion of the inference rule ACT , $\Gamma \mid \frac{c}{b : B, \Delta, a : A}$. By admissibility of right weakening, $\Gamma; l : A \vdash b :$

B, Δ . Now, $(|A|, 0, |c|) < (|A|, 0, |u|)$, because $|c| < |u| = 1 + |c|$. By IH(5), the above equations define the command $[l/a]c$ and $\Gamma \frac{[l/a]c}{b} : B, \Delta$. Hence, $\mu b^B.[l/a]c$ is defined. Applying the inference rule ACT , $\Gamma \vdash \mu b^B.[l/a]c : B \mid \Delta$.

□

Now, we are in conditions to prove the admissibility of a right contraction that is symmetric to $LCTR$, in LKT . Notice that $[l/a]c$, $[l/a]l'$ and $[l/a]t$ are always defined, by simultaneous recursion on c , l' and t , by the equations of Theorem 4.1, for the particular case $l = b$ (since only the case $[u :: l/a]\langle \lambda x^A.t \mid a \rangle$ calls a different operation, namely $@$).

Proposition 4.3. The following rules are admissible in LKT :

$$\frac{\Gamma \frac{c}{\Delta, b : A, a : A}}{\Gamma \frac{[b/a]c}{\Delta, b : A}} \quad \frac{\Gamma; l : B \vdash \Delta, b : A, a : A}{\Gamma; [b/a]l : B \vdash \Delta, b : A} \quad \frac{\Gamma \vdash t : B \mid \Delta, b : A, a : A}{\Gamma \vdash [b/a]t : B \mid \Delta, b : A}$$

Proof. By simultaneous induction on $\Gamma \frac{c}{\Delta, b : A, a : A}$ and $\Gamma; l : B \vdash \Delta, b : A, a : A$, and $\Gamma \vdash t : B \mid \Delta, b : A, a : A$. □

Proposition 4.4 (Admissibility of Right Contraction in LKT). Let $\text{rctr}_{a,A}(t)$ be defined by:

$$\begin{aligned} \text{rctr}_{a,A}(\lambda y^B.u) &= \mu a^A.\langle \lambda y^B.u \mid a \rangle \\ \text{rctr}_{a,A}(\mu b^A.u) &= \mu a^A.[a/b]c \end{aligned}$$

Then, the following rule is admissible in LKT :

$$\frac{\Gamma \vdash t : A \mid a : A, \Delta}{\Gamma \vdash \text{rctr}_{a,A}(t) : A \mid \Delta}$$

Proof. The proof is by case analysis of A and invertibility of RI and ACT .

- Case $t = \lambda y^B.u$. Hence $A = B \supset C$. We invert RI to obtain the premiss. Then, we have that

$$\frac{\Gamma, y : B \vdash u : C \mid a : B \supset C, \Delta}{\Gamma \frac{\langle \lambda y^B.u \mid a \rangle}{a : B \supset C, \Delta}} RCTR \quad \frac{\Gamma \vdash \mu a^{B \supset C}.\langle \lambda y^B.u \mid a \rangle : B \supset C \mid \Delta}{\Gamma \vdash \text{rctr}_{a,A}(\lambda y^B.u) : B \supset C \mid \Delta} ACT$$

- Case $t = \mu b^A.c$. We invert ACT to obtain the premiss. Then, we have that

$$\frac{\Gamma \frac{c}{b : A, a : A, \Delta}}{\Gamma \frac{[a/b]c}{a : A, \Delta}} \text{Prop. 4.3} \quad \frac{\Gamma \vdash \mu a^A.[a/b]c : A \mid \Delta}{\Gamma \vdash \text{rctr}_{a,A}(\mu b^A.c) : A \mid \Delta} ACT$$

□

4.3 Proof Search in LKT

Proof search in LKT proceeds by bottom-up application of inference rules and obeys the **focusing** discipline.

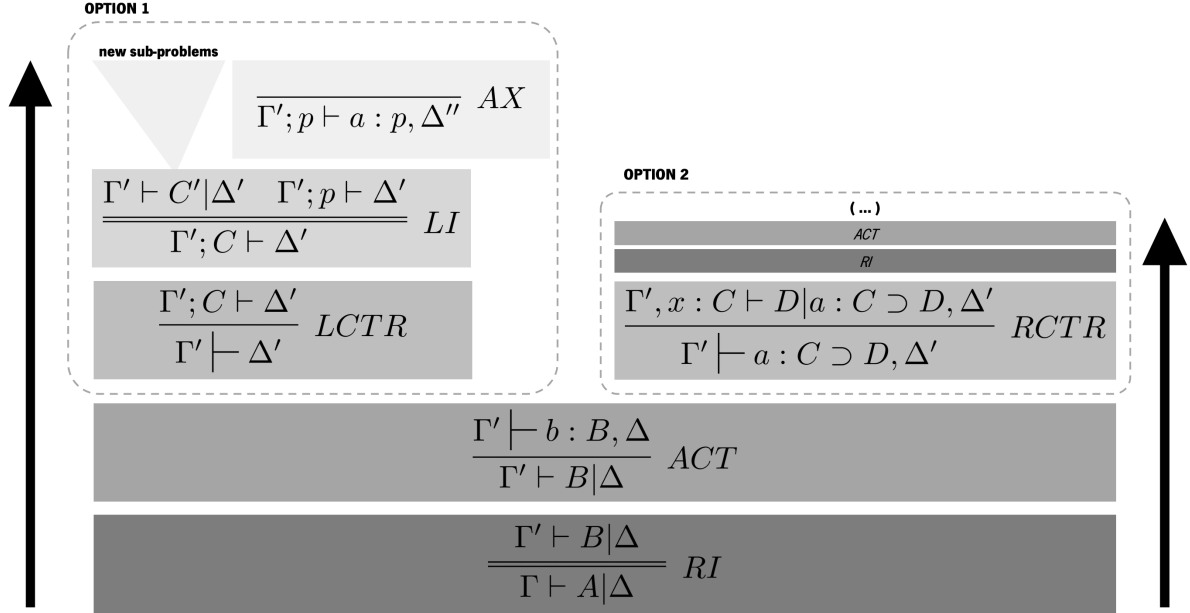


Figure 3: Focusing in LKT

We start by inverting RI to decompose implications in the right-hand side and this phase is explicitly finished by one application of ACT . Next there is an alternative: either we focus on a formula in the left-hand side, through rule $LCTR$, and, as long as the focus formula is an implication, we keep the focus on the succedent of the implication, as we apply LI – this is very much as what happens in LJT ; or we pick an implication in the right-hand side, decide this formula results from an introduction with contraction, and return to the inversion phase – this is the new option offered by rule $RCTR$.

Example 4.3. Keeping with Peirce's Law, $((p \supset q) \supset p) \supset p$, from the previous examples, we show an example of proof search in LKT described informally. Bottom-up applications of the ∂ -inference rules are signaled by the symbol $\uparrow$, Γ contains the assumption $(x : (p \supset q) \supset p)$ and Δ contains the alternative conclusions $(b : p)$ and $(a : ((p \supset q) \supset p) \supset p)$.

$$\frac{}{\vdash ((p \supset q) \supset p) \supset p} \partial \uparrow \quad \longrightarrow \quad \frac{\frac{}{\vdash a : ((p \supset q) \supset p) \supset p} \partial \uparrow}{\vdash ((p \supset q) \supset p) \supset p} ACT$$

$\downarrow$

[illegible]

We now represent this process as a rewriting system acting on partial proof terms. Thus, the system LKT with proof terms is grown to LKT_{∂} with partial proof terms.

Definition 4.9 (Formal Sequents of LKT_{∂}). LKT_{∂} handles three kinds of formal sequents: formal proof term sequents, $\underline{\Gamma \vdash A \mid \Delta}$, which are the underlined version of proof term sequents, ranged over by $\underline{\sigma}$; formal proof list sequent, $\underline{\Gamma; A \vdash \Delta}$, which are the underlined version of proof list sequents, ranged over by $\underline{\tau}$; and formal command sequents, $\underline{\Gamma \vdash \! \! \vdash \! \! \Delta}$, which are the underlined version of command sequents, ranged over by ζ .

Definition 4.10 (Proof Terms of LKT_∂). The proof terms of LKT_∂ are generated by the following

grammar:

$$\begin{aligned}
\text{(Partial proof terms)} \quad t, u &::= \lambda x^A.t \mid \mu a^A.c \mid \underline{\sigma} \\
\text{(Partial proof lists)} \quad l &::= a \mid \text{abort} \mid u :: l \mid \underline{\tau} \\
\text{(Partial commands)} \quad c &::= \langle x \mid l \rangle \mid \langle \lambda x^A.t \mid a \rangle \mid \underline{\zeta}
\end{aligned}$$

where $\underline{\sigma}$ ranges over formal proof term sequents, $\underline{\tau}$ ranges over formal proof list sequents and $\underline{\zeta}$ ranges over formal command sequents.

The typing system of LKT_∂ handles partial sequents, which, together with the inference rules, yield partial derivations.

Definition 4.11 (Partial Sequents of LKT_∂). LKT_∂ handles partial proof term sequents $\Xi \Vdash t : \sigma$, partial proof list sequents $\Xi \Vdash l : \tau$, and partial command sequents $\Xi \Vdash c : \zeta$. Here, Ξ ranges over lists of formal sequents, and such lists are written $[\underline{\sigma}, \underline{\tau}, \underline{\tau}', \underline{\zeta}]$ for instance, with ϵ denoting the empty list, as before.

Definition 4.12 (Inference Rules of LKT_∂). The set of inference rules of LKT_∂ is defined as:

$$\begin{aligned}
&\overline{[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma} \partial \quad \overline{[\underline{\tau}] \Vdash \underline{\tau} : \tau} \partial \quad \overline{[\underline{\zeta}] \Vdash \underline{\zeta} : \zeta} \partial \\
&\overline{\epsilon \Vdash a : (\Gamma; p \vdash a : p, \Delta)} AX \quad \overline{\epsilon \Vdash \text{abort} : (\Gamma; \perp \vdash \Delta)} L\perp \quad \frac{\Xi \Vdash c : (\Gamma \vdash a : A, \Delta)}{\Xi \Vdash \mu a^A.c : (\Gamma \vdash A \mid \Delta)} ACT \\
&\frac{\Xi \Vdash t : (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)}{\Xi \Vdash \langle \lambda x^A.t \mid a \rangle : (\Gamma \vdash a : A \supset B, \Delta)} RCTR \quad \frac{\Xi \Vdash l : (\Gamma, x : A; A \vdash \Delta)}{\Xi \Vdash \langle x \mid l \rangle : (\Gamma, x : A \vdash \Delta)} LCTR \\
&\frac{\Xi \Vdash t : (\Gamma, x : A \vdash B \mid \Delta)}{\Xi \Vdash \lambda x^A.t : (\Gamma \vdash A \supset B \mid \Delta)} RI \quad \frac{\Xi_1 \Vdash u : (\Gamma \vdash A \mid \Delta) \quad \Xi_2 \Vdash l : (\Gamma; B \vdash \Delta)}{\Xi_1 @ \Xi_2 \Vdash u :: l : (\Gamma; A \supset B \vdash \Delta)} LI
\end{aligned}$$

As in LJT_∂ (see Section 2.3), there is a partial inference rule for each inference rule of LKT ; and, furthermore, three partial rules. Here, the rules ACT , $RCTR$, $LCTR$ and RI are just modified presentations of the rules of LKT ; the rules AX and $L\perp$ say a and abort represent proof terms, without further proof obligations, for $\Gamma; p \vdash a : p, \Delta$ and $\Gamma; \perp \vdash \Delta$, respectively; and rule LI concatenates the proof obligations of the two branches. The first ∂ -rule say that $\underline{\sigma}$ will be converted to a proof of σ , once the proof obligation $\underline{\sigma}$ is fulfilled (similarly for $\underline{\tau}$ and $\underline{\zeta}$).

We are therefore in a position to prove the first result for this system, which states that derivability in LKT is partial derivability from the empty list of proof obligations.

Proposition 4.5 (Conservativity).

1. LKT derives $\Gamma \vdash t : A \mid \Delta$ iff LKT_{∂} derives $\epsilon \Vdash t : (\Gamma \vdash A \mid \Delta)$.
2. LKT derives $\Gamma; l : A \vdash \Delta$ iff LKT_{∂} derives $\epsilon \Vdash l : (\Gamma; A \vdash \Delta)$.
3. LKT derives $\Gamma \vdash^c \Delta$ iff LKT_{∂} derives $\epsilon \Vdash c : (\Gamma \vdash^c \Delta)$.

Proof. The “only if” statements are proved by simultaneous induction on the derivations of $\Gamma \vdash t : A \mid \Delta$, $\Gamma; l : A \vdash \Delta$ and $\Gamma \vdash^c \Delta$.

- Case AX . Suppose $\Gamma; a : p \vdash a : p, \Delta$. By the AX rule, $\epsilon \Vdash a : (\Gamma; p \vdash a : p, \Delta)$ in LKT_{∂} .
- Case $L\perp$. Suppose $\Gamma; \text{abort} : \perp \vdash \Delta$. By the $L\perp$ rule, $\epsilon \Vdash \text{abort} : (\Gamma; \perp \vdash \Delta)$ in LKT_{∂} .
- Case ACT . Suppose $\Gamma \vdash \mu a^A.c : A \mid \Delta$ is obtained from $\Gamma \vdash^c a : A, \Delta$. By IH, $\epsilon \Vdash c : (\Gamma \vdash^c a : A, \Delta)$. Applying the ACT rule, $\epsilon \Vdash \mu a^A.c : (\Gamma \vdash A \mid \Delta)$ in LKT_{∂} .
- Case $LCTR$. Suppose $\Gamma, x : A \vdash^{(x \mid l)} \Delta$ is obtained from $\Gamma, x : A; l : A \vdash \Delta$. By IH, $\epsilon \Vdash l : (\Gamma, x : A; A \vdash \Delta)$. Applying the $LCTR$ rule, $\epsilon \Vdash \langle x \mid l \rangle : (\Gamma, x : A \vdash^c \Delta)$ in LKT_{∂} .
- Case $RCTR$. Suppose $\Gamma \vdash^{(\lambda x^A.t \mid a)} a : A \supset B, \Delta$ is obtained from $\Gamma, x : A \vdash t : B \mid a : A \supset B, \Delta$. By IH, $\epsilon \Vdash t : (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)$. Applying the $RCTR$ rule, $\epsilon \Vdash \langle \lambda x^A.t \mid a \rangle : (\Gamma \vdash^c a : A \supset B, \Delta)$ in LKT_{∂} .
- Case RI . Suppose $\Gamma \vdash \lambda x^A.t : A \supset B \mid \Delta$ is obtained from $\Gamma, x : A \vdash t : B \mid \Delta$. By IH, $\epsilon \Vdash t : (\Gamma, x : A \vdash B \mid \Delta)$. Applying the RI rule, $\epsilon \Vdash \lambda x^A.t : (\Gamma \vdash A \supset B \mid \Delta)$ in LKT_{∂} .
- Case LI . Suppose $\Gamma; u :: l : A \supset B \vdash \Delta$ is obtained from $\Gamma \vdash u : A \mid \Delta$ and $\Gamma; l : B \vdash \Delta$. By IH, $\epsilon \Vdash u : (\Gamma \vdash A \mid \Delta)$ and $\epsilon \Vdash l : (\Gamma; B \vdash \Delta)$. Applying the LI rule, $\epsilon \Vdash u :: l : (\Gamma; A \supset B \vdash \Delta)$ in LKT_{∂} .

For the “if” statements, we prove:

- (1) If LKT_{∂} derives $\Xi \Vdash t : (\Gamma \vdash A \mid \Delta)$, with $t \in LKT$, then $\Xi = \epsilon$ and LKT derives $\Gamma \vdash t : A \mid \Delta$.
- (2) If LKT_{∂} derives $\Xi \Vdash l : (\Gamma; A \vdash \Delta)$, with $l \in LKT$, then $\Xi = \epsilon$ and LKT derives $\Gamma; l : A \vdash \Delta$.
- (3) If LKT_{∂} derives $\Xi \Vdash c : (\Gamma \vdash^c \Delta)$, with $c \in LKT$, then $\Xi = \epsilon$ and LKT derives $\Gamma \vdash^c \Delta$.

The proof is by simultaneous induction on the derivations of $\Xi \Vdash t : (\Gamma \vdash A \mid \Delta)$, $\Xi \Vdash l : (\Gamma; A \vdash \Delta)$ and $\Xi \Vdash c : (\Gamma \vdash \Delta)$.

- Case ∂ . Suppose $[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma$. Since $\underline{\sigma} \notin LKT$, the statement is trivially true. Similar reasoning applies to $[\underline{\tau}] \Vdash \underline{\tau} : \tau$ and $[\underline{\zeta}] \Vdash \underline{\zeta} : \zeta$.
- Case AX . Suppose $\epsilon \Vdash a : (\Gamma; p \vdash a : p, \Delta)$, with $a \in LKT$. Applying the AX rule, $\Gamma; a : p \vdash a : p, \Delta$ in LKT .
- Case $L\perp$. Suppose $\epsilon \Vdash \text{abort} : (\Gamma; \perp \vdash \Delta)$, with $\text{abort} \in LKT$. Applying the $L\perp$ rule, $\Gamma; \text{abort} : \perp \vdash \Delta$ in LKT .
- Case ACT . Suppose $\Xi \Vdash \mu a^A.c : (\Gamma \vdash A \mid \Delta)$ is obtained from $\Xi \Vdash c : (\Gamma \vdash a : A, \Delta)$, with $\mu a^A.c \in LKT$. Then $c \in LKT$. By IH, $\Xi = \epsilon$ and $\Gamma \vdash^c a : A, \Delta$. Applying the ACT rule, $\Gamma \vdash \mu a^A.c : A \mid \Delta$ in LKT .
- Case $LCTR$. Suppose $\Xi \Vdash \langle x \mid l \rangle : (\Gamma, x : A \vdash \Delta)$ is obtained from $\Xi \Vdash l : (\Gamma, x : A; A \vdash \Delta)$, with $\langle x \mid l \rangle \in LKT$. Then $l \in LKT$. By IH, $\Xi = \epsilon$ and $\Gamma, x : A; l : A \vdash \Delta$. Applying the $LCTR$ rule, $\Gamma, x : A \vdash^{\langle x \mid l \rangle} \Delta$ in LKT .
- Case $RCTR$. Suppose $\Xi \Vdash \langle \lambda x^A.t \mid a \rangle : (\Gamma \vdash a : A \supset B, \Delta)$ is obtained from $\Xi \Vdash t : (\Gamma, x : A \vdash B \mid \Delta)$, with $\langle \lambda x^A.t \mid a \rangle \in LKT$. Then $t \in LKT$. By IH, $\Xi = \epsilon$ and $\Gamma, x : A \vdash t : B \mid a : A \supset B, \Delta$. Applying the $RCTR$ rule, $\Gamma \vdash^{\langle \lambda x^A.t \mid a \rangle} a : A \supset B, \Delta$ in LKT .
- Case RI . Suppose $\Xi \Vdash \lambda x^A.t : (\Gamma \vdash A \supset B \mid \Delta)$ is obtained from $\Xi \Vdash t : (\Gamma, x : A \vdash B \mid \Delta)$, with $\lambda x^A.t \in LKT$. Then, $t \in LKT$. By IH, $\Xi = \epsilon$ and $\Gamma, x : A \vdash t : B \mid \Delta$. Applying the RI rule, $\Gamma \vdash \lambda x^A.t : A \supset B \mid \Delta$ in LKT .
- Case LI . Suppose $\Xi \Vdash u :: l : (\Gamma; A \supset B \vdash \Delta)$ is obtained from $\Xi_1 \Vdash u : (\Gamma \vdash A \mid \Delta)$ and $\Xi_2 \Vdash l : (\Gamma; B \vdash \Delta)$, with $\Xi = \Xi_1 @ \Xi_2$ and $u :: l \in LKT$. Then, $u, l \in LKT$. By IH, $\Xi_1 = \Xi_2 = \epsilon$ and $\Gamma \vdash u : A \mid \Delta$ and $\Gamma; l : B \vdash \Delta$. Hence, $\Xi = \epsilon @ \epsilon = \epsilon$. Applying the LI rule, $\Gamma; u :: l : A \supset B \vdash \Delta$ in LKT .

□

Proof obligations can be fulfilled by reducing t according to reduction rules given in the following definition.

Definition 4.13 (Reduction Rules of LKT_∂). The reduction rules for proof search in LKT are the following:

$$\begin{array}{ll}
(SIR) & \frac{\Gamma \vdash a : A \supset B, \Delta}{\Gamma \vdash a : A \supset B, \Delta} \rightarrow \langle \lambda x^A. (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta) \mid a \rangle \\
(KIR) & \frac{\Gamma \vdash A \supset B \mid \Delta}{\Gamma \vdash A \supset B \mid \Delta} \rightarrow \lambda x^A. (\Gamma, x : A \vdash B \mid \Delta) \\
(FIR) & \frac{\Gamma \vdash A \mid \Delta}{\Gamma \vdash A \mid \Delta} \rightarrow \mu a^A. (\Gamma \vdash a : A, \Delta) \\
(SFL) & \frac{\Gamma, x : A \vdash \Delta}{\Gamma, x : A \vdash \Delta} \rightarrow \langle x \mid (\Gamma, x : A; A \vdash \Delta) \rangle \\
(KFL) & \frac{\Gamma; A \supset B \vdash \Delta}{\Gamma; A \supset B \vdash \Delta} \rightarrow (\Gamma \vdash A \mid \Delta) :: (\Gamma; B \vdash \Delta) \\
(FFL_{atm}) & \frac{\Gamma; p \vdash a : p, \Delta}{\Gamma; p \vdash a : p, \Delta} \rightarrow a \\
(FFL_{ab}) & \frac{\Gamma; \perp \vdash \Delta}{\Gamma; \perp \vdash \Delta} \rightarrow \text{abort}
\end{array}$$

These reduction rules formalize the verbal explanation of the search procedure in the first paragraphs of this section. There is a reduction rule for each inference rule of the deductive system and their names stand for: start inversion right (SIR), keep inversion right (KIR), finish inversion right (FIR), start focus left (SFL), keep focus left (KFL), finish focus left – atomic (FFL_{atm}) and absurdity (FFL_{ab}). Since we implement bottom-up proof search, in the reduction rule the *redex* is a formal sequent corresponding to the conclusion of the inference rule and in the *contractum* we find the formal sequents corresponding to the premises.

We now explain how the reduction rules presented in Definition 4.13 describe the focusing proof search in LKT . Recall the rules in Definitions 4.5 and 4.8, and the diagram in Fig. 3. The goal is to prove the sequent $\Gamma \vdash A \mid \Delta$.

- As long as the selected conclusion A is an implication, we can apply the RI rule bottom-up to decompose the implication in the right-hand side formula. In this setup, the inversion phase is implemented by the KIR rule: the redex sequent evolves to a more determined proof term, reflecting the application of RI .
- The inversion phase ends with an application of the ACT rule, in which the formula A loses its distinguished position as the selected conclusion, and a sequent without any formula on the focus or any selected conclusion is obtained. This is implemented by the reduction rule FIR , which brings the search problem to the form $\Gamma \vdash a : A, \Delta$.
- Now, there is an alternative: we can return to the inversion-right phase by applying the inference rule $RCTR$ and picking an implication on the right-hand side that results from an introduction with a contraction — this is implemented by the reduction rule SIR ; or we start the focus-left phase by

applying the inference rule *LCTR* and selecting a formula B to put on the focus on the left-hand side. The latter is implemented by the reduction rule *SFL*.

- As long as the formula B in the left focus is not atomic or the absurdity, we apply *LI* bottom-up. This is implemented by the reduction rule *KFL*, which keeps the focus on the right premiss, and launches a new proof search problem in the left premiss. The redex sequent evolves to a more determined proof list.
- If, in the right premiss, we ever reach a sequent $\Gamma; p \vdash a : p, \Delta$, with the focus matching the type of a co-variable on the right-hand side, or a sequent $\Gamma; \perp \vdash \Delta$, with the absurdity on the focus, this branch of the search succeeds, and we may produce the proof list a or abort, respectively. The end of the left-focusing phase is then determined by one of the reduction rules *FFL_{atm}* or *FFL_{ab}*.

There are multiple points of non-determinism: the *FIR* rule, in which we abandon the selected conclusion instead of continuing the inversion phase; the *SIR* rule, where there is a choice of the selected implication to be introduced with contraction; the *SFL* rule, where we choose the formula to focus on the left-hand side; and the *FFL_{atm}* rule, where we choose a co-variable with the same type as the focus.

The reduction rules determine three different binary relations: one on partial proof terms, one on partial proof lists, and one on partial commands. We use $\rightarrow$ to denote the compatible closure of these binary relations and $\twoheadrightarrow$ to denote the reflexive and transitive closure of $\rightarrow$. The reduction rules also determine a relation $\Xi \rightarrow \Xi'$: if the redex of a rule is in Ξ , in Ξ' the redex is replaced by the formal sequents of the *contractum*. A formal definition of $\Xi \rightarrow \Xi'$ is analogous to what was done in the previous chapters.

Example 4.4. The formalization of the proof search in the Example 4.3, with the reduction rules of Definition 4.13, is presented.

$$\begin{array}{ll}
& \vdash ((p \supset q) \supset p) \supset p \mid \\
\rightarrow_{FIR} & \mu a^{((p \supset q) \supset p) \supset p}. \langle \vdash a : ((p \supset q) \supset p) \supset p \rangle \\
\rightarrow_{SIR} & \mu a^{((p \supset q) \supset p) \supset p}. \langle \lambda x^{(p \supset q) \supset p}. (\underline{x : (p \supset q) \supset p \vdash p \mid a : ((p \supset q) \supset p) \supset p}) \mid a \rangle \\
\rightarrow_{FIR} & \mu a^{((p \supset q) \supset p) \supset p}. \langle \lambda x^{(p \supset q) \supset p}. \mu b^p. (\underline{\Gamma \vdash b : p, a : ((p \supset q) \supset p) \supset p}) \mid a \rangle \\
\rightarrow_{SFL} & \mu a^{((p \supset q) \supset p) \supset p}. \langle \lambda x^{(p \supset q) \supset p}. \mu b^p. \langle x \mid (\underline{\Gamma; (p \supset q) \supset p \vdash \Delta}) \rangle \mid a \rangle \\
\rightarrow_{KFL} & \mu a^{((p \supset q) \supset p) \supset p}. \langle \lambda x^{(p \supset q) \supset p}. \mu b^p. \langle x \mid ((\underline{\Gamma \vdash p \supset q \mid \Delta}) :: (\underline{\Gamma; p \vdash \Delta})) \rangle \mid a \rangle \\
\rightarrow_{FFL} & \mu a^{((p \supset q) \supset p) \supset p}. \langle \lambda x^{(p \supset q) \supset p}. \mu b^p. \langle x \mid ((\underline{\Gamma \vdash p \supset q \mid \Delta}) :: b) \rangle \mid a \rangle \\
\rightarrow & (...)
\end{array}$$

Again, Γ contains the assumption $(x : (p \supset q) \supset p)$ and Δ contains the alternative conclusions $(b : p)$ and $(a : ((p \supset q) \supset p) \supset p)$.

We now present two more technical results.

Proposition 4.6 (Record of search).

1. If $\Xi \Vdash t : \sigma$ then $\underline{\sigma} \rightarrow t$.
2. If $\Xi \Vdash l : \tau$ then $\underline{\tau} \rightarrow l$.
3. If $\Xi \Vdash c : \zeta$ then $\underline{\zeta} \rightarrow c$.

Proof. By simultaneous induction on the derivations of $\Xi \Vdash t : \sigma$, $\Xi \Vdash l : \tau$ and $\Xi \Vdash c : \zeta$.

- Case ∂ . Suppose $[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma$. Then $\underline{\sigma} \rightarrow \underline{\sigma}$, because of the reflexivity of $\rightarrow$. Similar reasoning applies to $[\underline{\tau}] \Vdash \underline{\tau} : \tau$ and $[\underline{\zeta}] \Vdash \underline{\zeta} : \zeta$.
- Case AX . Suppose $\epsilon \Vdash a : (\Gamma; p \vdash a : p, \Delta)$. Then $\underline{\Gamma; p \vdash a : p, \Delta} \rightarrow a$, by FFL_{atm} .
- Case $L\perp$. Suppose $\epsilon \Vdash \text{abort} : (\Gamma; \perp \vdash \Delta)$. Then $\underline{\Gamma; \perp \vdash \Delta} \rightarrow \text{abort}$, by FFL_{ab} .
- Case ACT . Suppose $\Xi \Vdash \mu a^A.c : (\Gamma \vdash A \mid \Delta)$ is obtained from $\Xi \Vdash c : (\Gamma \vdash a : A, \Delta)$. Then $\underline{\Gamma \vdash A \mid \Delta} \rightarrow \mu a^A.(\underline{\Gamma \vdash a : A, \Delta}) \rightarrow \mu a^A.c$, where the first step is by FIR and the rest of the reduction is by IH.
- Case $LCTR$. Suppose $\Xi \Vdash \langle x \mid l \rangle : (\Gamma, x : A \vdash \Delta)$ is obtained from $\Xi \Vdash l : (\Gamma, x : A; A \vdash \Delta)$. Then $\underline{\Gamma, x : A \vdash \Delta} \rightarrow \langle x \mid (\underline{\Gamma, x : A; A \vdash \Delta}) \rangle \rightarrow \langle x \mid l \rangle$, where the first step is by SFL and the rest of reduction is by IH.
- Case $RCTR$. Suppose $\Xi \Vdash \langle \lambda x^A.t \mid a \rangle : (\Gamma \vdash a : A \supset B, \Delta)$ is obtained from $\Xi \Vdash t : (\Gamma, x : A \vdash B \mid \Delta)$. Then $\underline{\Gamma \vdash a : A \supset B, \Delta} \rightarrow \langle \lambda x^A.(\underline{\Gamma, x : A \vdash B \mid \Delta}) \mid a \rangle \rightarrow \langle \lambda x^A.t \mid a \rangle$, where the first step is by SIR and the rest of reduction is by IH.
- Case RI . Suppose $\Xi \Vdash \lambda x^A.t : (\Gamma \vdash A \supset B \mid \Delta)$ is obtained from $\Xi \Vdash t : (\Gamma, x : A \vdash B \mid \Delta)$. Then, $\underline{\Gamma \vdash A \supset B \mid \Delta} \rightarrow \lambda x^A.(\underline{\Gamma, x : A \vdash B \mid \Delta}) \rightarrow \lambda x^A.t$, where the first step is by KIR and the rest of the reduction is by IH.
- Case LI . Suppose $\Xi \Vdash u :: l : (\Gamma; A \supset B \vdash \Delta)$ is obtained from $\Xi_1 \Vdash u : (\Gamma \vdash A \mid \Delta)$ and $\Xi_2 \Vdash l : (\Gamma; B \vdash \Delta)$. Then, $\underline{\Gamma; A \supset B \vdash \Delta} \rightarrow (\underline{\Gamma \vdash A \mid \Delta}) :: (\underline{\Gamma; B \vdash \Delta}) \rightarrow u :: l$, where the first step is by KFL and the rest of the reduction is by IH.

□

Proposition 4.7 (Subject reduction for LKT_∂).

1. If $\Xi \Vdash t : \sigma$ and $t \rightarrow t'$, then there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash t' : \sigma$.
2. If $\Xi \Vdash l : \tau$ and $l \rightarrow l'$, then there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash l' : \tau$.
3. If $\Xi \Vdash c : \zeta$ and $c \rightarrow c'$, then there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash c' : \zeta$.

Proof. By simultaneous induction on $t \rightarrow t'$, $l \rightarrow l'$ and $c \rightarrow c'$.

Base Cases:

- Case *SIR*. Suppose $c = \Gamma \vdash a : A \supset B, \Delta \rightarrow \langle \lambda x^A. (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta) \mid a \rangle = c'$ and $\Xi \Vdash c : \zeta$. By the third ∂ -rule, $\Xi = [\Gamma \vdash a : A \supset B, \Delta]$ and $\zeta = \Gamma \vdash a : A \supset B, \Delta$. By *SIR*, $\Xi \rightarrow \Xi'$, with $\Xi' = [\Gamma, x : A \vdash B \mid a : A \supset B, \Delta]$. We now prove $\Xi' \Vdash \langle \lambda x^A. (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta) \mid a \rangle : (\Gamma \vdash a : A \supset B, \Delta)$. By the first ∂ -rule, $\Xi' \Vdash \Gamma, x : A \vdash B \mid a : A \supset B, \Delta : (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)$. Applying the inference rule *RCTR*, $\Xi' \Vdash \langle \lambda x^A. (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta) \mid a \rangle : (\Gamma \vdash a : A \supset B, \Delta)$.
- Case *KIR*. Suppose $t = \Gamma \vdash A \supset B \mid \Delta \rightarrow \lambda x^A. (\Gamma, x : A \vdash B \mid \Delta) = t'$ and $\Xi \Vdash t : \sigma$. By the first ∂ -rule, $\Xi = [\Gamma \vdash A \supset B \mid \Delta]$ and $\sigma = \Gamma \vdash A \supset B \mid \Delta$. By *KIR*, $\Xi \rightarrow \Xi'$, with $\Xi' = [\Gamma, x : A \vdash B \mid \Delta]$. We now prove $\Xi' \Vdash \lambda x^A. (\Gamma, x : A \vdash B \mid \Delta) : (\Gamma \vdash A \supset B \mid \Delta)$. By the first ∂ -rule, $\Xi' \Vdash \Gamma, x : A \vdash B \mid \Delta : (\Gamma, x : A \vdash B \mid \Delta)$. Applying the *RI* rule, $\Xi' \Vdash \lambda x^A. (\Gamma, x : A \vdash B \mid \Delta) : (\Gamma \vdash A \supset B \mid \Delta)$.
- Case *FIR*. Suppose $t = \Gamma \vdash A \mid \Delta \rightarrow \mu a^A. (\Gamma \vdash a : A, \Delta) = t'$ and $\Xi \Vdash t : \sigma$. By the first ∂ -rule, $\Xi = [\Gamma \vdash A \mid \Delta]$ and $\sigma = \Gamma \vdash A \mid \Delta$. By *FIR*, $\Xi \rightarrow \Xi'$, with $\Xi' = [\Gamma \vdash a : A, \Delta]$. We now prove $\Xi' \Vdash \mu a^A. (\Gamma \vdash a : A, \Delta) : (\Gamma \vdash A \mid \Delta)$. By the third ∂ -rule, $\Xi' \Vdash \Gamma \vdash a : A, \Delta : (\Gamma \vdash a : A, \Delta)$. Applying the *ACT* rule, $\Xi' \Vdash \mu a^A. (\Gamma \vdash a : A, \Delta) : (\Gamma \vdash A \mid \Delta)$.
- Case *SFL*. Suppose $c = \Gamma, x : A \vdash \Delta \rightarrow \langle x \mid (\Gamma, x : A; A \vdash \Delta) \rangle = c'$ and $\Xi \Vdash c : \zeta$. By the third ∂ -rule, $\Xi = [\Gamma, x : A \vdash \Delta]$ and $\zeta = \Gamma, x : A \vdash \Delta$. By *SFL*, $\Xi \rightarrow \Xi'$, with $\Xi' = [\Gamma, x : A; A \vdash \Delta]$. We now prove $\Xi' \Vdash \langle x \mid (\Gamma, x : A; A \vdash \Delta) \rangle : (\Gamma, x : A \vdash \Delta)$. By the second ∂ -rule, $\Xi' \Vdash \Gamma, x : A; A \vdash \Delta : (\Gamma, x : A; A \vdash \Delta)$. Applying the *LCTR* rule, $\Xi' \Vdash \langle x \mid (\Gamma, x : A; A \vdash \Delta) \rangle : (\Gamma, x : A \vdash \Delta)$.

- Case KFL . Suppose $l = \underline{\Gamma; A \supset B \vdash \Delta} \rightarrow (\Gamma \vdash A \mid \Delta) :: (\Gamma; B \vdash \Delta) = l'$ and $\Xi \Vdash l : \tau$. By the second ∂ -rule, $\Xi = [\underline{\Gamma; A \supset B \vdash \Delta}]$ and $\tau = \Gamma; A \supset B \vdash \Delta$. By KFL , $\Xi \rightarrow \Xi'$, with $\Xi' = [\underline{\Gamma \vdash A \mid \Delta; \Gamma; B \vdash \Delta}]$. We now prove $\Xi' \Vdash (\Gamma \vdash A \mid \Delta) :: (\Gamma; B \vdash \Delta) : (\Gamma; A \supset B \vdash \Delta)$. By the first ∂ -rule, $[\underline{\Gamma \vdash A \mid \Delta}] \Vdash \underline{\Gamma \vdash A \mid \Delta} : (\Gamma \vdash A \mid \Delta)$ and, by the second ∂ -rule, $[\underline{\Gamma; B \vdash \Delta}] \Vdash \underline{\Gamma; B \vdash \Delta} : (\Gamma; B \vdash \Delta)$. Applying the LI rule, $\Xi' \Vdash (\Gamma \vdash A \mid \Delta) :: (\Gamma; B \vdash \Delta) : (\Gamma; A \supset B \vdash \Delta)$.
- Case FFL_{atm} . Suppose $l = \underline{\Gamma; p \vdash a : p, \Delta} \rightarrow a = l'$ and $\Xi \Vdash l : \tau$. By the second ∂ -rule, $\Xi = [\underline{\Gamma; p \vdash a : p, \Delta}]$ and $\tau = \Gamma; p \vdash a : p, \Delta$. By FFL_{atm} , $\Xi \rightarrow \epsilon$. We now prove $\epsilon \Vdash a : (\Gamma; p \vdash a : p, \Delta)$, by just applying the AX rule.
- Case FFL_{ab} . Suppose $l = \underline{\Gamma; \perp \vdash \Delta} \rightarrow \text{abort} = l'$ and $\Xi \Vdash l : \tau$. By the second ∂ -rule, $\Xi = [\underline{\Gamma; \perp \vdash \Delta}]$ and $\tau = \Gamma; \perp \vdash \Delta$. By FFL_{ab} , $\Xi \rightarrow \epsilon$. We now prove $\epsilon \Vdash \text{abort} : (\Gamma; \perp \vdash \Delta)$, by just applying the $L\perp$ rule.

Inductive Cases:

- Suppose $\lambda x^A.t \rightarrow \lambda x^A.t'$, with $t \rightarrow t'$ and $\Xi \Vdash \lambda x^A.t : (\Gamma \vdash A \supset B \mid \Delta)$. By inversion of the inference rule RI , $\Xi \Vdash t : (\Gamma, x : A \vdash B \mid \Delta)$. By IH, there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash t' : (\Gamma, x : A \vdash B \mid \Delta)$. Applying the RI rule, $\Xi' \Vdash \lambda x^A.t' : (\Gamma \vdash A \supset B \mid \Delta)$.
- Suppose $\mu a^A.c \rightarrow \mu a^A.c'$, with $c \rightarrow c'$ and $\Xi \Vdash \mu a^A.c : (\Gamma \vdash A \mid \Delta)$. By inversion of the inference rule ACT , $\Xi \Vdash c : (\Gamma \vdash a : A, \Delta)$. By IH, there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash c' : (\Gamma \vdash a : A, \Delta)$. Applying the ACT rule, $\Xi' \Vdash \mu a^A.c' : (\Gamma \vdash A \mid \Delta)$.
- Suppose $\langle x \mid l \rangle \rightarrow \langle x \mid l' \rangle$, with $l \rightarrow l'$ and $\Xi \Vdash \langle x \mid l \rangle : (\Gamma, x : A \vdash \Delta)$. By inversion of the inference rule $LCTR$, $\Xi \Vdash l : (\Gamma, x : A; A \vdash \Delta)$. By IH, there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash l' : (\Gamma, x : A; A \vdash \Delta)$. Applying the $LCTR$ rule, $\Xi' \Vdash \langle x \mid l' \rangle : (\Gamma, x : A \vdash \Delta)$.
- Suppose $\langle \lambda x^A.t \mid a \rangle \rightarrow \langle \lambda x^A.t' \mid a \rangle$, with $t \rightarrow t'$ and $\Xi \Vdash \langle \lambda x^A.t \mid a \rangle : (\Gamma \vdash a : A \supset B, \Delta)$. By inversion of the inference rule $RCTR$, $\Xi \Vdash t : (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)$. By IH, there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash t' : (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)$. Applying the $RCTR$ rule, $\Xi' \Vdash \langle \lambda x^A.t' \mid a \rangle : (\Gamma \vdash a : A \supset B, \Delta)$.
- Suppose $u :: l \rightarrow u' :: l$, with $u \rightarrow u'$ and $\Xi \Vdash u :: l : (\Gamma; A \supset B \vdash \Delta)$. By the inversion of the inference rule LI , $\Xi_1 \Vdash u : (\Gamma \vdash A \mid \Delta)$ and $\Xi_2 \Vdash l : (\Gamma; B \vdash \Delta)$, and $\Xi = \Xi_1 @ \Xi_2$. By IH, there is Ξ'_1 such that $\Xi_1 \rightarrow \Xi'_1$ and $\Xi'_1 \Vdash u' : (\Gamma \vdash A \mid \Delta)$. Applying the LI rule, $\Xi'_1 @ \Xi_2 \Vdash u' :: l : (\Gamma; A \supset B \vdash \Delta)$. Moreover, $\Xi_1 @ \Xi_2 \rightarrow \Xi'_1 @ \Xi_2$.

- Suppose $u :: l \rightarrow u :: l'$, with $l \rightarrow l'$ and $\Xi \Vdash u :: l : (\Gamma; A \supset B \vdash \Delta)$. By the inversion of the inference rule LI , $\Xi_1 \Vdash u : (\Gamma \vdash A \mid \Delta)$ and $\Xi_2 \Vdash l : (\Gamma; B \vdash \Delta)$, and $\Xi = \Xi_1 @ \Xi_2$. By IH, there is Ξ'_2 such that $\Xi_2 \rightarrow \Xi'_2$ and $\Xi'_2 \Vdash l' : (\Gamma; B \vdash \Delta)$. Applying the LI rule, $\Xi_1 @ \Xi'_2 \Vdash u :: l' : (\Gamma; A \supset B \vdash \Delta)$. Moreover, $\Xi_1 @ \Xi_2 \rightarrow \Xi_1 @ \Xi'_2$.

□

From these, we arrive at the main result of this subsection, which can be roughly described like this: if $\sigma = \Gamma \vdash A \mid \Delta$, then proof search finds a proof t for σ iff $\underline{\sigma} \rightarrow t$.

Theorem 4.2 (Proof search as normalization).

1. For all $t \in LKT$: LKT derives $\Gamma \vdash t : A \mid \Delta$ iff $\underline{\sigma} \rightarrow t$, where $\underline{\sigma} = \underline{\Gamma \vdash A \mid \Delta}$.
2. For all $l \in LKT$: LKT derives $\Gamma; l : A \vdash \Delta$ iff $\underline{\tau} \rightarrow l$, where $\underline{\tau} = \underline{\Gamma; A \vdash \Delta}$.
3. For all $c \in LKT$: LKT derives $\Gamma \vdash^c \Delta$ iff $\underline{\zeta} \rightarrow c$, where $\underline{\zeta} = \underline{\Gamma \vdash^c \Delta}$.

Proof. The “only if” statements follow from conservativity and record of search. As to the “if” statement, let us prove the first. Suppose $\underline{\sigma} \rightarrow t$, where $\underline{\sigma} = \underline{\Gamma \vdash A \mid \Delta}$. Notice $[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma$. By subject reduction, $\Xi \Vdash t : \sigma$, for some Ξ , such that $[\underline{\sigma}] \rightarrow \Xi$. But t is total, so $\Xi = \epsilon$. By conservativity, LKT derives $\Gamma \vdash t : A \mid \Delta$. The other “if” statements are proved in a similar way. □

4.4 Natural Deduction NKT

NKT is formulated as a system of natural deduction with multiple conclusions, in the tradition of Parigot's classical natural deduction and the associated $\lambda\mu$ -calculus [49]. Additionally, it may be regarded as the classical counterpart of NJT introduced in Chapter 3.

The natural deduction NKT handles three kinds of sequents.

Definition 4.14 (Sequents of NKT). NKT handles sequents $\Gamma \vdash A \mid \Delta$ and $\Gamma \triangleright A \mid \Delta$ and $\Gamma \vdash^c \Delta$, ranged over by σ , ρ and ζ , respectively. In the first form, the sequent has an active formula, A ; in the second kind, the formula A to the right of $\triangleright$ is the focus of the sequent; and in the latter, all conclusions are passive. Following the ideas of Parigot's natural deduction [49], the sequents include alternative conclusions, collected in Δ .

Definition 4.15 (Inference Rules of NKT). The inference rules of NKT are defined as:

$$\begin{array}{c}
\frac{\Gamma, x : A \vdash B \mid \Delta}{\Gamma \vdash A \supset B \mid \Delta} I\supset \quad \frac{\Gamma \vdash a : A, \Delta}{\Gamma \vdash A \mid \Delta} ACT \\
\\
\frac{\Gamma \triangleright \perp \mid \Delta}{\Gamma \vdash \Delta} E\perp \quad \frac{\Gamma \triangleright p \mid a : p, \Delta}{\Gamma \vdash a : p, \Delta} PSS \quad \frac{\Gamma, x : A \vdash B \mid a : A \supset B, \Delta}{\Gamma \vdash a : A \supset B, \Delta} CTR \\
\\
\frac{}{\Gamma, x : A \triangleright A \mid \Delta} A \quad \frac{\Gamma \triangleright A \supset B \mid \Delta \quad \Gamma \vdash A \mid \Delta}{\Gamma \triangleright B \mid \Delta} E\supset
\end{array}$$

These rules are the axiom A , the introduction and elimination rules ($I\supset$, $E\supset$) for implication, the elimination of absurdity ($E\perp$) and three structural rules: activation (ACT), deactivation (PSS) and contraction (CTR). In a derivation, the formulas in the focus constitute the elimination part of a branch, starting with an assumption, containing a sequence of formulas that figure as the main premisses of inferences $E\supset$, down to an atom p or to $\perp$. The coercion to a sequent $\Gamma \vdash A \mid \Delta$ requires two steps: first, the atom p is made passive by rule PSS , or $\perp$ is eliminated by rule $E\perp$; second, one of the alternative conclusions in Δ is made active by rule ACT . The introduction part of the branch thus started can be grown downwards by applications of $I\supset$, or through a kind of *detour* made possible by rule CTR , where $A \supset B$ is implicitly introduced and immediately made passive and contracted with $a : A \supset B$ present in the set of alternative conclusions; in order to return to the introduction mode, ACT must be employed, but there is no guarantee that $a : A \supset B$ will be the activated formula.

Example 4.5. Let $((p \supset q) \supset p) \supset p$ be Peirce's Law. The derivation of Peirce's Law in NKT is represented as:

$$\begin{array}{c}
\frac{}{x : (p \supset q) \supset p, y : p \triangleright p \mid c : q, \Delta} A \\
\frac{}{x : (p \supset q) \supset p, y : p \vdash c : q, \Delta} PSS \\
\frac{}{x : (p \supset q) \supset p, y : p \vdash q \mid \Delta} ACT \\
\frac{}{x : (p \supset q) \supset p \triangleright (p \supset q) \supset p \mid \Delta} A \\
\frac{}{x : (p \supset q) \supset p \vdash p \supset q \mid \Delta} I\supset \\
\frac{}{x : (p \supset q) \supset p \triangleright p \mid b : p, a : ((p \supset q) \supset p) \supset p} E\supset \\
\frac{}{x : (p \supset q) \supset p \vdash b : p, a : ((p \supset q) \supset p) \supset p} PSS \\
\frac{}{x : (p \supset q) \supset p \vdash p \mid a : ((p \supset q) \supset p) \supset p} ACT \\
\frac{}{x : (p \supset q) \supset p \vdash p \mid a : ((p \supset q) \supset p) \supset p} CTR \\
\frac{}{\vdash a : ((p \supset q) \supset p) \supset p} ACT \\
\frac{}{\vdash ((p \supset q) \supset p) \supset p} ACT
\end{array}$$

NKT can also be understood as an extension of NJT [25] with “multiple conclusions”. To see this, think of the sequents $\Gamma \vdash A$ and $\Gamma \triangleright A$ of NJT as, respectively, the sequents $\Gamma \vdash A \mid \cdot$ and $\Gamma \triangleright A \mid \cdot$ of NKT , where $\cdot$ is the empty Δ . The inference rules of NKT , when acting on these restricted sequents, define deductions as in NJT : erase Δ in rules $I\supset$ and $E\supset$; the axiom concludes $\Gamma, x : A \triangleright A \mid \cdot$; rule CTR does not apply, as its premiss is beyond the considered restrictions; and rules PSS and ACT and the third form of sequents are hidden inside the following derivation:

$$\frac{\frac{\frac{\Gamma \triangleright p \mid \cdot}{\Gamma \triangleright p \mid a : p} W}{\Gamma \vdash a : p} PSS}{\Gamma \vdash p \mid \cdot} ACT$$

which simulates the inclusion of derivability of $\Gamma \triangleright p$ in derivability of $\Gamma \vdash p$ enjoyed by NJT .

We now extend NKT to include proof terms.

Definition 4.16 (Proof Terms of NKT). Proof terms M, N , neutral terms H and statements S of NKT are defined by the following grammar:

$$\begin{aligned} \text{(Proof terms)} \quad M, N &::= \lambda x^A.M \mid \mu a^A.S \\ \text{(Neutral terms)} \quad H &::= x \mid HN \\ \text{(Statements)} \quad S &::= \text{abort}(H) \mid a(H) \mid a(\lambda x^A.M) \end{aligned}$$

NKT with proof terms handles three forms of total sequents.

Definition 4.17 (Total Sequents of NKT with proof terms). NKT with proof terms handles proof term sequents $\Gamma \vdash M : A \mid \Delta$, neutral term sequents $\Gamma \triangleright H : A \mid \Delta$ and statement sequents $\Gamma \vdash^S \Delta$.

Definition 4.18 (Inference Rules of NKT with proof terms). The set of inference rules of NKT with proof terms is defined as:

$$\begin{aligned} &\frac{\Gamma, x : A \vdash M : B \mid \Delta}{\Gamma \vdash \lambda x^A.M : A \supset B \mid \Delta} I\supset \quad \frac{\Gamma \vdash^S a : A, \Delta}{\Gamma \vdash \mu a^A.S : A \mid \Delta} ACT \\ &\frac{\Gamma \triangleright H : \perp \mid \Delta}{\Gamma \vdash^{\text{abort}(H)} \Delta} E\perp \quad \frac{\Gamma \triangleright H : p \mid a : p, \Delta}{\Gamma \vdash^{a(H)} a : p, \Delta} PSS \quad \frac{\Gamma, x : A \vdash M : B \mid a : A \supset B, \Delta}{\Gamma \vdash^{a(\lambda x^A.M)} a : A \supset B, \Delta} CTR \\ &\frac{}{\Gamma, x : A \triangleright x : A \mid \Delta} A \quad \frac{\Gamma \triangleright H : A \supset B \mid \Delta \quad \Gamma \vdash N : A \mid \Delta}{\Gamma \triangleright HN : B \mid \Delta} E\supset \end{aligned}$$

Notice the following invariant.

Proposition 4.8 (Invariant of NKT). If $\Gamma \vdash^S \Delta, b : \perp$ is derivable, then b is not free in S , and $\Gamma \vdash^S \Delta$ is derivable. Similarly for M and H .

Proof. By simultaneous induction in the derivations of $\Gamma \vdash M : A \mid \Delta$, $\Gamma \triangleright H : A \mid \Delta$, and $\Gamma \vdash^S \Delta$.

- Case A . Suppose $\Gamma, x : A \triangleright x : A \mid \Delta, b : \perp$. We have that b is not free in x and, applying the inference A rule, $\Gamma, x : A \triangleright x : A \mid \Delta$.
- Case $E\perp$. Suppose $\Gamma \vdash^{\text{abort}(H)} \Delta, b : \perp$ is obtained from $\Gamma \triangleright H : \perp \mid \Delta, b : \perp$. By IH, b is not free in H and $\Gamma \triangleright H : \perp \mid \Delta$. Then, b is not free in $\text{abort}(H)$ and, applying the inference rule $E\perp$, $\Gamma \vdash^{\text{abort}(H)} \Delta$.
- Case ACT . Suppose $\Gamma \vdash \mu a^A.S : A \mid \Delta, b : \perp$ is obtained from $\Gamma \vdash^S a : A, \Delta, b : \perp$. By IH, b is not free in S , and $\Gamma \vdash^S a : A, \Delta$. Then, b is not free in $\mu a^A.S$ and, applying the inference rule ACT , $\Gamma \vdash \mu a^A.S : A \mid \Delta$.
- Case PSS . Suppose $\Gamma \vdash^{a(H)} a : p, \Delta, b : \perp$ is obtained from $\Gamma \triangleright H : p \mid a : p, \Delta, b : \perp$. By IH, b is not free in H and $\Gamma \triangleright H : p \mid a : p, \Delta$. Then, b is not free in $a(H)$, given that $b \neq a$, because they always have different types. Applying the inference rule CTR , $\Gamma \vdash^{a(H)} a : p, \Delta$.
- Case CTR . Suppose $\Gamma \vdash^{a(\lambda x^A.M)} a : A \supset B, \Delta, b : \perp$ is obtained from $\Gamma, x : A \vdash M : B \mid a : A \supset B, \Delta, b : \perp$. By IH, b is not free in M and $\Gamma, x : A \vdash M : B \mid a : A \supset B, \Delta$. Then, b is not free in $a(\lambda x^A.M)$, given that $b \neq a$, because they always have different types. Applying the inference rule CTR , $\Gamma \vdash^{a(\lambda x^A.M)} a : A \supset B, \Delta$.
- Case $I\supset$. Suppose $\Gamma \vdash \lambda x^A.M : A \supset B \mid \Delta, b : \perp$ is obtained from $\Gamma, x : A \vdash M : B \mid \Delta, b : \perp$. By IH, b is not free in M and $\Gamma, x : A \vdash M : B \mid \Delta$. Then, b is not free in $\lambda x^A.M$ and, applying the inference rule $I\supset$, $\Gamma \vdash \lambda x^A.M : A \supset B \mid \Delta$.
- Case $E\supset$. Suppose $\Gamma \triangleright HN : B \mid \Delta, b : \perp$ is obtained from $\Gamma \triangleright H : A \supset B \mid \Delta, b : \perp$ and $\Gamma \vdash N : A \mid \Delta, b : \perp$. By IH, b is not free in H and in N , and $\Gamma \triangleright H : A \supset B \mid \Delta$ and $\Gamma \vdash N : A \mid \Delta$. Then, b is not free in HN and, applying the inference rule $E\supset$, $\Gamma \triangleright HN : B \mid \Delta$.

□

Example 4.6. Adding proof terms to the derivation of Example 4.5, we have the following derivation in NKT with proof terms:

$$\begin{array}{c}
\frac{x : (p \supset q) \supset p, y : p \triangleright y : p \mid c : q, \Delta}{PSS} A \\
\frac{x : (p \supset q) \supset p, y : p \mid \frac{b(y)}{c : q, \Delta}}{ACT} \\
\frac{x : (p \supset q) \supset p, y : p \vdash \mu c^q.b(y) : q \mid \Delta}{I\supset} ACT \\
\frac{x : (p \supset q) \supset p \triangleright x : (p \supset q) \supset p \mid \Delta}{E\supset} A \\
\frac{x : (p \supset q) \supset p \vdash \lambda y^p.\mu c^q.b(y) : p \mid b : p, a : ((p \supset q) \supset p) \supset p}{PSS} \\
\frac{x : (p \supset q) \supset p \mid \frac{b(x(\lambda y^p.\mu c^q.b(y)))}{b : p, a : ((p \supset q) \supset p) \supset p}}{ACT} \\
\frac{x : (p \supset q) \supset p \vdash \mu b^p.b(x(\lambda y^p.\mu c^q.b(y))) : p \mid a : ((p \supset q) \supset p) \supset p}{CTR} \\
\frac{\mid \frac{a(\lambda x^{((p \supset q) \supset p)}.\mu b^p.b(x(\lambda y^p.\mu c^q.b(y))))}{a : ((p \supset q) \supset p) \supset p}}{ACT} \\
\frac{\vdash \mu a^{((p \supset q) \supset p) \supset p}.a(\lambda x^{((p \supset q) \supset p)}.\mu b^p.b(x(\lambda y^p.\mu c^q.b(y)))) : ((p \supset q) \supset p) \supset p \mid}{ACT}
\end{array}$$

where $\Delta = \{b : p, a : ((p \supset q) \supset p) \supset p\}$. Thus, the proof term $\mu a^{((p \supset q) \supset p) \supset p}. a(\lambda x^{((p \supset q) \supset p)}. \mu b^p. b(x(\lambda y^p. \mu c^q. b(y))))$ encodes the derivation of the sequent $\vdash ((p \supset q) \supset p) \supset p$, representing the Peirce's Law.

The generalization of rule *PSS* to arbitrary formulas is admissible.

Proposition 4.9 (Generalization of PSS). Let $\text{pass}(a, A, H)$ be the statement defined by recursion on A as follows:

$$\begin{aligned} \text{pass}(a, p, H) &= a(H) \\ \text{pass}(a, \perp, H) &= \text{abort}(H) \\ \text{pass}(a, A \supset B, H) &= a(\lambda x^A. \mu b^B. \text{pass}(b, B, H(\mu a'^A. \text{pass}(a', A, x)))) \end{aligned}$$

Then the following rules are admissible in NKT :

$$\frac{\Gamma \triangleright H : A \mid a : A, \Delta}{\Gamma \mid \frac{\text{pass}(a, A, H)}{} a : A, \Delta} PASS \qquad \frac{\Gamma \triangleright H : A \mid a : A, \Delta}{\Gamma \vdash \mu a^A. \text{pass}(a, A, H) : A \mid \Delta}$$

Proof. The first admissible rule is proved by case analysis of A :

- Case $A = p$. Suppose $\Gamma \triangleright H : p \mid a : p, \Delta$. Applying the inference rule PSS , $\Gamma \mid \frac{a(H)}{a : p, \Delta}$. Then, $\text{pass}(a, p, H) = a(H)$.
- Case $A = \perp$. Suppose $\Gamma \triangleright H : \perp \mid a : \perp, \Delta$. Applying the inference rule $E\perp$, $\Gamma \mid \frac{\text{abort}(H)}{a : \perp, \Delta}$. Then, $\text{pass}(a, \perp, H) = \text{abort}(H)$.
- Case $A = B \supset C$. Suppose $\Gamma \triangleright H : B \supset C \mid a : B \supset C, \Delta$. Then,

$$\begin{array}{c}
\frac{\Gamma \triangleright H : B \supset C \mid a : B \supset C, \Delta}{\Gamma, x : B \triangleright H : B \supset C \mid c : C, a : B \supset C, \Delta} W \quad (*) \\
\frac{\Gamma, x : B \triangleright H(\mu b^B.\text{pass}(b, B, x)) : C \mid c : C, a : B \supset C, \Delta}{\Gamma, x : B \mid \frac{\text{pass}(c, C, H(\mu b^B.\text{pass}(b, B, x)))}{c : C, a : B \supset C, \Delta}} E\supset \\
\frac{\Gamma, x : B \mid \frac{\text{pass}(c, C, H(\mu b^B.\text{pass}(b, B, x)))}{c : C, a : B \supset C, \Delta}}{\Gamma, x : B \vdash \mu c^C.\text{pass}(c, C, H(\mu b^B.\text{pass}(b, B, x))) : C \mid a : B \supset C, \Delta} ACT \\
\frac{\Gamma \mid \frac{a(\lambda x^B.\mu c^C.\text{pass}(c, C, H(\mu b^B.\text{pass}(b, B, x))))}{a : B \supset C, \Delta}}{\Gamma \mid a : B \supset C, \Delta} CTR
\end{array},$$

where $(*)$ the following derivation:

$$\begin{array}{c}
\frac{\Gamma, x : B \triangleright x : B \mid b : B, c : C, a : B \supset C, \Delta}{\Gamma, x : B \mid \frac{\text{pass}(b, B, x)}{b : B, c : C, a : B \supset C, \Delta}} A \\
\frac{\Gamma, x : B \mid \frac{\text{pass}(b, B, x)}{b : B, c : C, a : B \supset C, \Delta}}{\Gamma, x : B \vdash \mu b^B.\text{pass}(b, B, x) : B \mid c : C, a : B \supset C, \Delta} PASS* \\
\frac{\Gamma, x : B \vdash \mu b^B.\text{pass}(b, B, x) : B \mid c : C, a : B \supset C, \Delta}{\Gamma, x : B \vdash \mu b^B.\text{pass}(b, B, x) : B \mid c : C, a : B \supset C, \Delta} ACT
\end{array}.$$

In the previous derivations, $PASS*$ represents an application of the admissible rule $PASS$ by induction hypothesis.

The second admissible rule is an immediate consequence of the first, since:

$$\begin{array}{c}
\frac{\Gamma \triangleright H : A \mid a : A, \Delta}{\Gamma \mid \frac{\text{pass}(a, A, H)}{a : A, \Delta}} PASS \\
\frac{\Gamma \mid \frac{\text{pass}(a, A, H)}{a : A, \Delta}}{\Gamma \vdash \mu a^A.\text{pass}(a, A, H) : A \mid \Delta} ACT
\end{array}.$$

□

Making use of the previous proposition, we prove that the rule of double negation is also admissible in NKT , which, to some extent, demonstrates that sequents with multiple conclusions in the style of Parigot [49] replace the need for a double negation or reduction to absurdity inference rule in our natural deduction system.

Proposition 4.10 (Double negation elimination). NKT derives $y : (A \supset \perp) \supset \perp \vdash A \mid$.

Proof.

$$\begin{array}{c}
\frac{y : (A \supset \perp) \supset \perp, x : A \triangleright x : A \mid b : \perp, a : A}{y : (A \supset \perp) \supset \perp, x : A \mid \frac{\text{pass}(a, A, x)}{b : \perp, a : A}} A \\
\frac{y : (A \supset \perp) \supset \perp, x : A \mid \frac{\text{pass}(a, A, x)}{b : \perp, a : A}}{y : (A \supset \perp) \supset \perp, x : A \vdash \mu b^\perp.\text{pass}(a, A, x) : \perp \mid a : A} PASS \\
\frac{y : (A \supset \perp) \supset \perp, x : A \vdash \mu b^\perp.\text{pass}(a, A, x) : \perp \mid a : A}{y : (A \supset \perp) \supset \perp \vdash \lambda x^A.\mu b^\perp.\text{pass}(a, A, x) : A \supset \perp \mid a : A} ACT \\
(*) \quad \frac{y : (A \supset \perp) \supset \perp \vdash \lambda x^A.\mu b^\perp.\text{pass}(a, A, x) : A \supset \perp \mid a : A}{y : (A \supset \perp) \supset \perp \triangleright y(\lambda x^A.\mu b^\perp.\text{pass}(a, A, x)) : \perp \mid a : A} I\supset \\
\frac{y : (A \supset \perp) \supset \perp \triangleright y(\lambda x^A.\mu b^\perp.\text{pass}(a, A, x)) : \perp \mid a : A}{y : (A \supset \perp) \supset \perp \mid \frac{\text{abort}(y(\lambda x^A.\mu b^\perp.\text{pass}(a, A, x)))}{a : A}} E\supset \\
\frac{y : (A \supset \perp) \supset \perp \mid \frac{\text{abort}(y(\lambda x^A.\mu b^\perp.\text{pass}(a, A, x)))}{a : A}}{y : (A \supset \perp) \supset \perp \vdash \mu a^A.(\text{abort}(y(\lambda x^A.\mu b^\perp.\text{pass}(a, A, x)))) : A \mid} E\perp \\
\frac{y : (A \supset \perp) \supset \perp \vdash \mu a^A.(\text{abort}(y(\lambda x^A.\mu b^\perp.\text{pass}(a, A, x)))) : A \mid}{y : (A \supset \perp) \supset \perp \vdash \mu a^A.(\text{abort}(y(\lambda x^A.\mu b^\perp.\text{pass}(a, A, x)))) : A \mid} ACT
\end{array}$$

where $(*)$ is the proof:

$$\frac{}{y : (A \supset \perp) \supset \perp \triangleright y : (A \supset \perp) \supset \perp \mid a : A} A.$$

□

Weakening rules are admissible in NKT with proof terms.

Proposition 4.11 (Admissibility of Left Weakening in NKT). The left weakening rules

$$\frac{\Gamma \vdash M : B \mid \Delta}{\Gamma, x : A \vdash M : B \mid \Delta} \quad \frac{\Gamma \triangleright H : B \mid \Delta}{\Gamma, x : A \triangleright H : B \mid \Delta} \quad \frac{\Gamma \mid^S \Delta}{\Gamma, x : A \mid^S \Delta}$$

are admissible in NKT .

Proof. By a simultaneous induction on the derivations of $\Gamma \vdash M : B \mid \Delta$, $\Gamma \triangleright H : B \mid \Delta$, and $\Gamma \mid^S \Delta$.

- Case A . Suppose $\Gamma, y : B \triangleright y : B \mid \Delta$. Applying the A rule, $\Gamma, y : B, x : A \triangleright y : B \mid \Delta$.
- Case $E\perp$. Suppose $\Gamma \mid^{\text{abort}(H)} \Delta$ is obtained from $\Gamma \triangleright H : \perp \mid \Delta$. By IH, $\Gamma, x : A \triangleright H : \perp \mid \Delta$. Applying the $E\perp$ rule, $\Gamma, x : A \mid^{\text{abort}(H)} \Delta$.
- Case ACT . Suppose $\Gamma \vdash \mu b^B.S : B \mid \Delta$ is obtained from $\Gamma \mid^S b : B, \Delta$. By IH, $\Gamma, x : A \mid^S b : B, \Delta$. Applying the ACT rule, $\Gamma, x : A \vdash \mu b^B.S : B \mid \Delta$.
- Case PSS . Suppose $\Gamma \mid^{\text{b}(H)} a : p, \Delta$ is obtained from $\Gamma \triangleright H : p \mid a : p, \Delta$. By IH, $\Gamma, x : A \triangleright H : p \mid a : p, \Delta$. Applying the PSS rule, $\Gamma, x : A \mid^{\text{b}(H)} b : p, \Delta$.
- Case CTR . Suppose $\Gamma \mid^{\text{b}(\lambda y^B.M)} b : B \supset C, \Delta$ is obtained from $\Gamma, y : B \vdash M : C \mid b : B \supset C, \Delta$. By IH, $\Gamma, y : B, x : A \vdash M : C \mid b : B \supset C, \Delta$. Applying the CTR rule, $\Gamma, x : A \mid^{\text{b}(\lambda y^B.M)} b : B \supset C, \Delta$.
- Case $I\supset$. Suppose $\Gamma \vdash \lambda y^B.M : B \supset C \mid \Delta$ is obtained from $\Gamma, y : B \vdash M : C \mid \Delta$. By IH, $\Gamma, y : B, x : A \vdash M : C \mid \Delta$. Applying the $I\supset$ rule, $\Gamma, x : A \vdash \lambda y^B.M : B \supset C \mid \Delta$.
- Case $E\supset$. Suppose $\Gamma \triangleright HN : C \mid \Delta$ is obtained from $\Gamma \vdash N : B \mid \Delta$ and $\Gamma \triangleright H : B \supset C \mid \Delta$. By IH, $\Gamma, x : A \vdash N : B \mid \Delta$ and $\Gamma, x : A \triangleright H : B \supset C \mid \Delta$. Applying the $E\supset$ rule, $\Gamma, x : A \triangleright HN : C \mid \Delta$.

□

Proposition 4.12 (Admissibility of Right Weakening in NKT). The right weakening rules

$$\frac{\Gamma \vdash M : B \mid \Delta}{\Gamma \vdash M : B \mid \Delta, a : A} \quad \frac{\Gamma \triangleright H : B \mid \Delta}{\Gamma \triangleright H : B \mid \Delta, a : A} \quad \frac{\Gamma \mid^S \Delta}{\Gamma \mid^S \Delta, a : A}$$

are admissible in NKT .

Proof. By a simultaneous induction on the derivations of $\Gamma \vdash M : B \mid \Delta$, $\Gamma \triangleright H : B \mid \Delta$, and $\Gamma \mid^S \Delta$.

- Case A . Suppose $\Gamma, y : B \triangleright y : B \mid \Delta$. Applying the A rule, $\Gamma, y : B \triangleright y : B \mid \Delta, a : A$.
- Case $E\perp$. Suppose $\Gamma \mid^{\text{abort}(H)} \Delta$ is obtained from $\Gamma \triangleright H : \perp \mid \Delta$. By IH, $\Gamma \triangleright H : \perp \mid \Delta, a : A$. Applying the $E\perp$ rule, $\Gamma \mid^{\text{abort}(H)} \Delta, a : A$.
- Case ACT . Suppose $\Gamma \vdash \mu b^B.S : B \mid \Delta$ is obtained from $\Gamma \mid^S b : B, \Delta$. By IH, $\Gamma \mid^S b : B, \Delta, a : A$. Applying the ACT rule, $\Gamma \vdash \mu b^B.S : B \mid \Delta, a : A$.
- Case PSS . Suppose $\Gamma \mid^{b(H)} b : p, \Delta$ is obtained from $\Gamma \triangleright H : p \mid b : p, \Delta$. By IH, $\Gamma \triangleright H : p \mid b : p, \Delta, a : A$. Applying the PSS rule, $\Gamma \mid^{b(H)} b : p, \Delta, a : A$.
- Case CTR . Suppose $\Gamma \mid^{b(\lambda y^B.M)} b : B \supset C, \Delta$ is obtained from $\Gamma, y : B \vdash M : C \mid b : B \supset C, \Delta$. By IH, $\Gamma, y : B \vdash M : C \mid b : B \supset C, \Delta, a : A$. Applying the CTR rule, $\Gamma \mid^{b(\lambda y^B.M)} b : B \supset C, \Delta, a : A$.
- Case $I\supset$. Suppose $\Gamma \vdash \lambda y^B.M : B \supset C \mid \Delta$ is obtained from $\Gamma, y : B \vdash M : C \mid \Delta$. By IH, $\Gamma, y : B \vdash M : C \mid \Delta, a : A$. Applying the $I\supset$ rule, $\Gamma \vdash \lambda y^B.M : B \supset C \mid \Delta, a : A$.
- Case $E\supset$. Suppose $\Gamma \triangleright HN : C \mid \Delta$ is obtained from $\Gamma \vdash N : B \mid \Delta$ and $\Gamma \triangleright H : B \supset C \mid \Delta$. By IH, $\Gamma \vdash N : B \mid \Delta, a : A$ and $\Gamma \triangleright H : B \supset C \mid \Delta, a : A$. Applying the $E\supset$ rule, $\Gamma \triangleright HN : C \mid \Delta, a : A$.

□

Proposition 4.13. The following rules are admissible in NKT :

$$\frac{\Gamma \vdash M : B \mid b : A, a : A, \Delta}{\Gamma \vdash [a/b]M : B \mid a : A, \Delta} \quad \frac{\Gamma \triangleright H : B \mid b : A, a : A, \Delta}{\Gamma \triangleright [a/b]H : B \mid a : A, \Delta} \quad \frac{\Gamma \mid^S b : A, a : A, \Delta}{\Gamma \mid^{[a/b]S} a : A, \Delta}$$

Proof. By simultaneous induction in $\Gamma \vdash M : B \mid b : A, a : A, \Delta$, and $\Gamma \triangleright H : B \mid b : A, a : A, \Delta$, and $\Gamma \mid^S b : A, a : A, \Delta$. □

For the benefit of the next theorem, and as done in Chapter 3 in the context of system NJT , we introduce extended neutral terms of NKT .

Definition 4.19 (Extended Neutral Terms of NKT). An extended neutral term of NKT is either a neutral term H , or a term considered as a neutral term, called a head term and denoted $hd(M)$. Extended neutral terms are ranged over by $\mathbb{H}$.

Next we introduce the auxiliary form of sequent for extended neutral terms.

Definition 4.20 (Extended Neutral Term Sequents of NKT). An extended neutral term sequent of NKT is an expression of the form $\Gamma \blacktriangleright \mathbb{H} : A$, which is an abbreviation defined by the following two rules:

$$\frac{\Gamma \triangleright H : A \mid \Delta}{\Gamma \blacktriangleright H : A \mid \Delta} \quad \frac{\Gamma \vdash M : A \mid \Delta}{\Gamma \blacktriangleright hd(M) : A \mid \Delta}$$

Given $\mathbb{H}$, we define the statements $a(\mathbb{H})$ and $\text{abort}(\mathbb{H})$.

Proposition 4.14 (Extension of $a(H)$ and $\text{abort}(H)$ to extended neutral terms in NKT). Let $a(\mathbb{H})$ and $\text{abort}(\mathbb{H})$ be recursively defined as:

$$\begin{aligned} a(\mathbb{H}) &= a(H) & \text{abort}(\mathbb{H}) &= \text{abort}(H) \\ a(hd(\mu b^A.S)) &= [a/b]S & \text{abort}(hd(\mu b^\perp.S)) &= S \\ a(hd(\lambda x^C.M)) &= a(\lambda x^C.M) \end{aligned}$$

The following rules are admissible in NKT_∂ :

$$\frac{\Gamma \blacktriangleright hd(M) : A \mid a : A, \Delta}{\Gamma \mid \frac{a(hd(M))}{a : A, \Delta}} \quad \frac{\Gamma \blacktriangleright \mathbb{H} : p \mid a : p, \Delta}{\Gamma \mid \frac{a(\mathbb{H})}{a : p, \Delta}} \quad \frac{\Gamma \blacktriangleright \mathbb{H} : \perp \mid \Delta}{\Gamma \mid \frac{\text{abort}(\mathbb{H})}{\Delta}} .$$

Proof. The first admissible rule is proved by case analysis of M :

- Case $M = \lambda x^C.M$. Suppose $\Gamma \blacktriangleright hd(\lambda x^C.M) : A \mid a : A, \Delta$. We prove $\Gamma \mid \frac{a(\lambda x^C.M)}{a : A, \Delta} a : A, \Delta$, given that $a(hd(\lambda x^C.M)) = a(\lambda x^C.M)$. By Definition 4.20, $\Gamma \vdash \lambda x^C.M : A \mid a : A, \Delta$ and $A = C \supset D$. Now, by inversion of the inference rule $I\supset$, $\Gamma, x : C \vdash M : D \mid a : C \supset D, \Delta$. Applying the inference rule CTR , $\Gamma \mid \frac{a(\lambda x^C.M)}{a : A, \Delta} a : A, \Delta$.
- Case $M = \mu b^A.S$. Suppose $\Gamma \blacktriangleright hd(\mu b^A.S) : A \mid a : A, \Delta$. We prove $\Gamma \mid \frac{[a/b]S}{a : A, \Delta} a : A, \Delta$, given that $a(hd(\mu b^A.S)) = [a/b]S$. By Definition 4.20, $\Gamma \vdash \mu b^A.S : A \mid a : A, \Delta$. Now,

by inversion of the inference rule ACT , $\Gamma \vdash^S b : A, a : A, \Delta$. By Proposition 4.13, we have $\Gamma \vdash^{[a/b]S} a : A, \Delta$.

The second admissible rule is proved by case analysis of $\mathbb{H}$:

- Case $\mathbb{H} = H$. Suppose $\Gamma \blacktriangleright H : p \mid a : p, \Delta$. We prove $\Gamma \frac{}{a : p, \Delta} \frac{a(H)}{a : p, \Delta}$, given that $a(\lfloor H \rfloor) = a(H)$. By Definition 4.20, $\Gamma \triangleright H : p \mid a : p, \Delta$. Applying the inference rule PSS , $\Gamma \frac{}{a : p, \Delta} \frac{a(H)}{a : p, \Delta}$.
- Case $\mathbb{H} = hd(M)$. Particular case of the first admissible rule, with $A = p$.

The third admissible rule is proved by case analysis of $\mathbb{H}$:

- Case $\mathbb{H} = H$. Suppose $\Gamma \blacktriangleright H : \perp \mid \Delta$. We prove $\Gamma \mid \frac{}{\text{abort}(H)} \Delta$, given that $\text{abort}(\llbracket H \rrbracket) = \text{abort}(H)$. By Definition 4.20, $\Gamma \triangleright H : \perp \mid \Delta$. Applying the inference rule $E\perp$, $\Gamma \mid \frac{}{\text{abort}(H)} \Delta$.
- Case $\mathbb{H} = hd(\mu b^\perp.S)$. Suppose $\Gamma \blacktriangleright hd(\mu b^\perp.S) : \perp \mid \Delta$. We prove $\Gamma \mid \frac{S}{} \Delta$, given that $\text{abort}(\llbracket hd(\mu b^\perp.S) \rrbracket) = S$. By Definition 4.20, $\Gamma \vdash \mu b^\perp.S : \perp \mid \Delta$. Now, by inversion of the inference rule ACT , $\Gamma \mid \frac{S}{} b : \perp, \Delta$. Then, by Proposition 4.8, b is not free in S and $\Gamma \mid \frac{S}{} \Delta$ is derivable.

Theorem 4.3 (Admissible rules of NKT). Consider the equations

$$\begin{array}{ll}
[N/x](\lambda y^C.M) &= \lambda y^C.[N/x]M & (\lambda y^A.M)@N &= [N/y]M \\
[N/x](\mu a^B.S) &= \mu a^B.[N/x]S & (\mu a^{A \supset B}.S)@N &= \mu a^B.[@N/a]S \\
\\
[N/x](\textbf{abort}(H)) &= \textbf{abort}([N/x]H) & [@N/a]\textbf{abort}(H) &= \textbf{abort}([@N/a]H) \\
[N/x](a(H)) &= a([N/x]H) & [@N/a](c(H)) &= c([@N/a]H), \quad (c \neq a) \\
[N/x](a(\lambda y^C.M)) &= a(\lambda y^C.[N/x]M) & [@N/a](a(\lambda x^A.M)) &= a(hd([N/x]([@N/a]M))) \\
&& & [@N/a](c(\lambda x^A.M)) = c(\lambda x^A.[@N/a]M), \quad (c \neq a) \\
\\
[N/x]x &= hd(N) \\
[N/x]y &= y \quad (y \neq x) & [@N/a]x &= x \\
[N/x](HN') &= ([N/x]H)@([N/x]N') & [@N/a](HN') &= ([@N/a]H)([@N/a]N') \\
\\
H@N &= HN & [@N/a](\lambda y^C.M) &= \lambda y^C.[@N/a]M \\
(hd(M))@N &= hd(M@N) & [@N/a](\mu c^C.S) &= \mu c^C.[@N/a]S.
\end{array}$$

In NKT one has:

- (1) For all A, M : if $\Gamma \vdash N : A \mid \Delta$ and $\Gamma, x : A \vdash M : B \mid \Delta$, then the equations define the proof term $[N/x]M$ and $\Gamma \vdash [N/x]M : B \mid \Delta$.
- (2) For all A, S : if $\Gamma \vdash N : A \mid \Delta$ and $\Gamma, x : A \vdash^S \Delta$, then the equations define the statement $[N/x]S$ and $\Gamma \vdash^{[N/x]S} \Delta$.
- (3) For all A, H : if $\Gamma \vdash N : A \mid \Delta$ and $\Gamma, x : A \triangleright H : B \mid \Delta$, then the equations define the extended neutral term $[N/x]H$ and $\Gamma \blacktriangleright [N/x]H : B \mid \Delta$.
- (4) For all $A, B, \mathbb{H}$: if $\Gamma \blacktriangleright \mathbb{H} : A \supset B \mid \Delta$ and $\Gamma \vdash N : A \mid \Delta$, then the equations define the extended neutral term $\mathbb{H}@N$ and $\Gamma \blacktriangleright \mathbb{H}@N : B \mid \Delta$.
- (5) For all A, B, M : if $\Gamma \vdash M : A \supset B \mid \Delta$ and $\Gamma \vdash N : A \mid \Delta$, then the equations define the proof term $M@N$ and $\Gamma \vdash M@N : B \mid \Delta$.
- (6) For all A, B, S : if $\Gamma \vdash^S a : A \supset B, \Delta$ and $\Gamma \vdash N : A \mid \Delta$, then the equations define the statement $[@N/a]S$ and $\Gamma \vdash^{[@N/a]S} a : B, \Delta$.
- (7) For all A, B, H : if $\Gamma \triangleright H : C \mid a : A \supset B, \Delta$ and $\Gamma \vdash N : A \mid \Delta$, then the equations define the neutral term $[@N/a]H$ and $\Gamma \triangleright [@N/a]H : C \mid a : B, \Delta$.
- (8) For all A, B, M : if $\Gamma \vdash M : C \mid a : A \supset B, \Delta$ and $\Gamma \vdash N : A \mid \Delta$, then the equations define the proof term $[@N/a]M$ and $\Gamma \vdash [@N/a]M : C \mid a : B, \Delta$.

As a consequence, the following inference rules:

$$\frac{\Gamma \vdash N : A \mid \Delta \quad \Gamma, x : A \vdash M : B \mid \Delta}{\Gamma \vdash [N/x]M : B \mid \Delta} \quad (1) \qquad \frac{\Gamma \vdash N : A \mid \Delta \quad \Gamma, x : A \vdash^S \Delta}{\Gamma \vdash^{[N/x]S} \Delta} \quad (2)$$

$$\frac{\Gamma \vdash N : A \mid \Delta \quad \Gamma, x : A \triangleright H : B \mid \Delta}{\Gamma \blacktriangleright [N/x]H : B \mid \Delta} \quad (3) \qquad \frac{\Gamma \blacktriangleright \mathbb{H} : A \supset B \mid \Delta \quad \Gamma \vdash N : A \mid \Delta}{\Gamma \blacktriangleright \mathbb{H}@N : B \mid \Delta} \quad (4)$$

$$\frac{\Gamma \vdash M : A \supset B \mid \Delta \quad \Gamma \vdash N : A \mid \Delta}{\Gamma \vdash M@N : B \mid \Delta} \quad (5) \qquad \frac{\Gamma \vdash^S a : A \supset B, \Delta \quad \Gamma \vdash N : A \mid \Delta}{\Gamma \vdash^{[@N/a]S} a : B, \Delta} \quad (6)$$

$$\frac{\Gamma \triangleright H : C \mid a : A \supset B, \Delta \quad \Gamma \vdash N : A \mid \Delta}{\Gamma \triangleright [@N/a]H : C \mid a : B, \Delta} \quad (7) \qquad \frac{\Gamma \vdash M : C \mid a : A \supset B, \Delta \quad \Gamma \vdash N : A \mid \Delta}{\Gamma \vdash [@N/a]M : C \mid a : B, \Delta} \quad (8)$$

are admissible in NKT .

Proof. For all A , M , S and H , the size $|A|$ was already defined, and the sizes $|M|$, $|S|$ and $|H|$ are defined by simultaneous structural recursion on M , S and H as follows: $|\lambda x^A.M| = 1 + |M|$, $|\mu a^A.S| = 1 + |S|$, $|\text{abort}(H)| = 1 + |H|$, $|a(H)| = 1 + |H|$, $a(\lambda x^A.M) = 2 + |M|$, $|x| = 1$ and $|HN| = |H| + |N|$. We also put $|hd(M)| = 1 + |M|$. The proof of items (1) to (8) is made by simultaneous induction on the lexicographically ordered pairs: $(|A|, |M|)$, $(|A|, |S|)$, $(|A|, |H|)$, $(f(\mathbb{H}, A, B), |\mathbb{H}|)$, $(|A| + |B|, |M|)$, $(|A| + |B|, |S|)$, $(|A| + |B|, |H|)$ and $(|A| + |B|, |M|)$, where $f(H, A, B) = 0$ and $f(hd(M), A, B) = |A| + |B|$.

In (1),

- Case $M = \lambda y^C.M'$. Suppose $\Gamma \vdash N : A \mid \Delta$ and $\Gamma, x : A \vdash M : B \mid \Delta$. We show that the above equations define the proof term $\lambda y^C.[N/x]M'$ and that $\Gamma \vdash \lambda y^C.[N/x]M' : B \mid \Delta$. By inversion of the inference rule $I\supset$, $B = C \supset D$ and $\Gamma, x : A, y : C \vdash M' : D \mid \Delta$. By admissibility of weakening, $\Gamma, y : C \vdash N : A \mid \Delta$. Now, $(|A|, |M'|) < (|A|, |M|)$, because $|M'| < |M| = 1 + |M'|$. By IH(1), the above equations define the proof term $[N/x]M'$ and $\Gamma, y : C \vdash [N/x]M' : D \mid \Delta$. Hence, $\lambda y^C.[N/x]M'$ is defined. Applying the inference rule $I\supset$, $\Gamma \vdash \lambda y^C.[N/x]M' : C \supset D \mid \Delta$.
- Case $M = \mu b^B.S$. Suppose $\Gamma \vdash N : A \mid \Delta$ and $\Gamma, x : A \vdash M : B \mid \Delta$. We show that the above equations define the proof term $\mu b^B.[N/x]S$ and that $\Gamma \vdash \mu b^B.[N/x]S : B \mid \Delta$. By inversion of the inference rule ACT , $\Gamma, x : A \vdash^S b : B, \Delta$. By admissibility of weakening, $\Gamma \vdash N : A \mid b : B, \Delta$. Now $(|A|, |S|) < (|A|, |\mu b^B.S|)$, because $|S| < |\mu b^B.S| = 1 + |S|$. By IH(2), the above equations define the statement $[N/x]S$ and $\Gamma \vdash^{[N/x]S} b : B, \Delta$. Hence, $\mu b^B.[N/x]S$ is defined. Applying the inference rule ACT , $\Gamma \vdash \mu b^B.[N/x]S : B \mid \Delta$.

In (2),

- Case $S = \text{abort}(H)$. Suppose $\Gamma \vdash N : A \mid \Delta$ and $\Gamma, x : A \vdash^S \Delta$. We show that the above equations define the statement $\text{abort}([N/x]H)$ and that $\Gamma \vdash^{\text{abort}([N/x]H)} \Delta$. By inversion of the inference rule $E\perp$, $\Gamma, x : A \triangleright H : \perp \mid \Delta$. Now, $(|A|, |H|) < (|A|, |\text{abort}(H)|)$, because $|H| < |\text{abort}(H)| = 1 + |H|$. By IH(3), the above equations define the extended neutral term $[N/x]H$ and $\Gamma \triangleright [N/x]H : \perp \mid \Delta$. hence, $\text{abort}([N/x]H)$ is defined. Applying an admissible inference rule, $\Gamma \vdash^{\text{abort}([N/x]H)} \Delta$.
- Case $S = a(H)$. Suppose $\Gamma \vdash N : A \mid \Delta$ and $\Gamma, x : A \vdash^S \Delta$. We show that the above equations define the statement $a([N/x]H)$ and that $\Gamma \vdash^{a([N/x]H)} \Delta$. By inversion of the inference rule

PSS , $A = p$ and $\Gamma, x : A \triangleright H : p \mid \Delta$, with $a : p \in \Delta$. Now $(|A|, |H|) < (|A|, |a(H)|)$, because $|H| < |a(H)| = 1 + |H|$. By IH(3), the above equations define the extended neutral term $[N/x]H$ and $\Gamma \blacktriangleright [N/x]H : p \mid \Delta$. Hence, $a([N/x]H)$ is defined. Applying an admissible inference rule, $\Gamma \frac{a([N/x]H)}{\Delta}$.

- Case $S = a(\lambda y^C.M)$. Suppose $\Gamma \vdash N : A \mid \Delta$ and $\Gamma, x : A \frac{S}{\Delta}$. We show that the above equations define the statement $a(\lambda y^C.[N/x]M)$ and that $\Gamma \frac{a(\lambda y^C.[N/x]M)}{\Delta}$. By inversion of the inference rule CTR , $a : C \supset D \in \Delta$ and $\Gamma, x : A, y : C \vdash M : D \mid \Delta$. By admissibility of weakening, $\Gamma, y : C \vdash N : A \mid \Delta$. Now, $(|A|, |M|) < (|A|, |a(\lambda y^C.M)|)$, because $|M| < |a(\lambda y^C.M)| = 2 + |M|$. By IH(1), $\Gamma, y : C \vdash [N/x]M : D \mid \Delta$. Hence, $a(\lambda y^C.[N/x]M)$ is defined. Applying the inference rule CTR , $\Gamma \frac{a(\lambda y^C.[N/x]M)}{\Delta}$.

In (3),

- Case $H = x$. Suppose $\Gamma \vdash N : A \mid \Delta$ and $\Gamma, x : A \triangleright H : B \mid \Delta$. We show that $\Gamma \blacktriangleright hd(N) : B \mid \Delta$. Applying the A rule, $B = A$. By Definition 4.20, $\Gamma \blacktriangleright hd(N) : B \mid \Delta$.
- Case $H = y$, with $y \neq x$. Suppose $\Gamma \vdash N : A \mid \Delta$ and $\Gamma, x : A \triangleright H : B \mid \Delta$. Since $y : B \in \Gamma$, applying the A rule, $\Gamma \triangleright y : B \mid \Delta$. By Definition 4.20, $\Gamma \blacktriangleright y : B \mid \Delta$.
- Case $H = H'N'$. Suppose $\Gamma \vdash N : A \mid \Delta$ and $\Gamma, x : A \triangleright H : B \mid \Delta$. We show that the above equations define the extended neutral term $([N/x]H')@([N/x]N')$ and that $\Gamma \blacktriangleright ([N/x]H')@([N/x]N') : B \mid \Delta$. By inversion of the inference rule $E\supset$, $\Gamma, x : A \vdash N' : C \mid \Delta$ and $\Gamma, x : A \triangleright H' : C \supset B \mid \Delta$. Now, $(|A|, |N'|) < (|A|, |H|)$, because $|N'| < |H| = |N'| + |H'|$. By IH(1), the above equations define the proof term $[N/x]N'$ and $\Gamma, x : A \vdash [N/x]N' : C \mid \Delta$. Also, $(|A|, |H'|) < (|A|, |H|)$, because $|H'| < |H| = |N'| + |H'|$. By IH(4), the above equations define the extended neutral term $[N/x]H'$ and $\Gamma \blacktriangleright [N/x]H' : C \supset B \mid \Delta$. Now, we have two cases: if $[N/x]H'$ is a head term, then $H' = x$ and $A = C \supset B$ and $(f([N/x]H', C, D), |[N/x]H'|) < (|A|, |H|)$, because $f([N/x]H', C, D) = |C| + |B| < |A| = 1 + |C| + |B|$; otherwise $(f([N/x]H', C, D), |[N/x]H'|) < (|A|, |H|)$, because $f([N/x]H', C, D) = 0 < |A|$. So, by IH(3), the above equations define the extended neutral term $([N/x]H')@([N/x]N')$ and $\Gamma \blacktriangleright ([N/x]H')@([N/x]N') : B \mid \Delta$.

In (4),

- Case $\mathbb{H} = H$. Suppose $\Gamma \blacktriangleright \mathbb{H} : A \supset B \mid \Delta$ and $\Gamma \vdash N : A \mid \Delta$. We have that $H@N = HN$ is defined. We show that $\Gamma \blacktriangleright HN : B \mid \Delta$. By Definition 4.20, $\Gamma \triangleright H : A \supset B \mid \Delta$. Applying

the inference rule $E\supset, \Gamma \triangleright HN : B \mid \Delta$. By Definition 4.20, $\Gamma \blacktriangleright HN : B \mid \Delta$.

- Case $\mathbb{H} = hd(M)$. Suppose $\Gamma \blacktriangleright \mathbb{H} : A \supset B \mid \Delta$ and $\Gamma \vdash N : A \mid \Delta$. We show that the above equations define the extended neutral term $hd(M@N)$ and that $\Gamma \blacktriangleright hd(M@N) : B \mid \Delta$. Definition 4.20, $\Gamma \vdash M : A \supset B \mid \Delta$. Now, $(|A| + |B|, |M|) < (f(\mathbb{H}, A, B), \mathbb{H})$, because $|M| < |\mathbb{H}| = 1 + |M|$. By IH(5), the above equations define the proof term $M@N$ and $\Gamma \vdash M@N : B \mid \Delta$. Hence, $hd(M@N)$ is defined. By Definition 4.20, $\Gamma \blacktriangleright hd(M@N) : B \mid \Delta$.

In (5),

- Case $M = \lambda y^A.M'$. Suppose $\Gamma \vdash M : A \supset B \mid \Delta$ and $\Gamma \vdash N : A \mid \Delta$. We show that the above equations define the proof term $[N/y]M'$ and that $\Gamma \vdash [N/y]M' : B \mid \Delta$. By inversion of the inference rule $I\supset, \Gamma, y : A \vdash M' : B \mid \Delta$. Now, $(|A|, |M'|) < (|A| + |B|, |M|)$, because $|A| < |A| + |B|$. By IH(1), the above equations define the proof term $[N/y]M'$ and $\Gamma \vdash [N/y]M' : B \mid \Delta$.
- Case $M = \mu a^{A \supset B}.S$. Suppose $\Gamma \vdash M : A \supset B \mid \Delta$ and $\Gamma \vdash N : A \mid \Delta$. We show that the above equations define the proof term $\mu a^B.[@N/a]S$ and that $\Gamma \vdash \mu a^B.[@N/a]S : B \mid \Delta$. By inversion of the inference rule $ACT, \Gamma \vdash^S a : A \supset B, \Delta$. Now $(|A| + |B|, |S|) < (|A| + |B|, |M|)$, because $|S| < |M| = 1 + |S|$. By IH(6), the above equations define the statement $[@N/a]S$ and $\Gamma \vdash^{[@N/a]S} a : B, \Delta$. Hence, $\mu a^B.[@N/a]S$ is defined. Applying the inference rule $ACT, \Gamma \vdash \mu a^B.[@N/a]S : B \mid \Delta$.

In (6),

- Case $S = \text{abort}(H)$. Suppose $\Gamma \vdash^S a : A \supset B, \Delta$ and $\Gamma \vdash N : A \mid \Delta$. We show that the above equations define the statement $\text{abort}([@N/a]H)$ and that $\Gamma \vdash^{\text{abort}([@N/a]H)} a : B, \Delta$. By inversion of the inference rule $E\perp, \Gamma \triangleright H : \perp \mid a : A \supset B, \Delta$. Now, $(|A| + |B|, |H|) < (|A| + |B|, |S|)$, because $|H| < |S| = 1 + |H|$. By IH(7), the above equations define the neutral term $[@N/a]H$ and $\Gamma \triangleright [@N/a]H : \perp \mid a : B, \Delta$. Hence, $\text{abort}([@N/a]H)$ is defined. Applying the inference rule $E\perp, \Gamma \vdash^{\text{abort}([@N/a]H)} a : B, \Delta$.
- Case $S = c(H)$, with $c \neq a$. Suppose $\Gamma \vdash^S a : A \supset B, \Delta$ and $\Gamma \vdash N : A \mid \Delta$. We show that the above equations define the statement $c([@N/a]H)$ and that $\Gamma \vdash^{c([@N/a]H)} a : B, \Delta$. By inversion of the inference rule $PSS, c : p \in \Delta$ and $\Gamma \triangleright H : p \mid a : A \supset B, \Delta$. Now, $(|A| + |B|, |H|) < (|A| + |B|, |S|)$, because $|H| < |S| = 1 + |H|$. By IH(7), the

above equations define the neutral term $[@N/a]H$ and $\Gamma \triangleright [@N/a]H : p \mid a : B, \Delta$. Hence, $c([@N/a]H)$ is defined. Applying the inference rule PSS , $\Gamma \mid \frac{c([@N/a]H)}{a : B, \Delta}$.

- Case $S = a(\lambda x^A.M)$. Suppose $\Gamma \mid \frac{S}{a : A \supset B, \Delta}$ and $\Gamma \vdash N : A \mid \Delta$. We show that the above equations define the statement $a(hd([N/x]([@N/a]M)))$ and that $\Gamma \mid \frac{a(hd([N/x]([@N/a]M)))}{a : B, \Delta}$. By inversion of the inference rule CTR , $\Gamma, x : A \vdash M : B \mid a : A \supset B, \Delta$. By admissibility of weakening, $\Gamma, x : A \vdash N : A \mid \Delta$. Now, $(|A| + |B|, |M|) < (|A| + |B|, |S|)$, because $|M| < |S| = 2 + |M|$. By IH(8), the above equations define the proof term $[@N/a]M$ and $\Gamma, x : A \vdash [@N/a]M : B \mid a : B, \Delta$. By admissibility of weakening, $\Gamma \vdash N : A \mid a : B, \Delta$. Now, $(|A|, |M|) < (|A| + |B|, |S|)$, because $|A| < |A| + |B|$. By IH(1), the above equations define the proof term $[N/x]([@N/a]M)$ and $\Gamma \vdash [N/x]([@N/a]M) : B \mid a : B, \Delta$. By Definition 4.20, $\Gamma \blacktriangleright hd([N/x]([@N/a]M)) : B \mid a : B, \Delta$. Hence, $a(hd([N/x]([@N/a]M)))$ is defined. By Proposition 4.14, $\Gamma \mid \frac{a(hd([N/x]([@N/a]M)))}{a : B, \Delta}$.
- Case $S = c(\lambda x^A.M)$, with $c \neq a$. Suppose $\Gamma \mid \frac{S}{a : A \supset B, \Delta}$ and $\Gamma \vdash N : A \mid \Delta$. We show that the above equations define the statement $c(\lambda x^A.[@N/a]M)$ and that $\Gamma \mid \frac{c(\lambda x^A.[@N/a]M)}{a : B, \Delta}$. By inversion of the inference rule CTR , $c : A \supset C \in \Delta$ and $\Gamma, x : A \vdash M : B \mid a : A \supset B, \Delta$. By admissibility of weakening, $\Gamma, x : A \vdash N : A \mid \Delta$. Now, $(|A| + |B|, |M|) < (|A| + |B|, |S|)$, because $|M| < |S| = 2 + |M|$. By IH(8), the above equations define the proof term $[@N/a]M$ and $\Gamma, x : A \vdash [@N/a]M : C \mid a : B, \Delta$. Hence, $c(\lambda x^A.[@N/a]M)$ is defined. Applying the inference rule CTR , $\Gamma \mid \frac{c(\lambda x^A.[@N/a]M)}{a : B, \Delta}$.

In (7),

- Case $H = x$. Suppose $\Gamma \triangleright H : C \mid a : A \supset B, \Delta$ and $\Gamma \vdash N : A \mid \Delta$. We show that $\Gamma \triangleright x : C \mid a : B, \Delta$. By the A rule, $x : C \in \Gamma$. Then, applying the same rule, $\Gamma \triangleright x : C \mid a : B, \Delta$.
- Case $H = H'N'$. Suppose $\Gamma \triangleright H : C \mid a : A \supset B, \Delta$ and $\Gamma \vdash N : A \mid \Delta$. We show that the above equations define the neutral term $([@N/a]H')([@N/a]N')$ and that $\Gamma \triangleright ([@N/a]H')([@N/a]N') : C \mid a : B, \Delta$. By inversion of the inference rule $E\supset$, $\Gamma \vdash N' : D \mid a : A \supset B, \Delta$ and $\Gamma \triangleright H' : D \supset C \mid a : A \supset B, \Delta$. Now, $(|A| + |B|, |H'|) < (|A| + |B|, |H|)$, because $|H'| < |H| = |H'| + |N'|$. By IH(7), the above equations define the neutral term $[@N/a]H'$ and $\Gamma \triangleright [@N/a]H' : D \supset C \mid a : B, \Delta$. Also, $(|A| + |B|, |N'|) < (|A| + |B|, |H|)$, because $|N'| < |H| = |H'| + |N'|$. By IH(8), the above equations define the proof term $[@N/a]N'$ and $\Gamma \vdash [@N/a]N' : D \mid a : B, \Delta$. Hence, $([@N/a]H')([@N/a]N')$ is defined. Applying the inference rule $E\supset$, $\Gamma \triangleright ([@N/a]H')([@N/a]N') : C \mid a : B, \Delta$.

In (8),

- Case $M = \lambda y^D.M'$. Suppose $\Gamma \vdash M : C \mid a : A \supset B, \Delta$ and $\Gamma \vdash N : A \mid \Delta$. We show that the above equations define the proof term $\lambda y^D.[@N/a]M'$ and that $\Gamma \vdash \lambda y^D.[@N/a]M' : C \mid a : B, \Delta$. By inversion of the inference rule $I\supset$, $C = D \supset E$ and $\Gamma, y : D \vdash M : E \mid a : A \supset B, \Delta$. By admissibility of weakening, $\Gamma, y : D \vdash N : A \mid \Delta$. Now, $(|A| + |B|, |M'|) < (|A| + |B|, |M|)$, because $|M'| < |M| = 1 + |M'|$. By IH(8), the above equations define the proof term $[@N/a]M'$ and $\Gamma, y : D \vdash [@N/a]M' : E \mid a : B, \Delta$. Hence, $\lambda y^D.[@N/a]M'$ is defined. Applying the inference rule $I\supset$, $\Gamma \vdash \lambda y^D.[@N/a]M' : D \supset E \mid a : B, \Delta$.
- Case $M = \mu c^C.S$. Suppose $\Gamma \vdash M : C \mid a : A \supset B, \Delta$ and $\Gamma \vdash N : A \mid \Delta$. We show that the above equations define the proof term $\mu c^C.[@N/a]S$ and that $\Gamma \vdash \mu c^C.[@N/a]S : C \mid a : B, \Delta$. By inversion of the inference rule ACT , $\Gamma \vdash^S c : C, a : A \supset B, \Delta$. By admissibility of weakening, $\Gamma \vdash N : A \mid \Delta, c : C$. Now, $(|A| + |B|, |S|) < (|A| + |B|, |M|)$, because $|S| < |M| = 1 + |S|$. By IH(6), the above equations define the statement $[@N/a]S$ and $\Gamma \vdash^{[@N/a]S} a : B, c : C, \Delta$. Hence, $\mu c^C.[@N/a]S$ is defined. Applying the inference rule ACT , $\Gamma \vdash \mu c^C.[@N/a]S : C \mid a : B, \Delta$.

□

The constructor $a(M)$ of the $\lambda\mu$ -calculus is split in NKT into $a(M)$ and $a(H)$, with M even restricted to a λ -abstraction. This has an impact on the operation $[@N/a]_-$. It is a “structural substitution” [49, 3], but it only looks for occurrences of a of the form $a(\lambda x^A.M)$. For this reason, the typing rules (6), (7) and (8) from Theorem 4.3 are in fact elimination rules for alternative implications.

4.5 Proof search in NKT

Proof search in NKT extends the process in NJT , therefore also following the discipline of the intercalation calculus. First, as in focusing, implications in the right-hand side are decomposed by inversion of rule $I\supset$. This phase is explicitly stopped by an application of rule ACT , producing a certain sequent $\Gamma \vdash \Delta$. Next there is an alternative: either we change direction and decompose implications by top-down application of rule $E\supset$, starting from an assumption chosen in Γ by rule A , until we compute $\perp$ or an atom p (the latter we hope to see in Δ) – this is very much like in NJT ; or we pick an implication in Δ , decide this formula results from an introduction with contraction, and return to the inversion phase – this is the new option offered by rule CTR .

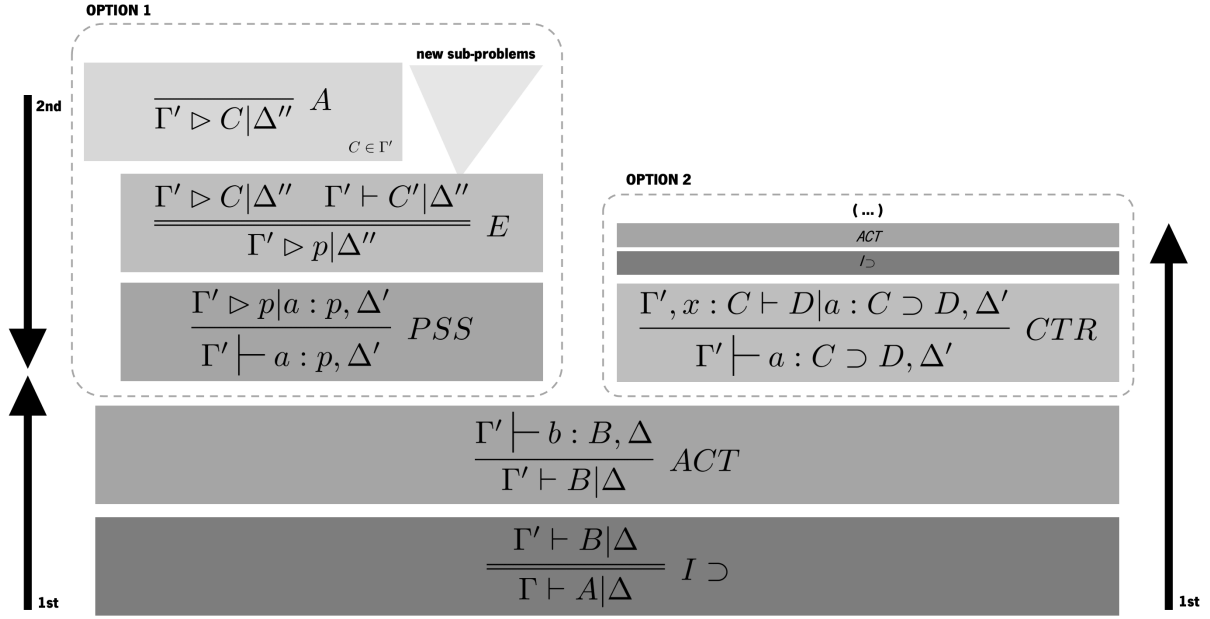


Figure 4: Intercalation in NKT

Example 4.7. Keeping with Peirce's Law, $((p \supset q) \supset p) \supset p$, from the previous examples, we have an example of proof search in LKT described informally. Applications of the ∂ -inference rules are signaled by the symbols $\uparrow$ and $\downarrow$; Γ contains the assumption $(x : (p \supset q) \supset p)$ and Δ contains the alternative conclusions $(b : p)$ and $(a : ((p \supset q) \supset p) \supset p)$.

$$\begin{array}{ccc}
 \overline{\vdash ((p \supset q) \supset p) \supset p} \partial \uparrow & \longrightarrow & \overline{\vdash a : ((p \supset q) \supset p) \supset p} \partial \uparrow \\
 & & \vdash ((p \supset q) \supset p) \supset p \text{ ACT} \\
 & & \downarrow \\
 \overline{\Gamma \vdash b : p, a : ((p \supset q) \supset p) \supset p} \partial \uparrow & & \overline{x : (p \supset q) \supset p \vdash p | a : ((p \supset q) \supset p) \supset p} \partial \uparrow \\
 \Gamma \vdash p | a : ((p \supset q) \supset p) \supset p \text{ ACT} & & \vdash a : ((p \supset q) \supset p) \supset p \text{ CTR} \\
 \vdash a : ((p \supset q) \supset p) \supset p \text{ CTR} & & \vdash a : ((p \supset q) \supset p) \supset p \text{ ACT} \\
 \vdash ((p \supset q) \supset p) \supset p \text{ ACT} & \longleftarrow & \vdash ((p \supset q) \supset p) \supset p \text{ ACT} \\
 & & \downarrow
 \end{array}$$

$$\begin{array}{c}
\downarrow \\
\frac{\frac{\frac{\Gamma \triangleright (p \supset q) \supset p \mid \Delta}{\Gamma \vdash b : p, a : ((p \supset q) \supset p) \supset p} \partial \downarrow}{\Gamma \vdash p \mid a : ((p \supset q) \supset p) \supset p} ACT}{\vdash a : ((p \supset q) \supset p) \supset p} CTR}{\vdash ((p \supset q) \supset p) \supset p \mid} ACT \quad \longrightarrow \quad \frac{\frac{\frac{\Gamma \triangleright (p \supset q) \supset p \mid \Delta}{\Gamma \vdash p \mid b : p, a : ((p \supset q) \supset p) \supset p} \partial \downarrow}{\Gamma \vdash p \mid a : ((p \supset q) \supset p) \supset p} ACT}{\vdash a : ((p \supset q) \supset p) \supset p} CTR}{\vdash ((p \supset q) \supset p) \supset p \mid} ACT \\
\downarrow \\
\frac{\frac{\frac{\Gamma \triangleright (p \supset q) \supset p \mid \Delta}{\Gamma \vdash b : p, a : ((p \supset q) \supset p) \supset p} \partial \downarrow}{\Gamma \vdash p \mid a : ((p \supset q) \supset p) \supset p} ACT}{\vdash a : ((p \supset q) \supset p) \supset p} CTR}{\vdash ((p \supset q) \supset p) \supset p \mid} ACT \quad \longleftarrow \quad \frac{\frac{\frac{\Gamma \triangleright (p \supset q) \supset p \mid \Delta}{\Gamma \vdash p \mid b : p, a : ((p \supset q) \supset p) \supset p} \partial \downarrow}{\Gamma \vdash p \mid a : ((p \supset q) \supset p) \supset p} ACT}{\vdash a : ((p \supset q) \supset p) \supset p} CTR}{\vdash ((p \supset q) \supset p) \supset p \mid} ACT \\
(\dots)
\end{array}$$

To formalize this process, we extend NKT to NKT_∂ , in a similar way as we extended LKT to LKT_∂ before in this chapter.

Definition 4.21 (Formal Sequents of NKT_∂). NKT_∂ handles formal proof term sequents, $\Gamma \vdash A \mid \Delta$, ranged over $\underline{\sigma}$; formal neutral term sequents, $\Gamma \triangleright A \mid \Delta$, ranged over $\underline{\rho}$; and formal statement sequents, $\Gamma \vdash \Delta$, ranged over $\underline{\zeta}$.

Definition 4.22 (Proof Terms of NKT_∂). Partial proof terms M , neutral terms H , and statements S are generated as follows:

$$\begin{array}{ll}
\text{(Partial proof terms)} & M, N ::= \lambda x^A.M \mid \mu a^A.S \mid \underline{\sigma} \\
\text{(Partial neutral terms)} & H ::= x \mid HN \\
\text{(Partial statements)} & S ::= \text{abort}(H) \mid a(H) \mid a(\lambda x^A.M) \mid \underline{\zeta} \mid \text{ab}(H, \underline{\rho})
\end{array}$$

where $\underline{\sigma}$ ranges over formal proof term sequents; $\underline{\rho}$ ranges over formal neutral term sequents; and $\underline{\zeta}$ ranges over formal statement sequents. As in NJT_∂ , formal neutral sequents $\underline{\rho}$ are not partial neutral terms, but, together with a partial neutral term H in a structure of the form $\text{ab}(H, \underline{\rho})$, they are partial statements instead. Contrary to LJT_∂ and LKT_∂ , in the systems of proof search in natural deduction, it is not immediate to infer in which syntactic class the formal sequents end up.

The typing system of NKT_{∂} handles three forms of partial sequents.

Definition 4.23 (Partial Sequents of NKT_{∂}). NKT_{∂} handles partial proof term sequents $\Xi \Vdash M : \sigma$, partial neutral term sequents $\Xi \Vdash H : \rho$, and partial statement sequents $\Xi \Vdash S : \zeta$. Lists of formal sequents are ranged over by Ξ , and written $[\underline{\sigma}, \underline{\rho}, \underline{\rho}', \underline{\zeta}]$ for instance.

Definition 4.24 (Inference Rules of NKT_{∂}). The set of inference rules of NKT_{∂} is defined as:

$$\begin{array}{c}
\frac{}{[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma} \partial \quad \frac{\Xi \Vdash H : \rho \quad \rho = (\Gamma \triangleright A \mid \Delta)}{\Xi @ [\underline{\rho}] \Vdash \text{ab}(H, \underline{\rho}) : (\Gamma \vdash \Delta)} \partial \quad \frac{}{[\underline{\zeta}] \Vdash \underline{\zeta} : \zeta} \partial \\
\\
\frac{}{\epsilon \Vdash x : (\Gamma, x : A \triangleright A \mid \Delta)} A \quad \frac{\Xi \Vdash H : (\Gamma \triangleright \perp \mid \Delta)}{\Xi \Vdash \text{abort}(H) : (\Gamma \vdash \Delta)} E_{\perp} \quad \frac{\Xi \Vdash S : (\Gamma \vdash a : A, \Delta)}{\Xi \Vdash \mu a^A.S : (\Gamma \vdash A \mid \Delta)} ACT \\
\\
\frac{\Xi \Vdash M : (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)}{\Xi \Vdash a(\lambda x^A.M) : (\Gamma \vdash a : A \supset B, \Delta)} CTR \quad \frac{\Xi \Vdash H : (\Gamma \triangleright p \mid a : p, \Delta)}{\Xi \Vdash a(H) : (\Gamma \vdash a : p, \Delta)} PSS \\
\\
\frac{\Xi \Vdash M : (\Gamma, x : A \vdash B \mid \Delta)}{\Xi \Vdash \lambda x^A.M : (\Gamma \vdash A \supset B \mid \Delta)} I_{\supset} \quad \frac{\Xi_1 \Vdash H : (\Gamma \triangleright A \supset B \mid \Delta) \quad \Xi_2 \Vdash N : (\Gamma \vdash A \mid \Delta)}{\Xi_1 @ \Xi_2 \Vdash HN : (\Gamma \triangleright B \mid \Delta)} E_{\supset}
\end{array}$$

As in LKT , there is a partial inference rule for each inference rule of NKT , and also three ∂ -rules, which collect the proof obligations. Notice the increased complexity of the second ∂ -rule, which says that $\text{ab}(H, \underline{\rho})$ is the gap between the sequent $\rho = (\Gamma \triangleright A \mid \Delta)$, obtained according to H , and the sequent $\Gamma \vdash \Delta$. Unlike NJT_{∂} , proof obligations here are formal sequents.

We now prove the result of conservativity, i.e., derivability is partial derivability from the empty list of proof obligations.

Proposition 4.15 (Conservativity).

1. NKT derives $\Gamma \vdash M : A \mid \Delta$ iff NKT_{∂} derives $\epsilon \Vdash M : (\Gamma \vdash A \mid \Delta)$.
2. NKT derives $\Gamma \triangleright H : A \mid \Delta$ iff NKT_{∂} derives $\epsilon \Vdash H : (\Gamma \triangleright A \mid \Delta)$.
3. NKT derives $\Gamma \vdash^S \Delta$ iff NKT_{∂} derives $\epsilon \Vdash S : (\Gamma \vdash \Delta)$.

Proof. The "only if" statements are proved by simultaneous induction on the derivations of $\Gamma \vdash M : A \mid \Delta$, $\Gamma \triangleright H : A \mid \Delta$ and $\Gamma \vdash^S \Delta$.

- Case A . Suppose $\Gamma, x : A \triangleright x : A \mid \Delta$. Since x is total, applying the inference rule A , $\epsilon \Vdash x : (\Gamma, x : A \triangleright A \mid \Delta)$ in NKT_{∂} .
- Case $E\perp$. Suppose $\Gamma \frac{\text{abort}(H)}{} \Delta$ is obtained from $\Gamma \triangleright H : \perp \mid \Delta$. By IH, $\epsilon \Vdash H : (\Gamma \triangleright \perp \mid \Delta)$. Applying the inference rule $E\perp$, $\epsilon \Vdash \text{abort}(H) : (\Gamma \mid \Delta)$ in NKT_{∂} .
- Case ACT . Suppose $\Gamma \vdash \mu a^A.S : A \mid \Delta$ is obtained from $\Gamma \frac{S}{} a : A, \Delta$. By IH, $\epsilon \Vdash S : (\Gamma \mid a : A, \Delta)$. Applying the inference rule ACT , $\epsilon \Vdash \mu a^A.S : (\Gamma \vdash A \mid \Delta)$ in NKT_{∂} .
- Case PSS . Suppose $\Gamma \frac{a(H)}{} a : p, \Delta$ is obtained from $\Gamma \triangleright H : p \mid a : p, \Delta$. By IH, $\epsilon \Vdash H : (\Gamma \triangleright p \mid a : p, \Delta)$. Applying the inference rule PSS , $\epsilon \Vdash a(H) : (\Gamma \mid a : p, \Delta)$ in NKT_{∂} .
- Case CTR . Suppose $\Gamma \frac{a(\lambda x^A.M)}{} a : A \supset B, \Delta$ is obtained from $\Gamma, x : A \vdash M : B \mid a : A \supset B, \Delta$. By IH, $\epsilon \Vdash M : (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)$. Applying the inference rule CTR , $\epsilon \Vdash a(\lambda x^A.M) : (\Gamma \mid a : A \supset B, \Delta)$ in NKT_{∂} .
- Case $I\supset$. Suppose $\Gamma \vdash \lambda x^A.M : A \supset B \mid \Delta$ is obtained from $\Gamma, x : A \vdash M : B \mid \Delta$. By IH, $\epsilon \Vdash M : (\Gamma, x : A \vdash B \mid \Delta)$. Applying the inference rule $I\supset$, $\epsilon \Vdash \lambda x^A.M : (\Gamma \vdash A \supset B \mid \Delta)$ in NKT_{∂} .
- Case $E\supset$. Suppose $\Gamma \triangleright HN : B \mid \Delta$ is obtained from $\Gamma \triangleright H : A \supset B \mid \Delta$ and $\Gamma \vdash N : A \mid \Delta$. By IH, $\epsilon \Vdash H : (\Gamma \triangleright A \supset B \mid \Delta)$ and $\epsilon \Vdash N : (\Gamma \vdash A \mid \Delta)$. Applying the inference rule $E\supset$, $\epsilon \Vdash HN : (\Gamma \triangleright B \mid \Delta)$ in NKT_{∂} .

For the “if” statements, we prove:

- (1) If NKT_{∂} derives $\Xi \Vdash M : (\Gamma \vdash A \mid \Delta)$, with $M \in NKT$, then $\Xi = \epsilon$ and NKT derives $\Gamma \vdash M : A \mid \Delta$.
- (2) If NKT_{∂} derives $\Xi \Vdash H : (\Gamma \triangleright A \mid \Delta)$, with $H \in NKT$, then $\Xi = \epsilon$ and NKT derives $\Gamma \triangleright H : A \mid \Delta$.
- (3) If NKT_{∂} derives $\Xi \Vdash S : (\Gamma \mid \Delta)$, with $S \in NKT$, then $\Xi = \epsilon$ and NKT derives $\Gamma \frac{S}{} \Delta$.

The proof is by simultaneous induction on the derivations of $\Xi \Vdash M : (\Gamma \vdash A)$ and $\Xi \Vdash H : (\Gamma \triangleright A)$.

- Case ∂ . Suppose $[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma$. Since $\underline{\sigma} \notin NKT$, the statement is trivially true. Similar reasoning applies to $\Xi @ [\underline{\rho}] \Vdash \text{ab}(H, \underline{\rho}) : (\Gamma \mid \Delta)$ and $[\underline{\zeta}] \Vdash \underline{\zeta} : \zeta$.

- Case A . Suppose $\epsilon \Vdash x : (\Gamma, x : A \triangleright A \mid \Delta)$, with $x \in NKT$. Applying the inference rule A , $\Gamma, x : A \triangleright x : A \mid \Delta$ in NKT .
- Case $E\perp$. Suppose $\Xi \Vdash \text{abort}(H) : (\Gamma \vdash \Delta)$ is obtained from $\Xi \Vdash H : (\Gamma \triangleright \perp \mid \Delta)$, with $\text{abort}(H) \in NKT$. Then $H \in NKT$. By IH, $\Xi = \epsilon$ and $\Gamma \triangleright H : \perp \mid \Delta$. Applying the inference rule $E\perp$, $\Gamma \vdash \frac{\text{abort}(H)}{} \Delta$ in NKT .
- Case ACT . Suppose $\Xi \Vdash \mu a^A.S : (\Gamma \vdash A \mid \Delta)$ is obtained from $\Xi \Vdash S : (\Gamma \vdash a : A, \Delta)$, with $\mu a^A.S \in NKT$. Then $S \in NKT$. By IH, $\Xi = \epsilon$ and $\Gamma \vdash \frac{S}{a : A, \Delta}$. Applying the inference rule ACT , $\Gamma \vdash \mu a^A.S : A \mid \Delta$ in NKT .
- Case PSS . Suppose $\Xi \Vdash a(H) : (\Gamma \vdash a : p, \Delta)$ is obtained from $\Xi \Vdash H : (\Gamma \triangleright p \mid a : p, \Delta)$, with $a(H) \in NKT$. Then $H \in NKT$. By IH, $\Xi = \epsilon$ and $\Gamma \triangleright H : p \mid a : p, \Delta$. Applying the inference rule PSS , $\Gamma \vdash \frac{a(H)}{a : p, \Delta}$ in NKT .
- Case CTR . Suppose $\Xi \Vdash a(\lambda x^A.M) : (\Gamma \vdash a : A \supset B, \Delta)$ is obtained from $\Xi \Vdash M : (\Gamma, x : A \vdash B \mid \Delta)$, with $a(\lambda x^A.M) \in NKT$. Then $M \in NKT$. By IH, $\Xi = \epsilon$ and $\Gamma, x : A \vdash M : B \mid a : A \supset B, \Delta$. Applying the inference rule CTR , $\Gamma \vdash \frac{a(\lambda x^A.M)}{a : A \supset B, \Delta}$ in NKT .
- Case $I\supset$. Suppose $\Xi \Vdash \lambda x^A.M : (\Gamma \vdash A \supset B \mid \Delta)$ is obtained from $\Xi \Vdash M : (\Gamma, x : A \vdash B \mid \Delta)$, with $\lambda x^A.M \in NKT$. Then $M \in NKT$. By IH, $\Xi = \epsilon$ and $\Gamma, x : A \vdash M : B \mid \Delta$. Applying the inference rule $I\supset$, $\Gamma \vdash \lambda x^A.M : A \supset B \mid \Delta$ in NKT .
- Case $E\supset$. Suppose $\Xi \Vdash HN : (\Gamma \triangleright B \mid \Delta)$ is obtained from $\Xi_1 \Vdash H : (\Gamma \triangleright A \supset B \mid \Delta)$ and $\Xi_2 \Vdash N : (\Gamma \vdash A \mid \Delta)$, with $HN \in NKT$. Then $\Xi = \Xi_1 @ \Xi_2$ and $N, H \in NKT$. By IH, $\Xi_1 = \Xi_2 = \epsilon$ and $\Gamma \triangleright H : A \supset B \mid \Delta$ and $\Gamma \vdash N : A \mid \Delta$. Hence, $\Xi = \epsilon @ \epsilon = \epsilon$. Applying the inference rule $E\supset$, $\Gamma \triangleright HN : B \mid \Delta$ in NKT .

□

The search process in NKT is formalized with the rules in the following definitions, according to which proof obligations can be fulfilled by reducing M .

Definition 4.25 (Reduction Rules of NKT_∂). The reduction rules for proof search in NKT are:

$$\begin{array}{ll}
(SIR) & \frac{\Gamma \vdash a : A \supset B, \Delta}{\Gamma \vdash a : A \supset B, \Delta} \rightarrow a(\lambda x^A.(\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)) \\
(KIR) & \frac{\Gamma \vdash A \supset B \mid \Delta}{\Gamma \vdash A \supset B \mid \Delta} \rightarrow \lambda x^A.(\Gamma, x : A \vdash B \mid \Delta) \\
(FIR) & \frac{\Gamma \vdash A \mid \Delta}{\Gamma \vdash A \mid \Delta} \rightarrow \mu a^A.(\Gamma \vdash a : A, \Delta) \\
(SFD) & \frac{\Gamma, x : A \vdash \Delta}{\Gamma, x : A \vdash \Delta} \rightarrow \text{ab}(x, (\Gamma, x : A \supset A \mid \Delta)) \\
(KFD) & \text{ab}(H, (\Gamma \supset A \supset B \mid \Delta)) \rightarrow \text{ab}(H(\Gamma \vdash A \mid \Delta), (\Gamma \supset B \mid \Delta)) \\
(FFD_{atm}) & \text{ab}(H, (\Gamma \supset p \mid a : p, \Delta)) \rightarrow a(H) \\
(FFD_{abs}) & \text{ab}(H, (\Gamma \supset \perp \mid \Delta)) \rightarrow \text{abort}(H)
\end{array}$$

Each reduction rule is in correspondence with an inference rule. The names of the reduction rules stand for: start inversion right (SIR), keep inversion right (KIR), finish inversion right (FIR), start focus down (SFD), keep focus down (KFD), finish focus down – atomic (FFD_{atm}) and absurdity (FFD_{abs}). Notice how $a(H)$ is generated when finishing the focus down phase, while $a(\lambda x^A.M)$ signals the start of the right inversion phase. Notice also that a redex is not necessarily a formal sequent.

We now explain how the reduction rules presented in Definition 4.25 describe bidirectional proof search in NKT . Recall the rules in Definitions 4.15 and 4.18 and the diagram in Figure 4. Our goal is to prove $\Gamma \vdash A \mid \Delta$.

- As long as the selected conclusion A is an implication, we apply $I\supset$ bottom-up to decompose the implication in the right-hand side formula. The inversion phase is implemented by the reduction rule KIR : the redex sequent evolves to a more determined proof term, reflecting the bottom-up application of $I\supset$.
- The inversion phase ends with an application of the ACT rule, in which the formula A loses its distinguished position as the selected conclusion, and a sequent without any formula on the focus or any selected conclusion is obtained. This is implemented by the reduction rule FIR , which brings the search problem to the form $\Gamma' \vdash a : A, \Delta$.
- Now, there is an alternative: we can return to the inversion-right phase by applying the inference rule CTR and picking an implication on the right-hand side that results from an introduction with a contraction – this is implemented by the reduction rule SIR ; or we leave the sequent $\Gamma' \vdash a : A, \Delta$ on hold and change the direction of the search, starting the focus-down phase. For the second option, we apply the inference rule A , which selects a formula B in Γ' to put in the focus, that is, to the right of the symbol $\supset$. This is implemented by the reduction rule SFD and converts our search problem to the form $\text{ab}(H, \Gamma' \supset B \mid \Delta)$.

- As long as the formula B in the focus is not atomic or the absurdity, we apply $E\supset$ in the top-down direction. This is implemented by the reduction rule KFD , which simplifies the focus, evolves the neutral term H , and launches a new proof search problem in the right premiss of $E\supset$.
- If, in the left premiss, we ever reach an atomic focus matching the type of a co-variable on the right-hand side, or the absurdity, this branch of the search succeeds, and we can conclude the proof by applying the PSS rule, reconnecting to the derivation produced before, and left on hold, in the inversion phase. In the reduction system, the end of the top-down focusing phase is then determined by one of the reduction rules FFD_{atm} or FFD_{abs} .

The non-determinism of the process is due to the FIR rule, in which we abandon the selected conclusion instead of continuing the inversion phase; the SIR rule, where there is a choice of the selected formula to be introduced by contraction; and the SFD rule, where we choose the formula in Γ to focus down.

Example 4.8. The formalization of the proof search in the Example 4.7, with the reduction rules of Definition 4.25 is presented. Again, Γ contains the assumption $(x : (p \supset q) \supset p)$ and Δ contains the alternative conclusions $(b : p)$ and $(a : ((p \supset q) \supset p) \supset p)$.

$$\begin{array}{ll}
& \vdash ((p \supset q) \supset p) \supset p \mid \\
\rightarrow_{FIR} & \mu a^{((p \supset q) \supset p) \supset p}. (\underbrace{\vdash a : ((p \supset q) \supset p) \supset p}_{\text{...}}) \\
\rightarrow_{SIR} & \mu a^{((p \supset q) \supset p) \supset p}. a(\lambda x^{(p \supset q) \supset p}. \underbrace{(x : (p \supset q) \supset p \vdash p \mid a : ((p \supset q) \supset p) \supset p)}_{\text{...}})) \\
\rightarrow_{FIR} & \mu a^{((p \supset q) \supset p) \supset p}. a(\lambda x^{(p \supset q) \supset p}. \mu b^p. (\underbrace{\Gamma \vdash b : p, a : ((p \supset q) \supset p) \supset p}_{\text{...}})) \\
\rightarrow_{SFD} & \mu a^{((p \supset q) \supset p) \supset p}. a(\lambda x^{(p \supset q) \supset p}. \mu b^p. (\text{ab}(x, (\Gamma \triangleright (p \supset q) \supset p \mid \Delta)))) \\
\rightarrow_{KFD} & \mu a^{((p \supset q) \supset p) \supset p}. a(\lambda x^{(p \supset q) \supset p}. \mu b^p. (\text{ab}(x(\Gamma \vdash p \supset q \mid \Delta), (\Gamma \triangleright p \mid \Delta)))) \\
\rightarrow_{FFD_{atm}} & \mu a^{((p \supset q) \supset p) \supset p}. a(\lambda x^{(p \supset q) \supset p}. \mu b^p. (b(x(\Gamma \vdash p \supset q \mid \Delta)))) \\
\rightarrow & (...)
\end{array}$$

As for LKT_∂ , we prove the following results of Record of Search and Subject Reduction for NKT_∂ .

Proposition 4.16 (Record of search).

1. If $\Xi \Vdash M : \sigma$ then $\underline{\sigma} \twoheadrightarrow M$.
2. If $\Xi \Vdash S : \zeta$ then $\underline{\zeta} \twoheadrightarrow S$.
3. If $\Xi \Vdash H : (\Gamma \triangleright A \mid \Delta)$ then $\underline{\Gamma \vdash \Delta} \twoheadrightarrow \text{ab}(H, (\Gamma \triangleright A \mid \Delta))$.

Proof. By simultaneous induction on the derivations of $\Xi \Vdash M : \sigma$, $\Xi \Vdash S : \zeta$ and $\Xi \Vdash H : (\Gamma \triangleright A \mid \Delta)$.

- Case of first ∂ -rule. Suppose $[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma$. Then, $\underline{\sigma} \rightarrow \underline{\sigma}$, because of the reflexivity of $\rightarrow$. Similar reasoning applies to $[\underline{\zeta}] \Vdash \underline{\zeta} : \zeta$.
- Case of second ∂ -rule. Suppose $\Xi' \Vdash \text{ab}(H, (\Gamma \triangleright A \mid \Delta)) : (\Gamma \vdash \Delta)$ is obtained from $\Xi \Vdash H : (\Gamma \triangleright A)$. Then $\underline{\Gamma \vdash \Delta} \rightarrow \text{ab}(H, (\Gamma \triangleright A \mid \Delta))$, by IH.
- Case A . Suppose $\epsilon \Vdash x : (\Gamma, x : A \triangleright A \mid \Delta)$. Then, $\underline{\Gamma, x : A \vdash \Delta} \rightarrow \text{ab}(x, (\Gamma, x : A \triangleright A \mid \Delta))$, by SFD .
- Case $E\perp$. Suppose $\Xi \Vdash \text{abort}(H) : (\Gamma \vdash \Delta)$ is obtained from $\Xi \Vdash H : (\Gamma \triangleright \perp \mid \Delta)$. Then, $\underline{\Gamma \vdash \Delta} \rightarrow \text{ab}(H, (\Gamma \triangleright \perp \mid \Delta)) \rightarrow \text{abort}(H)$, where the last step is by FFD_{abs} and the rest of the reduction is by IH.
- Case ACT . Suppose $\Xi \Vdash \mu a^A.S : (\Gamma \vdash A \mid \Delta)$ is obtained from $\Xi \Vdash S : (\Gamma \vdash a : A, \Delta)$. Then, $\underline{\Gamma \vdash A \mid \Delta} \rightarrow \mu a^A.(\underline{\Gamma \vdash a : A, \Delta}) \rightarrow \mu a^A.S$, where the first step is by FIR and the rest of the reduction is by IH.
- Case PSS . Suppose $\Xi \Vdash a(H) : (\Gamma \vdash a : p, \Delta)$ is obtained from $\Xi \Vdash H : (\Gamma \triangleright p \mid a : p, \Delta)$. Then, $\underline{\Gamma \vdash a : p, \Delta} \rightarrow \text{ab}(H, (\Gamma \triangleright p \mid a : p, \Delta)) \rightarrow a(H)$, where the last step is by FFD_{atm} and the rest of the reduction is by IH.
- Case CTR . Suppose $\Xi \Vdash a(\lambda x^A.M) : (\Gamma \vdash a : A \supset B, \Delta)$ is obtained from $\Xi \Vdash M : (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)$. Then, $\underline{\Gamma \vdash a : A \supset B, \Delta} \rightarrow a(\lambda x^A.(\underline{\Gamma, x : A \vdash B \mid a : A \supset B, \Delta})) \rightarrow a(\lambda x^A.M)$, where the first step is by SIR and the rest of the reduction is by IH.
- Case $I\supset$. Suppose $\Xi \Vdash \lambda x^A.M : (\Gamma \vdash A \supset B \mid \Delta)$ is obtained from $\Xi \Vdash M : (\Gamma, x : A \vdash B \mid \Delta)$. Then, $\underline{\Gamma \vdash A \supset B \mid \Delta} \rightarrow \lambda x^A.(\underline{\Gamma, x : A \vdash B \mid \Delta}) \rightarrow \lambda x^A.M$, where the first step is by KIR and the rest of the reduction is by IH.
- Case $E\supset$. Suppose $\Xi_1 @ \Xi_2 \Vdash HN : (\Gamma \triangleright B \mid \Delta)$ is obtained from $\Xi_1 \Vdash H : (\Gamma \triangleright A \supset B \mid \Delta)$ and $\Xi_2 \Vdash N : (\Gamma \vdash A \mid \Delta)$. Then $\underline{\Gamma \vdash \Delta} \rightarrow \text{ab}(H, (\Gamma \triangleright A \supset B \mid \Delta)) \rightarrow \text{ab}(H(\underline{\Gamma \vdash A \mid \Delta}), (\Gamma \triangleright B \mid \Delta)) \rightarrow \text{ab}(HN, (\Gamma \triangleright B \mid \Delta))$, where the middle step is by KFD and the rest of the reduction is by IH.

□

Proposition 4.17 (Subject reduction for NKT_{∂}).

1. If $\Xi \Vdash M : \sigma$ and $M \rightarrow M'$, then there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash M' : \sigma$.
2. If $\Xi \Vdash S : \zeta$ and $S \rightarrow S'$, then there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash S' : \zeta$.
3. If $\Xi \Vdash H : \rho$ and $H \rightarrow H'$, then there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash H' : \rho$.

Proof. By simultaneous induction on $M \rightarrow M'$, $H \rightarrow H'$ and $S \rightarrow S'$.

Base Cases:

- Case *SIR*. Suppose $S = \frac{\Gamma \vdash a : A \supset B, \Delta}{a(\lambda x^A.(\Gamma, x : A \vdash B \mid a : A \supset B, \Delta))} = S'$ and $\Xi \Vdash S : \zeta$. By the third ∂ -rule, $\Xi = [\frac{\Gamma \vdash a : A \supset B, \Delta}{a(\lambda x^A.(\Gamma, x : A \vdash B \mid a : A \supset B, \Delta))}]$ and $\zeta = \Gamma \vdash a : A \supset B, \Delta$. By *SIR*, $\Xi \rightarrow \Xi'$, with $\Xi' = [\frac{\Gamma, x : A \vdash B \mid a : A \supset B, \Delta}{a(\lambda x^A.(\Gamma, x : A \vdash B \mid a : A \supset B, \Delta))}]$. We now prove $\Xi' \Vdash a(\lambda x^A.(\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)) : (\Gamma \vdash a : A \supset B, \Delta)$. By the first ∂ -rule, $\Xi' \Vdash \frac{\Gamma, x : A \vdash B \mid a : A \supset B, \Delta}{a(\lambda x^A.(\Gamma, x : A \vdash B \mid a : A \supset B, \Delta))} : (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)$. Applying the inference rule *CTR*, $\Xi' \Vdash a(\lambda x^A.(\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)) : (\Gamma \vdash a : A \supset B, \Delta)$.
- Case *KIR*. Suppose $M = \frac{\Gamma \vdash A \supset B \mid \Delta}{\lambda x^A.(\Gamma, x : A \vdash B \mid \Delta)} = M'$ and $\Xi \Vdash M : \sigma$. By the first ∂ -rule, $\Xi = [\frac{\Gamma \vdash A \supset B \mid \Delta}{\lambda x^A.(\Gamma, x : A \vdash B \mid \Delta)}]$ and $\sigma = \Gamma \vdash A \supset B \mid \Delta$. By *KIR*, $\Xi \rightarrow \Xi'$, with $\Xi' = [\frac{\Gamma, x : A \vdash B \mid \Delta}{\lambda x^A.(\Gamma, x : A \vdash B \mid \Delta)}]$. We now prove $\Xi' \Vdash \lambda x^A.(\Gamma, x : A \vdash B \mid \Delta) : (\Gamma \vdash A \supset B \mid \Delta)$. By the first ∂ -rule, $\Xi' \Vdash \frac{\Gamma, x : A \vdash B \mid \Delta}{\lambda x^A.(\Gamma, x : A \vdash B \mid \Delta)} : (\Gamma, x : A \vdash B \mid \Delta)$. Applying the inference rule *I $\supset$* , $\Xi' \Vdash \lambda x^A.(\Gamma, x : A \vdash B \mid \Delta) : (\Gamma \vdash A \supset B \mid \Delta)$.
- Case *FIR*. Suppose $M = \frac{\Gamma \vdash A \mid \Delta}{\mu a^A.(\Gamma \vdash a : A, \Delta)} = M'$ and $\Xi \Vdash M : \sigma$. By the first ∂ -rule, $\Xi = [\frac{\Gamma \vdash A \mid \Delta}{\mu a^A.(\Gamma \vdash a : A, \Delta)}]$ and $\sigma = \Gamma \vdash A \mid \Delta$. By *FIR*, $\Xi \rightarrow \Xi'$, with $\Xi' = [\frac{\Gamma \vdash a : A, \Delta}{\mu a^A.(\Gamma \vdash a : A, \Delta)}]$. We now prove $\Xi' \Vdash \mu a^A.(\Gamma \vdash a : A, \Delta) : (\Gamma \vdash A \mid \Delta)$. By the third ∂ -rule, $\Xi' \Vdash \frac{\Gamma \vdash a : A, \Delta}{\mu a^A.(\Gamma \vdash a : A, \Delta)} : (\Gamma \vdash a : A, \Delta)$. Applying the inference rule *ACT*, $\Xi' \Vdash \mu a^A.(\Gamma \vdash a : A, \Delta) : (\Gamma \vdash A \mid \Delta)$.
- Case *SFD*. Suppose $S = \frac{\Gamma, x : A \vdash \Delta}{\text{ab}(x, (\Gamma, x : A \triangleright A \mid \Delta))} = S'$ and $\Xi \Vdash S : \zeta$. By the third ∂ -rule, $\Xi = [\frac{\Gamma, x : A \vdash \Delta}{\text{ab}(x, (\Gamma, x : A \triangleright A \mid \Delta))}]$ and $\zeta = \Gamma, x : A \vdash \Delta$. By *SFD*, $\Xi \rightarrow \Xi'$, with $\Xi' = [\frac{\Gamma, x : A \triangleright A \mid \Delta}{\text{ab}(x, (\Gamma, x : A \triangleright A \mid \Delta))}]$. We now prove $\Xi' \Vdash \text{ab}(x, (\Gamma, x : A \triangleright A \mid \Delta)) : (\Gamma, x : A \vdash \Delta)$. By the inference rule *A, $\epsilon \Vdash x : (\Gamma, x : A \triangleright A \mid \Delta)$* . Applying the the second ∂ -rule, $\Xi' \Vdash \text{ab}(x, (\Gamma, x : A \triangleright A \mid \Delta)) : (\Gamma, x : A \vdash \Delta)$.
- Case *KFD*. Suppose $S = \text{ab}(H, (\Gamma \triangleright A \supset B \mid \Delta)) \rightarrow \text{ab}(H(\Gamma \vdash A \mid \Delta), (\Gamma \triangleright B \mid \Delta)) = S'$ and $\Xi \Vdash S : \zeta$. By the second ∂ -rule, $\Xi = \Xi_1 @ [\frac{\Gamma \triangleright A \supset B \mid \Delta}{\text{ab}(H(\Gamma \vdash A \mid \Delta), (\Gamma \triangleright B \mid \Delta))}]$ and $\zeta = \Gamma \vdash$

Δ . By KFD , $\Xi \rightarrow \Xi'$, with $\Xi' = \Xi_1 @ [\Gamma \vdash A \mid \Delta, \Gamma \triangleright B \mid \Delta]$. We now prove $\Xi' \Vdash \text{ab}(H(\Gamma \vdash A \mid \Delta), (\Gamma \triangleright B \mid \Delta)) : (\Gamma \vdash \Delta)$. By inversion of the inference rule, $\Xi_1 \Vdash H : (\Gamma \triangleright A \supset B \mid \Delta)$ and, by the first ∂ -rule, $[\Gamma \vdash A \mid \Delta] \Vdash \Gamma \vdash A \mid \Delta : (\Gamma \vdash A \mid \Delta)$. Applying the inference rule $E\supset$, $\Xi_1 @ [\Gamma \vdash A \mid \Delta] \Vdash H(\Gamma \vdash A \mid \Delta) : (\Gamma \triangleright B \mid \Delta)$. Applying the second ∂ -rule, $\Xi' \Vdash \text{ab}(H(\Gamma \vdash A \mid \Delta), (\Gamma \triangleright B \mid \Delta)) : (\Gamma \vdash \Delta)$.

- Case FFD_{atm} . Suppose $S = \text{ab}(H, (\Gamma \triangleright p \mid a : p, \Delta)) \rightarrow a(H) = S'$ and $\Xi \Vdash S : \zeta$. By the second ∂ -rule, $\Xi = \Xi_1 @ [\Gamma \triangleright p \mid a : p, \Delta]$ and $\zeta = \Gamma \vdash \Delta$. By FFD_{atm} , $\Xi \rightarrow \Xi_1$. We now prove $\Xi_1 \Vdash a(H) : (\Gamma \vdash \Delta)$. By inversion of the inference rule, $\Xi_1 \Vdash H : (\Gamma \triangleright p \mid a : p, \Delta)$. Applying the inference rule PSS , $\Xi_1 \Vdash a(H) : (\Gamma \vdash \Delta)$.
- Case FFD_{abs} . Suppose $S = \text{ab}(H, (\Gamma \triangleright \perp \mid \Delta)) \rightarrow \text{abort}(H) = S'$ and $\Xi \Vdash S : \zeta$. By the second ∂ -rule, $\Xi = \Xi_1 @ [\Gamma \triangleright \perp \mid \Delta]$ and $\zeta = \Gamma \vdash \Delta$. By FFD_{abs} , $\Xi \rightarrow \Xi_1$. We now prove $\Xi \Vdash \text{abort}(H) : (\Gamma \vdash \Delta)$. By inversion of the inference rule, $\Xi_1 \Vdash H : (\Gamma \triangleright \perp \mid \Delta)$. Applying the inference rule $E\perp$, $\Xi \Vdash \text{abort}(H) : (\Gamma \vdash \Delta)$.

Inductive Cases:

- Suppose $\lambda x^A.M \rightarrow \lambda x^A.M'$, with $M \rightarrow M'$ and $\Xi \Vdash \lambda x^A.M : (\Gamma \vdash A \supset B \mid \Delta)$. By inversion of the inference rule $I\supset$, $\Xi \Vdash M : (\Gamma, x : A \vdash B \mid \Delta)$. By IH, there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash M' : (\Gamma, x : A \vdash B \mid \Delta)$. Applying the inference rule $I\supset$, $\Xi' \Vdash \lambda x^A.M' : (\Gamma \vdash A \supset B \mid \Delta)$.
- Suppose $\mu a^A.S \rightarrow \mu a^A.S'$, with $S \rightarrow S'$ and $\Xi \Vdash \mu a^A.S : (\Gamma \vdash A \mid \Delta)$. By inversion of the inference rule ACT , $\Xi \Vdash S : (\Gamma \vdash a : A, \Delta)$. By IH, there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash S' : (\Gamma \vdash a : A, \Delta)$. Applying the inference rule ACT , $\Xi' \Vdash \mu a^A.S' : (\Gamma \vdash A \mid \Delta)$.
- Suppose $HN \rightarrow HN'$, with $N \rightarrow N'$ and $\Xi \Vdash HN : (\Gamma \triangleright B \mid \Delta)$. By inversion of the inference rule $E\supset$, $\Xi_1 \Vdash H : (\Gamma \triangleright A \supset B \mid \Delta)$ and $\Xi_2 \Vdash N : (\Gamma \vdash A \mid \Delta)$, and $\Xi = \Xi_1 @ \Xi_2$. By IH, there is Ξ'_2 such that $\Xi_2 \rightarrow \Xi'_2$ and $\Xi'_2 \Vdash N' : (\Gamma \vdash A \mid \Delta)$. Applying the inference rule $E\supset$, $\Xi_1 @ \Xi'_2 \Vdash HN' : (\Gamma \triangleright B \mid \Delta)$. Moreover $\Xi_1 @ \Xi_2 \rightarrow \Xi_1 @ \Xi'_2$.
- Suppose $HN \rightarrow H'N$, with $H \rightarrow H'$ and $\Xi \Vdash HN : (\Gamma \triangleright B \mid \Delta)$. By inversion of the inference rule $E\supset$, $\Xi_1 \Vdash H : (\Gamma \triangleright A \supset B \mid \Delta)$ and $\Xi_2 \Vdash N : (\Gamma \vdash A \mid \Delta)$, and $\Xi = \Xi_1 @ \Xi_2$. By IH, there is Ξ'_1 such that $\Xi_1 \rightarrow \Xi'_1$ and $\Xi'_1 \Vdash H' : (\Gamma \triangleright A \supset B \mid \Delta)$. Applying the inference rule $E\supset$, $\Xi'_1 @ \Xi_2 \Vdash H'N : (\Gamma \triangleright B \mid \Delta)$. Moreover $\Xi_1 @ \Xi_2 \rightarrow \Xi'_1 @ \Xi_2$.

- Suppose $\text{abort}(H) \rightarrow \text{abort}(H')$, with $H \rightarrow H'$ and $\Xi \Vdash \text{abort}(H) : (\Gamma \vdash \Delta)$. By inversion of the inference rule $E\perp$, $\Xi \Vdash H : (\Gamma \triangleright \perp \mid \Delta)$. By IH, there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash H' : (\Gamma \triangleright \perp \mid \Delta)$. Applying the inference rule $E\perp$, $\Xi' \Vdash \text{abort}(H') : (\Gamma \vdash \Delta)$.
- Suppose $a(H) \rightarrow a(H')$, with $H \rightarrow H'$ and $\Xi \Vdash a(H) : (\Gamma \vdash a : p, \Delta)$. By inversion of the inference rule PSS , $\Xi \Vdash H : (\Gamma \triangleright p \mid a : p, \Delta)$. By IH, there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash H' : (\Gamma \triangleright p \mid a : p, \Delta)$. Applying the inference rule PSS , $\Xi' \Vdash a(H') : (\Gamma \vdash a : p, \Delta)$.
- Suppose $a(\lambda x^A.M) \rightarrow a(\lambda x^A.M')$, with $M \rightarrow M'$ and $\Xi \Vdash a(\lambda x^A.M) : (\Gamma \vdash a : A \supset B, \Delta)$. By inversion of the inference rule CTR , $\Xi \Vdash M : (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)$. By IH, there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash M' : (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)$. Applying the inference rule CTR , $\Xi' \Vdash a(\lambda x^A.M') : (\Gamma \vdash a : A \supset B, \Delta)$.
- Suppose $\text{ab}(H, \underline{\rho}) \rightarrow \text{ab}(H', \underline{\rho})$, with $H \rightarrow H'$ and $\Xi @ [\underline{\rho}] \Vdash \text{ab}(H, \underline{\rho}) : (\Gamma \vdash \Delta)$. By inversion of the ∂ -rule, $\Xi \Vdash H : \underline{\rho}$. By IH, there is Ξ' such that $\Xi \rightarrow \Xi'$ and $\Xi' \Vdash H' : \underline{\rho}$. Applying the second ∂ -rule, $\Xi' @ [\underline{\rho}] \Vdash \text{ab}(H', \underline{\rho}) : (\Gamma \vdash \Delta)$.

□

Finally, we reach the main result of this subsection.

Theorem 4.4 (Proof search as normalization).

1. For all $M \in NKT$: NKT derives $\Gamma \vdash M : A \mid \Delta$ iff $\Gamma \vdash \underline{A \mid \Delta} \rightarrow M$.
2. For all $S \in NKT$: NKT derives $\Gamma \vdash^S \Delta$ iff $\Gamma \vdash \underline{\Delta} \rightarrow S$.
3. For all $H \in NKT$: NKT derives $\Gamma \triangleright H : A \mid \Delta$ iff $\Gamma \vdash \underline{\Delta} \rightarrow \text{ab}(H, (\Gamma \triangleright A \mid \Delta))$.

Proof. The “only if” statements follow from conservativity and record of search. As to the “if” statement, let us prove the third. Suppose $\zeta \rightarrow \text{ab}(H, (\Gamma \triangleright A \mid \Delta))$, where $\zeta = \Gamma \vdash \underline{\Delta}$. Notice $[\zeta] \Vdash \zeta : \zeta$. By subject reduction, $\Xi \Vdash \text{ab}(H, (\Gamma \triangleright A \mid \Delta)) : \zeta$, for some Ξ , such that $[\zeta] \rightarrow \Xi$. Since H is total, $\Xi = [\Gamma \triangleright A \mid \Delta]$. By the invertibility of the second ∂ -rule, $\epsilon \Vdash H : (\Gamma \triangleright A \mid \Delta)$. By conservativity, NKT derives $\Gamma \triangleright H : A \mid \Delta$. The other “if” statements are proved in a similar way. □

4.6 Focusing versus Bidirectionality

In this section, we prove that the correspondence between focusing in LJT and intercalation in NJT presented in Section 3.4 can be lifted to classical logic, through the systems LKT_∂ and NKT_∂ .

First, we present the bijection between LKT_∂ -terms and NKT_∂ -terms.

Definition 4.26 (Translations Θ and Ψ). Let us name as $L\text{-Terms}$, $Lists$ and $Commands$ the three sets of expressions defined simultaneously by the grammar of LKT_∂ , and analogously use the names $N\text{-Terms}$, $Neutrals$ and $Statements$ for NKT_∂ . The map $\Theta : L\text{-Terms} \rightarrow N\text{-Terms}$ is defined together with $\Theta : Neutrals \times Lists \rightarrow Statements$ and $\Theta : Commands \rightarrow Statements$ by simultaneous recursion on $t \in L\text{-Terms}$, $l \in Lists$ and $c \in Commands$. The map $\Psi : N\text{-Terms} \rightarrow L\text{-Terms}$ is defined together with $\Psi : Neutrals \times Lists \rightarrow Commands$ and $\Psi : Statements \rightarrow Commands$ by simultaneous recursion on $M \in N\text{-Terms}$, $H \in Neutrals$ and $S \in Statements$, as:

$$\begin{aligned}
\Theta(\lambda x^A.t) &= \lambda x^A.\Theta(t) & \Psi(\lambda x^A.M) &= \lambda x^A.\Psi(M) \\
\Theta(\mu a^A.c) &= \mu a^A.\Theta(c) & \Psi(\mu a^A.A) &= \mu a^A.\Psi(S) \\
\Theta(\underline{\sigma}) &= \underline{\sigma} & \Psi(\underline{\sigma}) &= \underline{\sigma} \\
\\
\Theta(H, a) &= a(H) & \Psi(x, l) &= \langle x \mid l \rangle \\
\Theta(H, \text{abort}) &= \text{abort}(H) & \Psi(HN, l) &= \Psi(H, \Psi(N) :: l) \\
\Theta(H, u :: l) &= \Theta(H\Theta(u), l) & \\
\Theta(H, (\Gamma; A \vdash \Delta)) &= \text{ab}(H, (\Gamma \triangleright A \mid \Delta)) & \Psi(\text{abort}(H)) &= \Psi(H, \text{abort}) \\
\\
\Theta(\langle x \mid l \rangle) &= \Theta(x, l) & \Psi(a(H)) &= \Psi(H, a) \\
\Theta(\langle \lambda x^A.t \mid a \rangle) &= a(\lambda x^A.\Theta(t)) & \Psi(a(\lambda x^A.M)) &= \langle \lambda x^A.\Psi(M) \mid a \rangle \\
\Theta(\underline{\zeta}) &= \underline{\zeta} & \Psi(\text{ab}(H, (\Gamma \triangleright A \mid \Delta))) &= \Psi(H, (\Gamma; A \vdash \Delta)) \\
\Theta(\underline{\zeta}) &= \underline{\zeta} & \Psi(\underline{\zeta}) &= \underline{\zeta}
\end{aligned}$$

To help us prove the next theorem, we extend the definitions of Θ and Ψ to formal sequents $\underline{\sigma}$, $\underline{\tau}$, $\underline{\rho}$ and $\underline{\zeta}$, and to lists of proof obligations Ξ .

Definition 4.27 (Translations Θ and Ψ - Formal Sequents). The extension of the translations Θ and Ψ to formal sequents is defined as:

$$\begin{aligned}
\Theta(\underline{\sigma}) &= \underline{\sigma} & \Psi(\underline{\sigma}) &= \underline{\sigma} \\
\Theta(\underline{\Gamma; A \vdash \Delta}) &= \underline{\Gamma \triangleright A \mid \Delta} & \Psi(\underline{\Gamma \triangleright A \mid \Delta}) &= \underline{\Gamma; A \vdash \Delta} \\
\Theta(\underline{\zeta}) &= \underline{\zeta} & \Psi(\underline{\zeta}) &= \underline{\zeta}
\end{aligned}$$

Definition 4.28 (Translations Θ and Ψ - Lists of Proof Obligations). The extension of the translations Θ and Ψ to lists of proof obligations is defined as:

$$\begin{aligned}
\Theta(\epsilon) &= \epsilon & \Psi(\epsilon) &= \epsilon \\
\Theta([\underline{\sigma}] @ \Xi) &= [\Theta(\underline{\sigma})] @ \Theta(\Xi) & \Psi([\underline{\sigma}] @ \Xi) &= [\Psi(\underline{\sigma})] @ \Psi(\Xi) \\
\Theta([\underline{\tau}] @ \Xi) &= [\Theta(\underline{\tau})] @ \Theta(\Xi) & \Psi([\underline{\rho}] @ \Xi) &= [\Psi(\underline{\rho})] @ \Psi(\Xi) \\
\Theta([\underline{\zeta}] @ \Xi) &= [\Theta(\underline{\zeta})] @ \Theta(\Xi) & \Psi([\underline{\zeta}] @ \Xi) &= [\Psi(\underline{\zeta})] @ \Psi(\Xi)
\end{aligned}$$

The following theorem states that Θ and Ψ preserve typing and derivability between the two proof systems LKT_{∂} and NKT_{∂} .

Theorem 4.5 (Soundness).

1. If $\Xi \Vdash t : \sigma$ in LKT_{∂} then $\Theta(\Xi) \Vdash \Theta(t) : \sigma$ in NKT_{∂} .
2. If $\Xi \Vdash M : \sigma$ in NKT_{∂} then $\Psi(\Xi) \Vdash \Psi(M) : \sigma$ in LKT_{∂} .

Proof. The first item is proved together with:

- If $\Xi \Vdash l : \tau$ in LKT_{∂} , with $\tau = (\Gamma; A \vdash \Delta)$, and $\Xi' \Vdash H : (\Gamma \triangleright A \mid \Delta)$ in NKT_{∂} , then $\Xi' @ \Theta(\Xi) \Vdash \Theta(H, l) : (\Gamma \vdash \Delta)$ in NKT_{∂} ; and
- If $\Xi \Vdash c : \zeta$ in LKT_{∂} then $\Theta(\Xi) \Vdash \Theta(c) : \zeta$ in NKT_{∂} .

By simultaneous induction on the derivations of $\Xi \Vdash t : \sigma$, $\Xi \Vdash l : \tau$ and $\Xi \Vdash c : \zeta$.

- Case of first ∂ -rule. Suppose $[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma$ in LKT_{∂} . By the definitions of Θ , $\Theta(\underline{\sigma}) = \underline{\sigma}$ and $\Theta([\underline{\sigma}]) = [\underline{\sigma}]$. By the first ∂ -rule, $[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma$ in NKT_{∂} . Same reasoning applies to $\Xi \Vdash \underline{\zeta} : \zeta$.
- Case of second ∂ -rule. Suppose $[\underline{\tau}] \Vdash \underline{\tau} : \tau$ in LKT_{∂} , with $\underline{\tau} = \underline{\Gamma; A \vdash \Delta}$, and $\Xi' \Vdash H : (\Gamma \triangleright A \mid \Delta)$ in NKT_{∂} . By the definitions of Θ , $\Theta(H, \underline{\tau}) = \text{ab}(H, (\Gamma \triangleright A \mid \Delta))$ and $\Theta([\underline{\tau}]) = [\underline{\Gamma \triangleright A \mid \Delta}]$. By the second ∂ -rule, $\Xi' @ [\underline{\Gamma \triangleright A \mid \Delta}] \Vdash \text{ab}(H, (\Gamma \triangleright A \mid \Delta)) : (\Gamma \vdash \Delta)$ in NKT_{∂} .
- Case A . Suppose $\epsilon \Vdash a : (\Gamma; p \vdash a : p, \Delta)$ in LKT_{∂} and $\Xi' \Vdash H : (\Gamma \triangleright p \mid a : p, \Delta)$ in NKT_{∂} . By the definitions of Θ , $\Theta(H, a) = a(H)$ and $\Theta(\epsilon) = \epsilon$. Applying the inference rule PSS , $\Xi' \Vdash a(H) : (\Gamma \vdash a : p, \Delta)$ in NKT_{∂} .

- Case $L\perp$. Suppose $\epsilon \Vdash a : (\Gamma; \perp \vdash \Delta)$ in LKT_∂ and $\Xi' \Vdash H : (\Gamma \triangleright \perp \mid \Delta)$ in NKT_∂ . By the definitions of Θ , $\Theta(H, \text{abort}) = \text{abort}(H)$ and $\Theta(\epsilon) = \epsilon$. Applying the inference rule $E\perp$, $\Xi' \Vdash \text{abort}(H) : (\Gamma \vdash \Delta)$ in NKT_∂ .
- Case ACT . Suppose $\Xi \Vdash \mu a^A.c : (\Gamma \vdash A \mid \Delta)$ is obtained from $\Xi \Vdash c : (\Gamma \vdash a : A, \Delta)$ in LKT_∂ . By the definition of Θ , $\Theta(\mu a^A.c) = \mu a^A.\Theta(c)$. We now prove $\Theta(\Xi) \Vdash \mu a^A.\Theta(c) : (\Gamma \vdash A \mid \Delta)$. By IH, $\Theta(\Xi) \Vdash \Theta(c) : (\Gamma \vdash a : A, \Delta)$. Applying the inference rule ACT , $\Theta(\Xi) \Vdash \mu a^A.\Theta(c) : (\Gamma \vdash A \mid \Delta)$ in NKT_∂ .
- Case $LCTR$. Suppose $\Xi \Vdash \langle x \mid l \rangle : (\Gamma, x : A \vdash \Delta)$ is obtained from $\Xi \Vdash l : (\Gamma, x : A; A \vdash \Delta)$ in LKT_∂ . By the definition of Θ , $\Theta(\langle x \mid l \rangle) = \Theta(x, l)$. We now prove $\Theta(\Xi) \Vdash \Theta(x, l) : (\Gamma, x : A \vdash \Delta)$. Applying the A rule, $\epsilon \Vdash x : (\Gamma, x : A \triangleright A \mid \Delta)$, in NKT_∂ . By IH, $\Theta(\Xi) \Vdash \Theta(x, l) : (\Gamma, x : A \vdash \Delta)$ in NKT_∂ .
- Case $RCTR$. Suppose $\Xi \Vdash \langle \lambda x^A.t \mid a \rangle : (\Gamma \vdash a : A \supset B, \Delta)$ is obtained from $\Xi \Vdash t : (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)$ in LKT_∂ . By the definition of Θ , $\Theta(\langle \lambda x^A.t \mid a \rangle) = a(\lambda x^A.\Theta(t))$. We now prove $\Theta(\Xi) \Vdash a(\lambda x^A.\Theta(t)) : (\Gamma \vdash a : A \supset B, \Delta)$. By IH, $\Theta(\Xi) \Vdash \Theta(t) : (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)$. Applying the inference rule CTR , $\Theta(\Xi) \Vdash a(\lambda x^A.\Theta(t)) : (\Gamma \vdash a : A \supset B, \Delta)$ in NKT_∂ .
- Case RI . Suppose $\Xi \Vdash \lambda x^A.t : (\Gamma \vdash A \supset B \mid \Delta)$ is obtained from $\Xi \Vdash t : (\Gamma, x : A \vdash B \mid \Delta)$ in LKT_∂ . By the definition of Θ , $\Theta(\lambda x^A.t) = \lambda x^A.\Theta(t)$. We now prove $\Theta(\Xi) \Vdash \lambda x^A.\Theta(t) : (\Gamma \vdash A \supset B \mid \Delta)$. By IH, $\Theta(\Xi) \Vdash \Theta(t) : (\Gamma, x : A \vdash B \mid \Delta)$. Applying the $I\supset$ rule, $\Theta(\Xi) \Vdash \lambda x^A.\Theta(t) : (\Gamma \vdash A \supset B \mid \Delta)$ in NKT_∂ .
- Case LI . Suppose $\Xi_1 @ \Xi_2 \Vdash u :: l : (\Gamma; A \supset B \vdash \Delta)$ is obtained from $\Xi_1 \Vdash u : (\Gamma \vdash A \mid \Delta)$ and $\Xi_2 \Vdash l : (\Gamma; B \vdash \Delta)$ in LKT_∂ , and $\Xi' \Vdash H : (\Gamma \triangleright A \supset B \mid \Delta)$ in NKT_∂ . By the definition of Θ , $\Theta(H, u :: l) = \Theta(H\Theta(u), l)$. We now prove $\Xi' @ \Theta(\Xi_1) @ \Theta(\Xi_2) \Vdash \Theta(H\Theta(u), l) : (\Gamma \vdash \Delta)$. By IH, $\Theta(\Xi_1) \Vdash \Theta(u) : (\Gamma \vdash A \mid \Delta)$ in NKT_∂ . Applying the $E\supset$ rule, $\Xi' @ \Theta(\Xi_1) \Vdash H\Theta(u) : (\Gamma \triangleright B \mid \Delta)$. By IH, $\Xi' @ \Theta(\Xi_1) @ \Theta(\Xi_2) \Vdash \Theta(H\Theta(u), l) : (\Gamma \vdash \Delta)$ in NKT_∂ .

The second item is proved together with:

- If $\Xi \Vdash H : \rho$, with $\rho = (\Gamma \triangleright A \mid \Delta)$, in NKT_∂ and $\Xi' \Vdash l : (\Gamma; A \vdash \Delta)$ in LKT_∂ then $\Psi(\Xi) @ \Xi' \Vdash \Psi(H, l) : (\Gamma \vdash \Delta)$ in LKT_∂ ; and

- If $\Xi \Vdash S : \zeta$ in NKT_{∂} then $\Psi(\Xi) \Vdash \Psi(S) : \zeta$ in NKT_{∂} .

By simultaneous induction on the derivations of $\Xi \Vdash M : \sigma$, $\Xi \Vdash H : \rho$ and $\Xi \Vdash S : \zeta$.

- Case of first ∂ -rule. Suppose $[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma$ in NKT_{∂} . By the definitions of Ψ , $\Psi(\underline{\sigma}) = \underline{\sigma}$ and $\Psi([\underline{\sigma}]) = [\underline{\sigma}]$. By the first ∂ -rule, $[\underline{\sigma}] \Vdash \underline{\sigma} : \sigma$ in LKT_{∂} . Same reasoning applies to $\Xi \Vdash \underline{\zeta} : \zeta$.
- Case of second ∂ -rule. Suppose $\Xi @ [\underline{\Gamma \triangleright A \mid \Delta}] \Vdash \text{ab}(H, (\underline{\Gamma \triangleright A \mid \Delta})) : (\Gamma \vdash \Delta)$ is obtained from $\Xi \Vdash H : (\Gamma \triangleright A \mid \Delta)$ in NKT_{∂} . By the definitions of Ψ , $\Psi(\text{ab}(H, (\underline{\Gamma \triangleright A \mid \Delta}))) = \Psi(H, (\underline{\Gamma; A \vdash \Delta}))$ and $\Psi(\Xi @ [\underline{\Gamma \triangleright A \mid \Delta}]) = \Psi(\Xi) @ [\underline{\Gamma; A \vdash \Delta}]$. We now prove $\Psi(\Xi) @ [\underline{\Gamma; A \vdash \Delta}] \Vdash \Psi(H, (\underline{\Gamma; A \vdash \Delta})) : (\Gamma \vdash \Delta)$. Applying the second ∂ -rule, $[\underline{\Gamma; A \vdash \Delta}] \Vdash \underline{\Gamma; A \vdash \Delta} : (\Gamma; A \vdash \Delta)$ in LKT_{∂} . By IH, $\Psi(\Xi) @ [\underline{\Gamma; A \vdash \Delta}] \Vdash \Psi(H, (\underline{\Gamma; A \vdash \Delta})) : (\Gamma \vdash \Delta)$ in LKT_{∂} .
- Case A . Suppose $\epsilon \Vdash x : (\Gamma, x : A \triangleright A \mid \Delta)$ in NKT_{∂} and $\Xi \Vdash l : (\Gamma, x : A; A \vdash \Delta)$ in LKT_{∂} . By the definitions of Ψ , $\Psi(x, l) = \langle x \mid l \rangle$ and $\Psi(\epsilon) = \epsilon$. Applying $LCTR$ rule, $\Xi \Vdash \langle x \mid l \rangle : (\Gamma, x : A \vdash \Delta)$ in LKT_{∂} .
- Case $E \perp$. Suppose $\Xi \Vdash \text{abort}(H) : (\Gamma \vdash \Delta)$ is obtained from $\Xi \Vdash H : (\Gamma \triangleright \perp \mid \Delta)$ in NKT_{∂} . By the definition of Ψ , $\Psi(\text{abort}(H)) = \Psi(H, \text{abort})$. We now prove $\Psi(\Xi) \Vdash \Psi(H, \text{abort}) : (\Gamma \vdash \Delta)$. Applying the inference rule $L \perp$, $\epsilon \Vdash \text{abort} : (\Gamma; \perp \vdash \Delta)$ in LKT_{∂} . By IH, $\Psi(\Xi) \Vdash \Psi(H, \text{abort}) : (\Gamma \vdash \Delta)$ in LKT_{∂} .
- Case ACT . Suppose $\Xi \Vdash \mu a^A.S : (\Gamma \vdash A \mid \Delta)$ is obtained from $\Xi \Vdash S : (\Gamma \vdash a : A, \Delta)$ in NKT_{∂} . By the definition of Ψ , $\Psi(\mu a^A.S) = \mu a^A.\Psi(S)$. We now prove $\Psi(\Xi) \Vdash \mu a^A.\Psi(S) : (\Gamma \vdash A \mid \Delta)$. By IH, $\Psi(\Xi) \Vdash \Psi(S) : (\Gamma \vdash a : A, \Delta)$. Applying the inference rule ACT , $\Psi(\Xi) \Vdash \mu a^A.\Psi(S) : (\Gamma \vdash A \mid \Delta)$ in LKT_{∂} .
- Case PSS . Suppose $\Xi \Vdash a(H) : (\Gamma \vdash a : p, \Delta)$ is obtained from $\Xi \Vdash H : (\Gamma \triangleright p \mid a : p, \Delta)$ in NKT_{∂} . By the definition of Ψ , $\Psi(a(H)) = \Psi(H, a)$. We now prove $\Psi(\Xi) \Vdash \Psi(H, a) : (\Gamma \vdash a : p, \Delta)$. Applying the AX rule, $\epsilon \Vdash a : (\Gamma; p \vdash a : p, \Delta)$, in LKT_{∂} . By IH, $\Psi(\Xi) \Vdash \Psi(H, a) : (\Gamma \vdash a : p, \Delta)$ in LKT_{∂} .
- Case CTR . Suppose $\Xi \Vdash a(\lambda x^A.M) : (\Gamma \vdash a : A \supset B, \Delta)$ is obtained from $\Xi \Vdash M : (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)$ in NKT_{∂} . By the definition of Ψ , $\Psi(a(\lambda x^A.M)) = \langle \lambda x^A.\Psi(M) \mid a \rangle$. We now prove $\Psi(\Xi) \Vdash \langle \lambda x^A.\Psi(M) \mid a \rangle : (\Gamma \vdash a : A \supset B, \Delta)$. By IH,

$\Psi(\Xi) \Vdash \Psi(M) : (\Gamma, x : A \vdash B \mid a : A \supset B, \Delta)$ in LKT_∂ . Applying the inference rule $RCTR$, $\Psi(\Xi) \Vdash \langle \lambda x^A. \Psi(M) \mid a \rangle : (\Gamma \vdash a : A \supset B, \Delta)$ in LKT_∂ .

- Case $I\supset$. Suppose $\Xi \Vdash \lambda x^A.M : (\Gamma \vdash A \supset B \mid \Delta)$ is obtained from $\Xi \Vdash M : (\Gamma, x : A \vdash B \mid \Delta)$ in NKT_∂ . By the definition of Ψ , $\Psi(\lambda x^A.M) = \lambda x^A. \Psi(M)$. We now prove $\Psi(\Xi) \Vdash \lambda x^A. \Psi(M) : (\Gamma \vdash A \supset B \mid \Delta)$. By IH, $\Psi(\Xi) \Vdash \Psi(M) : (\Gamma, x : A \vdash B \mid \Delta)$. Applying the RI rule, $\Psi(\Xi) \Vdash \lambda x^A. \Psi(M) : (\Gamma \vdash A \supset B \mid \Delta)$ in LKT_∂ .
- Case $E\supset$. Suppose $\Xi_1 @ \Xi_2 \Vdash HN : (\Gamma \triangleright B \mid \Delta)$ is obtained from $\Xi_1 \Vdash H : (\Gamma \triangleright A \supset B \mid \Delta)$ and $\Xi_2 \Vdash N : (\Gamma \vdash A \mid \Delta)$ in NKT_∂ and $\Xi \Vdash l : (\Gamma; B \vdash \Delta)$ in LKT_∂ . By the definition of Ψ , $\Psi(HN, l) = \Psi(H, \Psi(N) :: l)$. We now prove $\Psi(\Xi_1) @ \Psi(\Xi_2) @ \Xi \Vdash \Psi(H, \Psi(N) :: l) : (\Gamma \vdash \Delta)$. By IH, $\Psi(\Xi_2) \Vdash \Psi(N) : (\Gamma \vdash A \mid \Delta)$ in LKT_∂ . Applying LI rule, $\Psi(\Xi_2) @ \Xi \Vdash \Psi(N) :: l : (\Gamma; A \supset B \vdash \Delta)$. By IH, $\Psi(\Xi_1) @ \Psi(\Xi_2) @ \Xi \Vdash \Psi(H, \Psi(N) :: l) : (\Gamma \vdash \Delta)$ in LKT_∂ .

□

We now prove a lemma that will be useful in the proof of the most important and final theorem of this chapter.

Lemma 4.1.

1. If $l \rightarrow l'$ then $\Psi(H, l) \rightarrow \Psi(H, l')$.
2. If $H \rightarrow H'$ then $\Theta(H, l) \rightarrow \Theta(H', l)$.

Proof. The first item is proved by induction on H and the second item is proved by induction on l .

In (1),

- Case $H = x$. Suppose $l \rightarrow l'$. Then, $\Psi(x, l) = \langle x \mid l \rangle \rightarrow \langle x \mid l' \rangle = \Psi(x, l')$. The reduction step is by the assumption.
- Case $H = H'N$. Suppose $l \rightarrow l'$. Then $\Psi(H'N, l) = \Psi(H', \Psi(N) :: l) \rightarrow \Psi(H', \Psi(N) :: l') = \Psi(H'N, l')$. The reduction step is by IH, given that $\Psi(N) :: l \rightarrow \Psi(N) :: l'$.

In (2),

- Case $l = a$. Suppose $H \rightarrow H'$. Then, $\Theta(H, l) = \Theta(H, a) = a(H) \rightarrow a(H') = \Theta(H', a) = \Theta(H', l)$. The reduction step is by the assumption.

- Case $l = \text{abort}$. Suppose $H \rightarrow H'$. Then, $\Theta(H, l) = \Theta(H, \text{abort}) = \text{abort}(H) \rightarrow \text{abort}(H') = \Theta(H', \text{abort}) = \Theta(H', l)$. The reduction step is by the assumption.
- Case $l = u :: l'$. Suppose $H \rightarrow H'$. Then, $\Theta(H, l) = \Theta(H, u :: l') = \Theta(H\Theta(u), l') \rightarrow \Theta(H'\Theta(u), l') = \Theta(H', u :: l') = \Theta(H', l)$. The reduction step is by IH, given that $H\Theta(u) \rightarrow H'\Theta(u)$.
- Case $l = \underline{\Gamma; A \vdash \Delta}$. Suppose $H \rightarrow H'$. Then, $\Theta(H, l) = \Theta(H, \underline{\Gamma; A \vdash \Delta}) = \text{ab}(H, (\underline{\Gamma \triangleright A} \mid \underline{\Delta})) \rightarrow \text{ab}(H', (\underline{\Gamma \triangleright A} \mid \underline{\Delta})) = \Theta(H', \underline{\Gamma; A \vdash \Delta}) = \Theta(H', l)$. The reduction step is by the assumption.

□

Finally, the following theorem establishes the isomorphism between the terms and reductions of LKT_∂ and NKT_∂ .

Theorem 4.6 ($LKT_\partial \cong NKT_\partial$).

1. $\Theta(\Psi(M)) = M$ and $\Psi(\Theta(t)) = t$.
2. $t \rightarrow t'$ in LKT_∂ iff $\Theta(t) \rightarrow \Theta(t')$ in NKT_∂ .
3. $M \rightarrow M'$ in NKT_∂ iff $\Psi(M) \rightarrow \Psi(M')$ in LKT_∂ .

Proof. The proof of item (1) is also an extension of the proof found in [13]. We prove that $\Theta(\Psi(M)) = M$, together with $\Theta(\Psi(H, l)) = \Theta(H, l)$ and $\Theta(\Psi(S)) = S$, by simultaneous induction on M , H and S :

- Case $M = \lambda x^A.M'$.

$$\begin{aligned} \Theta(\Psi(\lambda x^A.M')) &= \Theta(\lambda x^A.\Psi(M')) \quad (\text{def. of } \Psi) \\ &= \lambda x^A.\Theta(\Psi(M')) \quad (\text{def. of } \Theta) \\ &= \lambda x^A.M' \quad (\text{IH}); \end{aligned}$$

- Case $M = \mu a^A.S$.

$$\begin{aligned} \Theta(\Psi(\mu a^A.S)) &= \Theta(\mu a^A.\Psi(S)) \quad (\text{def. of } \Psi) \\ &= \mu a^A.\Theta(\Psi(S)) \quad (\text{def. of } \Theta) \\ &= \mu a^A.S \quad (\text{IH}); \end{aligned}$$

- Case $M = \underline{\sigma}$.

$$\begin{aligned} \Theta(\Psi(\underline{\sigma})) &= \Theta(\underline{\sigma}) \quad (\text{def. of } \Psi) \\ &= \underline{\sigma} \quad (\text{def. of } \Theta); \end{aligned}$$

- Case $H = x$.

$$\begin{aligned}\Theta(\Psi(x, l)) &= \Theta(\langle x \mid l \rangle) \quad (\text{def. of } \Psi) \\ &= \Theta(x, l) \quad (\text{def. of } \Theta);\end{aligned}$$

- Case $H = H'N$.

$$\begin{aligned}\Theta(\Psi(H'N, l)) &= \Theta(\Psi(H', \Psi(N) :: l)) \quad (\text{def. of } \Psi) \\ &= \Theta(H', \Psi(N) :: l) \quad (\text{IH}) \\ &= \Theta(H'\Theta(\Psi(N)), l) \quad (\text{def. of } \Theta) \\ &= \Theta(H'N, l) \quad (\text{IH});\end{aligned}$$

- Case $S = \text{abort}(H)$.

$$\begin{aligned}\Theta(\Psi(\text{abort}(H))) &= \Theta(\Psi(H, \text{abort})) \quad (\text{def. of } \Psi) \\ &= \Theta(H, \text{abort}) \quad (\text{IH}) \\ &= \text{abort}(H) \quad (\text{def. of } \Theta);\end{aligned}$$

- Case $S = a(H)$.

$$\begin{aligned}\Theta(\Psi(a(H))) &= \Theta(\Psi(H, a)) \quad (\text{def. of } \Psi) \\ &= \Theta(H, a) \quad (\text{IH}) \\ &= a(H) \quad (\text{def. of } \Theta);\end{aligned}$$

- Case $S = a(\lambda x^A.M)$.

$$\begin{aligned}\Theta(\Psi(a(\lambda x^A.M))) &= \Theta(\langle \lambda x^A.\Psi(M) \mid a \rangle) \quad (\text{def. of } \Psi) \\ &= a(\lambda x^A.\Theta(\Psi(M))) \quad \text{def. of } \Theta \\ &= a(\lambda x^A.M) \quad (\text{IH});\end{aligned}$$

- Case $S = \text{ab}(H, (\Gamma \triangleright A \mid \Delta))$.

$$\begin{aligned}\Theta(\Psi(\text{ab}(H, (\Gamma \triangleright A \mid \Delta)))) &= \Theta(\Psi(H, (\Gamma; A \vdash \Delta))) \quad (\text{def. of } \Psi) \\ &= \Theta(H, (\Gamma; A \vdash \Delta)) \quad (\text{IH}) \\ &= \text{ab}(H, (\Gamma \triangleright A \mid \Delta)) \quad (\text{def. of } \Theta);\end{aligned}$$

- Case $S = \underline{\zeta}$.

$$\begin{aligned}\Theta(\Psi(\underline{\zeta})) &= \Theta(\underline{\zeta}) \quad (\text{def. of } \Psi) \\ &= \underline{\zeta} \quad (\text{def. of } \Theta).\end{aligned}$$

On the other hand, we prove $\Psi(\Theta(t)) = t$, together with $\Psi(\Theta(H, l)) = \Psi(H, l)$ and $\Psi(\Theta(c)) = c$, by simultaneous induction on t , l and c .

- Case $t = \lambda x^A.t'$.

$$\begin{aligned}
\Psi(\Theta(\lambda x^A.t')) &= \Psi(\lambda x^A.\Theta(t')) \quad (\text{def. of } \Theta) \\
&= \lambda x^A.\Psi(\Theta(t')) \quad (\text{def. of } \Psi) \\
&= \lambda x^A.t' \quad (\text{IH});
\end{aligned}$$

- Case $t = \mu a^A.c$.

$$\begin{aligned}
\Psi(\Theta(\mu a^A.c)) &= \Psi(\mu a^A.\Theta(c)) \quad (\text{def. of } \Theta) \\
&= \mu a^A.\Psi(\Theta(c)) \quad (\text{def. of } \Psi) \\
&= \mu a^A.c \quad (\text{IH});
\end{aligned}$$

- Case $t = \underline{\sigma}$.

$$\begin{aligned}
\Psi(\Theta(\underline{\sigma})) &= \Psi(\underline{\sigma}) \quad (\text{def. of } \Theta) \\
&= \underline{\sigma} \quad (\text{def. of } \Psi);
\end{aligned}$$

- Case $l = a$.

$$\begin{aligned}
\Psi(\Theta(H, a)) &= \Psi(a(H)) \quad (\text{def. of } \Theta) \\
&= \Psi(H, a) \quad (\text{def. of } \Psi);
\end{aligned}$$

- Case $l = \text{abort}$.

$$\begin{aligned}
\Psi(\Theta(H, \text{abort})) &= \Psi(\text{abort}(H)) \quad (\text{def. of } \Theta) \\
&= \Psi(H, \text{abort}) \quad (\text{def. of } \Psi);
\end{aligned}$$

- Case $l = u :: l'$.

$$\begin{aligned}
\Psi(\Theta(H, u :: l')) &= \Psi(\Theta(H\Theta(u), l')) \quad (\text{def. of } \Theta) \\
&= \Psi(H\Theta(u), l') \quad (\text{IH}) \\
&= \Psi(H, \Psi(\Theta(u)) :: l') \quad (\text{def. of } \Psi) \\
&= \Psi(H, u :: l') \quad (\text{IH});
\end{aligned}$$

- Case $l = \underline{\Gamma; A \vdash \Delta}$.

$$\begin{aligned}
\Psi(\Theta(H, \underline{\Gamma; A \vdash \Delta})) &= \Psi(\text{ab}(H, \underline{\Gamma \triangleright A \mid \Delta})) \quad (\text{def. of } \Theta) \\
&= \Psi(H, \underline{\Gamma; A \vdash \Delta}) \quad (\text{def. of } \Psi);
\end{aligned}$$

- Case $c = \langle x \mid l \rangle$.

$$\begin{aligned}
\Psi(\Theta(\langle x \mid l \rangle)) &= \Psi(\Theta(x, l)) \quad (\text{def. of } \Theta) \\
&= \Psi(x, l) \quad (\text{IH}) \\
&= \langle x \mid l \rangle \quad (\text{def. of } \Psi);
\end{aligned}$$

- Case $c = \langle \lambda x^A.t \mid a \rangle$.

$$\begin{aligned}
\Psi(\Theta(\langle \lambda x^A.t \mid a \rangle)) &= \Psi(a(\lambda x^A.\Theta(t))) && \text{(def. of } \Theta) \\
&= \langle \lambda x^A.\Psi(\Theta(t)) \mid a \rangle && \text{(def. of } \Psi) \\
&= \langle \lambda x^A.t \mid a \rangle && \text{(IH);}
\end{aligned}$$

- Case $c = \underline{\zeta}$.

$$\begin{aligned}
\Psi(\Theta(\underline{\zeta})) &= \Psi(\underline{\zeta}) && \text{(def. of } \Theta) \\
&= \underline{\zeta} && \text{(def. of } \Psi).
\end{aligned}$$

Using item (1), we obtain the “if” statements of items (2) and (3) from the “only if” statements: in item (3), suppose $\Psi(M) \rightarrow \Psi(M')$ in LKT_∂ . Then $M = \Theta(\Psi(M)) \rightarrow \Theta(\Psi(M')) = M'$, so $M \rightarrow M'$. Item (2) is proved analogously. It remains to prove the “only if” statements.

The “only if” statement of item (2) is proved together with:

- if $l \rightarrow l'$ in LKT_∂ then, for all H , $\Theta(H, l) \rightarrow \Theta(H, l')$ in NKT_∂ ; and
- if $c \rightarrow c'$ in LKT_∂ then $\Theta(c) \rightarrow \Theta(c')$ in NKT_∂ .

The proof is by simultaneous induction on $t \rightarrow t'$, $l \rightarrow l'$ and $c \rightarrow c'$.

Base Cases:

- Case *SIR*. Suppose $c = \underline{\Gamma \vdash a : A \supset B, \Delta} \rightarrow \langle \lambda x^A.(\underline{\Gamma, x : A \vdash B \mid a : A \supset B, \Delta}) \mid a \rangle = c'$. Then:

$$\begin{aligned}
\Theta(\underline{\Gamma \vdash a : A \supset B, \Delta}) &= \underline{\Gamma \vdash a : A \supset B, \Delta} && \text{(def. of } \Theta) \\
&\rightarrow a(\lambda x^A.(\underline{\Gamma, x : A \vdash B \mid a : A \supset B, \Delta})) && \text{(SIR)} \\
&= a(\lambda x^A.\Theta(\underline{\Gamma, x : A \vdash B \mid a : A \supset B, \Delta})) && \text{(def. of } \Theta) \\
&= \Theta(\langle \lambda x^A.(\underline{\Gamma, x : A \vdash B \mid a : A \supset B, \Delta}) \mid a \rangle) && \text{(def. of } \Theta);
\end{aligned}$$

- Case *KIR*. Suppose $t = \underline{\Gamma \vdash A \supset B \mid \Delta} \rightarrow \lambda x^A.(\underline{\Gamma, x : A \vdash B \mid \Delta}) = t'$. then

$$\begin{aligned}
\Theta(\underline{\Gamma \vdash A \supset B \mid \Delta}) &= \underline{\Gamma \vdash A \supset B \mid \Delta} && \text{(def. of } \Theta) \\
&\rightarrow \lambda x^A.(\underline{\Gamma, x : A \vdash B \mid \Delta}) && \text{(KIR)} \\
&= \lambda x^A.\Theta(\underline{\Gamma, x : A \vdash B \mid \Delta}) && \text{(def. of } \Theta) \\
&= \Theta(\lambda x^A.(\underline{\Gamma, x : A \vdash B \mid \Delta})) && \text{(def. of } \Theta);
\end{aligned}$$

- Case *FIR*. Suppose $t = \underline{\Gamma \vdash A \mid \Delta} \rightarrow \mu a^A.(\underline{\Gamma \vdash a : A, \Delta}) = t'$. Then

$$\begin{aligned}
\Theta(\underline{\Gamma \vdash A \mid \Delta}) &= \underline{\Gamma \vdash A \mid \Delta} && (\text{def. of } \Theta) \\
&\rightarrow \mu a^A.(\underline{\Gamma \vdash a : A, \Delta}) && (FIR) \\
&= \mu a^A.\Theta(\underline{\Gamma \vdash a : A, \Delta}) && (\text{def. of } \Theta) \\
&= \Theta(\mu a^A.(\underline{\Gamma \vdash a : A, \Delta})) && (\text{def. of } \Theta);
\end{aligned}$$

- Case *SFL*. Suppose $c = \underline{\Gamma, x : A \vdash \Delta} \rightarrow \langle x \mid (\underline{\Gamma, x : A; A \vdash \Delta}) \rangle = c'$. Then:

$$\begin{aligned}
\Theta(\underline{\Gamma, x : A \vdash \Delta}) &= \underline{\Gamma, x : A \vdash \Delta} && (\text{def. of } \Theta) \\
&\rightarrow \text{ab}(x, (\underline{\Gamma, x : A \triangleright A \mid \Delta})) && (SFD) \\
&= \Theta(x, (\underline{\Gamma, x : A; A \vdash \Delta})) && (\text{def. of } \Theta) \\
&= \Theta(\langle x \mid (\underline{\Gamma, x : A; A \vdash \Delta}) \rangle) && (\text{def. of } \Theta);
\end{aligned}$$

- Case *KFL*. Suppose $l = \underline{\Gamma; A \supset B \vdash \Delta} \rightarrow (\underline{\Gamma \vdash A \mid \Delta}) :: (\underline{\Gamma; B \vdash \Delta}) = l'$. Then:

$$\begin{aligned}
\Theta(H, (\underline{\Gamma; A \supset B \vdash \Delta})) &= \text{ab}(H, (\underline{\Gamma \triangleright A \supset B \mid \Delta})) && (\text{def. of } \Theta) \\
&\rightarrow \text{ab}(H(\underline{\Gamma \vdash A \mid \Delta}), (\underline{\Gamma \triangleright B \mid \Delta})) && (KFD) \\
&= \Theta(H\Theta(\underline{\Gamma \vdash A \mid \Delta}), (\underline{\Gamma; B \vdash \Delta})) && (\text{def. of } \Theta) \\
&= \Theta(H, ((\underline{\Gamma \vdash A \mid \Delta}) :: (\underline{\Gamma; B \vdash \Delta}))) && (\text{def. of } \Theta)
\end{aligned}$$

- Case *FFL_{atm}*. Suppose $l = \underline{\Gamma; p \vdash a : p, \Delta} \rightarrow a = l'$. Then:

$$\begin{aligned}
\Theta(H, (\underline{\Gamma; p \vdash a : p, \Delta})) &= \text{ab}(H, (\underline{\Gamma \triangleright p \mid a : p, \Delta})) && (\text{def. of } \Theta) \\
&\rightarrow a(H) && (FFD_{atm}) \\
&= \Theta(H, a) && (\text{def. of } \Theta)
\end{aligned}$$

- Case *FFL_{abs}*. Suppose $l = \underline{\Gamma; \perp \vdash \Delta} \rightarrow \text{abort} = l'$. Then:

$$\begin{aligned}
\Theta(H, \underline{\Gamma; \perp \vdash \Delta}) &= \text{ab}(H, (\underline{\Gamma \triangleright \perp \mid \Delta})) && (\text{def. of } \Theta) \\
&\rightarrow \text{abort}(H) && (FFD_{abs}) \\
&= \Theta(H, \text{abort}) && (\text{def. of } \Theta)
\end{aligned}$$

Inductive Cases:

- Suppose $\lambda x^A.t \rightarrow \lambda x^A.t'$, with $t \rightarrow t'$. Then

$$\begin{aligned}
\Theta(\lambda x^A.t) &= \lambda x^A.\Theta(t) && (\text{def. of } \Theta) \\
&\rightarrow \lambda x^A.\Theta(t') && (IH) \\
&= \Theta(\lambda x^A.t') && (\text{def. of } \Theta);
\end{aligned}$$

- Suppose $\mu a^A.c \rightarrow \mu a^A.c'$, with $c \rightarrow c'$. Then

$$\begin{aligned}\Theta(\mu a^A.c) &= \mu a^A.\Theta(c) \quad (\text{def. of } \Theta) \\ &\rightarrow \mu a^A.\Theta(c') \quad (\text{IH}) \\ &= \Theta(\mu a^A.c') \quad (\text{def. of } \Theta);\end{aligned}$$

- Suppose $\langle x \mid l \rangle \rightarrow \langle x \mid l' \rangle$, with $l \rightarrow l'$. Then

$$\begin{aligned}\Theta(\langle x \mid l \rangle) &= \Theta(x, l) \quad (\text{def. of } \Theta) \\ &\rightarrow \Theta(x, l') \quad (\text{IH}) \\ &= \Theta(\langle x \mid l' \rangle) \quad (\text{def. of } \Theta);\end{aligned}$$

- Suppose $\langle \lambda x^A.t \mid a \rangle \rightarrow \langle \lambda x^A.t' \mid a \rangle$, with $t \rightarrow t'$. Then

$$\begin{aligned}\Theta(\langle \lambda x^A.t \mid a \rangle) &= a(\lambda x^A.\Theta(t)) \quad (\text{def. of } \Theta) \\ &\rightarrow a(\lambda x^A.\Theta(t')) \quad (\text{IH}) \\ &= \Theta(\langle \lambda x^A.t' \mid a \rangle) \quad (\text{def. of } \Theta);\end{aligned}$$

- Suppose $u :: l \rightarrow u' :: l$, with $u \rightarrow u'$. Then

$$\begin{aligned}\Theta(H, u :: l) &= \Theta(H\Theta(u), l) \quad (\text{def. of } \Theta) \\ &\rightarrow \Theta(H\Theta(u'), l) \quad (\text{IH \& Lemma 4.1}) \\ &= \Theta(H, u' :: l) \quad (\text{def. of } \Theta);\end{aligned}$$

- Suppose $u :: l \rightarrow u :: l'$, with $l \rightarrow l'$. Then

$$\begin{aligned}\Theta(H, u :: l) &= \Theta(H\Theta(u), l) \quad (\text{def. of } \Theta) \\ &\rightarrow \Theta(H\Theta(u), l') \quad (\text{IH}) \\ &= \Theta(H, u :: l') \quad (\text{def. of } \Theta).\end{aligned}$$

The “only if” statement of item 3 is proved together with:

- if $H \rightarrow H'$ in NKT_∂ then, for all l , $\Psi(H, l) \rightarrow \Psi(H', l)$ in LJT_∂ ; and
- if $S \rightarrow S'$ in NKT_∂ then $\Psi(S) \rightarrow \Psi(S')$.

The proof is by simultaneous induction on $M \rightarrow M'$, $H \rightarrow H'$ and $S \rightarrow S'$.

Base Cases:

- Case *SIR*. Suppose $S = \underline{\Gamma \vdash a : A \supset B, \Delta} \rightarrow a(\lambda x^A.(\underline{\Gamma, x : A \vdash B} \mid a : A \supset B, \Delta)) = S'$. Then:

$$\begin{aligned}
\Psi(\underline{\Gamma \vdash a : A \supset B, \Delta}) &= \underline{\Gamma \vdash a : A \supset B, \Delta} && (\text{def. of } \Psi) \\
&\rightarrow \langle \lambda x^A.(\underline{\Gamma, x : A \vdash B} \mid a : A \supset B, \Delta) \mid a \rangle && (SIR) \\
&= \langle \lambda x^A.\Psi(\underline{\Gamma, x : A \vdash B} \mid a : A \supset B, \Delta) \mid a \rangle && (\text{def. of } \Psi) \\
&= \Psi(a(\lambda x^A.(\underline{\Gamma, x : A \vdash B} \mid a : A \supset B, \Delta))) && (\text{def. of } \Psi);
\end{aligned}$$

- Case *KIR*. Suppose $M = \underline{\Gamma \vdash A \supset B \mid \Delta} \rightarrow \lambda x^A.(\underline{\Gamma, x : A \vdash B} \mid \Delta) = M'$. Then:

$$\begin{aligned}
\Psi(\underline{\Gamma \vdash A \supset B \mid \Delta}) &= \underline{\Gamma \vdash A \supset B \mid \Delta} && (\text{def. of } \Psi) \\
&\rightarrow \lambda x^A.(\underline{\Gamma, x : A \vdash B} \mid \Delta) && (KIR) \\
&= \lambda x^A.\Psi(\underline{\Gamma, x : A \vdash B} \mid \Delta) && (\text{def. of } \Psi) \\
&= \Psi(\lambda x^A.(\underline{\Gamma, x : A \vdash B} \mid \Delta)) && (\text{def. of } \Psi);
\end{aligned}$$

- Case *FIR*. Suppose $M = \underline{\Gamma \vdash A \mid \Delta} \rightarrow \mu a^A.(\underline{\Gamma \vdash a : A, \Delta}) = M'$. Then:

$$\begin{aligned}
\Psi(\underline{\Gamma \vdash A \mid \Delta}) &= \underline{\Gamma \vdash A \mid \Delta} && (\text{def. of } \Psi) \\
&\rightarrow \mu a^A.(\underline{\Gamma \vdash a : A, \Delta}) && (FIR) \\
&= \mu a^A.\Psi(\underline{\Gamma \vdash a : A, \Delta}) && (\text{def. of } \Psi) \\
&= \Psi(\mu a^A.(\underline{\Gamma \vdash a : A, \Delta})) && (\text{def. of } \Psi);
\end{aligned}$$

- Case *SFD*. Suppose $S = \underline{\Gamma, x : A \vdash \Delta} \rightarrow \text{ab}(x, (\underline{\Gamma, x : A \supset A \mid \Delta})) = S'$. Then:

$$\begin{aligned}
\Psi(\underline{\Gamma, x : A \vdash \Delta}) &= \underline{\Gamma, x : A \vdash \Delta} && (\text{def. of } \Psi) \\
&\rightarrow \langle x \mid (\underline{\Gamma, x : A; A \vdash \Delta}) \rangle && (SFL) \\
&= \Psi(x, (\underline{\Gamma, x : A; A \vdash \Delta})) && (\text{def. of } \Psi) \\
&= \Psi(\text{ab}(x, (\underline{\Gamma, x : A \supset A \mid \Delta}))) && (\text{def. of } \Psi);
\end{aligned}$$

- Case *KFD*. Suppose $S = \text{ab}(H, (\underline{\Gamma \supset A \supset B \mid \Delta})) \rightarrow \text{ab}(H(\underline{\Gamma \vdash A \mid \Delta}), (\underline{\Gamma \supset B \mid \Delta})) = S'$. Then:

$$\begin{aligned}
\Psi(\text{ab}(H, (\underline{\Gamma \supset A \supset B \mid \Delta}))) &= \Psi(H, (\underline{\Gamma; A \supset B \vdash \Delta})) && (\text{def. of } \Psi) \\
&\rightarrow \Psi(H, (\underline{\Gamma \vdash A \mid \Delta} :: (\underline{\Gamma; B \vdash \Delta}))) && (\text{IH \& Lemma 4.1}) \\
&= \Psi(H, \Psi(\underline{\Gamma \vdash A \mid \Delta} :: (\underline{\Gamma; B \vdash \Delta}))) && (\text{def. of } \Psi) \\
&= \Psi(H(\underline{\Gamma \vdash A \mid \Delta}), (\underline{\Gamma; B \vdash \Delta})) && (\text{def. of } \Psi) \\
&= \Psi(\text{ab}(H(\underline{\Gamma \vdash A \mid \Delta}), (\underline{\Gamma \supset B \mid \Delta}))) && (\text{def. of } \Psi);
\end{aligned}$$

- Case FFD_{atm} . Suppose $S = \text{ab}(H, (\Gamma \triangleright p \mid a : p, \Delta)) \rightarrow a(H) = S'$. Then:

$$\begin{aligned}
\Psi(\text{ab}(H, (\Gamma \triangleright p \mid a : p, \Delta))) &= \Psi(H, (\Gamma; p \vdash a : p, \Delta)) \quad (\text{def. of } \Psi) \\
&\rightarrow \Psi(H, a) \quad (\text{IH \& Lemma 4.1}) \\
&= \Psi(a(H)) \quad (\text{by def. of } \Psi);
\end{aligned}$$

- Case FFD_{abs} . Suppose $S = \text{ab}(H, (\Gamma \triangleright \perp \mid \Delta)) \rightarrow \text{abort}(H) = S'$. Then:

$$\begin{aligned}
\Psi(\text{ab}(H, (\Gamma \triangleright \perp \mid \Delta))) &= \Psi(H, (\Gamma; \perp \vdash \Delta)) \quad (\text{def. of } \Psi) \\
&\rightarrow \Psi(H, \text{abort}) \quad (\text{IH \& Lemma 4.1}) \\
&= \Psi(\text{abort}(H)) \quad (\text{by def. of } \Psi).
\end{aligned}$$

Inductive Cases:

- Suppose $\lambda x^A.M \rightarrow \lambda x^A.M'$, with $M \rightarrow M'$. Then

$$\begin{aligned}
\Psi(\lambda x^A.M) &= \lambda x^A.\Psi(M) \quad (\text{def. of } \Psi) \\
&\rightarrow \lambda x^A.\Psi(M') \quad (\text{IH}) \\
&= \Psi(\lambda x^A.M') \quad (\text{def. of } \Psi);
\end{aligned}$$

- Suppose $\mu a^A.S \rightarrow \mu a^A.S'$, with $S \rightarrow S'$. Then

$$\begin{aligned}
\Psi(\mu a^A.S) &= \mu s^A.\Psi(S) \quad (\text{def. of } \Psi) \\
&\rightarrow \mu a^A.\Psi(S') \quad (\text{IH}) \\
&= \Psi(\mu a^A.S') \quad (\text{def. of } \Psi);
\end{aligned}$$

- Suppose $\text{abort}(H) \rightarrow \text{abort}(H')$, with $H \rightarrow H'$. Then

$$\begin{aligned}
\Psi(\text{abort}(H)) &= \Psi(H, \text{abort}) \quad (\text{def. of } \Psi) \\
&\rightarrow \Psi(H', \text{abort}) \quad (\text{IH}) \\
&= \Psi(\text{abort}(H')) \quad (\text{def. of } \Psi);
\end{aligned}$$

- Suppose $a(H) \rightarrow a(H')$, with $H \rightarrow H'$. Then

$$\begin{aligned}
\Psi(a(H)) &= \Psi(H, a) \quad (\text{def. of } \Psi) \\
&\rightarrow \Psi(H', a) \quad (\text{IH}) \\
&= \Psi(a(H')) \quad (\text{def. of } \Psi);
\end{aligned}$$

- Suppose $a(\lambda x^A.M) \rightarrow a(\lambda x^A.M')$, with $M \rightarrow M'$. Then

$$\begin{aligned}
\Psi(a(\lambda x^A.M)) &= \langle \lambda x^A.M \mid a \rangle \quad (\text{def. of } \Psi) \\
&\rightarrow \langle \lambda x^A.M' \mid a \rangle \quad (\text{IH}) \\
&= \Psi(a(\lambda x^A.M')) \quad (\text{def. of } \Psi);
\end{aligned}$$

- Suppose $\text{ab}(H, (\Gamma \triangleright A \mid \Delta)) \rightarrow \text{ab}(H', (\Gamma \triangleright A \mid \Delta))$, with $H \rightarrow H'$. Then

$$\begin{aligned}
\Psi(\text{ab}(H, (\Gamma \triangleright A \mid \Delta))) &= \Psi(H, (\Gamma; A \vdash \Delta)) && \text{(def. of } \Psi) \\
&\rightarrow \Psi(H', (\Gamma; A \vdash \Delta)) && \text{(IH)} \\
&= \Psi(\text{ab}(H', (\Gamma \triangleright A \mid \Delta))) && \text{(def. of } \Psi);
\end{aligned}$$

- Suppose $HN \rightarrow HN'$, with $N \rightarrow N'$. Then

$$\begin{aligned}
\Psi(HN, l) &= \Psi(H, \Psi(N) :: l) && \text{(def. of } \Psi) \\
&\rightarrow \Psi(H, \Psi(N') :: l) && \text{(IH \& Lemma 4.1)} \\
&= \Psi(HN', l) && \text{(def. of } \Psi);
\end{aligned}$$

- Suppose $HN \rightarrow H'N$, with $H \rightarrow H'$. Then

$$\begin{aligned}
\Psi(HN, l) &= \Psi(H, \Psi(N) :: l) && \text{(def. of } \Psi) \\
&\rightarrow \Psi(H', \Psi(N) :: l) && \text{(IH)} \\
&= \Psi(H'N, l) && \text{(def. of } \Psi).
\end{aligned}$$

□

The previous theorem says that reduction of $t \in LKT_\partial$ is isomorphic to reduction of $M = \Theta(t) \in NKT_\partial$. If we consider the particular case of $t = M = \underline{\sigma}$, and given that, according to Theorems 4.2 and 4.4, reduction in LKT_∂ (resp. NKT_∂) implements of $\underline{\sigma}$ proof search in LKT (resp. NKT), we may conclude:

Corollary 4.1. Proof search in LKT is isomorphic to proof search in NKT .

Chapter 5

An Intercalation Calculus with Partial Proof Terms

In the previous chapters, we introduced partial proof terms within focused proof search systems for sequent calculus and natural deduction, where right inversion and focus (left or down) phases are distinctly separated and proceed in a specific order.

In this chapter, we revisit the original formulation of the Intercalation Calculus (*IC*) for intuitionistic implicative logic, introduced by Sieg [55], in which the the bottom-up application of introduction inference rules from conclusion and the top-down application of elimination inference rules from the hypotheses are mixed in arbitrary order to yield normal derivations in ordinary natural deduction.

We confirm that partial proof terms can adequately represent this search procedure. Our formalization is a generalization of NJT_{∂} and the search procedure obtains total proof terms and the corresponding derivations in NJT . Therefore, the obtained derivations are η -expanded and, for this reason, we explain beforehand how to adapt *IC* to yield such kind of derivations.

5.1 Sieg's Intercalation Calculus

We start this section with the Intercalation Calculus, introduced by Sieg [55] and developed as a calculus of proof search in natural deduction.

Definition 5.1 (Sequents of Intercalation Calculus). A sequent of Intercalation Calculus is a triple of the form

$$\alpha ; \beta ? G$$

where α is a sequence of formulas, which are the available assumptions; β is a sequence of formulas obtained by eliminations from the formulas in α or β , and G is a formula representing the current goal.

Let $\alpha\beta$ be the concatenation of the sequences α and β . The search rules of Intercalation Calculus (*IC*-Rules) are defined as follows.

Definition 5.2 (IC-Rules). The introduction and elimination search rules of Intercalation Calculus are defined as follows:

$$\begin{array}{l} (\supset_{\uparrow}) \quad \alpha; \beta?A \supset B \rightarrow \alpha, A; \beta?B \\ (\supset_{\downarrow}) \quad \alpha; \beta?G \rightarrow \alpha; \beta?A \text{ AND } \alpha; \beta, B?G, \text{ if } A \supset B \in \alpha\beta \end{array}$$

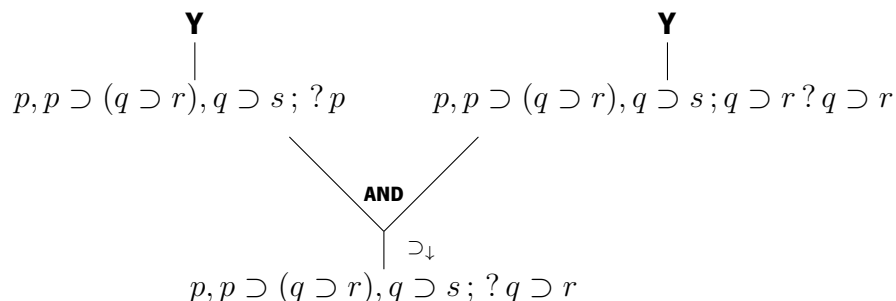
In the Intercalation Calculus, constructing a proof involves building a search tree whose nodes are sequents, and that enumerates all possible ways for closing the gap between the assumptions and the goal G by systematically applying all the suitable IC-Rules.

Definition 5.3 (Search Process of Intercalation Calculus). Given a current node $\alpha ; \beta ? G$, that is, a leaf of the search tree, the search process of the Intercalation Calculus is described in the following manner:

- If $G \in \alpha\beta$, the branch determined by the current node is closed with **Y**;
- If $G \notin \alpha\beta$: if every applicable rule leads to a sequent that is not new for the branch determined by the current sequent, the branch is closed with **N**;
- If $G \notin \alpha\beta$: if some applicable rule leads to a new sequent, the tree is extended at the current sequent node for all such rules by adding the appropriate rule and sequent nodes.

The search is successful for a sequent if at least one subtree with root on that sequent has all its branches closed with **Y**. Translating this subtree yields a normal derivation in natural deduction.

Example 5.1. Let $p, p \supset (q \supset r), q \supset s; ? q \supset r$ be a sequent of the Intercalation Calculus, where the goal is $q \supset r$ and the assumptions are $p, p \supset (q \supset p)$ and $q \supset s$. A well-succeed proof search for this sequent is represented by the following sub-tree of the search tree.



Notice how the rules are applied to build the derivation: we eliminate the hypothesis $p \supset (q \supset r)$ and obtain two sequent nodes in which the goal $G \in \alpha\beta$. We then close both branches with **Y**. The extracted normal derivation is

$$\frac{p \supset (q \supset r) \quad p}{q \supset r}$$

which is not η -expanded.

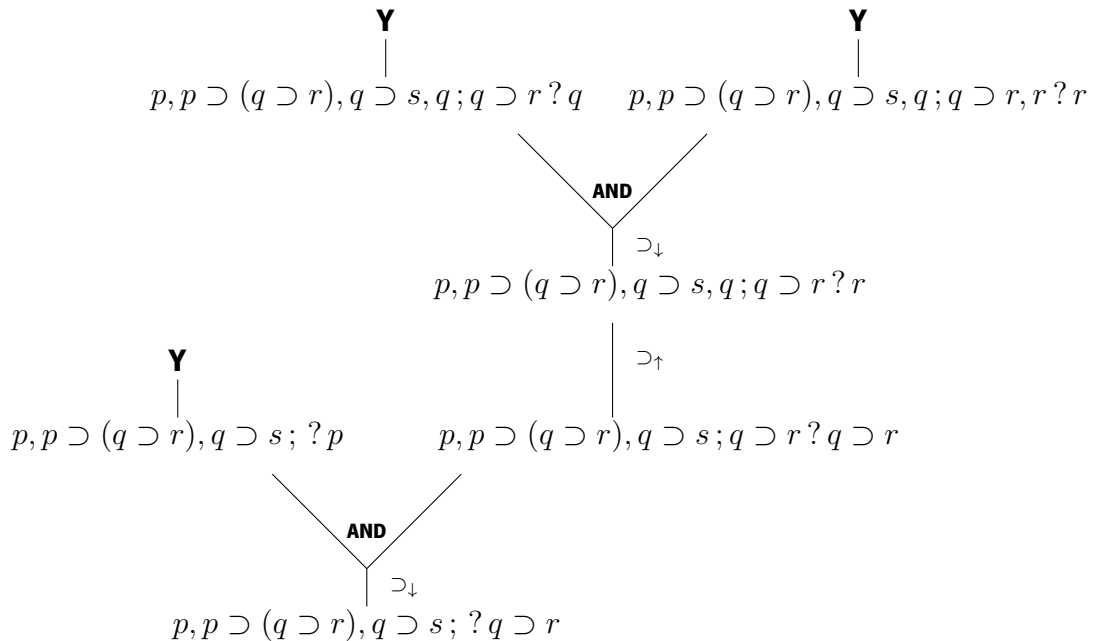
According to [55], normal natural deduction derivations in various styles can be easily obtained from a successful proof search sub-tree.

If we want to obtain η -expanded proofs, then we define a new proof search system, which retains the reduction rules of IC given in the Definition 5.2, but differs in its search process, which is described below.

Definition 5.4 (A Variation of the Search Process of IC). Given a current node $\alpha; \beta ? G$, that is, a leaf of the search tree, the search process of Intercalation Calculus for η -expanded proofs is described in the following manner:

- If G is atomic and $G \in \alpha\beta$, the branch determined by the current node is closed with **Y**;
- If G is not atomic or $G \notin \alpha\beta$: if every applicable rule leads to a sequent that is not new for the branch determined by the current sequent, the branch is closed with **N**;
- If G is not atomic or $G \notin \alpha\beta$: if some applicable rule leads to a new sequent, the tree is extended at the current sequent node for all such rules by adding the appropriate rule and sequent nodes.

Example 5.2. Consider the same sequent as in Example 5.1. A well-succeed proof search for this sequent, generalized by the variant of the search process, is represented by the following sub-tree of the search tree.



Notice the order in which the rules are applied to build the derivation: first, we eliminate the hypothesis $p \supset (q \supset r)$, then we invert the implication $q \supset r$ in the conclusion/goal side and finally, we eliminate the implication $q \supset r$, obtained as result of the first elimination.

An ordinary natural deduction derivation extracted from the search tree is:

$$\frac{\frac{p \supset (q \supset r) \quad p}{q \supset r} \quad [q]}{\frac{r}{q \supset r}}$$

For an easier comparison with NJT , this derivation can be presented as:

$$\frac{\frac{\frac{\Gamma, x : q \vdash p \supset (q \supset r)}{\Gamma, x : q \vdash q \supset r} Ax \quad \frac{\Gamma, x : q \vdash p}{\Gamma, x : q \vdash q} E\supset}{\frac{\Gamma, x : q \vdash r}{\Gamma \vdash q \supset r} I\supset} E\supset$$

where Γ contains the assumptions $(z : p)$, $(y : p \supset (q \supset r))$, and $(w : q \supset s)$.

Clearly, derivations in NJT can be obtained from derivations in ordinary natural deduction.

Example 5.3. The NJT derivation obtained from the derivation in ordinary natural deduction presented in Example 5.2 is:

$$\frac{\frac{\frac{\Gamma, x : q \triangleright p \supset (q \supset r)}{\Gamma, x : q \triangleright q \supset r} A \quad \frac{\frac{\Gamma, x : q \triangleright p}{\Gamma, x : q \vdash p} C \quad \frac{\Gamma, x : q \triangleright q}{\Gamma, x : q \vdash q} A}{\Gamma, x : q \triangleright r} E \quad \frac{\Gamma, x : q \triangleright r}{\Gamma, x : q \vdash r} C}{\Gamma \vdash q \supset r} I$$

where Γ is the same as in Example 5.2.

Intercalation Calculus obtains normal, ordinary natural deduction derivations, and these can be translated to NJT derivations. On the other hand NJT_∂ also obtains NJT derivations, by a restriction of the search procedure of this variation of Intercalation Calculus for η -expanded proofs. This suggests that Intercalation Calculus can be described as a generalization of NJT_∂ , using partial proof terms and the associated framework developed in the previous chapters.

5.2 The Intercalation Calculus NJT_∂

Proof search in NJT_∂ looks for derivations in the natural deduction system NJT , without employing goal-directed or focused strategies. As shown in Figure 5, implications are decomposed in no particular

order: the inference rule I is applied bottom-up, while implications in the context are decomposed top-down using rule A , immediately followed by an elimination E . Multiple implications in the context can be decomposed simultaneously, and the top-down and bottom-up phases can alternate, as is characteristic of Intercalation Calculus. Typically, the same atom p is computed both bottom-up and in at least one top-down decomposition, allowing the search to conclude with an application of rule C .

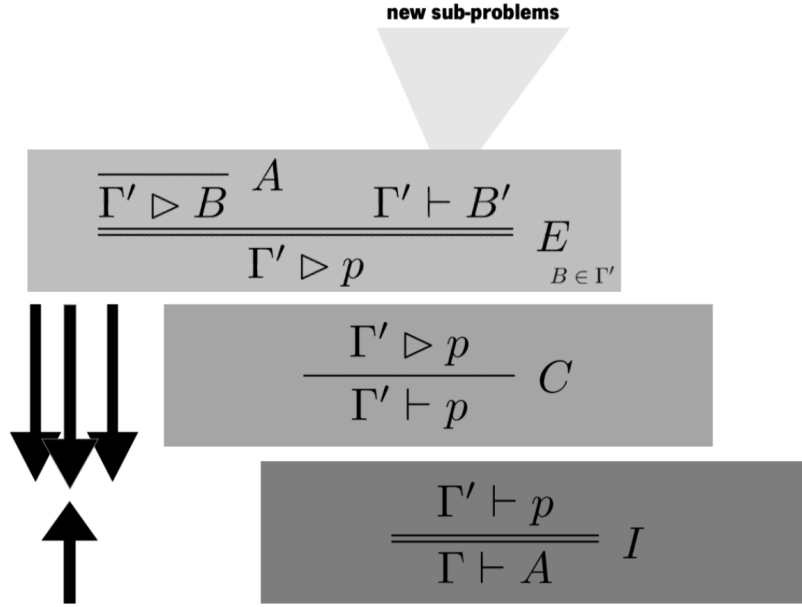


Figure 5: Intercalation Calculus NJT_∂ in NJT

We show three different sequences of proof search for the same sequent, described informally.

Example 5.4. Let $\Gamma \vdash q \supset r$ be a sequent of NJT , with $\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$.

We start by eliminating the hypothesis $p \supset (q \supset r)$; then we invert the formula $q \supset r$ on the right-hand side, and finally, we eliminate the implication $q \supset r$, obtained in the first step.

$$\begin{array}{c}
 \overline{\Gamma \vdash q \supset r} \quad \partial \\
 \downarrow \\
 \frac{\frac{\overline{\Gamma \triangleright p \supset (q \supset r)} \quad A}{\Gamma \triangleright q \supset r} \quad \frac{\overline{\Gamma \vdash p} \quad \partial}{\Gamma \vdash p} E}{\Gamma \vdash q \supset r} \quad \partial \\
 \downarrow
 \end{array}$$

$$\begin{array}{c}
\downarrow \\
\frac{\frac{\overline{\Gamma \triangleright p \supset (q \supset r)} \quad A}{\Gamma \triangleright q \supset r} \quad \frac{\overline{\Gamma \vdash p} \quad \partial}{\Gamma \vdash p} E}{\frac{\Gamma \triangleright q \supset r}{\Gamma, x : q \vdash r} \quad \partial}{\Gamma \vdash q \supset r} I \\
\downarrow \\
\frac{\frac{\overline{\Gamma, x : q \triangleright p \supset (q \supset r)} \quad A}{\Gamma, x : q \triangleright q \supset r} \quad \frac{\overline{\Gamma, x : q \vdash p} \quad \partial}{\Gamma, x : q \vdash p} E}{\frac{\Gamma, x : q \triangleright q \supset r}{\Gamma, x : q \vdash r} \quad \partial}{\Gamma \vdash q \supset r} I \quad \frac{\overline{\Gamma, x : q \vdash q} \quad \partial}{\Gamma, x : q \vdash q} E \\
\downarrow \\
\frac{\frac{\overline{\Gamma, x : q \triangleright p \supset (q \supset r)} \quad A}{\Gamma, x : q \triangleright q \supset r} \quad \frac{\overline{\Gamma, x : q \vdash p} \quad \partial}{\Gamma, x : q \vdash p} E}{\frac{\Gamma, x : q \triangleright q \supset r}{\Gamma, x : q \vdash r} \quad C}{\Gamma \vdash q \supset r} I \quad \frac{\overline{\Gamma, x : q \triangleright q} \quad A}{\Gamma, x : q \vdash q} C \quad \frac{\overline{\Gamma, x : q \vdash q} \quad \partial}{\Gamma, x : q \vdash q} E \\
\downarrow \\
\frac{\frac{\overline{\Gamma, x : q \triangleright p \supset (q \supset r)} \quad A}{\Gamma, x : q \triangleright q \supset r} \quad \frac{\overline{\Gamma, x : q \triangleright p} \quad A}{\Gamma, x : q \vdash p} C}{\frac{\Gamma, x : q \triangleright q \supset r}{\Gamma, x : q \vdash r} \quad E} \quad \frac{\overline{\Gamma, x : q \triangleright q} \quad A}{\Gamma, x : q \vdash q} C \quad \frac{\overline{\Gamma, x : q \vdash q} \quad \partial}{\Gamma, x : q \vdash q} E \\
\frac{\Gamma, x : q \triangleright q \supset r}{\Gamma, x : q \vdash r} C \quad \frac{\Gamma, x : q \vdash r}{\Gamma \vdash q \supset r} I
\end{array}$$

Example 5.5. Consider the same sequent as in Example 5.4. We start by inverting the formula $q \supset r$ on the right-hand side; then we eliminate the hypothesis $p \supset (q \supset r)$, and finally, we eliminate the implication $q \supset r$, obtained in the previous step.

$$\begin{array}{c}
\overline{\Gamma \vdash q \supset r} \quad \partial \\
\downarrow \\
\frac{\overline{\Gamma, x : q \vdash r} \quad \partial}{\Gamma \vdash q \supset r} I \\
\downarrow
\end{array}$$

$$\begin{array}{c}
\downarrow \\
\frac{\frac{\frac{\Gamma, x : q \triangleright p \supset (q \supset r)}{\Gamma, x : q \triangleright q \supset r} A \quad \frac{\Gamma, x : q \vdash p}{\Gamma, x : q \vdash q \supset r} \partial}{\Gamma, x : q \vdash q \supset r} E \\
\downarrow \\
\frac{\frac{\frac{\Gamma, x : q \triangleright p \supset (q \supset r)}{\Gamma, x : q \triangleright q \supset r} A \quad \frac{\Gamma, x : q \vdash p}{\Gamma, x : q \vdash q \supset r} \partial}{\Gamma, x : q \triangleright q \supset r} E \quad \frac{\Gamma, x : q \vdash q}{\Gamma, x : q \vdash q \supset r} \partial}{\Gamma, x : q \triangleright q \supset r} E \\
\downarrow \\
\frac{\frac{\frac{\Gamma, x : q \triangleright p \supset (q \supset r)}{\Gamma, x : q \triangleright q \supset r} A \quad \frac{\Gamma, x : q \vdash p}{\Gamma, x : q \vdash q \supset r} \partial}{\Gamma, x : q \triangleright q \supset r} E \quad \frac{\Gamma, x : q \vdash q}{\Gamma, x : q \vdash q \supset r} \partial}{\Gamma, x : q \triangleright q \supset r} E \\
\downarrow \\
\frac{\frac{\frac{\Gamma, x : q \triangleright p \supset (q \supset r)}{\Gamma, x : q \triangleright q \supset r} A \quad \frac{\frac{\frac{\Gamma, x : q \triangleright p}{\Gamma, x : q \vdash p} A \quad \frac{\Gamma, x : q \vdash p}{\Gamma, x : q \vdash q \supset r} C}{\Gamma, x : q \vdash q \supset r} E \quad \frac{\Gamma, x : q \triangleright q}{\Gamma, x : q \vdash q \supset r} A}{\Gamma, x : q \triangleright q \supset r} E \quad \frac{\Gamma, x : q \vdash q}{\Gamma, x : q \vdash q \supset r} C}{\Gamma, x : q \triangleright q \supset r} E \\
\downarrow \\
\frac{\frac{\frac{\Gamma, x : q \triangleright p \supset (q \supset r)}{\Gamma, x : q \triangleright q \supset r} A \quad \frac{\frac{\frac{\Gamma, x : q \triangleright p}{\Gamma, x : q \vdash p} A \quad \frac{\Gamma, x : q \vdash p}{\Gamma, x : q \vdash q \supset r} C}{\Gamma, x : q \vdash q \supset r} E \quad \frac{\Gamma, x : q \triangleright q}{\Gamma, x : q \vdash q \supset r} A}{\Gamma, x : q \triangleright q \supset r} E \quad \frac{\Gamma, x : q \vdash q}{\Gamma, x : q \vdash q \supset r} C}{\Gamma, x : q \triangleright q \supset r} E
\end{array}$$

Example 5.6. Consider the same sequent as in Example 5.4 and 5.5. We start by inverting the formula $q \supset r$ on the right-hand side; then we eliminate the hypothesis $p \supset (q \supset r)$. After that, we decide to start, in parallel, the elimination of the formula $q \supset s$. Finally, we eliminate the implication $q \supset r$, obtained in the first elimination step.

$$\begin{array}{c}
\overline{\Gamma \vdash q \supset r} \partial \\
\downarrow \\
\frac{\overline{\Gamma, x : q \vdash r} \partial}{\Gamma \vdash q \supset r} I \\
\downarrow
\end{array}$$

$$\begin{array}{c}
\downarrow \\
\frac{\frac{\frac{\Gamma, x : q \triangleright p \supset (q \supset r)}{\Gamma, x : q \triangleright q \supset r} A \quad \frac{\Gamma, x : q \vdash p}{\Gamma, x : q \vdash q \supset r} \partial}{\Gamma, x : q \vdash q \supset r} E \quad \frac{\Gamma, x : q \vdash r}{\Gamma, x : q \vdash q \supset r} I \\
\downarrow \\
\frac{\frac{\frac{\Gamma, x : q \triangleright p \supset (q \supset r)}{\Gamma, x : q \triangleright q \supset r} A \quad \frac{\Gamma, x : q \vdash p}{\Gamma, x : q \vdash q \supset r} \partial}{\Gamma, x : q \vdash q \supset r} E \quad \frac{\frac{\frac{\Gamma, x : q \triangleright q \supset s}{\Gamma, x : q \triangleright s} A \quad \frac{\Gamma, x : q \vdash q}{\Gamma, x : q \vdash q \supset s} \partial}{\Gamma, x : q \vdash q \supset r} \partial}{\Gamma, x : q \vdash q \supset r} I \\
\downarrow \\
\frac{\frac{\frac{\frac{\Gamma, x : q \triangleright p \supset (q \supset r)}{\Gamma, x : q \triangleright q \supset r} A \quad \frac{\Gamma, x : q \vdash p}{\Gamma, x : q \vdash q \supset r} \partial}{\Gamma, x : q \vdash q \supset r} E \quad \frac{\frac{\frac{\Gamma, x : q \vdash q}{\Gamma, x : q \vdash q \supset r} \partial}{\Gamma, x : q \vdash q \supset r} E \quad \frac{\frac{\frac{\Gamma, x : q \triangleright q \supset s}{\Gamma, x : q \triangleright s} A \quad \frac{\Gamma, x : q \vdash q}{\Gamma, x : q \vdash q \supset s} \partial}{\Gamma, x : q \vdash q \supset r} \partial}{\Gamma, x : q \vdash q \supset r} I \\
\downarrow \\
\frac{\frac{\frac{\frac{\Gamma, x : q \triangleright p \supset (q \supset r)}{\Gamma, x : q \triangleright q \supset r} A \quad \frac{\Gamma, x : q \vdash p}{\Gamma, x : q \vdash q \supset r} \partial}{\Gamma, x : q \vdash q \supset r} E \quad \frac{\frac{\Gamma, x : q \vdash q}{\Gamma, x : q \vdash q \supset r} \partial}{\Gamma, x : q \vdash q \supset r} E \quad \frac{\frac{\Gamma, x : q \vdash r}{\Gamma, x : q \vdash r} C \quad \frac{\Gamma, x : q \vdash r}{\Gamma, x : q \vdash r} I}{\Gamma \vdash q \supset r} I \\
\downarrow \\
\frac{\frac{\frac{\frac{\Gamma, x : q \triangleright p \supset (q \supset r)}{\Gamma, x : q \triangleright q \supset r} A \quad \frac{\frac{\frac{\Gamma, x : q \triangleright p}{\Gamma, x : q \vdash p} A \quad \frac{\Gamma, x : q \vdash p}{\Gamma, x : q \vdash p} C}{\Gamma, x : q \vdash p} E \quad \frac{\frac{\Gamma, x : q \triangleright q}{\Gamma, x : q \vdash q} A \quad \frac{\Gamma, x : q \vdash q}{\Gamma, x : q \vdash q} C}{\Gamma, x : q \vdash q} E}{\Gamma, x : q \vdash q \supset r} C \quad \frac{\Gamma, x : q \vdash r}{\Gamma, x : q \vdash r} I}{\Gamma \vdash q \supset r} I
\end{array}$$

Notice that the three sequences described in the previous examples only differ in the order of inference rule application, but produce the same final derivation in *NJT*.

The rest of this section formally describes the proof search process according to the variant of the Intercalation Calculus described before.

Definition 5.5 (Proof terms of $\mathbb{NJT}_\partial$). The proof terms of $\mathbb{NJT}_\partial$ are generated as follows:

$$\begin{array}{ll}
\text{(Partial proof terms)} & M, N ::= \lambda x^A.M \mid \text{app}(H) \mid \underline{\Sigma} \vdash A \\
\text{(Partial neutral terms)} & H ::= x \mid HN \\
\text{(Saturation sets)} & \Sigma ::= \emptyset \mid \Sigma \cup \{(H : A)\}
\end{array}$$

In each pair $(H : A)$ of Σ , H is the neutral term that represents an already obtained partial proof of the formula A .

Remark 5.1.

- Γ denotes again a finite set of $(x : A)$;
- Γ is a particular case of Σ ;
- $\Sigma \cup \{(H : A)\}$ is usually written $\Sigma, H : A$;
- $\underline{\Sigma} \vdash A$ is a formal sequent;
- $\underline{\Gamma} \vdash A$ is a particular case of $\underline{\Sigma} \vdash A$.

The terms of $\mathbb{NJT}_\partial$ can be interpreted as specific instances of the proof terms in $\mathbb{NJT}_\partial$. Specifically, $\underline{\sigma} = \underline{\Gamma} \vdash A$ constitutes a special case of $\underline{\Sigma} \vdash A$, where $\Sigma = \Gamma$. The proof term $\text{app}(H, (\underline{\Gamma} \triangleright A), p)$ appears to correspond to the proof $\underline{\Gamma}, H : B \vdash p$. However, this correspondence is not exact. As we will see, $\mathbb{NJT}_\partial$ maintains a record of all eliminations performed prior to obtaining the proof H of B . The proof term $\text{app}(H, (\underline{\Gamma} \triangleright A), p)$ corresponds more accurately to $\underline{\Sigma}, H : B \vdash p$, where Σ is derived from Γ through applications of the inference rule E .

The inference system of $\mathbb{NJT}_\partial$ handles two kinds of partial sequents. Notice that the proof obligations are entirely determined by the partial proof term, as they correspond to the formal sequents that occur within it. Consequently, in this chapter, we discard the list of proof obligations Ξ . This approach can be understood, in the context of natural deduction and proof theory, as analogous to abandoning natural deduction with sequents in favor of working with ordinary natural deduction using formulas only.

Definition 5.6 (Partial Sequents of $\mathbb{NJT}_\partial$). $\mathbb{NJT}_\partial$ handles partial proof terms sequents $\Vdash M : (\Gamma \vdash A)$ and partial saturation set sequents and $\Vdash \Sigma : \Gamma$.

The second form of sequents $\Vdash \Sigma : \Gamma$ indicates that the set Σ is derived through saturation, i.e., by applications of the inference rule E , starting from the context Γ .

Definition 5.7 (Inference Rules of $\mathbb{NJT}_\partial$). The set of inference rules of $\mathbb{NJT}_\partial$ is defined as:

$$\frac{\text{|| } \Sigma : \Gamma}{\text{|| } \underline{\Sigma \vdash A} : (\Gamma \vdash A)} \partial$$

$$\frac{}{\text{|| } \Gamma : \Gamma} A \quad \frac{\text{|| } \Sigma : \Gamma \quad (H : p) \in \Sigma}{\text{|| } \text{app}(H) : (\Gamma \vdash p)} C$$

$$\frac{\text{|| } M : (\Gamma, x : A \vdash B)}{\text{|| } \lambda x^A.M : (\Gamma \vdash A \supset B)} I \quad \frac{\text{|| } (\Sigma, H : A \supset B) : \Gamma \quad \text{|| } N : (\Gamma \vdash A)}{\text{|| } (\Sigma, HN : B) : \Gamma} E$$

Rule E has a proviso: if $H = x$, then $(H : A \supset B) \in \Sigma$.

Each inference rule of NJT has a corresponding rule in this system $\mathbb{NJT}_\partial$, along with a ∂ -rule. Rule ∂ says that, in the context of a set of assumptions Γ , whatever the goal A , we can create a gap from whatever the saturation of Γ . Rule I retains its standard form. Rule A says Γ is a saturation of Γ , and also that every assumption is ready to be main premiss of an elimination. Rule C states that if a derivation of an atom p represented by H is obtained in the context of a saturation set Σ of Γ , then we can derive $\Gamma \vdash p$ and discard the remaining formulas, and their corresponding neutral terms in Σ . Rule E adds the pair $(HN : B)$ to Σ , given that there is already a derivation of a saturation set Σ , and, in particular, of $A \supset B$ represented by H and a proof N of A . Notice that, if H is a variable, then $(H : A \supset B) \in \Sigma$; otherwise it may or may not be in Σ .

The deductive system of $\mathbb{NJT}_\partial$ gives origin to partial derivations.

Example 5.7. The partial derivation, after the first three steps of the informal sequence of proof search in the Example 5.6 is:

$$\frac{\frac{\text{|| } (\Gamma, x : q) : (\Gamma, x : q)}{\text{|| } (\Gamma, x : q) : (\Gamma, x : q)} A \quad \frac{\text{|| } (\Gamma, x : q) : (\Gamma, x : q) \quad \text{|| } \underline{\Gamma, x : q \vdash p} : (\Gamma, x : q \vdash p)}{\text{|| } (\Gamma, x : q, y(\Gamma, x : q \vdash p) : q \supset r) : (\Gamma, x : q)} \partial}{\text{|| } (\Gamma, x : q, y(\Gamma, x : q \vdash p) : q \supset r, w(\Gamma, x : q \vdash p) : s) : (\Gamma, x : q)} E \quad \frac{\text{|| } (\Gamma, x : q) : (\Gamma, x : q)}{\text{|| } \underline{\Gamma, x : q \vdash q} : (\Gamma, x : q \vdash q)} A \quad \frac{\text{|| } \underline{\Gamma, x : q \vdash q} : (\Gamma, x : q \vdash q)}{\text{|| } (\Gamma, x : q, y(\Gamma, x : q \vdash p) : q \supset r, w(\Gamma, x : q \vdash p) : s \vdash r) : (\Gamma, x : q \vdash r)} \partial}{\text{|| } \lambda x^q.(\Gamma, x : q, y(\Gamma, x : q \vdash p) : q \supset r, w(\Gamma, x : q \vdash p) : s \vdash r) : (\Gamma \vdash q \supset r)} I$$

where $\Gamma = \{z : p, y : p \supset (q \supset r), w : (q \supset s)\}$

Weakening rules are admissible in $\mathbb{NJT}_\partial$.

Proposition 5.1 (Admissibility of Weakening in $\mathbb{NJ}\mathbb{T}_\partial$). Let $\omega_{x:A}(M)$, $\omega_{x:A}(H)$, and $\omega_{x:A}(\Sigma)$ be defined by:

$$\begin{aligned}
\omega_{x:A}(\lambda y^B.M) &= \lambda y^B.\omega_{x:A}(M) \\
\omega_{x:A}(\text{app}(H)) &= \text{app}(\omega_{x:A}(H)) \\
\omega_{x:A}(\underline{\Sigma \vdash B}) &= \underline{\omega_{x:A}(\Sigma) \vdash B} \\
\omega_{x:A}(y) &= y \\
\omega_{x:A}(HN) &= (\omega_{x:A}(H))(\omega_{x:A}(N)) \\
\omega_{x:A}(\emptyset) &= (x : A) \\
\omega_{x:A}(\Sigma, H : B) &= \omega_{x:A}(\Sigma), \omega_{x:A}(H) : B
\end{aligned}$$

Then, the following rules are admissible in $\mathbb{NJ}\mathbb{T}_\partial$:

$$\frac{\vdash M : (\Gamma \vdash B)}{\vdash \omega_{x:A}(M) : (\Gamma, x : A \vdash B)} \quad \frac{\vdash \Sigma : \Gamma}{\vdash \omega_{x:A}(\Sigma) : \Gamma, x : A}$$

Proof. By induction on the derivations of $\vdash M : (\Gamma \vdash B)$ and $\vdash \Sigma : \Gamma$.

- **Case ∂ .** Suppose $\vdash \underline{\Sigma \vdash B} : (\Gamma \vdash B)$ is obtained from $\vdash \Sigma : \Gamma$. By IH, $\vdash \omega_{x:A}(\Sigma) : (\Gamma, x : A)$. Applying the ∂ -rule, $\vdash \underline{\omega_{x:A}(\Sigma) \vdash B} : (\Gamma, x : A \vdash B)$.
- **Case A .** Suppose $\vdash \Gamma : \Gamma$. By the A rule, $\vdash \omega_{x:A}(\Gamma) : (\Gamma, x : A)$, since $\omega_{x:A}(\Gamma, x : A) = \Gamma, x : A$.
- **Case C .** Suppose $\vdash \text{app}(H) : (\Gamma \vdash p)$ is obtained from $\vdash \Sigma : \Gamma$, with $(H : p) \in \Sigma$. By IH, $\vdash \omega_{x:A}(\Sigma) : (\Gamma, x : A)$. Thus, $(\omega_{x:A}(H) : p) \in \omega_{x:A}(\Sigma)$. Applying the inference rule C , $\vdash \text{app}(\omega_{x:A}(H)) : (\Gamma, x : A \vdash p)$.
- **Case I .** Suppose $\vdash \lambda y^B.M : (\Gamma \vdash B \supset C)$ is obtained from $\vdash M : (\Gamma, y : B \vdash C)$. By IH, $\vdash \omega_{x:A}(M) : (\Gamma, y : B, x : A \vdash C)$. Applying the inference rule I , $\vdash \lambda y^B.\omega_{x:A}(M) : (\Gamma, x : A \vdash B \supset C)$.
- **Case E .** Suppose $\vdash (\Sigma, HN : C) : \Gamma$ is obtained from $\vdash (\Sigma, H : B \supset C) : \Gamma$, and $\vdash N : (\Gamma \vdash B)$. By IH, $\vdash (\omega_{x:A}(\Sigma), \omega_{x:A}(H) : B \supset C) : \Gamma, x : A$, and $\vdash \omega_{x:A}(N) : (\Gamma, x : A \vdash B)$. Applying the inference rule E , $\vdash (\omega_{x:A}(\Sigma), \omega_{x:A}(H)\omega_{x:A}(N) : C) : (\Gamma, x : A)$.

□

Example 5.8. The partial derivation, and before updating the set of hypotheses, of the informal sequence of proof search in the Example 5.4 is:

$$\begin{array}{c}
\frac{\frac{\frac{\overline{\vdash \Gamma : \Gamma} \quad A}{\vdash \Gamma : \Gamma} \quad \frac{\overline{\vdash \Gamma : \Gamma} \quad A}{\vdash (\Gamma \vdash p) : (\Gamma \vdash p)} \quad \partial}{\vdash (\Gamma, y(\Gamma \vdash p) : q \supset r) : \Gamma} \quad E \\
\frac{\frac{\vdash \omega_{x:q}(\Gamma, y(\Gamma \vdash p) : q \supset r) : (\Gamma, x : q)}{\vdash \omega_{x:q}(\Gamma, y(\Gamma \vdash p) : q \supset r) \vdash r : (\Gamma, x : q \vdash r)} \quad \partial}{\vdash \lambda x^q. \omega_{x:q}(\Gamma, y(\Gamma \vdash p) : q \supset r) \vdash r : (\Gamma \vdash q \supset r)} \quad I
\end{array}$$

where $\Gamma = \{z : p, y : p \supset (q \supset r), w : (q \supset s)\}$

We prove, in the following proposition, that derivability in NJT is partial derivability in $\mathbb{NJT}_\partial$.

Proposition 5.2 (Conservativity).

- (1) NJT derives $\Gamma \vdash M : A$ iff $\mathbb{NJT}_\partial$ derives $\vdash M : (\Gamma \vdash A)$.
- (2) NJT derives $\Gamma \triangleright H : A$ iff $\mathbb{NJT}_\partial$ derives $\vdash (\Sigma, H : A) : \Gamma$, for some saturation set Σ .

Proof. The two “only if” statements are proved by a simultaneous induction on the derivations of $\Gamma \vdash M : A$ and $\Gamma \triangleright H : A$.

- Case A . Suppose $\Gamma, x : A \triangleright x : A$. Take $\Sigma = \Gamma$. Applying the A rule, $\vdash (\Sigma, x : A) : (\Gamma, x : A)$ in $\mathbb{NJT}_\partial$.
- Case C . Suppose $\Gamma \vdash \text{app}(H) : p$ is obtained from $\Gamma \triangleright H : p$. By IH, there is a saturation set Σ such that $\vdash \Sigma : \Gamma$, with $(H : p) \in \Sigma$. Applying the C rule, $\vdash \text{app}(H) : (\Gamma \vdash p)$ in $\mathbb{NJT}_\partial$.
- Case I . Suppose $\Gamma \vdash \lambda x^A. M : A \supset B$ is obtained from $\Gamma, x : A \vdash M : B$. By IH, $\vdash M : (\Gamma, x : A \vdash B)$. Applying the I rule, $\vdash \lambda x^A. M : (\Gamma \vdash A \supset B)$ in $\mathbb{NJT}_\partial$.
- Case E . Suppose $\Gamma \triangleright HN : B$ is obtained from $\Gamma \triangleright H : A \supset B$ and $\Gamma \vdash N : A$. By IH, there is a saturation set Σ such that $\vdash (\Sigma, H : A \supset B) : \Gamma$, and $\vdash N : (\Gamma \vdash A)$. Applying the E rule, $\vdash (\Sigma, HN : B) : \Gamma$ in $\mathbb{NJT}_\partial$.

For the “if” statements, we prove:

- (1) If $\mathbb{NJT}_\partial$ derives $\vdash M : (\Gamma \vdash A)$, with $M \in NJT$, then NJT derives $\Gamma \vdash M : A$.
- (2) If $\mathbb{NJT}_\partial$ derives $\vdash (\Sigma, H : A) : \Gamma$, with $H \in NJT$, then NJT derives $\Gamma \triangleright H : A$.

The proof is by simultaneous induction on the derivations of $\Vdash M : (\Gamma \vdash A)$ and $\Vdash \Sigma : \Gamma$.

- Case ∂ . Suppose $\Vdash \underline{\Sigma \vdash A} : (\Gamma \vdash A)$ is obtained from $\Vdash \Sigma : \Gamma$. Since $\underline{\Sigma \vdash A} \notin NJT$, the statement is trivially true.
- Case A . Suppose $\Vdash \Gamma : \Gamma$. For each $(x : A) \in \Gamma$, $x \in NJT$. Applying the A rule, $\Gamma \triangleright x : A$ in NJT .
- Case C . Suppose $\Vdash \text{app}(H) : (\Gamma \vdash p)$ is obtained from $\Vdash \Sigma : \Gamma$, with $(H : p) \in \Sigma$ and $\text{app}(H) \in NJT$. Then, $H \in NJT$. By IH, $\Gamma \triangleright H : p$. Applying the C rule, $\Gamma \vdash \text{app}(H) : p$ in NJT .
- Case I . Suppose $\Vdash \lambda x^A.M : (\Gamma \vdash A \supset B)$ is obtained from $\Vdash M : (\Gamma, x : A \vdash B)$, with $\lambda x^A.M \in NJT$. Then, $M \in NJT$. By IH, $\Gamma, x : A \vdash M : B$. Applying the I rule, $\Gamma \vdash \lambda x^A.M : A \supset B$ in NJT .
- Case E . Suppose $\Vdash (\Sigma, HN : B) : \Gamma$ is obtained from $\Vdash (\Sigma, H : A \supset B) : \Gamma$, and $\Vdash N : (\Gamma \vdash A)$. If $HN \in NJT$, then $H, N \in NJT$. By IH, $\Gamma \triangleright H : A \supset B$ and $\Gamma \vdash N : A$. Applying the E rule, $\Gamma \triangleright HN : B$ in NJT . If $(H' : C) \in \Sigma$ and $H' \in NJT$, we have, by IH, that $\Gamma \triangleright H' : C$ in NJT .

□

The following definition tells how to extract the context Γ which originated Σ .

Definition 5.8. $\Sigma \sqsubseteq \Gamma$ if $\Gamma = \{(x : A) \mid (x : A) \in \Sigma\}$.

Remark 5.2.

- If $\Sigma \sqsubseteq \Gamma$, then $\Sigma \supseteq \Gamma$ (but not conversely);
- If $\Vdash \Sigma : \Gamma$ then $\Sigma \sqsubseteq \Gamma$ (but not conversely).

Formal sequents are eliminated by reducing partial proof terms according to the reduction rules given in the following definition.

Definition 5.9 (Reduction Rules of NJT_∂). The reduction rules for proof search in NJT , following NJT_∂ , are the following:

$$\begin{array}{lll}
 (IR) & \underline{\Sigma \vdash A \supset B} & \rightarrow \lambda x^A. (\omega_{x:A}(\Sigma) \vdash B) \\
 (S) & \Sigma, H : A \supset B & \rightarrow \Sigma, H(\underline{\Gamma \vdash A}) : B, \Sigma \sqsubseteq \Gamma \\
 (FS) & \underline{\Sigma, H : p \vdash p} & \rightarrow \text{app}(H)
 \end{array}$$

Rule S has a proviso: if $H = x$, then $(H : A \supset B) \in \Sigma$.

Three reduction rules are defined: invert right (IR), saturation (S), and finish saturation (FS). Notice that the structure of the formal sequent and proof term $\Sigma \vdash A$ permits alternation between the right inversion and the saturation phases.

We now explain how the reduction rules presented in Definition 5.9 describe intercalation proof search in NJT . Recall the rules in Definitions 3.2 and 3.5 and the diagram in Figure 5. Our goal is to prove $\Gamma \vdash A$.

- As long as A is not atomic and the context contains hypotheses that are implications, either right inversion or context saturation may be applied in any order.
- A right inversion step consists of applying the inference rule I in a bottom-up manner to decompose the right-hand side formula, as implemented by the reduction rule IR . Notice that $\omega_{x:A}(\Sigma)$ does a global update that is not carried out by Sieg's rule $\supset_{\uparrow}$ in Definition 5.9.
- A saturation step consists of selecting either a formula in Γ , by applying the inference rule A , or a formula already obtain by saturation of the formulas in Γ , immediately followed by eliminating that formula using the inference rule E . The reduction rule S implements this process by adding the conclusion of the elimination and the corresponding neutral term to the context Σ , and initiating a search sub-problem corresponding to the right premiss of E .
- When the right-hand side formula becomes an atom p , and saturation has produced the same atom p along with a neutral term H in the context, the saturation phase can be concluded by applying the coercion rule C . The reduction rule FS implements this by collecting the relevant neutral term H , discharging the other neutral terms in Σ , and producing the proof term $\text{app}(H)$.
- The proof search terminates when no gaps remain between the hypotheses and the conclusion; that is, when the proof term obtained by applying the reduction rules is total.

The reduction rules are three distinct binary relations: two on partial proof terms M , and a third on Σ . The compatible closure of those binary relations, denoted $\rightarrow$ is a triple of binary relations, one on partial proof terms M , another on partial neutral terms H , and a third on Σ . The closure rules producing the compatible closure say: reduction on M is reduction in $\lambda x^A.M$ and in HM ; reduction on H is reduction in $\text{app}(H)$, in HN and in $\Sigma, H : A$; and reduction in Σ is reduction in $\Sigma \vdash A$. We use $\rightarrow$ to denote the reflexive and transitive closure of $\rightarrow$.

Example 5.9. Let $\Gamma \vdash q \supset r$ be a sequent of NJT , with $\Gamma = \{z : p, y : p \supset (q \supset r), w : q \supset s\}$.

We show the formalization of proof search for the sequence presented in the Example 5.4.

$$\begin{aligned}
& \frac{\Gamma \vdash q \supset r}{\rightarrow_S \quad \Gamma, y(\Gamma \vdash p) : q \supset r \vdash q \supset r} \\
& \rightarrow_{IR} \quad \lambda x^q. (\omega_{x:q}(\Gamma, y(\Gamma \vdash p) : q \supset r) \vdash r) \\
& = \quad \lambda x^q. (\Gamma, y(\Gamma, x : q \vdash p) : q \supset r, x : q \vdash r) \\
& \rightarrow_S \quad \lambda x^q. (\Gamma, y(\Gamma, x : q \vdash p) : q \supset r, x : q, (y(\Gamma, x : q \vdash p))(\Gamma, x : q \vdash q) : r \vdash r) \\
& \rightarrow_{FS} \quad \lambda x^q. \text{app}((y(\Gamma, x : q \vdash p))(\Gamma, x : q \vdash q)) \\
& \rightarrow_{FS} \quad \lambda x^q. \text{app}((y(\text{app}(z))) (\Gamma, x : q \vdash q)) \\
& \rightarrow_{FS} \quad \lambda x^q. \text{app}((y(\text{app}(z))) (\text{app}(x)))
\end{aligned}$$

Example 5.10. Consider the same sequent as in Example 5.9. We show the formalization of proof search for the sequence presented in the Example 5.5.

$$\begin{aligned}
& \frac{\Gamma \vdash q \supset r}{\rightarrow_{IR} \quad \lambda x^q. \Gamma, x : q \vdash r} \\
& \rightarrow_S \quad \lambda x^q. \Gamma, x : q, y(\Gamma, x : q \vdash p) : q \supset r \vdash r \\
& \rightarrow_S \quad \lambda x^q. \Gamma, x : q, y(\Gamma, x : q \vdash p) : q \supset r, (y(\Gamma, x : q \vdash p))(\Gamma, x : q \vdash q) : r \vdash r \\
& \rightarrow_{FS} \quad \lambda x^q. \text{app}((y(\Gamma, x : q \vdash p))(\Gamma, x : q \vdash q)) \\
& \rightarrow_{FS} \quad \lambda x^q. \text{app}((y(\text{app}(z))) (\Gamma, x : q \vdash q)) \\
& \rightarrow_{FS} \quad \lambda x^q. \text{app}((y(\text{app}(z))) (\text{app}(x)))
\end{aligned}$$

Example 5.11. Consider the same sequent as in Example 5.9 and 5.10. We show the formalization of proof search for the sequence presented in the Example 5.6.

$$\begin{aligned}
& \frac{\Gamma \vdash q \supset r}{\rightarrow_{IR} \quad \lambda x^q. \Gamma, x : q \vdash r} \\
& \rightarrow_S \quad \lambda x^q. \Gamma, x : q, y(\Gamma, x : q \vdash p) : q \supset r \vdash r \\
& \rightarrow_S \quad \lambda x^q. \Gamma, x : q, y(\Gamma, x : q \vdash p) : q \supset r, w(\Gamma, x : q \vdash q) : s \vdash r \\
& \rightarrow_S \quad \lambda x^q. \Gamma, x : q, y(\Gamma, x : q \vdash p) : q \supset r, w(\Gamma, x : q \vdash q) : s, (y(\Gamma, x : q \vdash p))(\Gamma, x : q \vdash q) : r \vdash r \\
& \rightarrow_{FS} \quad \lambda x^q. \text{app}((y(\Gamma, x : q \vdash p))(\Gamma, x : q \vdash q)) \\
& \rightarrow_{FS} \quad \lambda x^q. \text{app}((y(\text{app}(z))) (\Gamma, x : q \vdash q)) \\
& \rightarrow_{FS} \quad \lambda x^q. \text{app}((y(\text{app}(z))) (\text{app}(x)))
\end{aligned}$$

As for the proof search systems presented in the previous chapters, we prove the following results of Record of Search and Subject Reduction for $\mathbb{NJT}_\partial$.

Proposition 5.3 (Record of search).

- (1) If $\Vdash M : (\Gamma \vdash A)$, then $\underline{\Gamma \vdash A} \rightarrow M$.
- (2) If $\Vdash \Sigma : \Gamma$, then $\Gamma \rightarrow \Sigma$.

Proof. The proof is made by simultaneous induction on the derivations of $\Vdash M : (\Gamma \vdash A)$ and $\Vdash \Sigma : \Gamma$.

- Case ∂ -rule. Suppose $\Vdash \underline{\Sigma \vdash A} : (\Gamma \vdash A)$ is obtained from $\Vdash \Sigma : \Gamma$. By IH, $\Gamma \rightarrow \Sigma$. Then $\underline{\Gamma \vdash A} \rightarrow \underline{\Sigma \vdash A}$.
- Case A . Suppose $\Vdash \Gamma : \Gamma$. Then $\Gamma \rightarrow \Gamma$, because of the reflexivity of $\rightarrow$.
- Case C . Suppose $\Vdash \text{app}(H) : (\Gamma \vdash p)$ is obtained from $\Vdash \Sigma : \Gamma$, with $(H : p) \in \Sigma$. Then, $\underline{\Gamma \vdash p} \rightarrow \underline{\Sigma \vdash p} \rightarrow \text{app}(H)$, where the last reduction step is by the rule FS and the rest of the reduction is by IH.
- Case I . Suppose $\Vdash \lambda x^A.M : (\Gamma \vdash A \supset B)$ is obtained from $\Vdash M : (\Gamma, x : A \vdash B)$. Then, $\underline{\Gamma \vdash A \supset B} \rightarrow \lambda x^A.(\underline{\omega_{x:A}(\Gamma) \vdash B}) \rightarrow \lambda x^A.M$, where the first step is by IR and the rest of the reduction is by IH, since $\omega_{x:A}(\Gamma) = \Gamma, x : A$.
- Case E . Suppose $\Vdash (\Sigma, HN : B) : \Gamma$ is obtained from $\Vdash (\Sigma, H : A \supset B) : \Gamma$, and $\Vdash N : (\Gamma \vdash A)$. Then $\Gamma \rightarrow \Sigma, H : A \supset B \rightarrow \Sigma, H(\underline{\Gamma \vdash A}) : B \rightarrow \Sigma, HN : B$, where the second step is by the S rule and the rest of the reduction is by IH.

□

Proposition 5.4 (Subject reduction for $\mathbb{NJT}_\partial$).

1. If $\Vdash M : (\Gamma \vdash A)$ and $M \rightarrow M'$, then $\Vdash M' : (\Gamma \vdash A)$.
2. If $\Vdash \Sigma : \Gamma$ and $\Sigma \rightarrow \Sigma'$, then $\Vdash \Sigma' : \Gamma$.
3. If $\Vdash (\Sigma, H : A) : \Gamma$ and $H \rightarrow H'$, then $\Vdash (\Sigma, H' : A) : \Gamma$.

Proof. The proof is by simultaneous induction on $M \rightarrow M'$, $\Sigma \rightarrow \Sigma'$ and $H \rightarrow H'$, the latter with sub-induction on $\Vdash (\Sigma, H : A) : \Gamma$.

Base Cases:

- Case *IR*. Suppose $M = \underline{\Sigma \vdash A \supset B} \rightarrow \lambda x^A.(\underline{\omega_{x:A}(\Sigma) \vdash B}) = M'$ and $\Vdash M : (\Gamma \vdash C)$. By the ∂ -rule, $C = A \supset B$. We now prove $\Vdash \lambda x^A.(\underline{\omega_{x:A}(\Sigma) \vdash B}) : (\Gamma \vdash A \supset B)$. By the inversion of the ∂ -rule, $\Vdash \Sigma : \Gamma$ and, by the admissibility of weakening, $\Vdash \omega_{x:A}(\Sigma) : (\Gamma, x : A)$. Applying the ∂ -rule, $\Vdash \underline{\omega_{x:A}(\Sigma) \vdash B} : (\Gamma, x : A \vdash B)$. Applying the *I* rule, $\Vdash \lambda x^A.(\underline{\omega_{x:A}(\Sigma) \vdash B}) : (\Gamma \vdash A \supset B)$.
- Case *S*. Suppose $\Sigma, H : A \supset B \rightarrow \Sigma, H(\underline{\Gamma \vdash A}) : B = \Sigma'$, and $\Vdash (\Sigma, H : A \supset B) : \Gamma$. We now prove $\Vdash (\Sigma, H(\underline{\Gamma \vdash A}) : B) : \Gamma$. By *A*, $\Vdash \Gamma : \Gamma$. Applying the ∂ -rule, $\Vdash \underline{\Gamma \vdash A} : (\Gamma \vdash A)$. Applying the *E* rule, $\Vdash (\Sigma, H(\underline{\Gamma \vdash A}) : B) : \Gamma$.
- Case *FS*. Suppose $M = \underline{\Sigma, H : p \vdash p} \rightarrow \text{app}(H) = M'$ and $\Vdash M : (\Gamma \vdash C)$. By the ∂ -rule, $C = p$. We now prove $\Vdash \text{app}(H) : (\Gamma \vdash p)$. By the inversion of the ∂ -rule, $\Vdash (\Sigma, H : p) : \Gamma$. Applying the *C* rule, $\Vdash \text{app}(H) : (\Gamma \vdash p)$.

Inductive Cases:

- Suppose $\lambda x^A.M \rightarrow \lambda x^A.M'$, with $M \rightarrow M'$ and $\Vdash \lambda x^A.M : (\Gamma \vdash A \supset B)$. By the inversion of the inference rule *I*, $\Vdash M : (\Gamma, x : A \vdash B)$. By IH, $\Vdash M' : (\Gamma, x : A \vdash B)$. Applying the inference rule *I*, $\Vdash \lambda x^A.M' : (\Gamma \vdash A \supset B)$.
- Suppose $\text{app}(H) \rightarrow \text{app}(H')$, with $H \rightarrow H'$ and $\Vdash \text{app}(H) : (\Gamma \vdash p)$. By the inversion of the inference rule *C*, $\Vdash (\Sigma, H : p) : \Gamma$. By IH, $\Vdash (\Sigma, H' : p) : \Gamma$. Applying the inference rule *C*, $\Vdash \text{app}(H') : (\Gamma \vdash p)$.
- Suppose $\underline{\Sigma \vdash A} \rightarrow \underline{\Sigma' \vdash A}$, with $\Sigma \rightarrow \Sigma'$ and $\Vdash \underline{\Sigma \vdash A} : (\Gamma \vdash A)$. By the inversion of the ∂ -rule, $\Vdash \Sigma : \Gamma$. By IH, $\Vdash \Sigma' : \Gamma$. Applying the ∂ -rule, $\Vdash \underline{\Sigma' \vdash A} : (\Gamma \vdash A)$.
- Suppose $HN \rightarrow H'N$, with $H \rightarrow H'$ and $\Vdash (\Sigma, HN : B) : \Gamma$. If $(HN : B)$ was obtained from the last application of *E*, by inversion of the inference rule *E*, $\Vdash (\Sigma, H : A \supset B) : \Gamma$ and $\Vdash N : (\Gamma \vdash A)$. By IH, $\Vdash (\Sigma, H' : A \supset B) : \Gamma$. Applying the inference rule *E*, $\Vdash (\Sigma, H'N : B) : \Gamma$. Otherwise, $\Sigma = \Sigma', GM : D$, and, by inversion of the inference rule *E*, $\Vdash (\Sigma', HN : B, G : C \supset D) : \Gamma$ and $\Vdash M : (\Gamma \vdash C)$. By IH, $\Vdash (\Sigma', H'N : B, G : C \supset D) : \Gamma$. Applying the inference rule *E*, $\Vdash (\Sigma, H'N : B) : \Gamma$.
- Suppose $HN \rightarrow HN'$, with $N \rightarrow N'$ and $\Vdash (\Sigma, HN : B) : \Gamma$. If $(HN : B)$ was obtained from the last application of *E*, by inversion of the inference rule *E*, $\Vdash (\Sigma, H : A \supset B) : \Gamma$ and $\Vdash N : (\Gamma \vdash A)$. By IH, $\Vdash N' : (\Gamma \vdash A)$. Applying the inference rule *E*, $\Vdash (\Sigma, HN' : B) : \Gamma$.

Otherwise, $\Sigma = \Sigma', GM : D$, and, by inversion of the inference rule E , $\vdash (\Sigma', HN : B, G : C \supset D) : \Gamma$ and $\vdash M : (\Gamma \vdash C)$. By IH, $\vdash (\Sigma', HN' : B, G : C \supset D) : \Gamma$. Applying the inference rule E , $\vdash (\Sigma, HN' : B) : \Gamma$.

- Suppose $\Sigma, H : A \rightarrow \Sigma, H' : A$, with $H \rightarrow H'$ and $\vdash (\Sigma, H : A) : \Gamma$. By IH, $\vdash (\Sigma, H' : A) : \Gamma$.

□

Proposition 5.5 (“Typed” Saturation Rule). If $\vdash (\Sigma, H : A \supset B) : \Gamma$, then

$\Sigma, H : A \supset B \rightarrow \Sigma, H(\Sigma, H : A \supset B \vdash A) : B$, where: if $H = x$, then $(H : A \supset B) \in \Sigma$.

Proof. From $\vdash (\Sigma, H : A \supset B) : \Gamma$, we get $(\Sigma, H : A \supset B) \sqsubseteq \Gamma$. If $H = x$, then $(H : A \supset B) \in \Sigma$ and $\Sigma \sqsubseteq \Gamma$. Otherwise, by the definition of $\sqsubseteq$, we have that $\Sigma \sqsubseteq \Gamma$. By Record of Search, we get $\Gamma \rightarrow \Sigma, H : A \supset B$. Hence, we have $(\Gamma \vdash A) \rightarrow (\Sigma, H : A \supset B \vdash A)$. Then, $\Sigma, H : A \supset B \rightarrow \Sigma, H(\Gamma \vdash A) : B \rightarrow \Sigma, H(\Sigma, H : A \supset B \vdash A) : B$. □

Remark 5.3. This is how we recover Sieg’s $\supset_{\downarrow}$ -rule, from Definition 5.2.

Proof search succeeds in $\mathbb{NJT}_{\partial}$ if and only if the formal sequent normalizes. This result confirms that the methodology of partial proof terms adequately represents proof search in NJT following Intercalation.

Theorem 5.1 (Proof search as normalization).

1. For all $M \in NJT$: NJT derives $\Gamma \vdash M : A$ iff $\Gamma \vdash A \rightarrow M$.
2. For all $H \in NJT$: NJT derives $\Gamma \triangleright H : A$ iff $\Gamma \rightarrow \Sigma, H : A$, for some saturation set Σ .

Proof. For the first statement, suppose NJT derives $\Gamma \vdash M : A$. By Conservativity (Proposition 5.2), $\mathbb{NJT}_{\partial}$ derives $\vdash M : (\Gamma \vdash A)$. By Record of Search (Proposition 5.3), $\Gamma \vdash A \rightarrow M$. For the converse implication, suppose $\Gamma \vdash A \rightarrow M$. By the inference rule A and the ∂ -rules, $\vdash \Gamma \vdash A : (\Gamma \vdash A)$ is derivable in $\mathbb{NJT}_{\partial}$. By Subject Reduction (Proposition 5.4) and the assumption, $\vdash M : (\Gamma \vdash A)$. Since $M \in NJT$, by Conservativity, NJT derives $\Gamma \vdash M : A$.

For the second statement, suppose NJT derives $\Gamma \triangleright H : A$. By Conservativity, $\mathbb{NJT}_{\partial}$ derives $\vdash (\Sigma, H : A) : \Gamma$, for some saturation set Σ . By Record of Search, $\Gamma \rightarrow \Sigma, H : A$. For the converse implication, suppose $\Gamma \rightarrow \Sigma, H : A$. By the A rule, $\vdash \Gamma : \Gamma$ is derivable in $\mathbb{NJT}_{\partial}$. By Subject Reduction and the assumption, $\vdash (\Sigma, H : A) : \Gamma$. Since $H \in NJT$, by Conservativity, NJT derives $\Gamma \triangleright H : A$. □

Chapter 6

Conclusions and Future Work

6.1 Contributions

In this thesis, we proposed partial proof terms as a new methodology in the study of proof search. Partial proof terms extend the Curry-Howard representation of proofs by incorporating formal occurrences of sequents within proof terms. Such generalized proof terms represent incomplete derivations, encoding both what has already been proved and, through the presence of formal sequents, what remains to be proved. We used partial proof terms to define proof search procedures through rewrite rules, and developed typing systems that extend the original proof systems to accommodate partial sequents. The resulting rewriting systems characterize the logic's derivability relation, as formalized in the “proof search as normalization” theorems.

To demonstrate the relevance of the partial proof terms methodology, we conducted several case studies in propositional intuitionistic and classical logics. Although this work focuses on natural deduction, our methodology is initially introduced using the sequent calculus systems LJT and LKT . This choice is motivated by pedagogical clarity, since the notion of gap in these sequent calculi coincides with the notion of formal sequent.

Following the presentation of the methodology in sequent calculus, we extended it to focused proof search systems in natural deduction, specifically NJT_∂ and NKT_∂ , which respectively produce total proofs in NJT and NKT . We then applied partial proof terms to an adaptation of the Intercalation Calculus returning proofs in NJT , referred to as $\mathbb{NJT}_\partial$. The update and development of the formal definitions of these systems and their associated search procedures is a contribution. In particular, we highlight that $\mathbb{NJT}_\partial$ constitutes an update of Sieg's Intercalation Calculus [55], in several directions: integration with bidirectional natural deduction in both intuitionistic and classical cases; identification of focused fragments; adoption of “alternative conclusions” in the style of Parigot [49] for the classical case; and formalization of proof-search rules as rewriting rules on partial proof terms.

Using our tools, we established a theoretical result: the equivalence between focusing in sequent calculus and intercalation in natural deduction, for both intuitionistic and classical logics. This result addresses not only the static correspondence between normal and cut-free forms but also the dynamics of proof search, and it is interesting because it shows that proof search in natural deduction can match optimized proof search procedures in sequent calculus like focusing.

The proposed methodology reinforces the area of Idealized Proof Search, which deliberately excludes developments related to automation or practical tools such as interactive theorem provers, analogously to the study of abstract logic programming languages [39]. Within this context, partial proof terms provide a theoretical basis for the clarification of search procedures and objects, including the combination of cut elimination/normalization and proof search.

6.2 Related Work

The area of theorem proving has inspired the development of type theories with meta-variables and explicit substitutions [44, 45, 46]. These are more powerful and complex tools, operating at a different level of abstraction than partial proof terms: with them we can model the reuse of proofs, or the delay of choices in unification and proof search. Such consideration of optimization and effectiveness are obviously important, but do not belong to Idealized Proof Theory.

As already mentioned in the introduction, there also are important differences to “coinductive proof search” [21, 22, 24]. On the one hand, this approach does not provide yet a method for expressing proof search procedures. On the other hand, in this approach, we have fixed-point variables, typed with a sequent σ , allowing the representation of an entire search space. Here we are not interested in a finite and explicit representation of the whole search space, and content ourselves with an indirect representation as the reduction graphs of partial proof terms. This is liberating, since this attitude is the correct for the study of logics which do not admit such finite representation; and since some simplifications follow, which permit us to obtain the concept of partial proof term. The simplification is this: in “coinductive proof search”, sequents occur in the λ -terms of the calculus used for the finite representation of spaces; but they occur as the types of fixed-point variables, and the respective typing is governed by complex rules dealing with the phenomenon of “de-contraction” – a necessity, to obtain the representation of the whole space. Here, this can be simplified to plain occurrences of sequents in the proof terms, governed in the typing system by simple rules that just take note of which sequents occur in which terms.

Our work is particularly close, regarding motivations, to that on open derivation and open proof terms

reported in [33]. Open derivations are derivations in natural deduction possibly containing a generic “claim” inference, whereas our natural deduction partial derivations may contain partiality points generated by applications of the ∂ -rules. Here are the differences between the two approaches, beyond the fact that *op. cit.* employs meta-variables while we employ formal sequents, and the fact that *op. cit.* deals with natural deduction, while we also cover sequent calculi. Open derivations were developed directly for higher-order logic, as a formal tool to model, for example, the delay of choices in proof search or the reuse of proofs – but without an explicit formalization of a search process; whereas here we only gave details about propositional logic, but the modeling and study of proof search as rewriting is central. Open derivations are faithfully embedded in a type theory in order to capture the notion of proof state as a type-theoretic context – but no central role seems to be given to proof terms; whereas for us (partial) proof terms are the main artifact, proof states include a “record of search” provided by partial proof terms, and we do not insist so much in the distinction between a proof system and its “type-theoretic” version with proof terms.

Sieg and co-authors studied proof search directly in natural deduction, through the intercalation calculus [54, 55]. Here we distinguish between intercalation as a proof search strategy in natural deduction from the specific intercalation calculi developed in the cited papers. *NJT* and *NKT* follow the intercalation strategy, but differ greatly from the intercalation calculus. In fact, *NKT* is a striking version of the $\lambda\mu$ -calculus, with the typing rules for “structural substitution” [49, 3] flagrantly being elimination rules for implication. In [17] one finds another variant of $\lambda\mu$ and its *T*-fragment, in perfect correspondence, regarding cut-elimination vs normalization, with the $\bar{\lambda}\mu\tilde{\mu}$ -calculus and its *T*-fragment of [7], respectively. But the systems of [17], being aligned with those of [7], have a different syntactic organization. In addition, [17] is not a study of proof search.

We should also mention the work [48] on rewriting logic as a meta-logical framework for the specification of sequent calculi and the automation of meta-theory. This work is somehow related to ours, but sufficiently different. First, for Idealized Proof Search, the goal of automation is even less central than the goal of reflecting in theory the advances of theorem proving; second, we are not limited to bottom-up proof search, nor to sequent calculus, nor to propositional logic; third, we indeed propose term rewriting as a “framework”, but are not interested, so far, in distilling a single formal system that can act as a logical framework.

6.3 Prospect for Future Work

Partial Proof Terms may prove valuable for investigating a variety of topics within (Idealized) Proof Search, as well as related areas such as Proof Theory and Theorem Proving.

A first direction for future research involves extending the NJT_∂ system to classical logic, following the methodology previously applied to LJT_∂ and NJT_∂ . Furthermore, it is necessary to determine whether a correspondence exists between NJT_∂ and a sequent calculus system, and to conduct a comprehensive analysis of this relationship, including a possible generalization of the equivalence between intercalation and focusing.

Another line of research involves defining additional subsystems of NJT_∂ , such as ADC [40]. A further research question is whether, analogous to the fragment LJT for sequent calculus corresponding to the NJT , a natural deduction system, let's say NJQ , exists that corresponds to the fragment LJQ [10] of sequent calculus. Once these questions are answered, the respective search procedures can be defined using partial proof terms.

Future work also includes extending the application of partial proof terms to additional logical systems. For logics with additional connectives, such as the universal quantifier, significant challenges are not anticipated. Within the idealized proof search framework, obtaining finite reduction graphs for a formal sequent σ is not required. Similarly, there is no need to ensure finite branching through mechanisms such as cycle detection or to restrict the degree of nodes in these graphs. Therefore, it is unnecessary to constrain quantifier rules that may result in infinite branching.

Another potential direction is to use partial proof terms to analyze how proof search is related with cut elimination or normalization within partial derivations. This approach would treat the admissible cut rules of Theorems 2.1 and 4.1, and the admissible inference rules of Theorems 3.1 and 4.3, as primary rules in $\mathsf{LJT}_\partial/\mathsf{LKT}_\partial$ and in $\mathsf{NJT}_\partial/\mathsf{NKT}_\partial$, respectively. Partial proof terms could then be employed with formal sequents in two distinct rewriting processes: one eliminating occurrences of formal sequents for proof search, the other applying cut-elimination or normalization rules.

In the context of theorem proving, and following the ideas of Sieg [56], who argued that a human-centered approach is best achieved by grounding reasoning in natural deduction, the partial proof terms may facilitate the generation of human-readable outputs that more closely resemble the natural language proofs constructed by mathematicians. A promising starting point is the idea of partial proof terms as “records of search”.

Finally, an open question remains regarding the extent to which partial proof terms will be useful in a

less naive context of a “Less Idealized Proof Search”, where issues of control, termination, backtracking, and delaying are central.

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"We have not succeed in answering all our problems. The answers we have found only serve to raise a whole set of new questions. In some ways we feel we are as confused as ever, but we believe we are confused on a higher level and about more important things."

Earl C. Kelley